US012415547B2

(12) **United States Patent**
Raichelgauz et al.

(10) **Patent No.: US 12,415,547 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **SAFE TRANSFER BETWEEN MANNED AND AUTONOMOUS DRIVING MODES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 63 days.

(21) Appl. No.: **16/729,547**

(22) Filed: **Dec. 30, 2019**

(65) **Prior Publication Data**

US 2020/0133266 A1 Apr. 30, 2020

Related U.S. Application Data

(63) Continuation of application No. PCT/IB2019/058207, filed on Sep. 27, 2019.

(Continued)

(51) **Int. Cl.**

B60W 60/00 (2020.01)
B60W 30/09 (2012.01)

(Continued)

(52) **U.S. Cl.**

CPC ***B60W 60/0025*** (2020.02); ***G06F 18/23*** (2023.01); ***G06F 18/23213*** (2023.01);

(Continued)

(58) **Field of Classification Search**

CPC G08G 1/166; G08G 1/056; G08G 1/162; G08G 1/165; G08G 1/048; G08G 1/09626;

(Continued)

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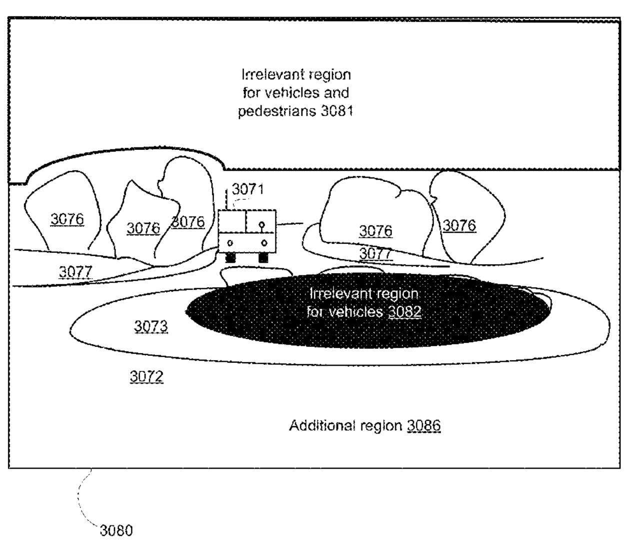
(57) **ABSTRACT**

A method for safe transfer between manned and autonomous driving modes, the method may include detecting, based on first sensed information sensed during a first period, a situation related to an environment of the autonomous vehicle; searching for one or more matching concepts of a group of reference concepts, to which the situation belongs, each reference concept of the group represents a plurality of situations and has a reference concept safety level; wherein for each reference concept of at least a sub-group of the group of reference concepts the safety level of the reference concept is based on a tested success level of at only some of the plurality of scenarios represented by the reference concept; and determining, based on an outcome of the searching, whether the vehicle is capable to safely autonomously drive through the environment.

21 Claims, 94 Drawing Sheets



Related U.S. Application Data

(60) Provisional application No. 62/747,147, filed on Oct. 18, 2018, provisional application No. 62/750,822, filed on Oct. 26, 2018, provisional application No. 62/827,112, filed on Mar. 31, 2019.

(51) **Int. Cl.**

B60W 30/095	(2012.01)
B60W 40/02	(2006.01)
B60W 40/06	(2012.01)
B60W 40/08	(2012.01)
B60W 50/14	(2020.01)
G05D 1/00	(2024.01)
G06F 18/23	(2023.01)
G06F 18/23213	(2023.01)
G06N 20/00	(2019.01)
G06V 10/20	(2022.01)
G06V 10/44	(2022.01)
G06V 20/58	(2022.01)
G06V 20/59	(2022.01)
G08G 1/048	(2006.01)
G08G 1/056	(2006.01)
G08G 1/16	(2006.01)
G05B 13/02	(2006.01)
G05D 1/225	(2024.01)
G05D 1/617	(2024.01)
G05D 1/695	(2024.01)
G05D 1/81	(2024.01)
G06F 18/22	(2023.01)
G06F 18/2431	(2023.01)
G06N 3/042	(2023.01)
G06N 3/08	(2023.01)
G06N 5/04	(2023.01)
G06V 10/75	(2022.01)
G06V 20/56	(2022.01)
G07C 5/02	(2006.01)
G08G 1/0962	(2006.01)
H04W 4/46	(2018.01)

(52) **U.S. Cl.**

CPC ***G06V 10/255*** (2022.01); ***G06V 10/454*** (2022.01); ***G06V 20/58*** (2022.01); ***G06V 20/597*** (2022.01); ***G08G 1/056*** (2013.01); ***G08G 1/162*** (2013.01); ***G08G 1/165*** (2013.01); ***G08G 1/166*** (2013.01); *B60W 30/09* (2013.01); *B60W 30/0956* (2013.01); *B60W 40/02* (2013.01); *B60W 40/06* (2013.01); *B60W 40/08* (2013.01); *B60W 2040/0872* (2013.01); *B60W 50/14* (2013.01); *B60W 60/0011* (2020.02); *B60W 60/0016* (2020.02); *B60W 60/0017* (2020.02); *B60W 60/0051* (2020.02); *B60W 2540/10* (2013.01); *B60W 2540/12* (2013.01); *B60W 2540/18* (2013.01); *B60W 2554/00* (2020.02); *B60W 2554/4023* (2020.02); *B60W 2554/4046* (2020.02); *B60W 2556/65* (2020.02); *G05B 13/029* (2013.01); *G05D 1/0044* (2013.01); *G05D 1/0061* (2013.01); *G05D 1/0088* (2013.01); *G05D 1/0214* (2013.01); *G05D 1/0293* (2013.01); *G05D 1/225* (2024.01); *G05D 1/617* (2024.01); *G05D 1/695* (2024.01); *G05D 1/81* (2024.01); *G06F 18/22* (2023.01); *G06F 18/2431* (2023.01); *G06N 3/042* (2023.01); *G06N 3/08* (2013.01); *G06N 5/04* (2013.01); *G06N 20/00* (2019.01); *G06T 2207/30261* (2013.01); *G06V 10/759* (2022.01); *G06V 20/56* (2022.01); *G06V 20/584* (2022.01); *G07C 5/02* (2013.01); *G08G 1/048* (2013.01); *G08G 1/09626* (2013.01); *H04W 4/46* (2018.02)

(58) **Field of Classification Search**

CPC G06K 9/00845; G06K 9/6218; G06K 9/6223; G06K 9/00791; G06K 9/00805; G06K 9/00825; G06K 9/6201; G06K 9/628; G06K 2009/6213; H04W 4/46; G06N 20/00; G06N 3/0427; G06N 3/08; G06N 5/04; B60W 60/0016; B60W 60/0025; B60W 2554/00; B60W 60/0017; B60W 2554/4023; B60W 60/0051; B60W 2556/65; B60W 60/0011; B60W 2554/4046; B60W 30/09; B60W 30/0956; B60W 40/02; B60W 40/06; B60W 40/08; B60W 50/14; B60W 2040/0872; B60W 2540/10; B60W 2540/12; B60W 2540/18; G05B 13/029; G05D 1/0044; G05D 1/0061; G05D 1/0088; G05D 1/0214; G05D 1/0293; G05D 2201/0213; G06T 2207/30261; G07C 5/02

See application file for complete search history.

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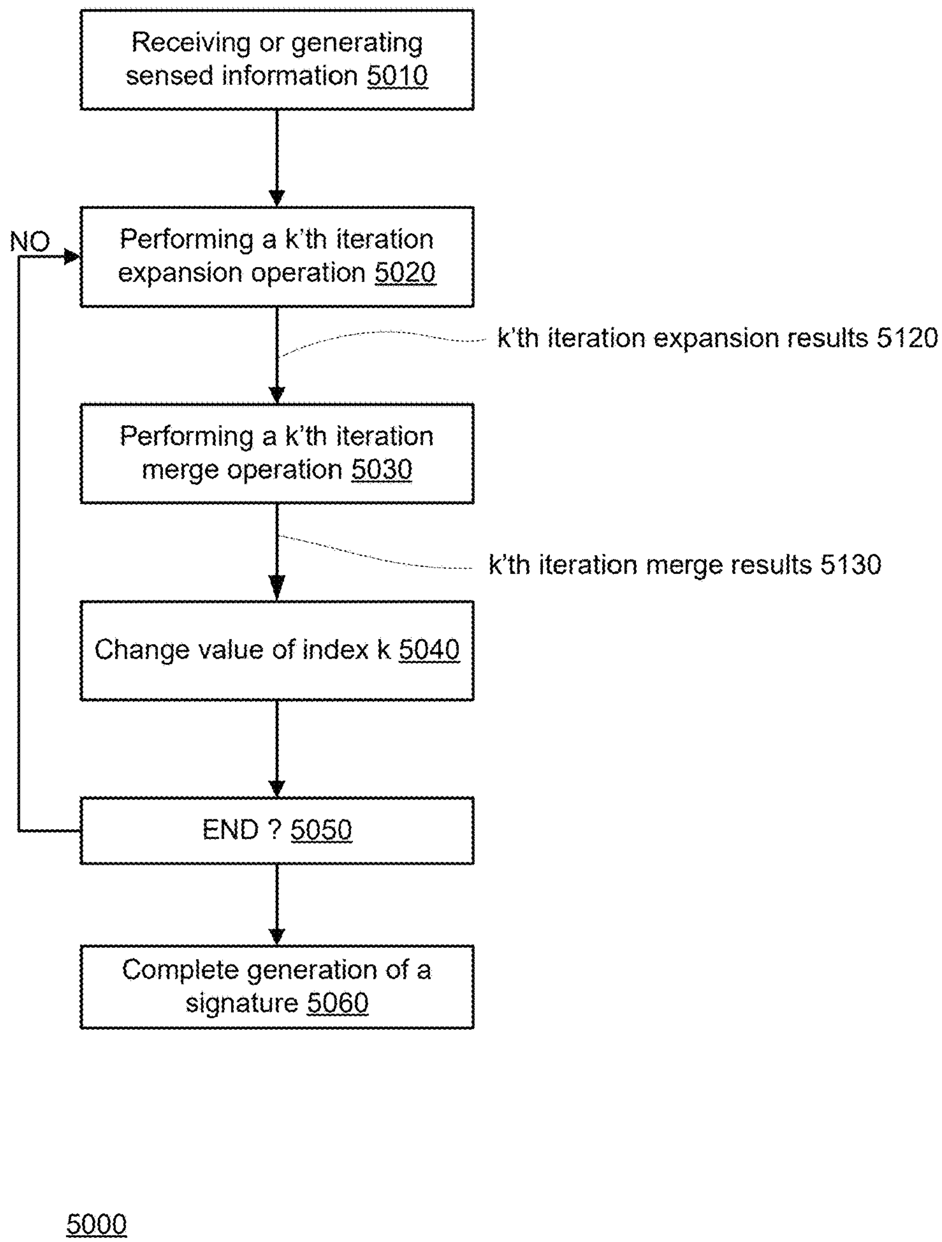

Receiving or generating sensed information 5010
NO
Performing a k'th iteration expansion operation 5020
k'th iteration expansion results 5120
Performing a k'th iteration merge operation 5030
k'th iteration merge results 5130
Change value of index k 5040
END ? 5050
Complete generation of a signature 5060
5000

FIG. 1A

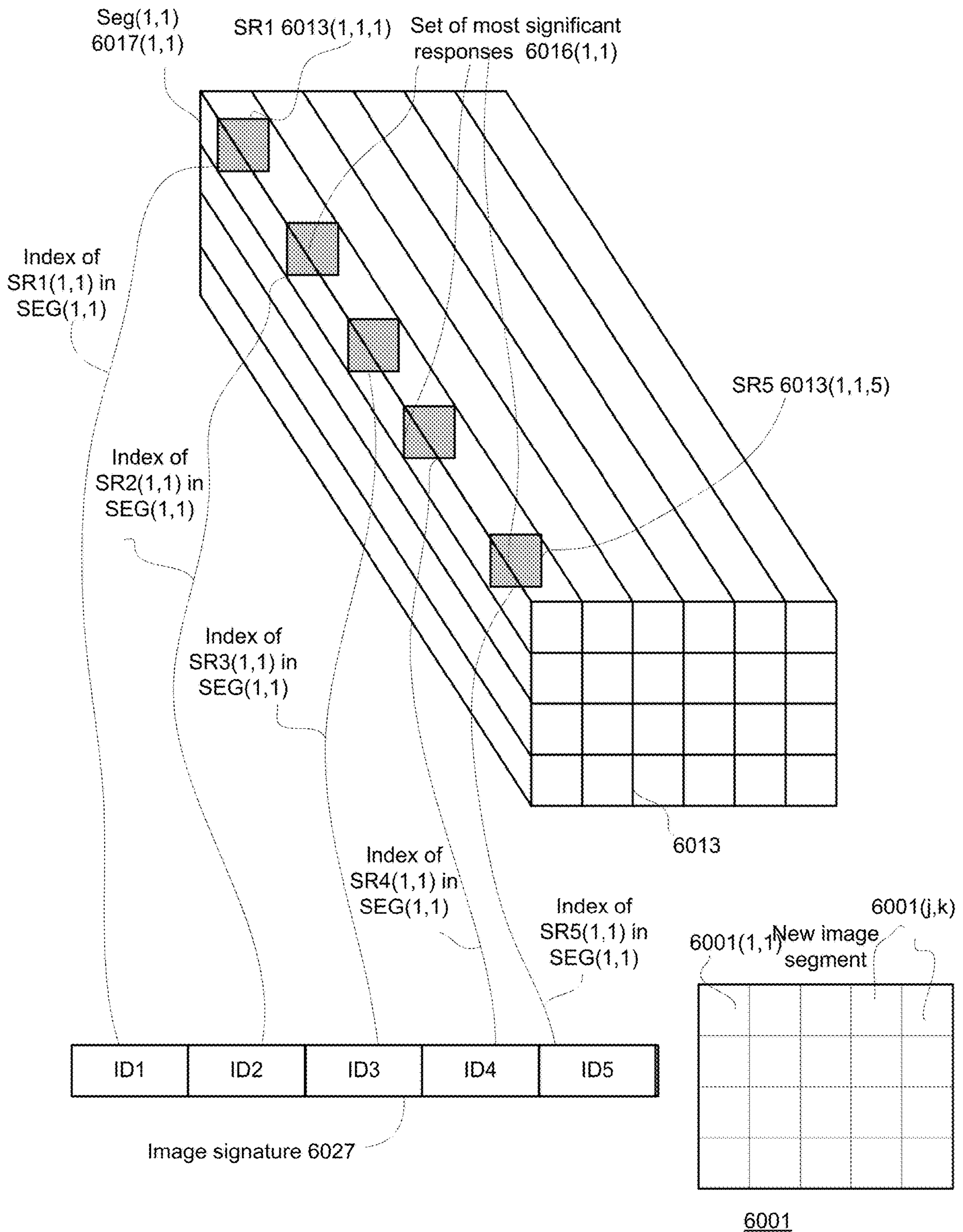

Seg(1,1)
6017(1,1)
SR1 6013(1,1,1)
Set of most significant
responses 6016(1,1)
Index of
SR1(1,1) in
SEG(1,1)
Index of
SR2(1,1) in
SEG(1,1)
Index of
SR3(1,1) in
SEG(1,1)
Index of
SR4(1,1) in
SEG(1,1)
Index of
SR5(1,1) in
SEG(1,1)
SR5 6013(1,1,5)
6013
6001(j,k)
6001(1,1)
New image
segment
ID1
ID2
ID3
ID4
ID5
Image signature 6027
6001

FIG. 1B

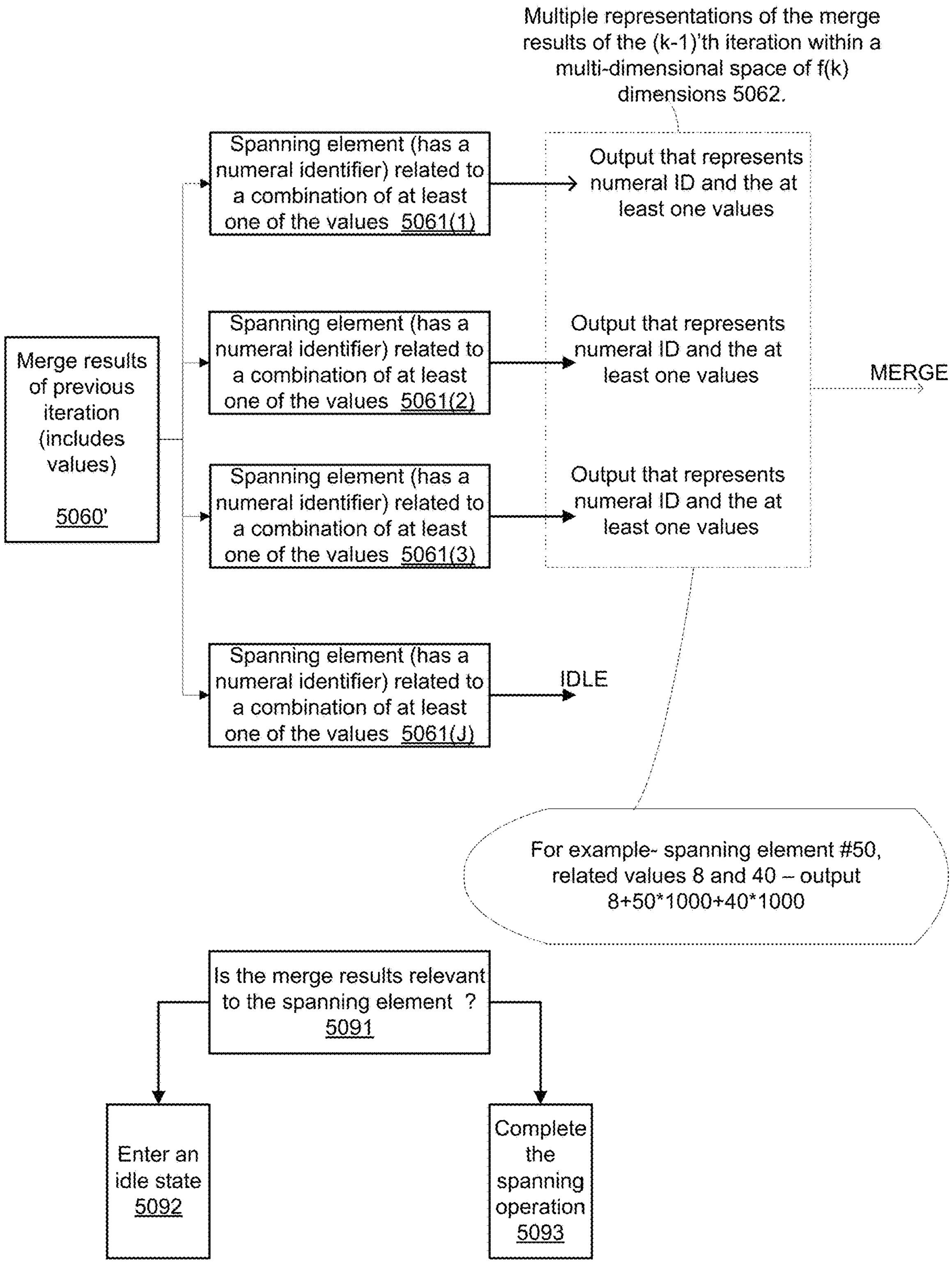

Multiple representations of the merge results of the (k-1)'th iteration within a multi-dimensional space of f(k) dimensions 5062.
Merge results of previous iteration (includes values) 5060'
Spanning element (has a numeral identifier) related to a combination of at least one of the values 5061(1)
Spanning element (has a numeral identifier) related to a combination of at least one of the values 5061(2)
Spanning element (has a numeral identifier) related to a combination of at least one of the values 5061(3)
Spanning element (has a numeral identifier) related to a combination of at least one of the values 5061(J)
Output that represents numeral ID and the at least one values
Output that represents numeral ID and the at least one values
Output that represents numeral ID and the at least one values
MERGE
IDLE
For example- spanning element #50, related values 8 and 40 – output 8+50*1000+40*1000
Is the merge results relevant to the spanning element ? 5091
Enter an idle state 5092
Complete the spanning operation 5093

FIG. 1C

Searching for overlaps between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the overlaps 5031

Determine to drop one or more region of interest, and dropping according to the determination 5032

Searching for relationships between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the relationship 5033

Searching for proximate regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the proximity 5034

Searching for relationships between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the relationship 5035

Merging and/or dropping k'th iteration regions of interest based on shape information related to the k'th iteration regions of interest 5036

5030

FIG. 1D

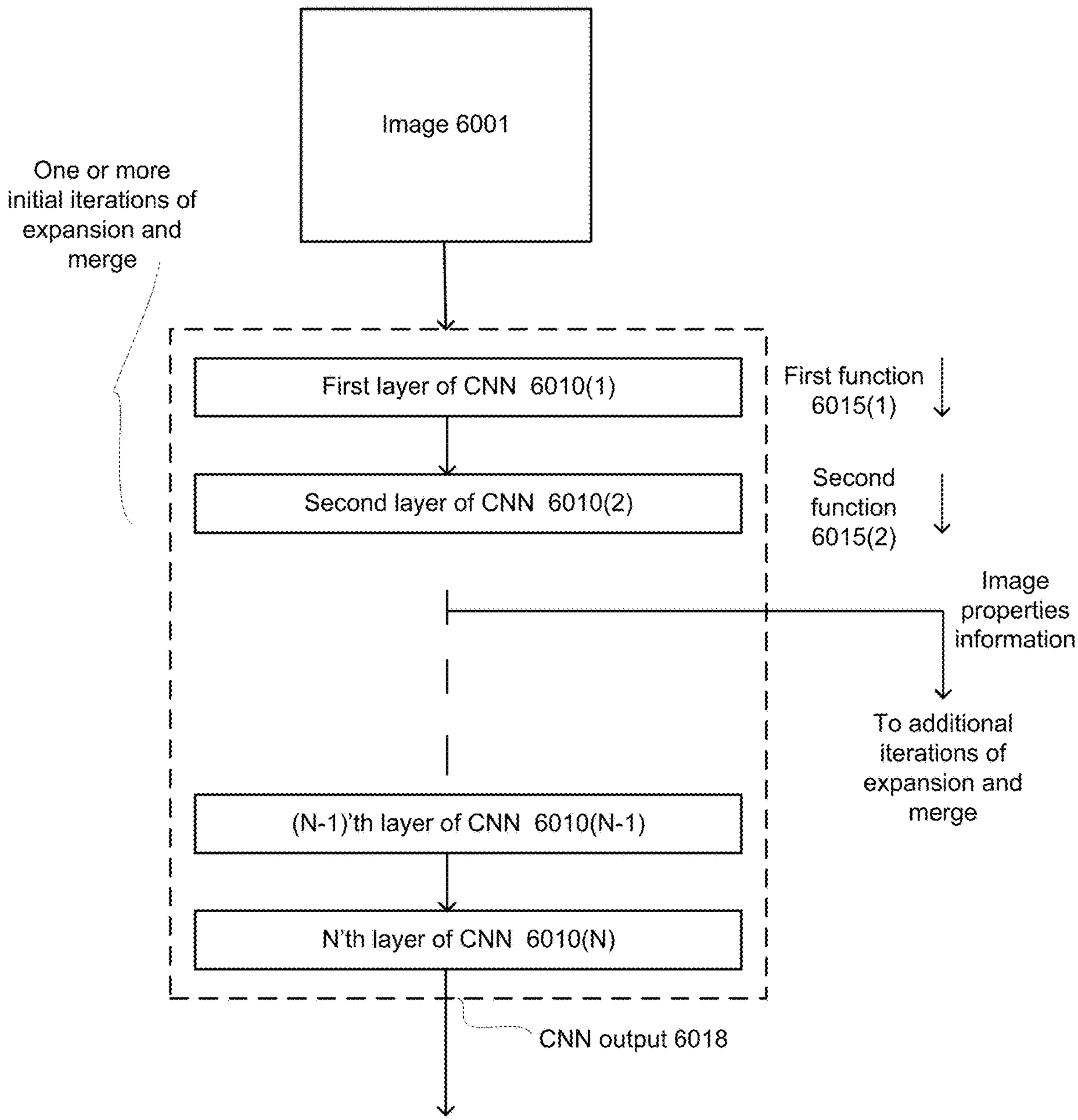


Image 6001
One or more initial iterations of expansion and merge
First layer of CNN 6010(1)
First function 6015(1)
Second layer of CNN 6010(2)
Second function 6015(2)
Image properties information
To additional iterations of expansion and merge
(N-1)'th layer of CNN 6010(N-1)
N'th layer of CNN 6010(N)
CNN output 6018

FIG. 1E

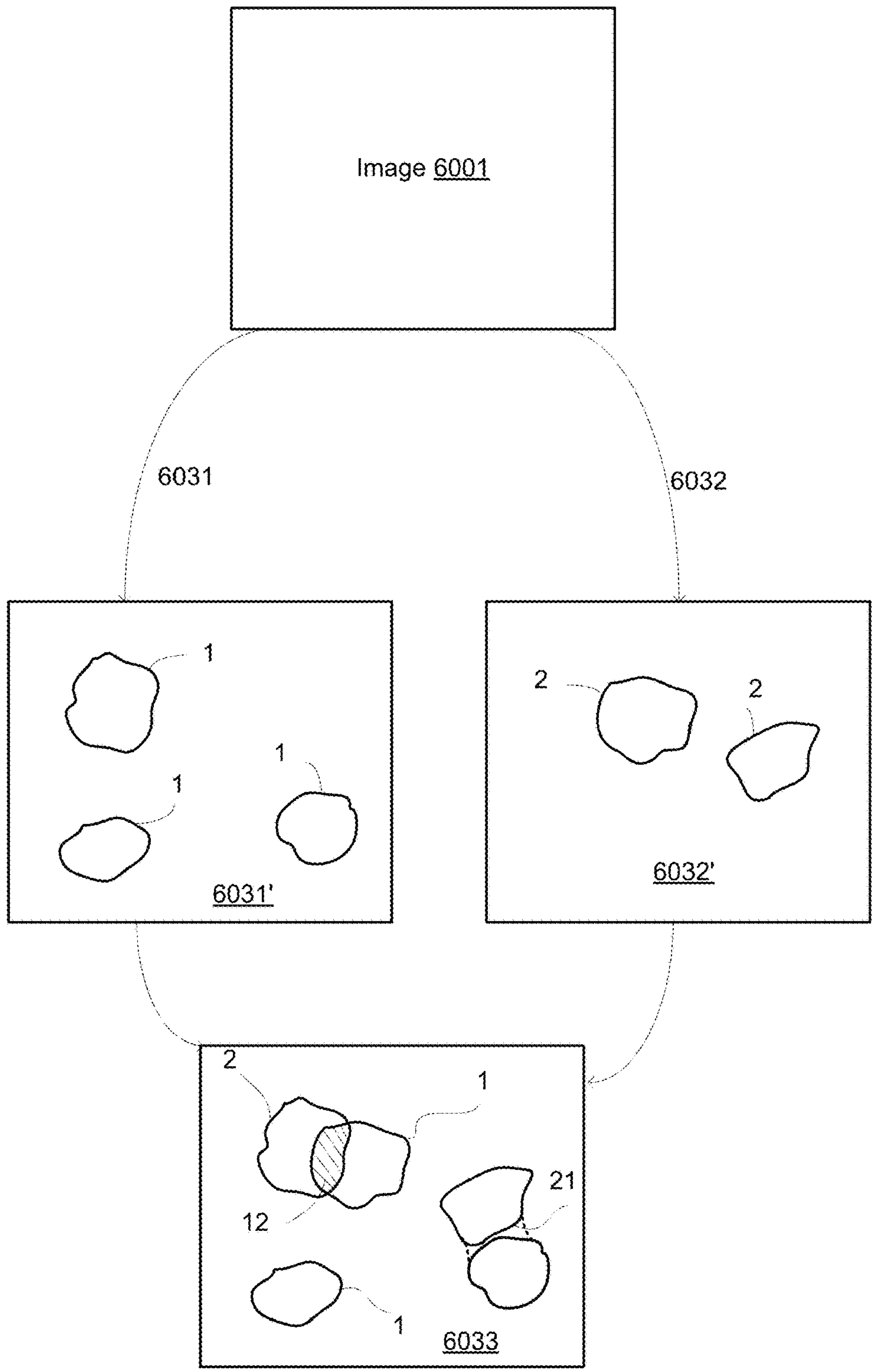


FIG. 1F

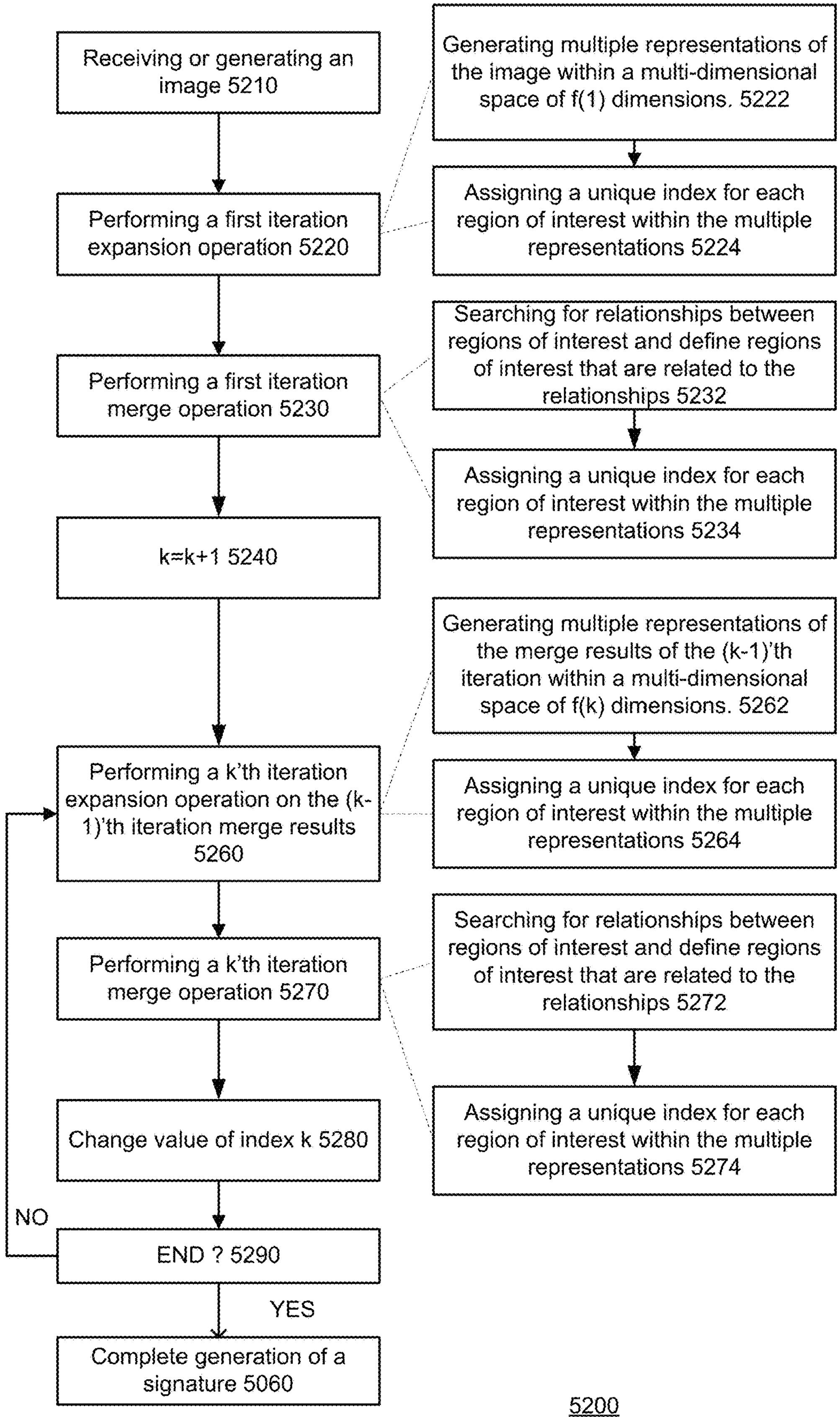
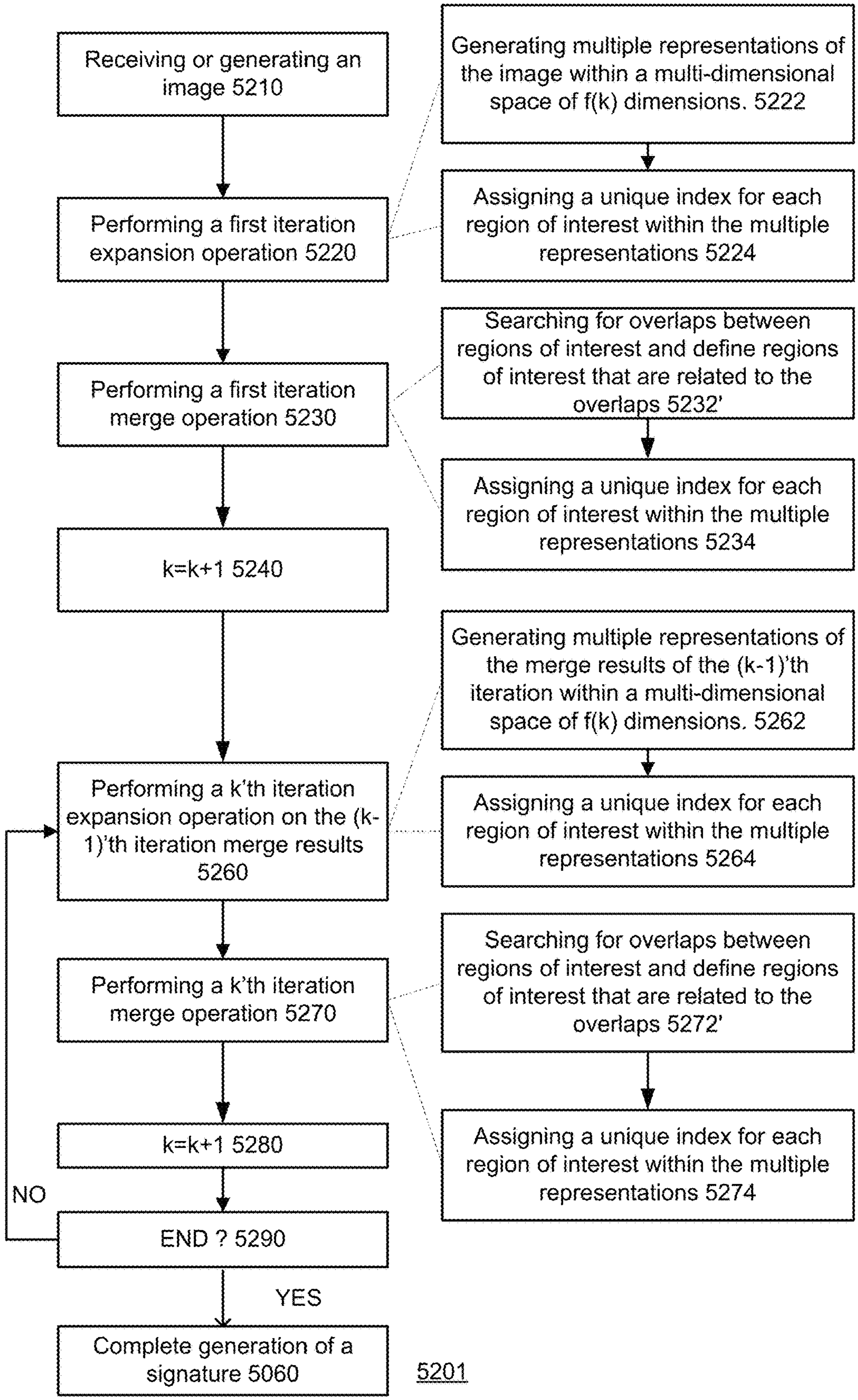
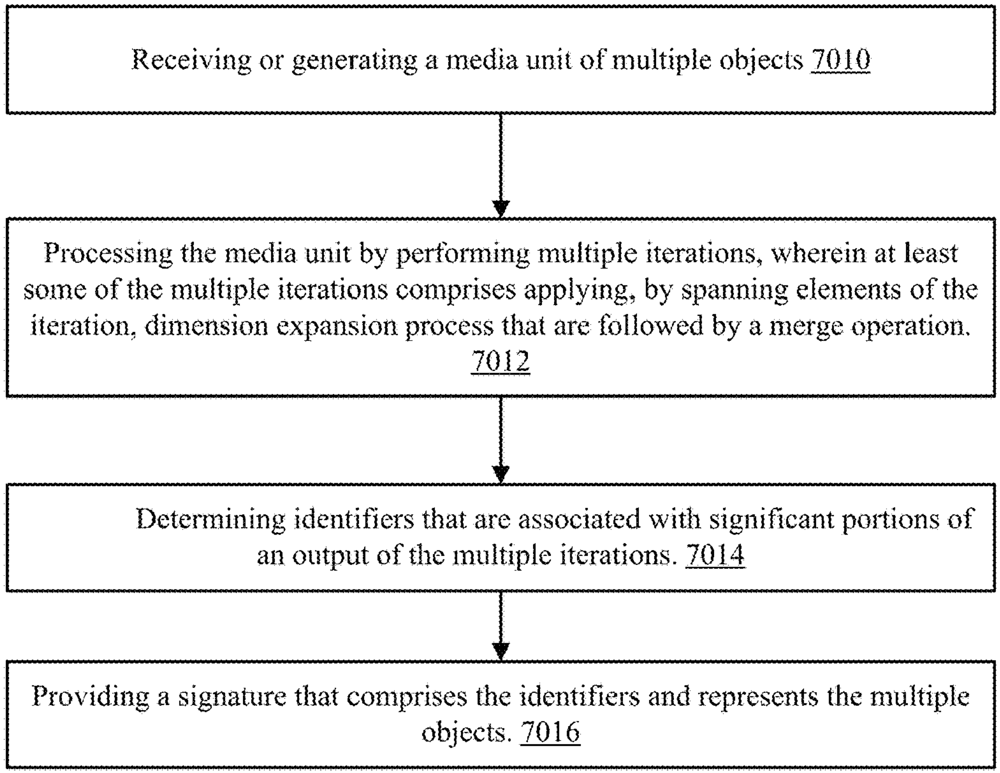

Receiving or generating an image 5210
Performing a first iteration expansion operation 5220
Performing a first iteration merge operation 5230
k=k+1 5240
Performing a k'th iteration expansion operation on the (k-1)'th iteration merge results 5260
Performing a k'th iteration merge operation 5270
Change value of index k 5280
END ? 5290
NO
YES
Complete generation of a signature 5060
Generating multiple representations of the image within a multi-dimensional space of f(1) dimensions. 5222
Assigning a unique index for each region of interest within the multiple representations 5224
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5232
Assigning a unique index for each region of interest within the multiple representations 5234
Generating multiple representations of the merge results of the (k-1)'th iteration within a multi-dimensional space of f(k) dimensions. 5262
Assigning a unique index for each region of interest within the multiple representations 5264
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5272
Assigning a unique index for each region of interest within the multiple representations 5274
5200

FIG. 1G



Receiving or generating an image 5210
Performing a first iteration expansion operation 5220
Performing a first iteration merge operation 5230
k=k+1 5240
Performing a k'th iteration expansion operation on the (k-1)'th iteration merge results 5260
Performing a k'th iteration merge operation 5270
k=k+1 5280
END ? 5290
NO
YES
Complete generation of a signature 5060
Generating multiple representations of the image within a multi-dimensional space of f(k) dimensions. 5222
Assigning a unique index for each region of interest within the multiple representations 5224
Searching for overlaps between regions of interest and define regions of interest that are related to the overlaps 5232'
Assigning a unique index for each region of interest within the multiple representations 5234
Generating multiple representations of the merge results of the (k-1)'th iteration within a multi-dimensional space of f(k) dimensions. 5262
Assigning a unique index for each region of interest within the multiple representations 5264
Searching for overlaps between regions of interest and define regions of interest that are related to the overlaps 5272'
Assigning a unique index for each region of interest within the multiple representations 5274
5201

FIG. 1H


Receiving or generating a media unit of multiple objects 7010
Processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation. 7012
Determining identifiers that are associated with significant portions of an output of the multiple iterations. 7014
Providing a signature that comprises the identifiers and represents the multiple objects. 7016

7000

FIG. 1I

Receiving or generating a media unit 7020

Processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation. 7022

Selecting, based on an output of the multiple iterations, media unit regions of interest that contributed to the output of the multiple iterations. 7024

Providing a hybrid representation, wherein the hybrid representation may include (a) shape information regarding shapes of the media unit regions of interest, and (b) a media unit signature that includes identifiers that identify the media unit regions of interest. 7026

Generating the shape information 7027

Compressing the shape information of the media unit to provide compressed shape information of the media unit 7028

7002

FIG. 1J

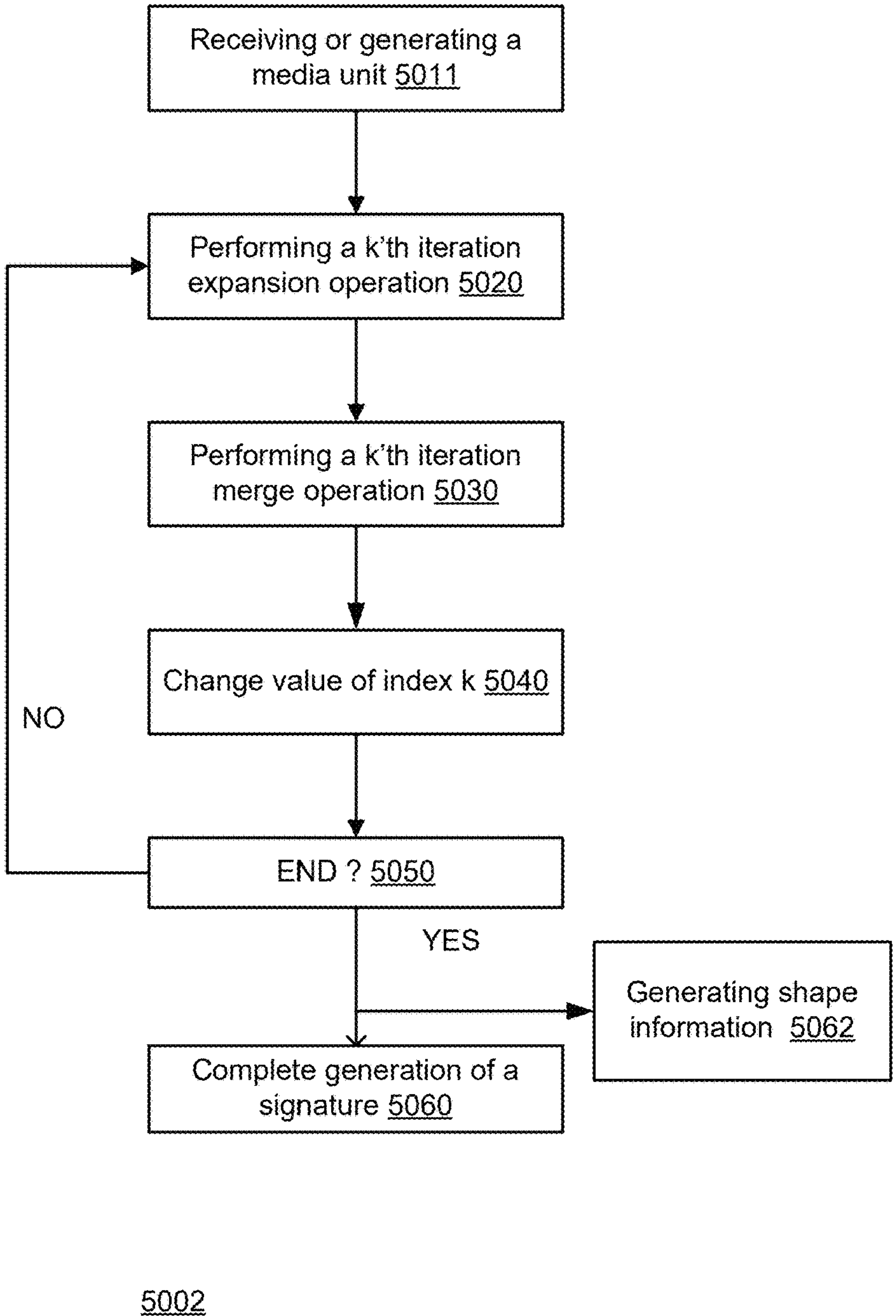
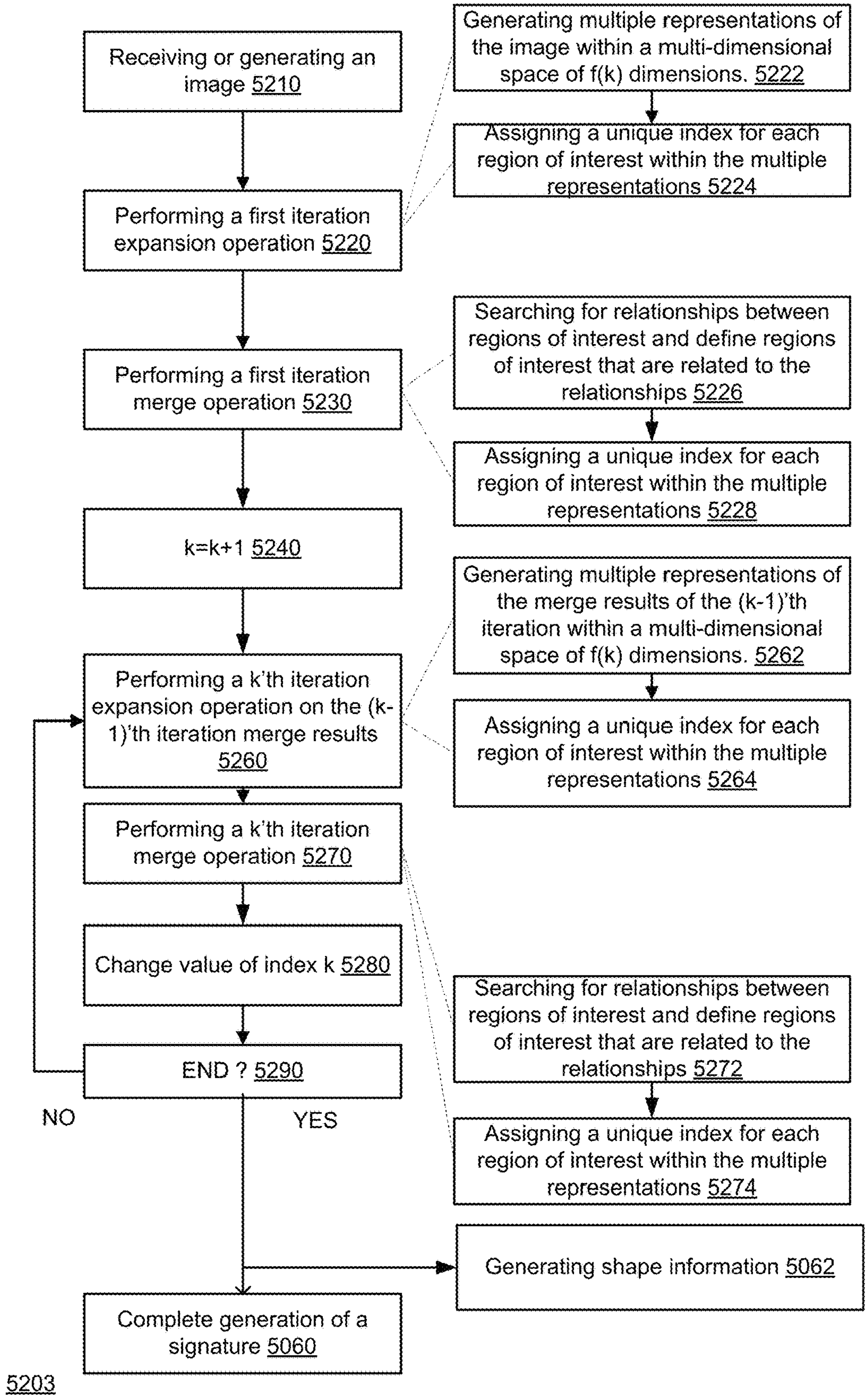

Receiving or generating a media unit 5011
Performing a k'th iteration expansion operation 5020
Performing a k'th iteration merge operation 5030
Change value of index k 5040
NO
END ? 5050
YES
Generating shape information 5062
Complete generation of a signature 5060
5002

FIG. 1K



Receiving or generating an image 5210
Performing a first iteration expansion operation 5220
Generating multiple representations of the image within a multi-dimensional space of f(k) dimensions. 5222
Assigning a unique index for each region of interest within the multiple representations 5224
Performing a first iteration merge operation 5230
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5226
Assigning a unique index for each region of interest within the multiple representations 5228
k=k+1 5240
Performing a k'th iteration expansion operation on the (k-1)'th iteration merge results 5260
Generating multiple representations of the merge results of the (k-1)'th iteration within a multi-dimensional space of f(k) dimensions. 5262
Assigning a unique index for each region of interest within the multiple representations 5264
Performing a k'th iteration merge operation 5270
Change value of index k 5280
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5272
END ? 5290
NO
YES
Assigning a unique index for each region of interest within the multiple representations 5274
Generating shape information 5062
Complete generation of a signature 5060
5203

FIG. 1L

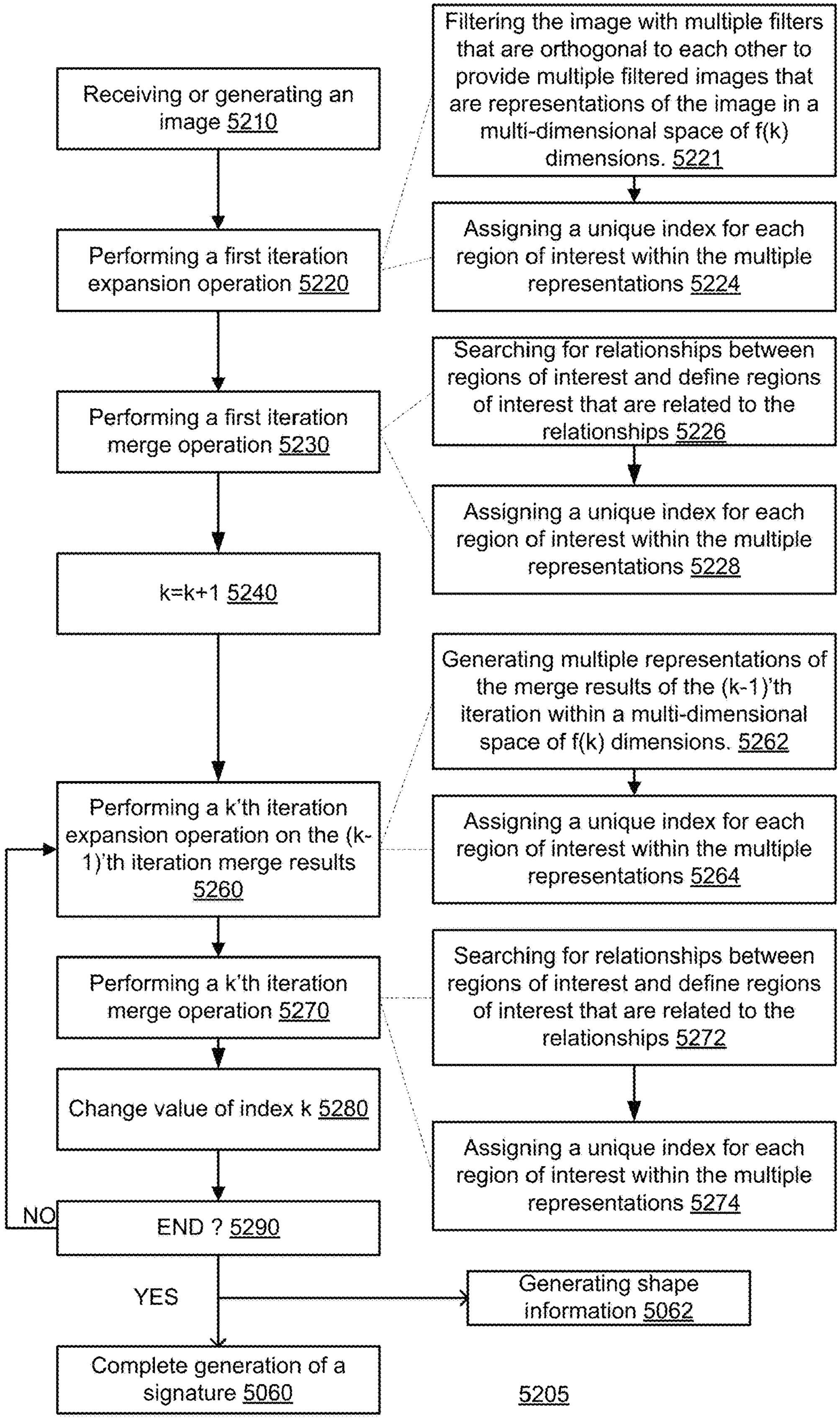
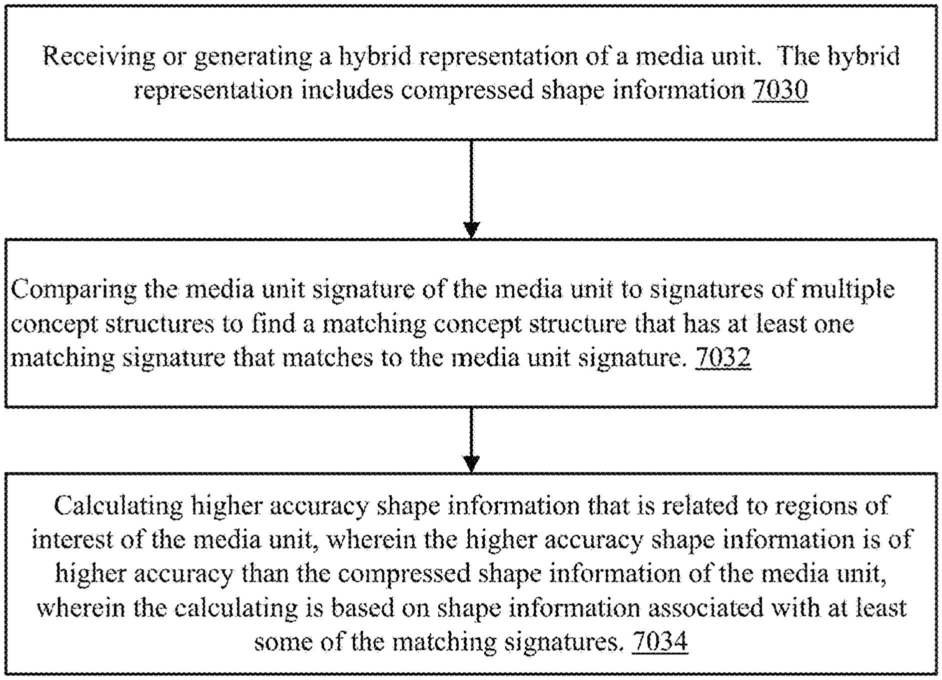

Receiving or generating an image 5210
Performing a first iteration expansion operation 5220
Performing a first iteration merge operation 5230
k=k+1 5240
Performing a k'th iteration expansion operation on the (k-1)'th iteration merge results 5260
Performing a k'th iteration merge operation 5270
Change value of index k 5280
NO
END ? 5290
YES
Complete generation of a signature 5060
Filtering the image with multiple filters that are orthogonal to each other to provide multiple filtered images that are representations of the image in a multi-dimensional space of f(k) dimensions. 5221
Assigning a unique index for each region of interest within the multiple representations 5224
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5226
Assigning a unique index for each region of interest within the multiple representations 5228
Generating multiple representations of the merge results of the (k-1)'th iteration within a multi-dimensional space of f(k) dimensions. 5262
Assigning a unique index for each region of interest within the multiple representations 5264
Searching for relationships between regions of interest and define regions of interest that are related to the relationships 5272
Assigning a unique index for each region of interest within the multiple representations 5274
Generating shape information 5062
5205

FIG. 1M


Receiving or generating a hybrid representation of a media unit. The hybrid representation includes compressed shape information 7030
Comparing the media unit signature of the media unit to signatures of multiple concept structures to find a matching concept structure that has at least one matching signature that matches to the media unit signature. 7032
Calculating higher accuracy shape information that is related to regions of interest of the media unit, wherein the higher accuracy shape information is of higher accuracy than the compressed shape information of the media unit, wherein the calculating is based on shape information associated with at least some of the matching signatures. 7034

7003

FIG. 1N

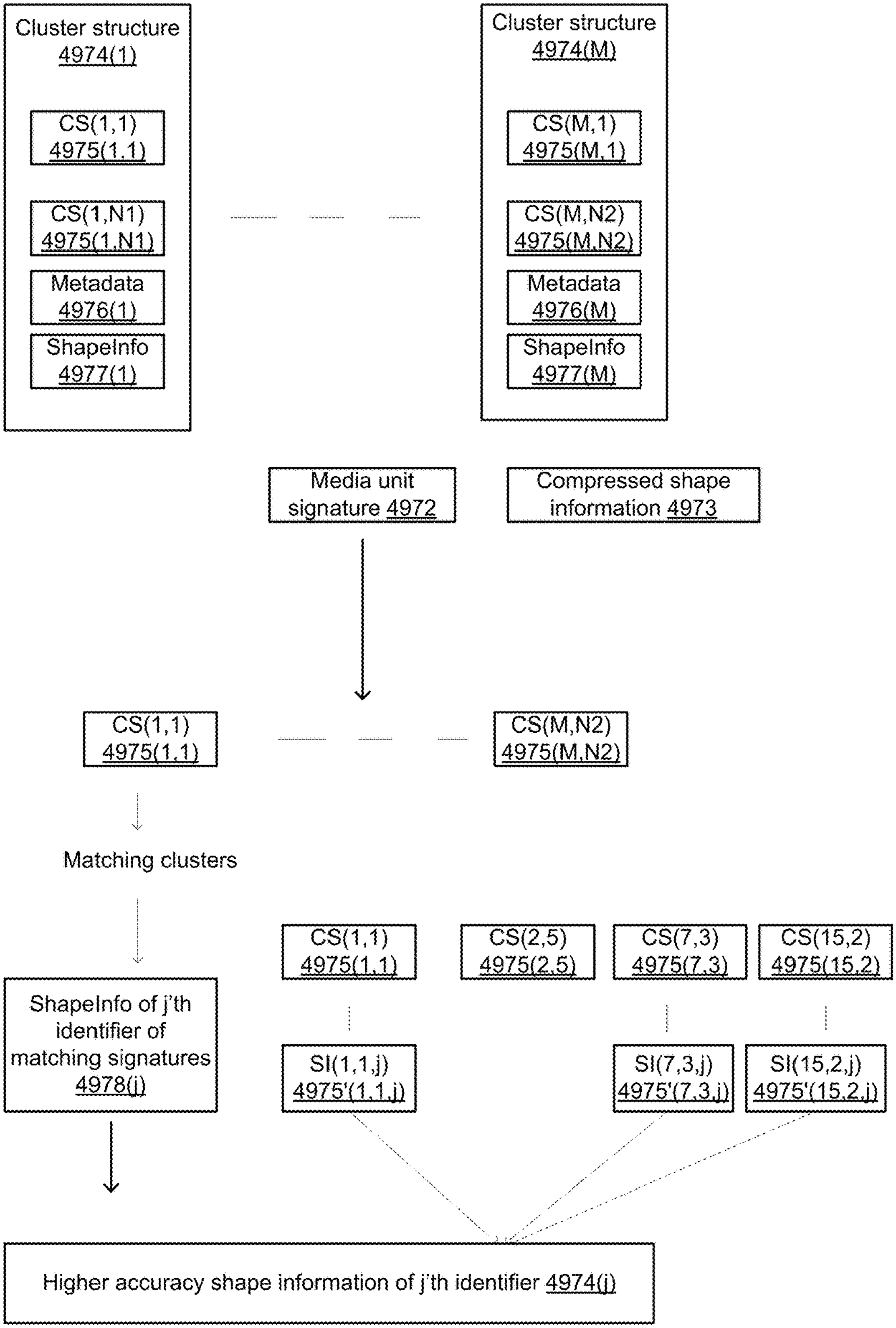

Cluster structure
4974(1)
CS(1,1)
4975(1,1)
CS(1,N1)
4975(1,N1)
Metadata
4976(1)
ShapeInfo
4977(1)
Cluster structure
4974(M)
CS(M,1)
4975(M,1)
CS(M,N2)
4975(M,N2)
Metadata
4976(M)
ShapeInfo
4977(M)
Media unit signature 4972
Compressed shape information 4973
CS(1,1)
4975(1,1)
CS(M,N2)
4975(M,N2)
Matching clusters
ShapeInfo of j'th identifier of matching signatures 4978(j)
CS(1,1)
4975(1,1)
CS(2,5)
4975(2,5)
CS(7,3)
4975(7,3)
CS(15,2)
4975(15,2)
SI(1,1,j)
4975'(1,1,j)
SI(7,3,j)
4975'(7,3,j)
SI(15,2,j)
4975'(15,2,j)
Higher accuracy shape information of j'th identifier 4974(j)

FIG. 1O

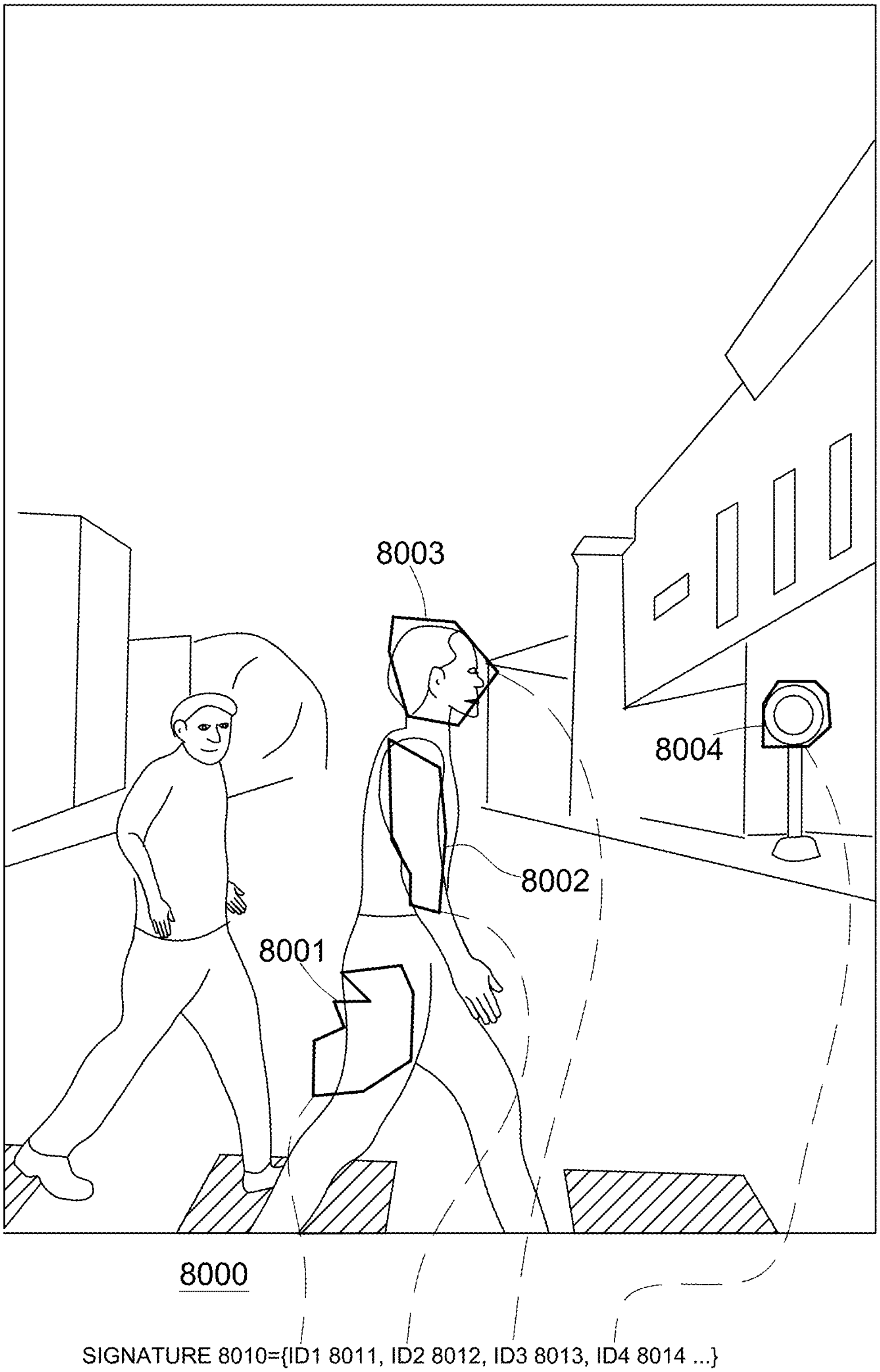

8003
8004
8002
8001
8000
SIGNATURE 8010={ID1 8011, ID2 8012, ID3 8013, ID4 8014 ...}

FIG. 1P

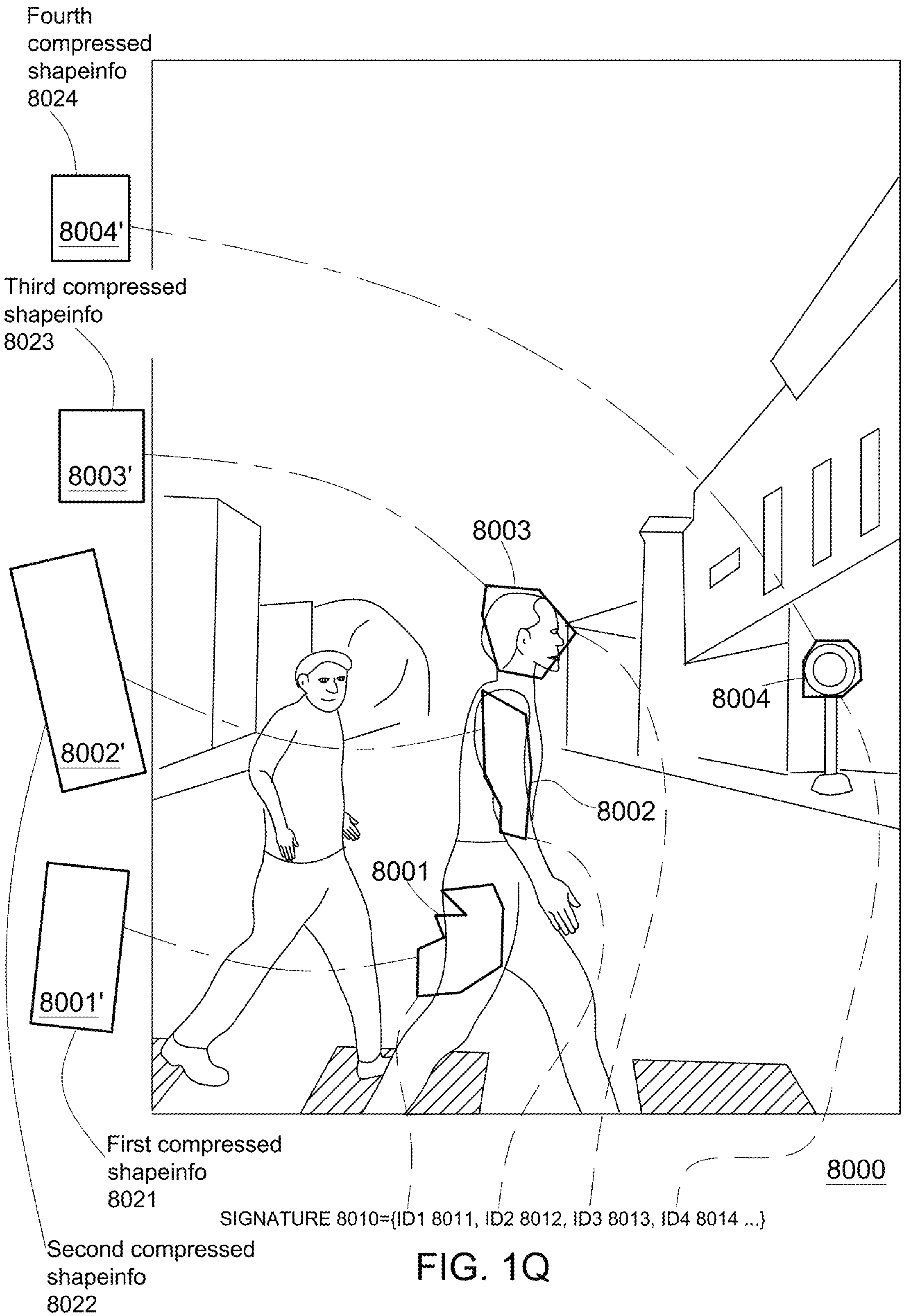

Fourth compressed shapeinfo 8024
8004'
Third compressed shapeinfo 8023
8003'
8002'
8001'
8003
8004
8002
8001
First compressed shapeinfo 8021
8000
SIGNATURE 8010={ID1 8011, ID2 8012, ID3 8013, ID4 8014 ...}
Second compressed shapeinfo 8022

FIG. 1Q

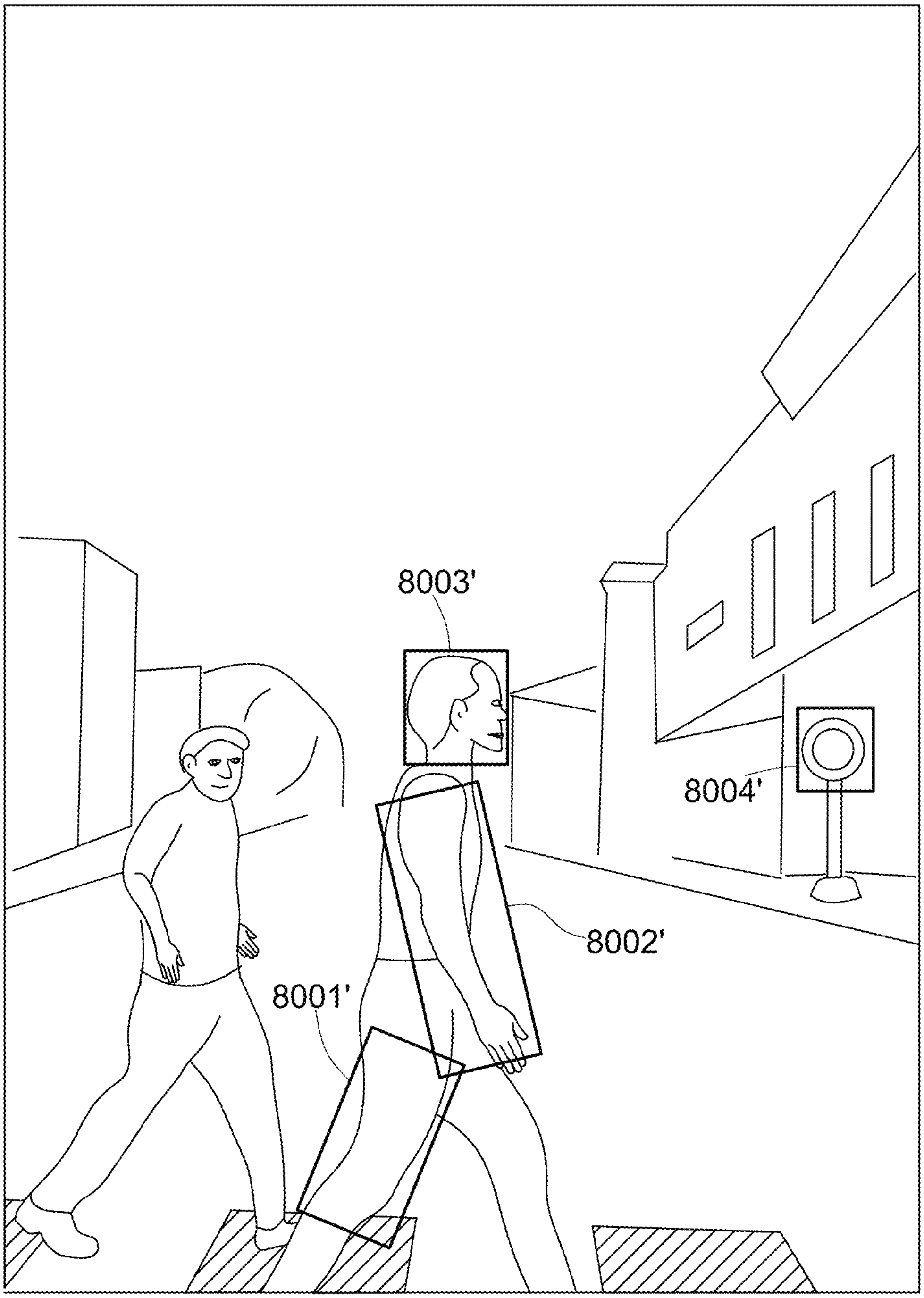


FIG. 1R

Receiving or generating a first image in which an object appears in a first scale and a second image in which the object appears in a second scale that differs from the first scale 8022

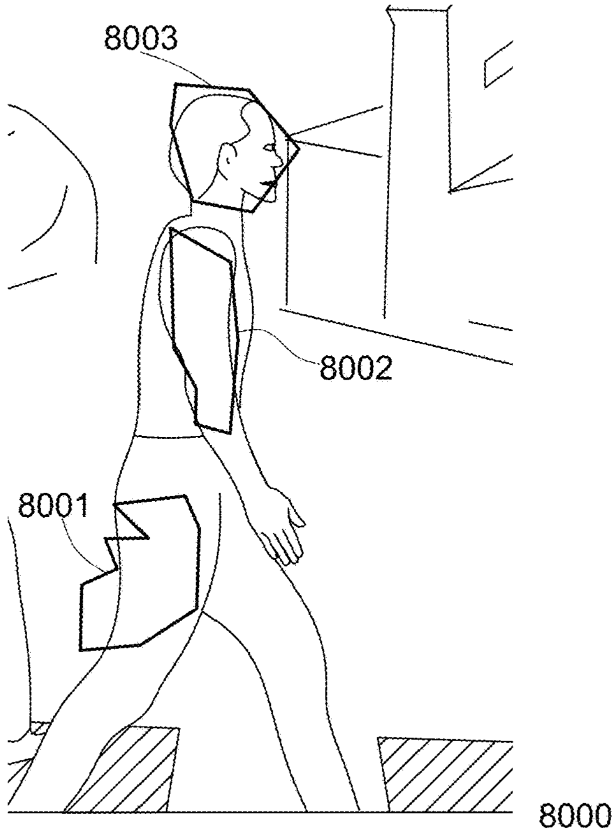
Generating a first image signature and a second image signature 8024

linking between the at least one certain first image identifier and the certain second image identifiers 8026

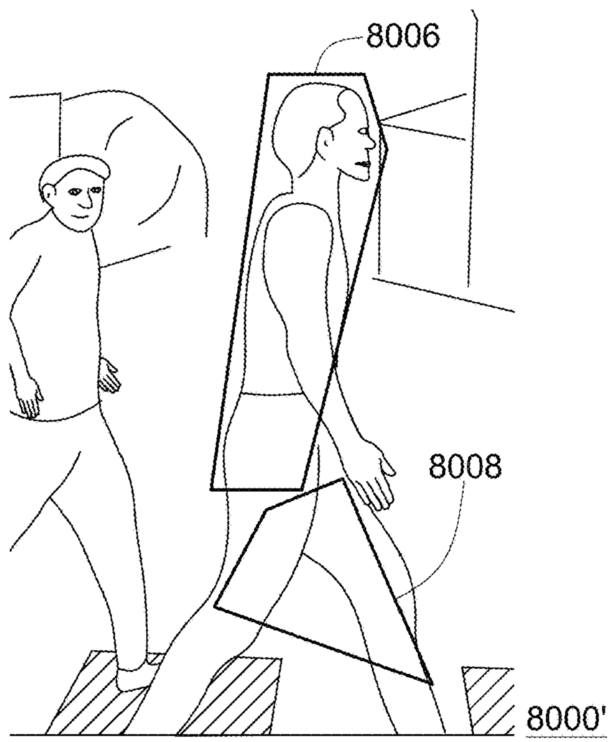
Determining whether an input image includes the object based, at least in part, on the linking 8028

8020

FIG. 1S

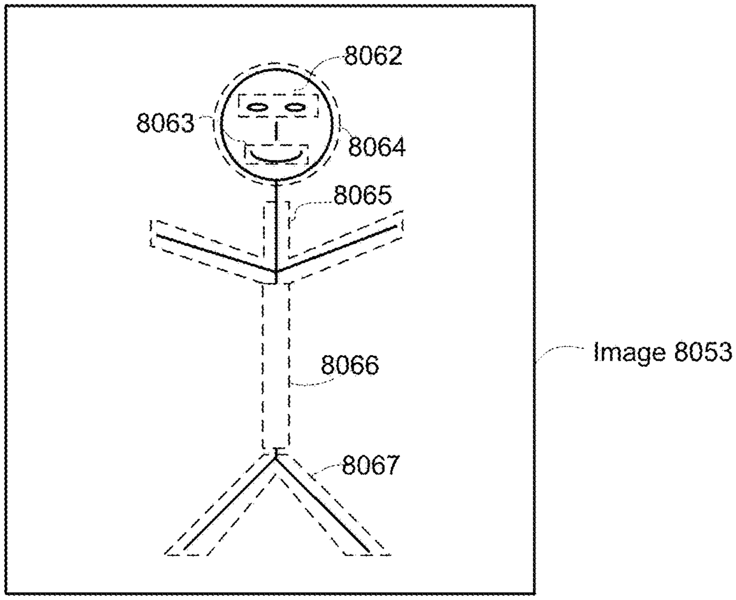


SIGNATURE 8010={ID1 8011,..., ID2 8012,..., ID3 8013,..., ID4 8014 ...}

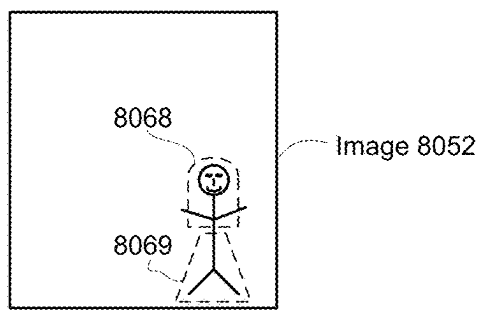


SIGNATURE 8010'={ID6 8016,..., ID8 8018,...}

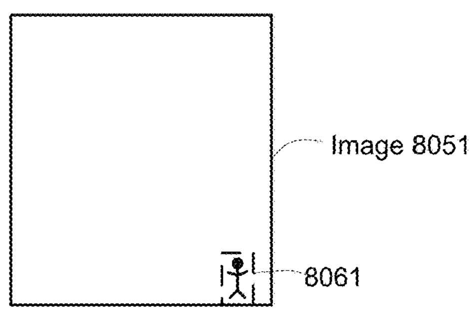
FIG. 2A



Signature 8053' = {ID62,ID63,…, ID64,.., ID65,…, ID66, ID67}

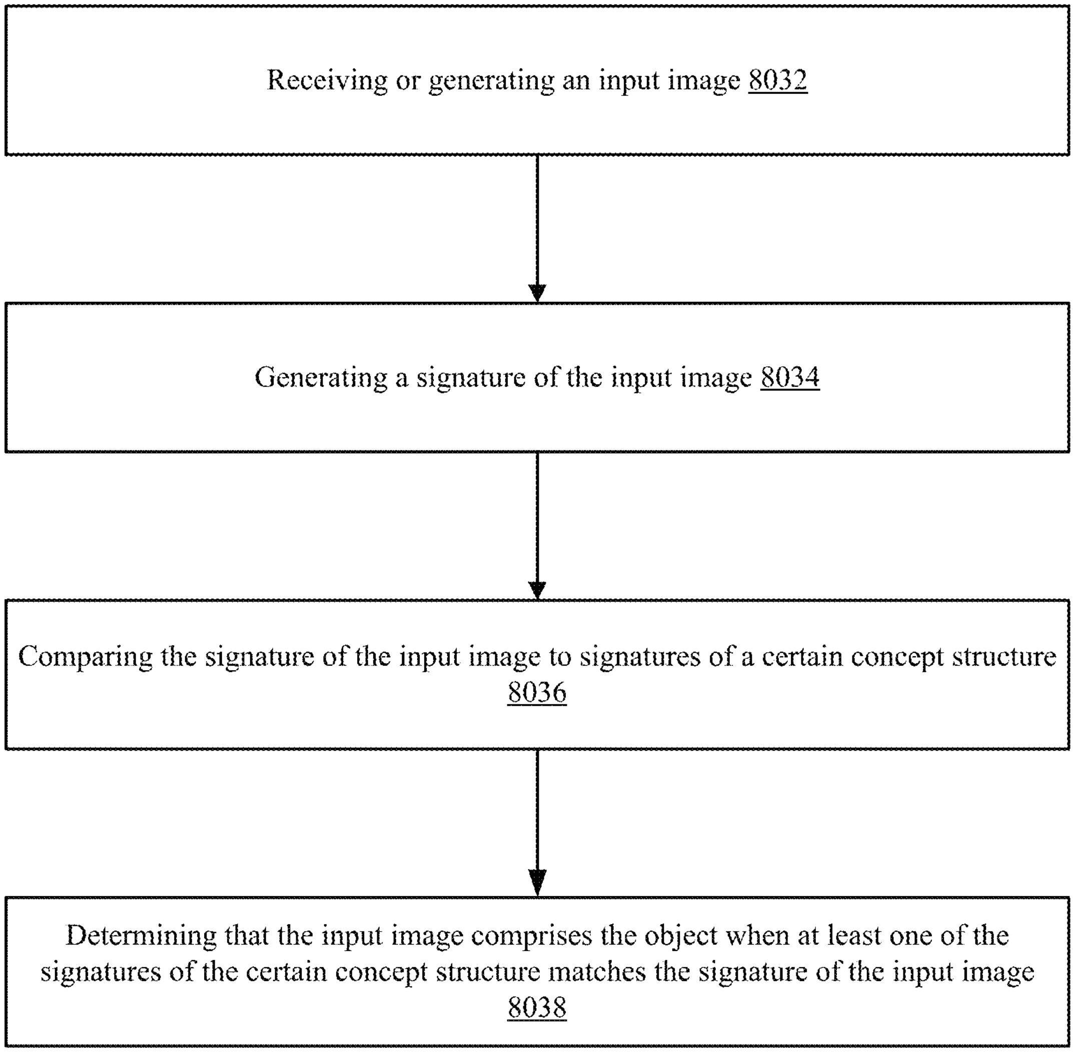


Signature 8052' = {…,ID68,.., ID69}



Signature 8051' = {...ID61…}

FIG. 2B


Receiving or generating an input image 8032
Generating a signature of the input image 8034
Comparing the signature of the input image to signatures of a certain concept structure 8036
Determining that the input image comprises the object when at least one of the signatures of the certain concept structure matches the signature of the input image 8038

8030

FIG. 2C

Receiving or generating an input image 8041

↓

Generating a signature of the input image. The signature of the input image comprises only some of the certain second image identifiers. The input image is of the second scale 8043

↓

Changing a scale of the input image to the first scale to a provide an amended input image 8045

↓

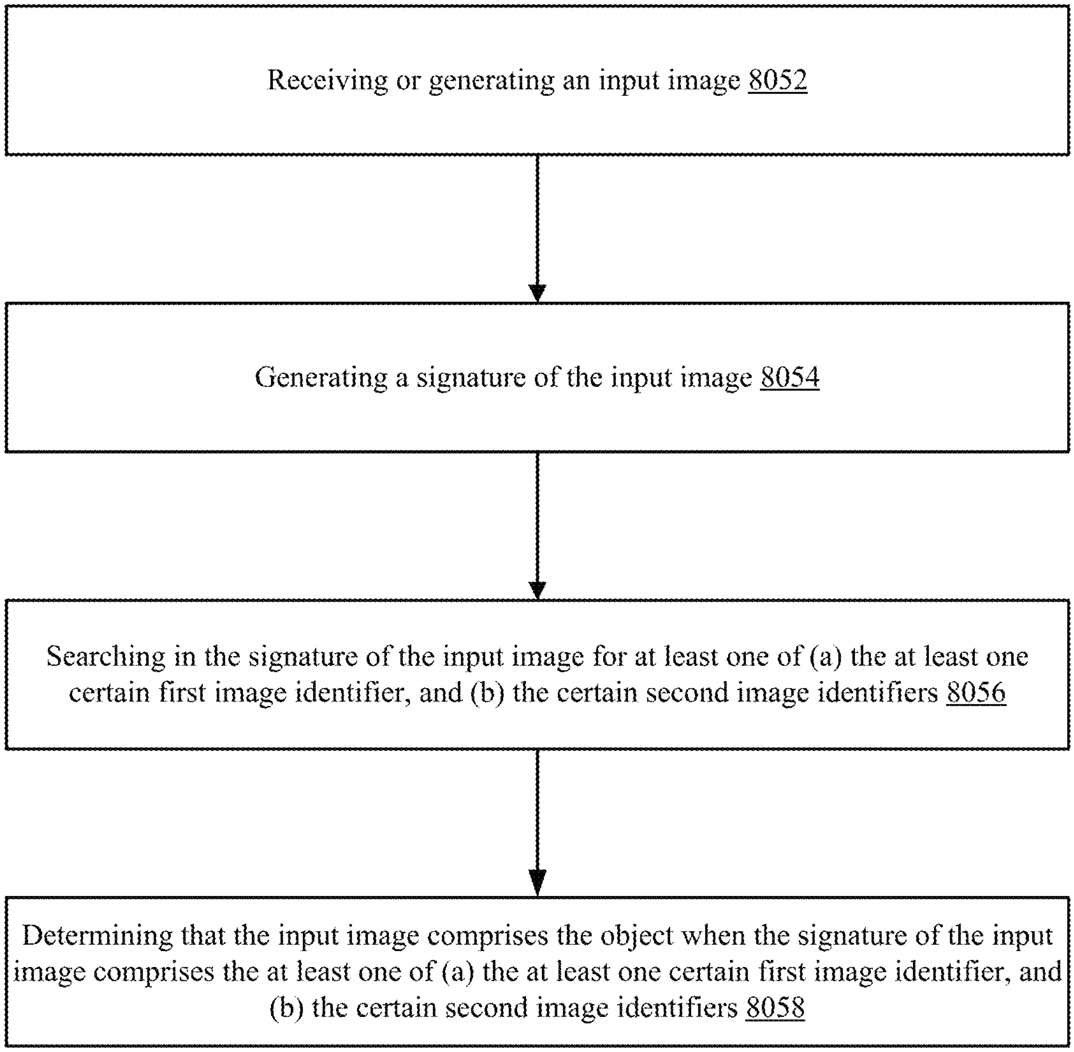
Generating a signature of the amended input image 8047

↓

Verifying that the input image comprises the object when the signature of the amended input image comprises the at least one certain first image identifier 8049

8040

FIG. 2D


Receiving or generating an input image 8052
Generating a signature of the input image 8054
Searching in the signature of the input image for at least one of (a) the at least one certain first image identifier, and (b) the certain second image identifiers 8056
Determining that the input image comprises the object when the signature of the input image comprises the at least one of (a) the at least one certain first image identifier, and (b) the certain second image identifiers 8058

8050

FIG. 2E

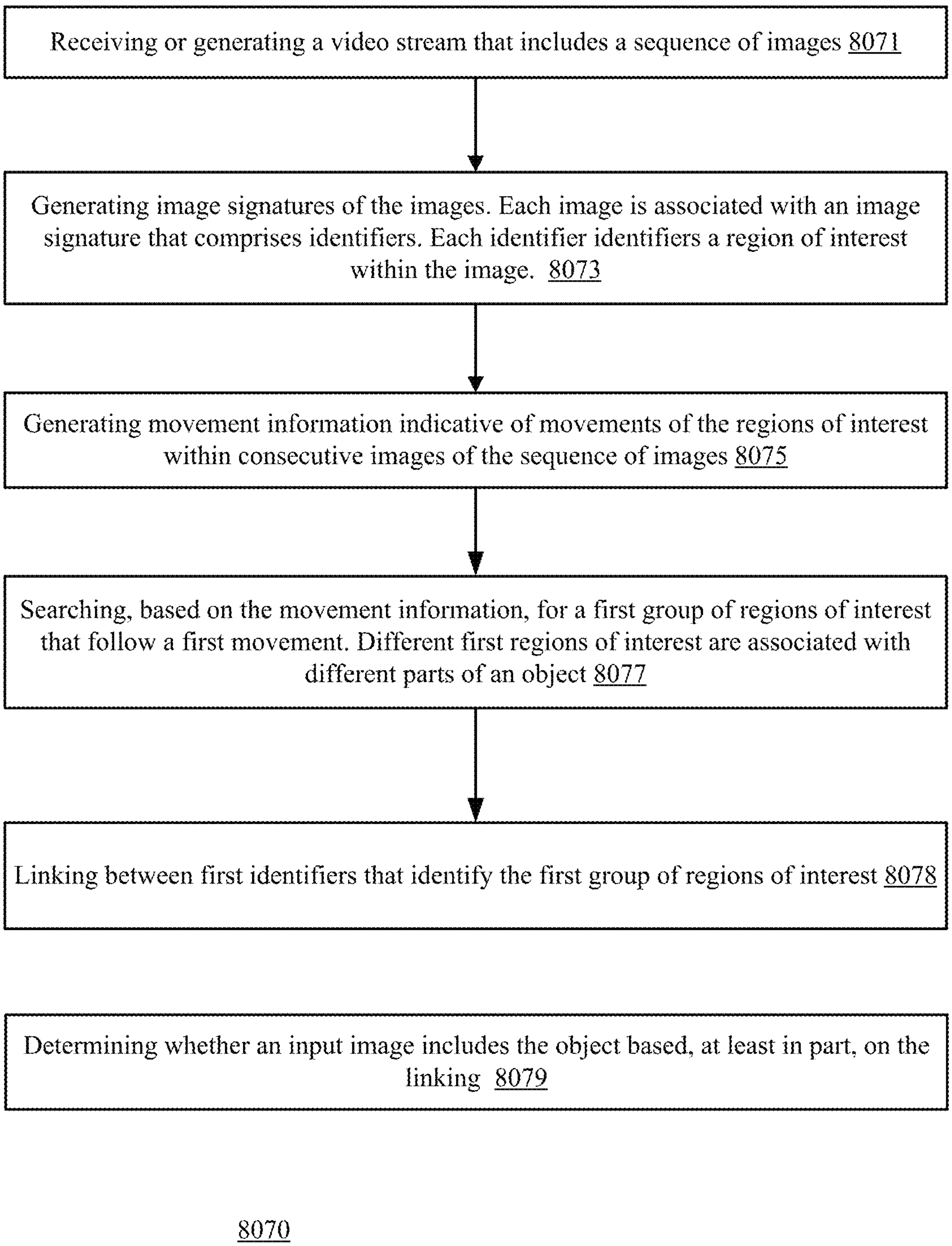
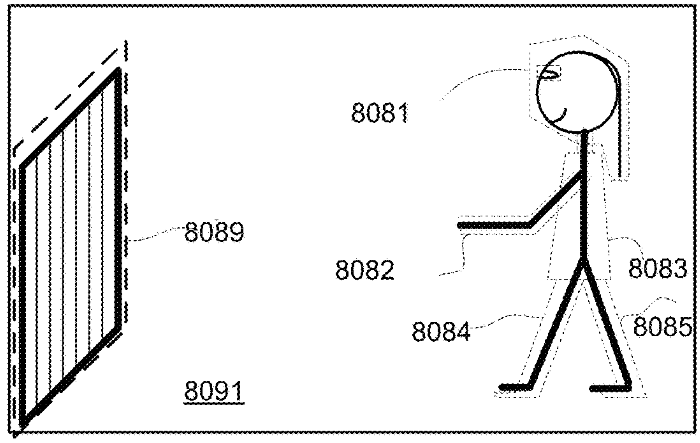
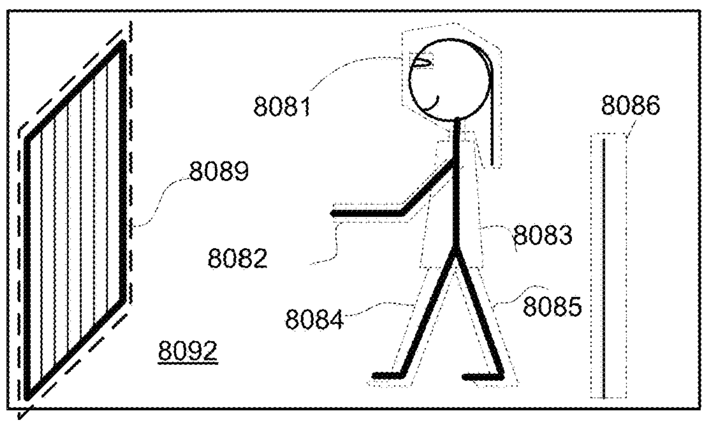

Receiving or generating a video stream that includes a sequence of images 8071
Generating image signatures of the images. Each image is associated with an image signature that comprises identifiers. Each identifier identifiers a region of interest within the image. 8073
Generating movement information indicative of movements of the regions of interest within consecutive images of the sequence of images 8075
Searching, based on the movement information, for a first group of regions of interest that follow a first movement. Different first regions of interest are associated with different parts of an object 8077
Linking between first identifiers that identify the first group of regions of interest 8078
Determining whether an input image includes the object based, at least in part, on the linking 8079
8070

FIG. 2F



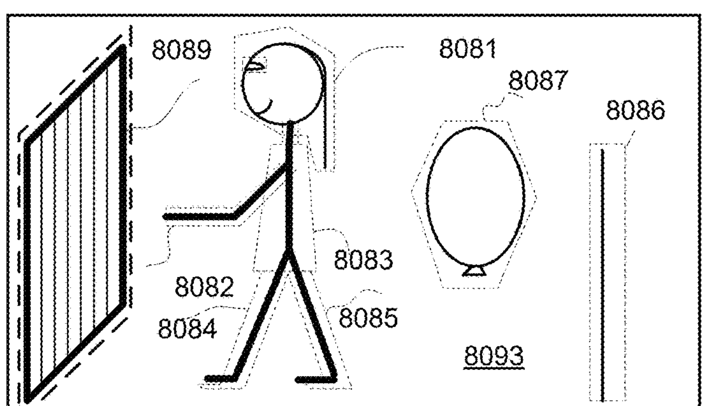
Signature 8091' = {ID81,ID82, ID83, ID84, ID85, ID89}

Location information 8091" = {L81, L82, L83, L84, L85, L89}



Signature 8092' = {ID81,ID82, ID83, ID84, ID85, ID86, ID89}

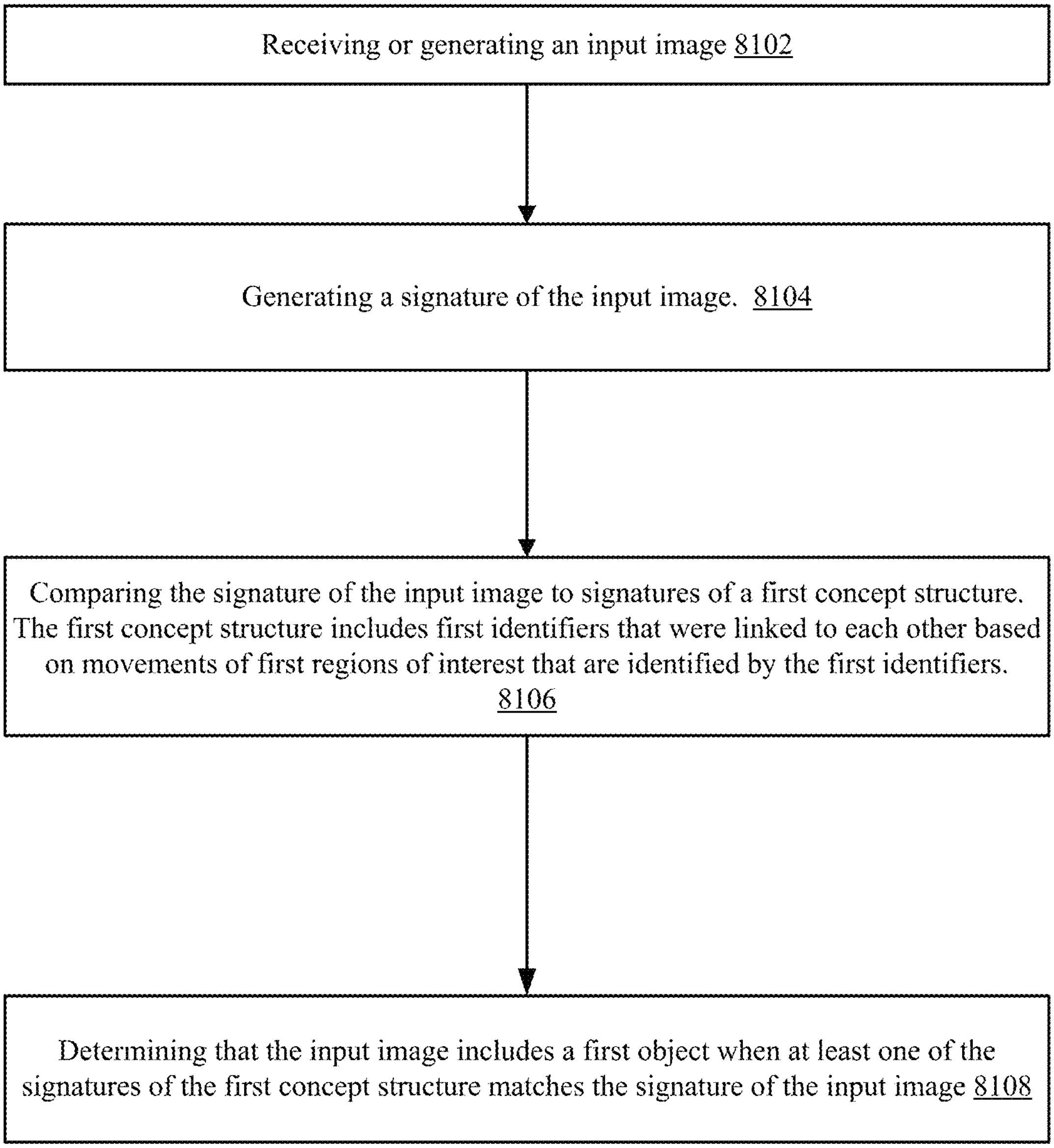
Location information 8092" = {L81, L82, L83, L84, L85, L86, L89}



Signature 8093' = {ID81,ID82, ID83, ID84, ID85, ID86, ID87, ID89}

Location information 8093" = {L81, L82, L83, L84, L85, L86, L87, L89}

FIG. 2G


Receiving or generating an input image 8102
Generating a signature of the input image. 8104
Comparing the signature of the input image to signatures of a first concept structure. The first concept structure includes first identifiers that were linked to each other based on movements of first regions of interest that are identified by the first identifiers. 8106
Determining that the input image includes a first object when at least one of the signatures of the first concept structure matches the signature of the input image 8108

8100

FIG. 2H

Receiving or generating an input image 8112

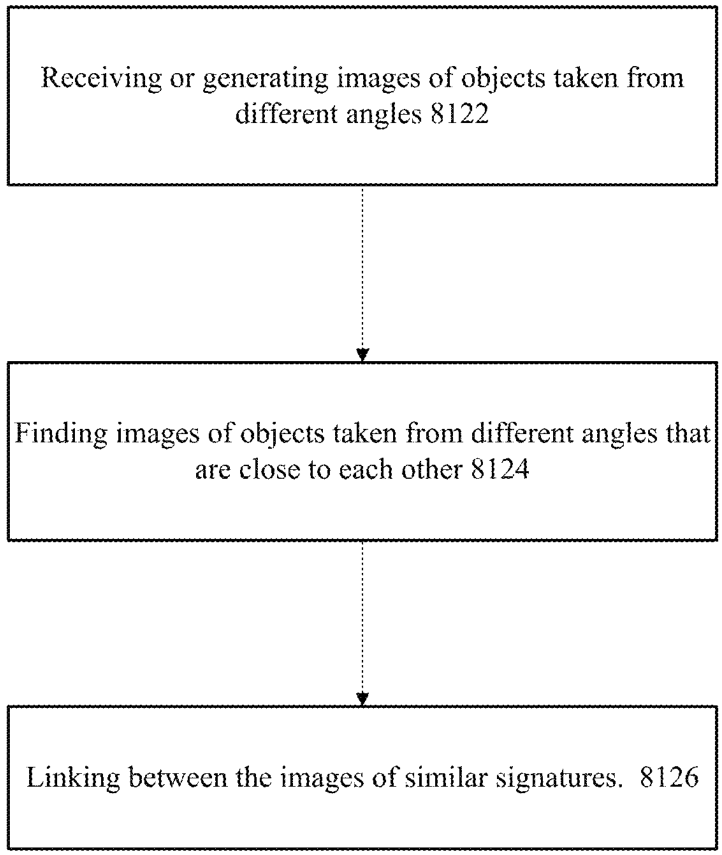
Generating a signature of the input image. 8114

Searching in the signature of the input image for at least one of the first identifiers 8116

Determining that the input image comprises the object when the signature of the input image comprises at least one of the first identifiers 8118

81`0

FIG. 2I


Receiving or generating images of objects taken from different angles 8122
Finding images of objects taken from different angles that are close to each other 8124
Linking between the images of similar signatures. 8126

8120

FIG. 2J

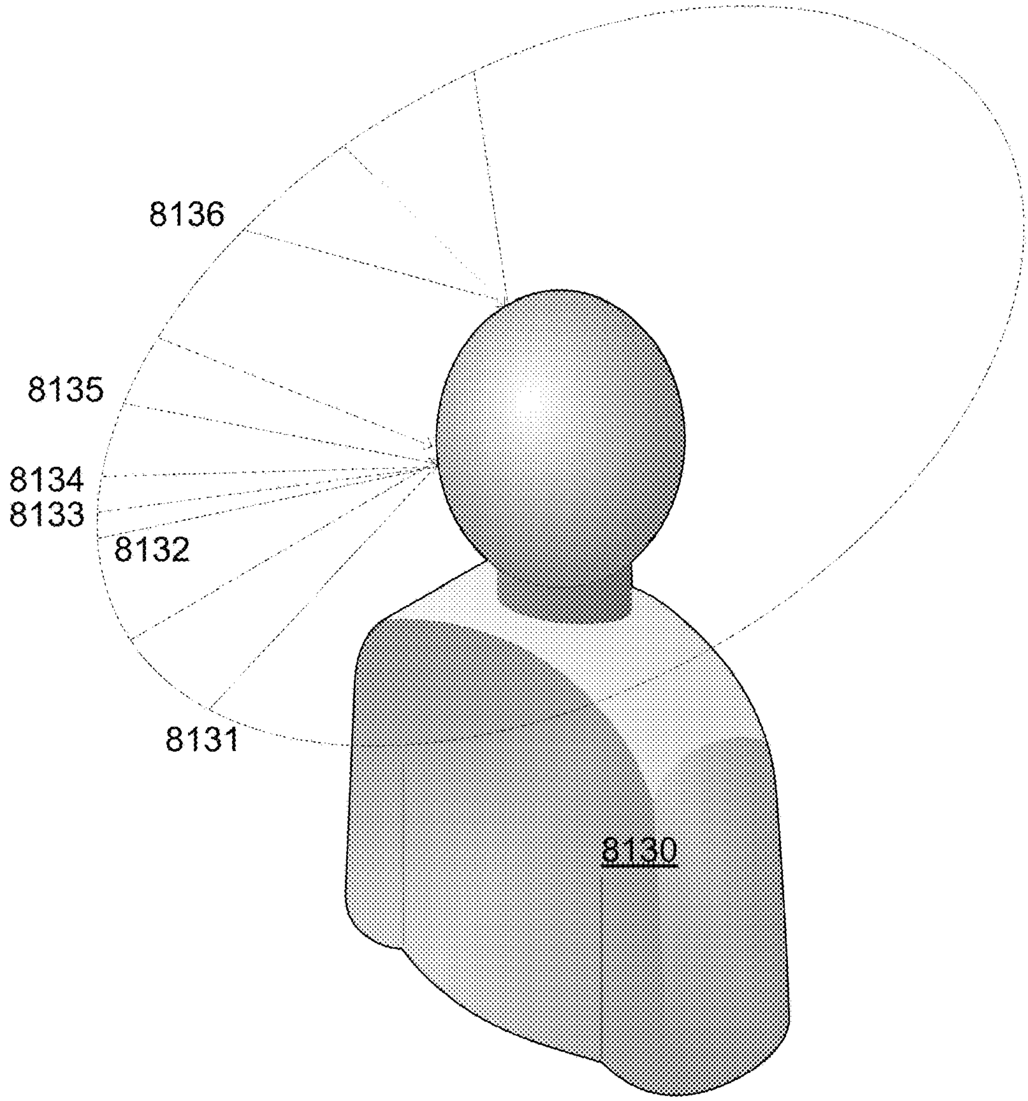


FIG. 2K

Receiving or generating signatures of a group of test images 8203

Calculating the object detection capability of the signature, for each signature matching criterion of the different signature matching criteria 8204

Selecting the signature matching criterion based on the object detection capabilities of the signature under the different signature matching criteria. 8206

Determining each signature matching criterion by evaluating object detection capabilities of the signature under different signature matching criteria 8202

Receiving an input image 8210

Generating a signature of the input image 8212

Comparing the signature of the input image to signatures of a concept structure. 8214

Determining whether the signature of the input image matches any of the signatures of the concept structure based on signature matching criteria, wherein each signature of the concept structure is associated within a signature matching criterion that is determined based on an object detection parameter of the signature 8216

Concluding that the input image comprises an object associated with the concept structure based on an outcome of the determining 8212

8200

FIG. 2L

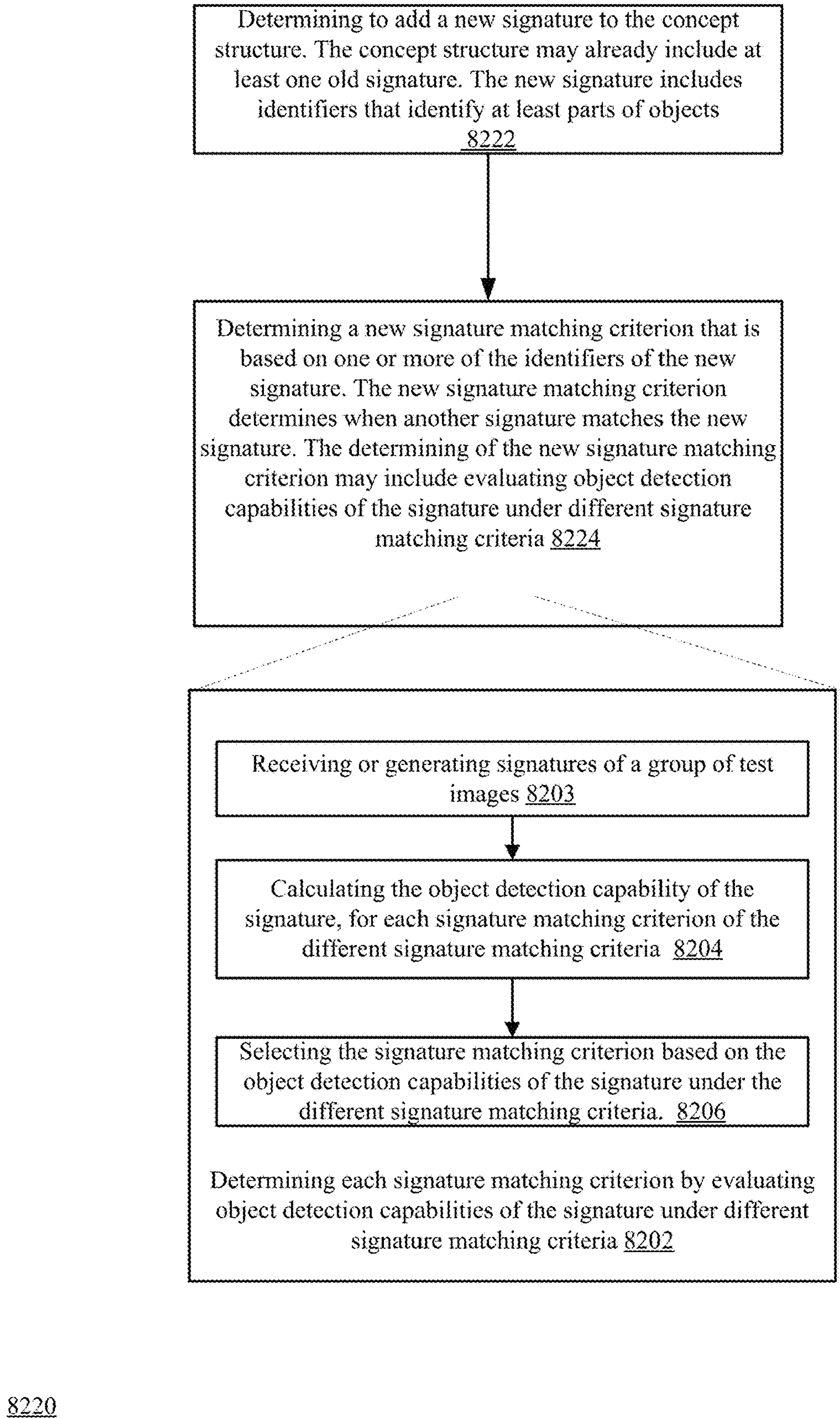

Determining to add a new signature to the concept structure. The concept structure may already include at least one old signature. The new signature includes identifiers that identify at least parts of objects 8222
Determining a new signature matching criterion that is based on one or more of the identifiers of the new signature. The new signature matching criterion determines when another signature matches the new signature. The determining of the new signature matching criterion may include evaluating object detection capabilities of the signature under different signature matching criteria 8224
Receiving or generating signatures of a group of test images 8203
Calculating the object detection capability of the signature, for each signature matching criterion of the different signature matching criteria 8204
Selecting the signature matching criterion based on the object detection capabilities of the signature under the different signature matching criteria. 8206
Determining each signature matching criterion by evaluating object detection capabilities of the signature under different signature matching criteria 8202
8220

FIG. 2M

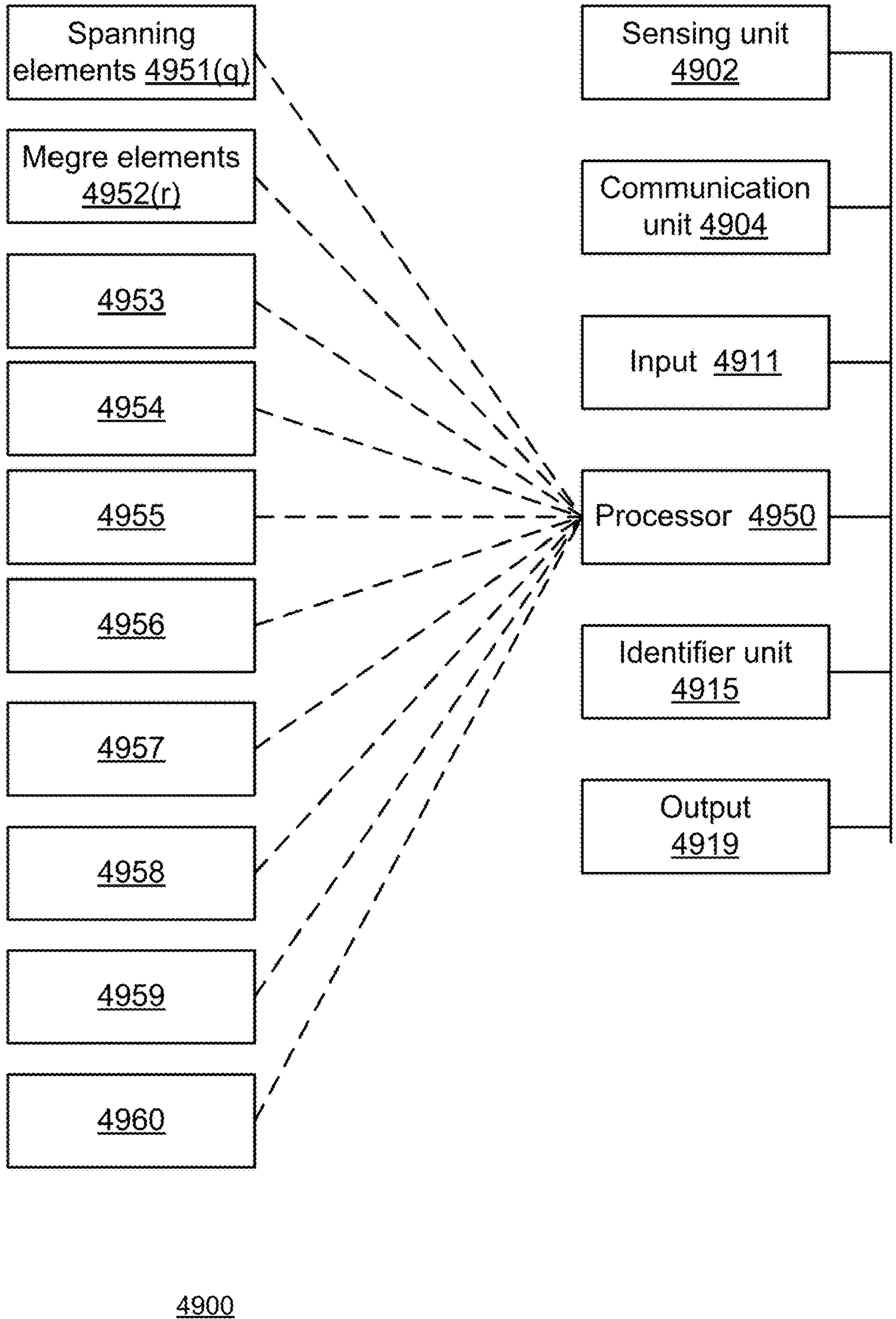

Spanning elements 4951(q)
Megre elements 4952(r)
4953
4954
4955
4956
4957
4958
4959
4960
Sensing unit 4902
Communication unit 4904
Input 4911
Processor 4950
Identifier unit 4915
Output 4919
4900

FIG. 2N

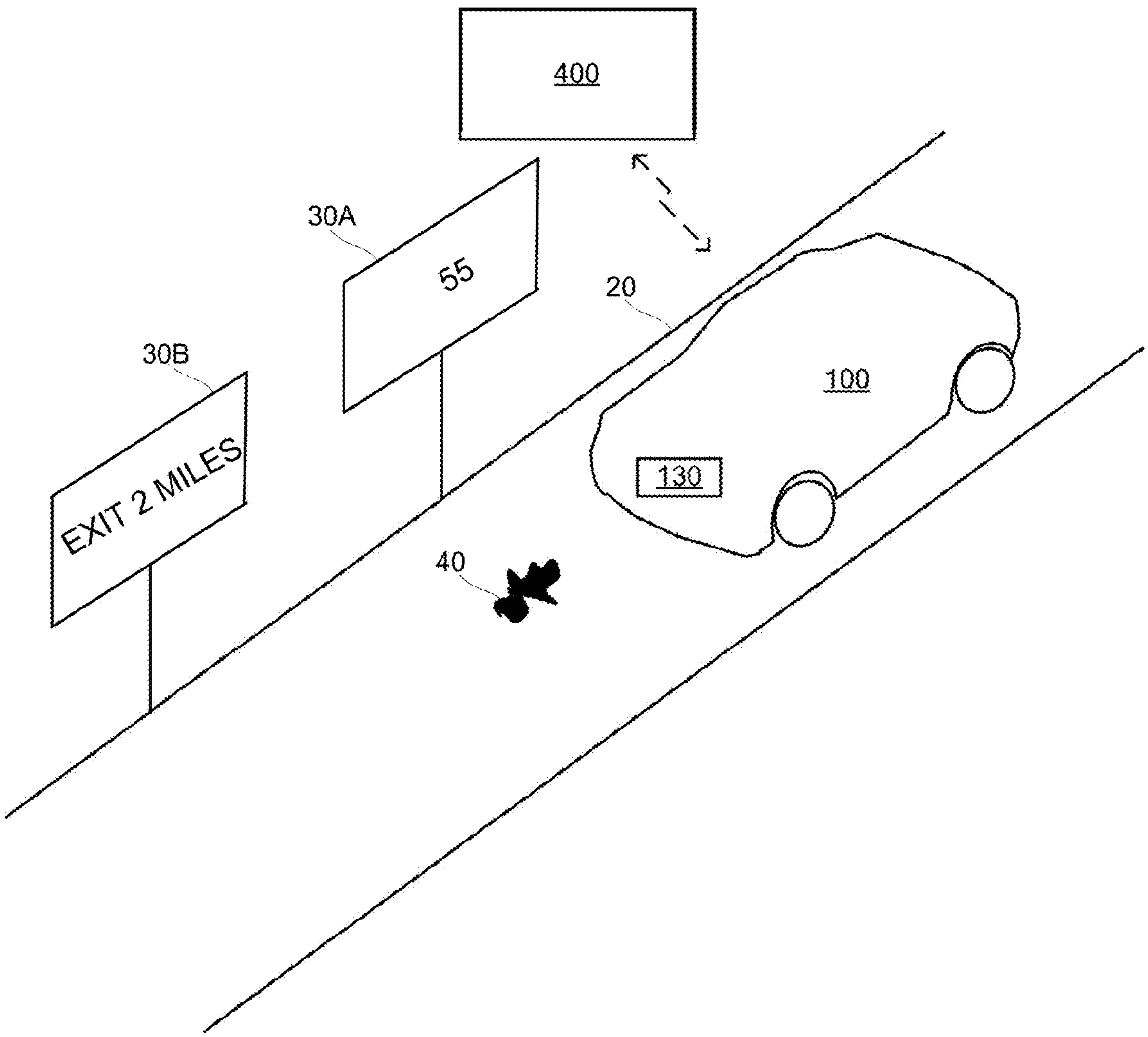
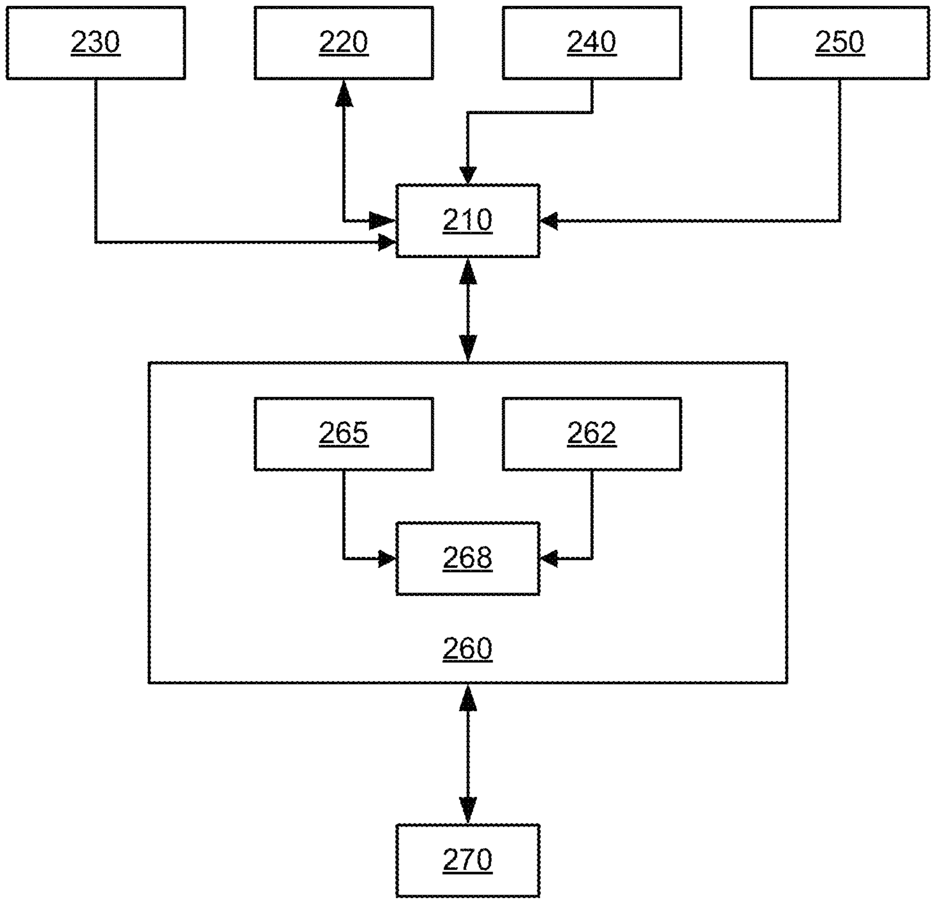


FIG. 3A



200

FIG. 3B

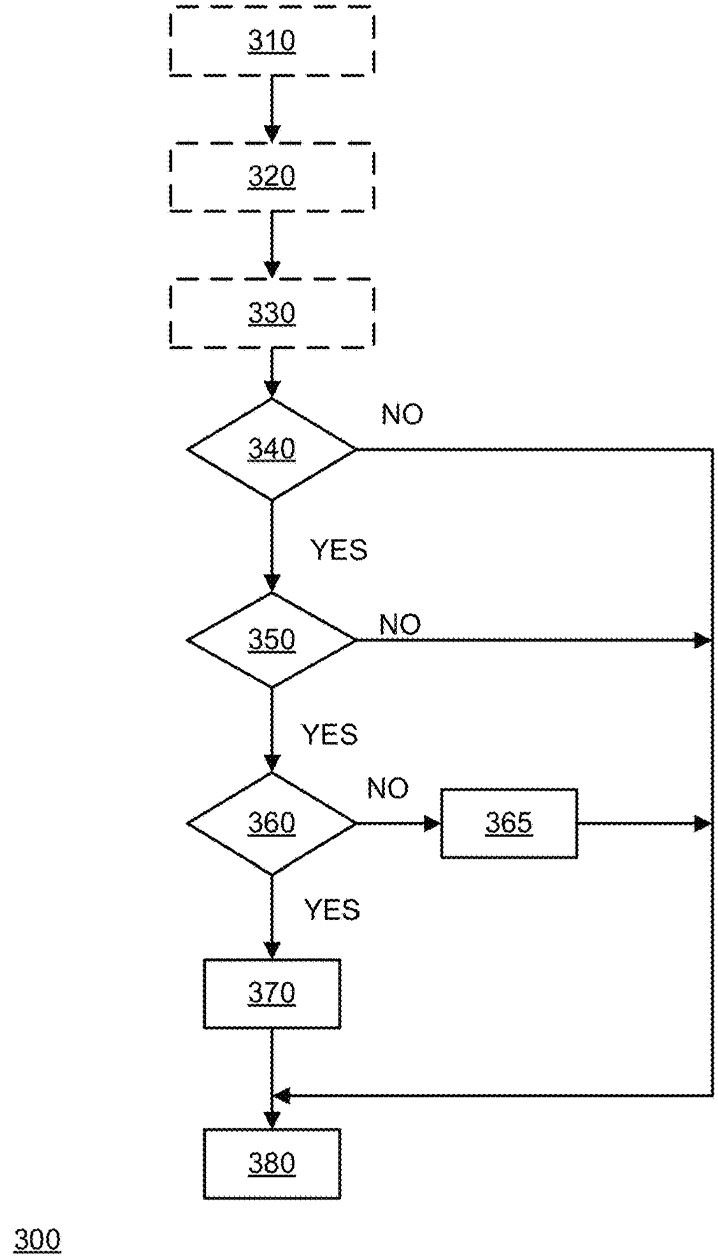
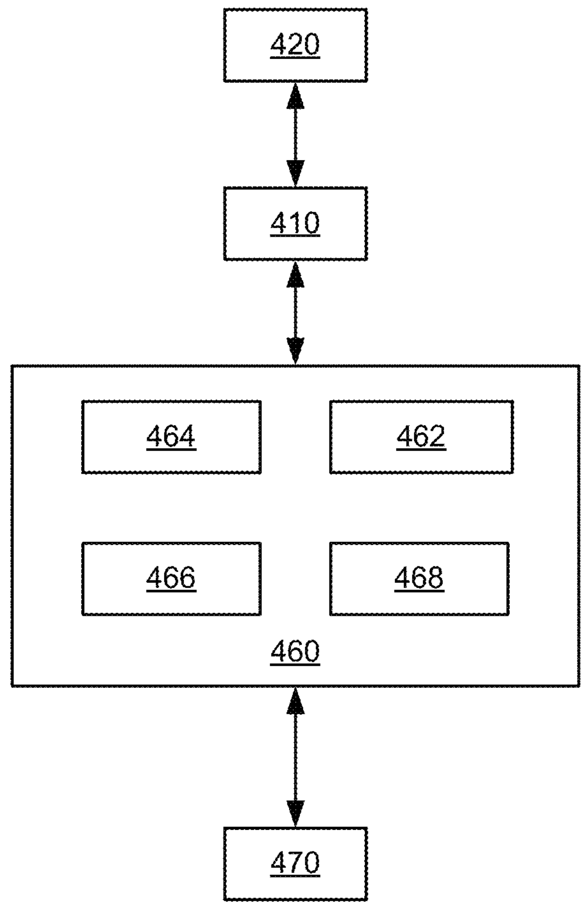


FIG. 3C



400

FIG. 4

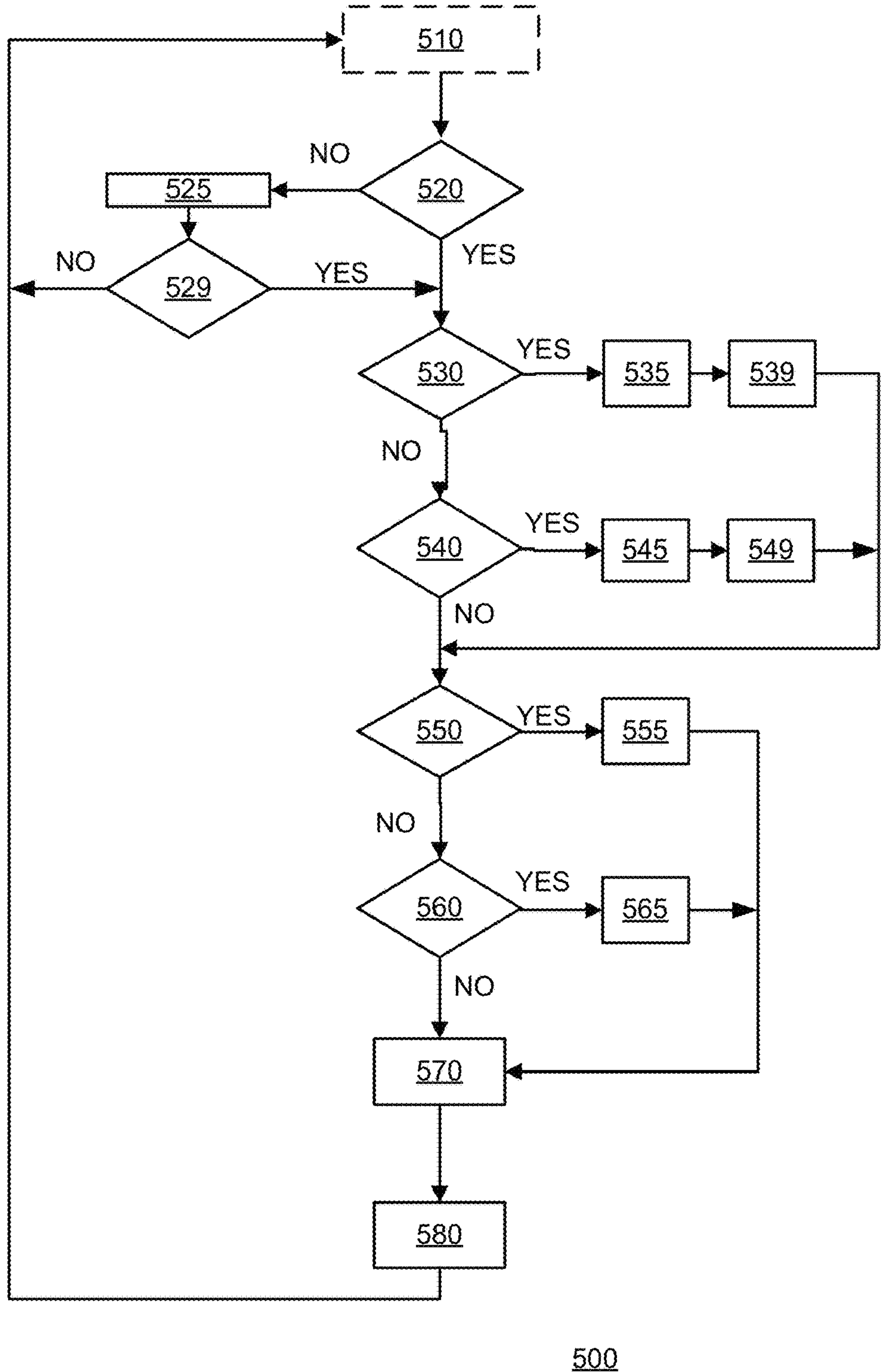
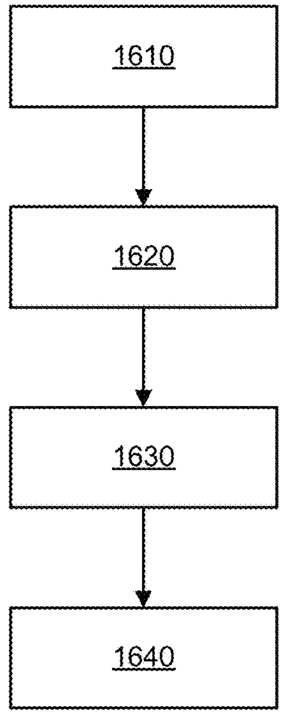

510
520
NO
525
YES
529
NO
YES
530
YES
535
539
NO
540
YES
545
549
NO
550
YES
555
NO
560
YES
565
NO
570
580
500

FIG. 5



1600

FIG. 6

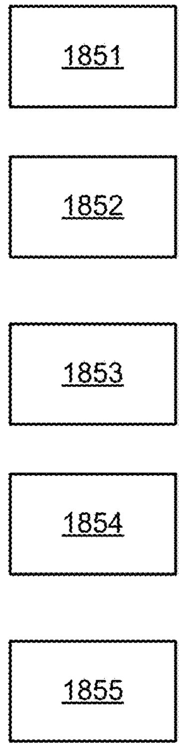


FIG. 7

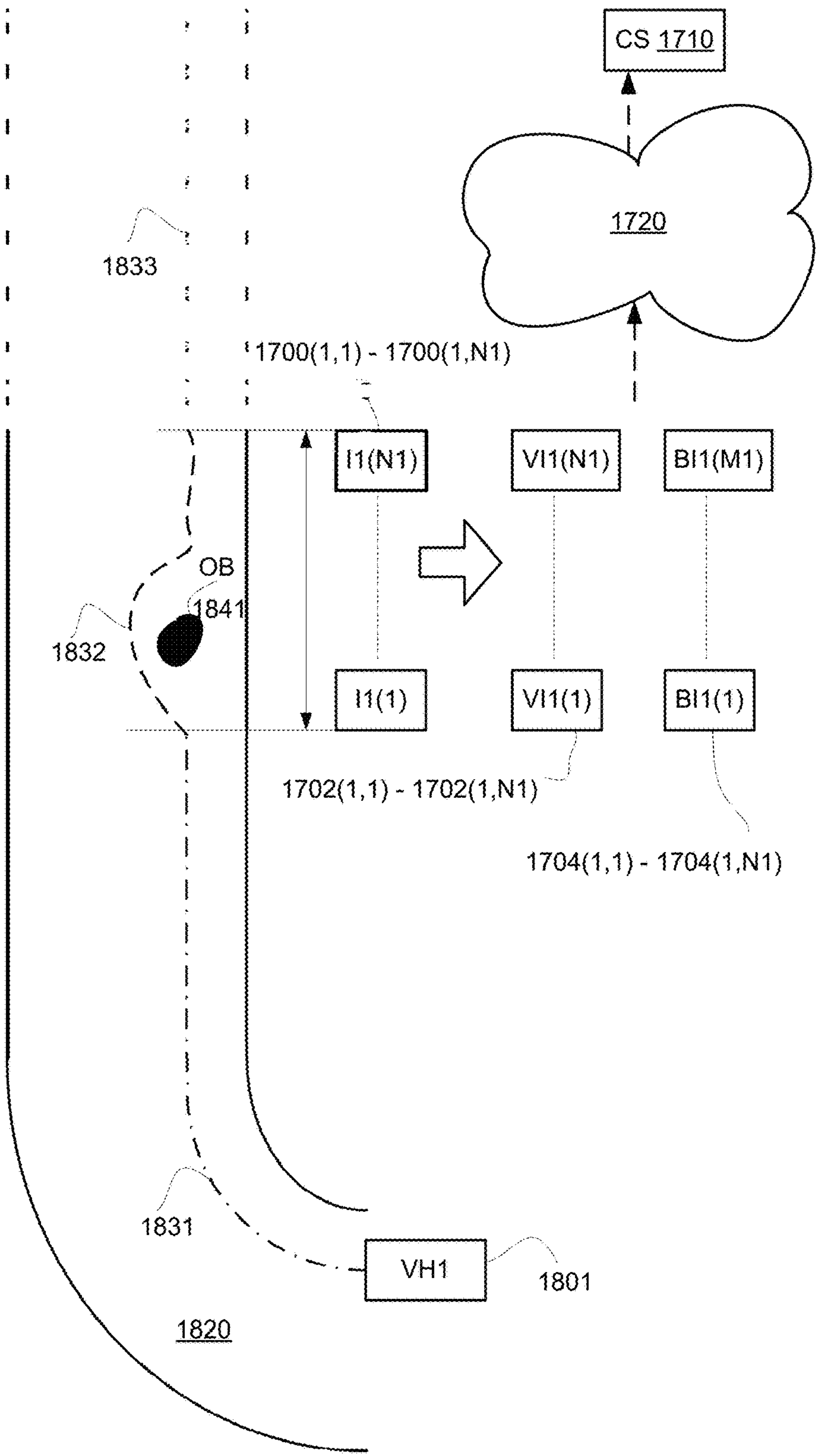

CS 1710
1720
1833
1700(1,1) - 1700(1,N1)
I1(N1)
VI1(N1)
BI1(M1)
OB
1841
1832
I1(1)
VI1(1)
BI1(1)
1702(1,1) - 1702(1,N1)
1704(1,1) - 1704(1,N1)
1831
VH1
1801
1820

FIG. 8

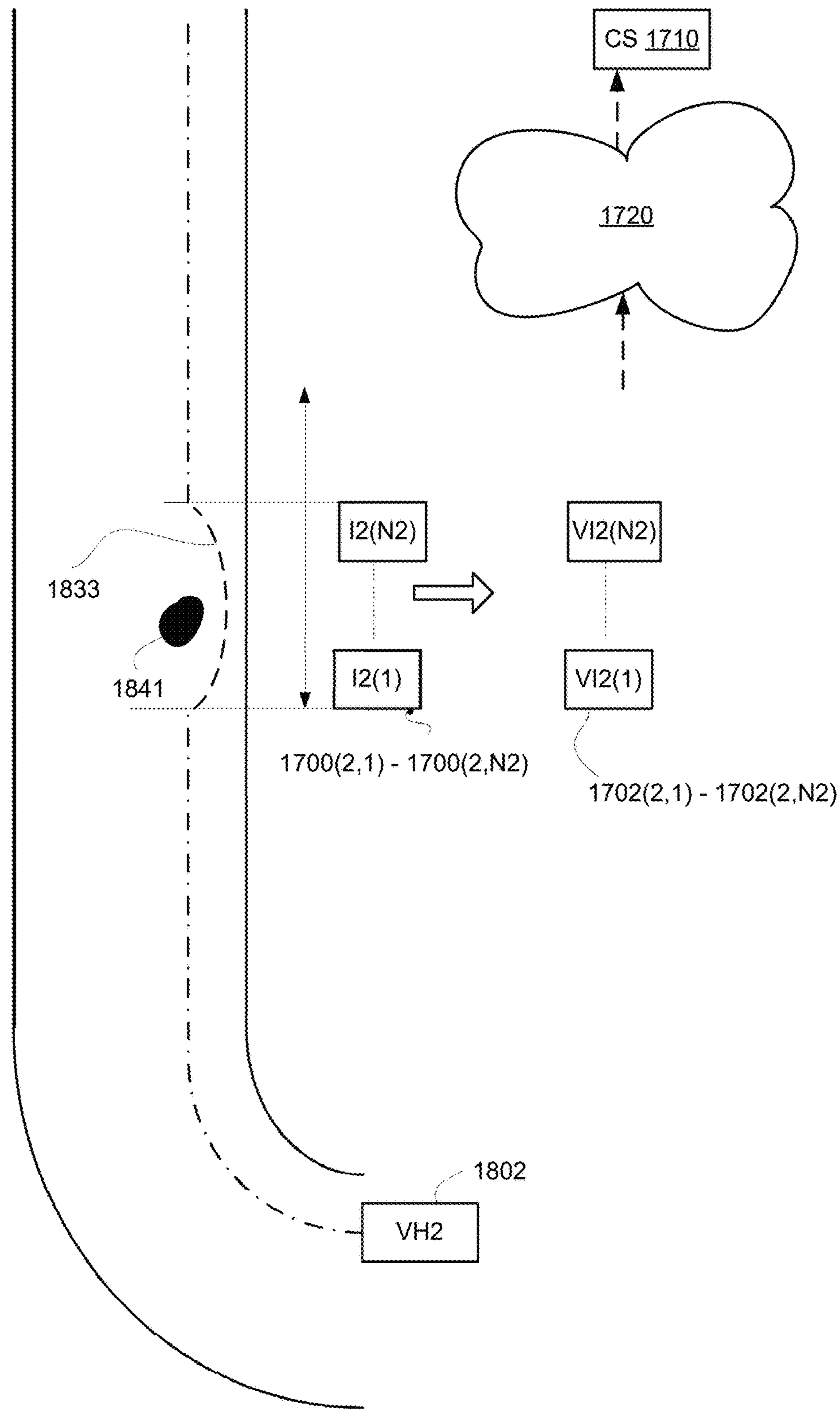

CS 1710
1720
I2(N2)
VI2(N2)
1833
1841
I2(1)
VI2(1)
1700(2,1) - 1700(2,N2)
1702(2,1) - 1702(2,N2)
1802
VH2

FIG. 9

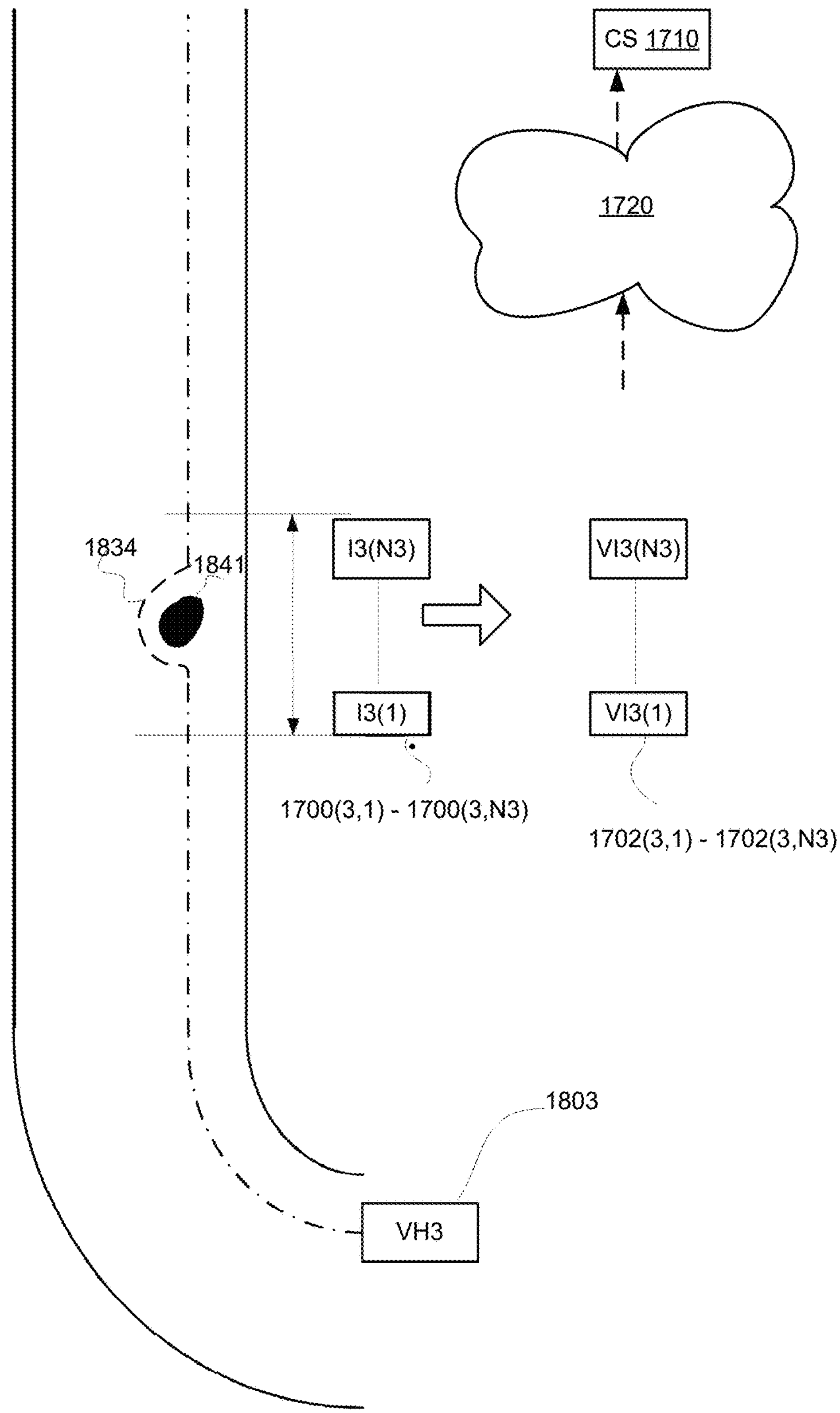

CS 1710
1720
1834
1841
I3(N3)
VI3(N3)
I3(1)
VI3(1)
1700(3,1) - 1700(3,N3)
1702(3,1) - 1702(3,N3)
1803
VH3

FIG. 10

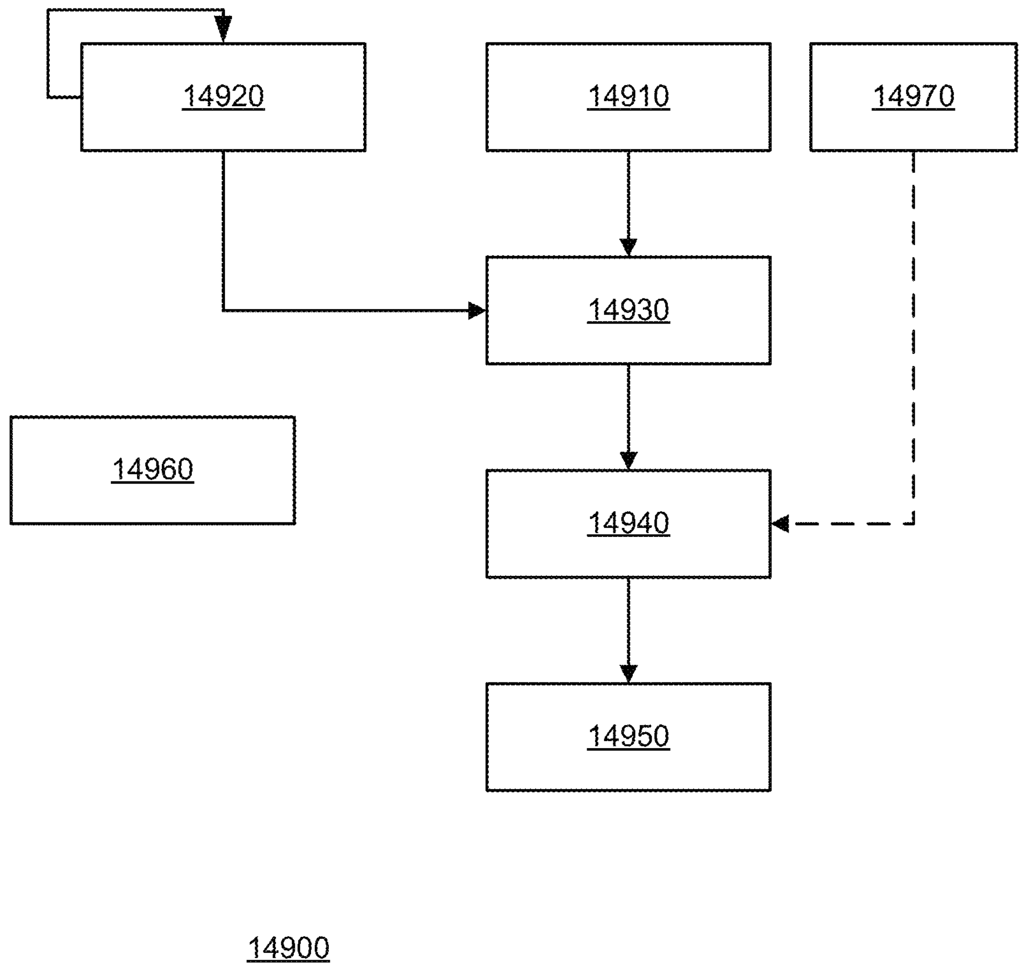
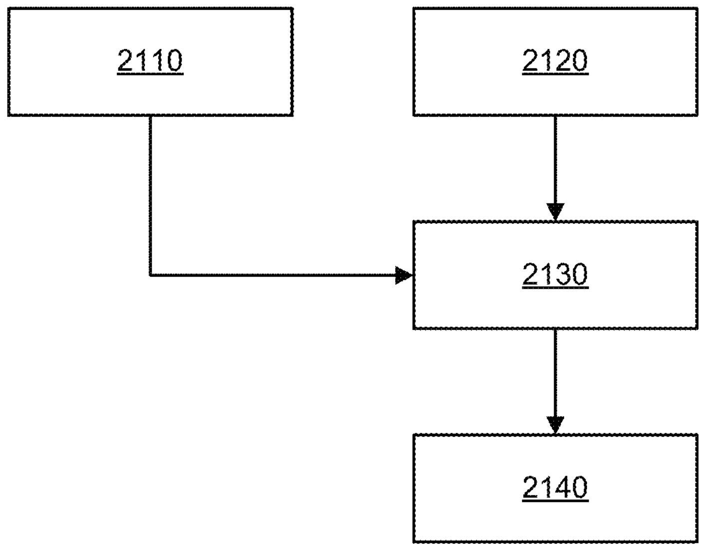


FIG. 11



2100

FIG. 12

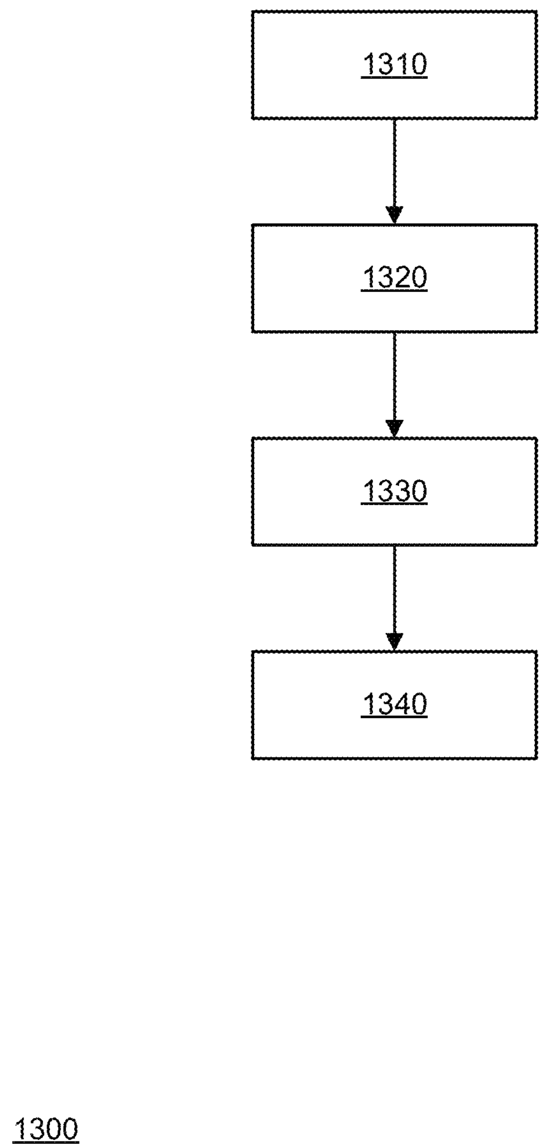


FIG. 13

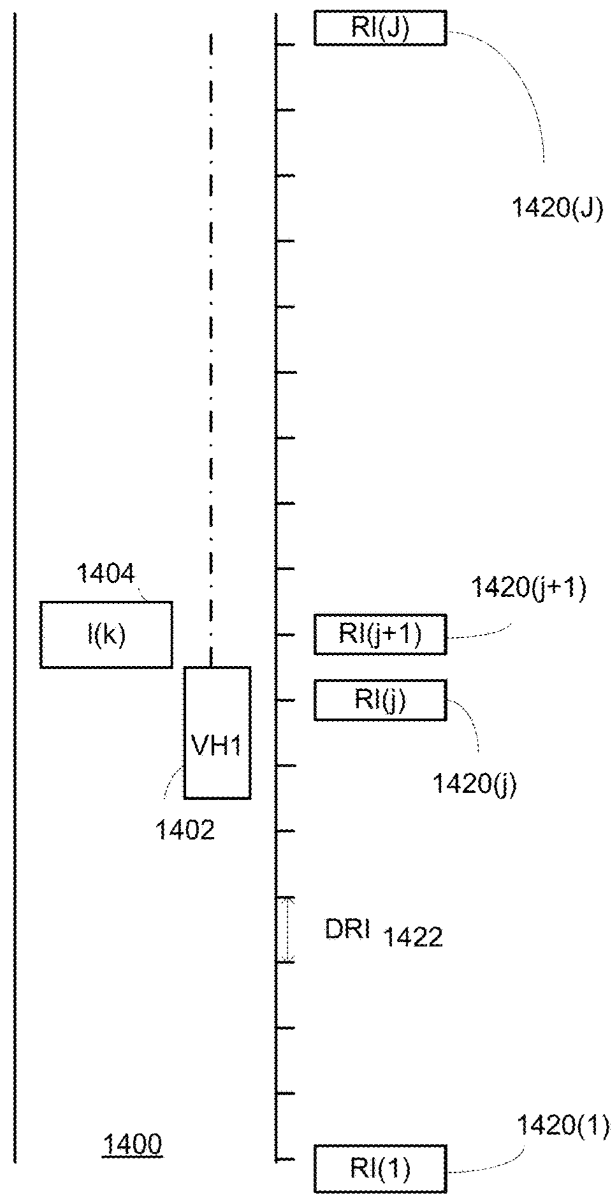

RI(J)
1420(J)
1404
I(k)
1420(j+1)
RI(j+1)
RI(j)
VH1
1420(j)
1402
DRI
1422
1420(1)
1400
RI(1)

FIG. 14

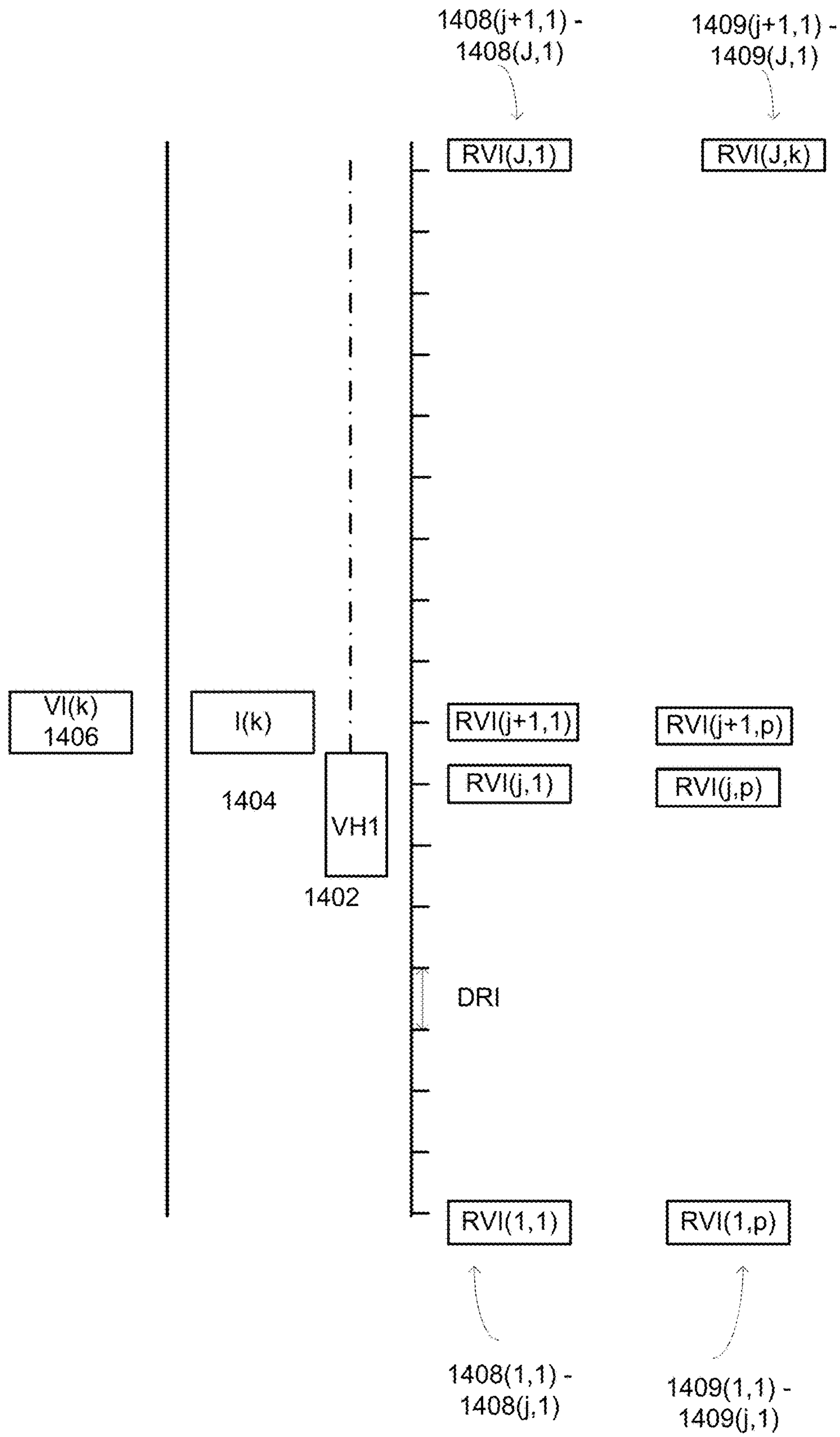

1408(j+1,1) -
1408(J,1)
1409(j+1,1) -
1409(J,1)
RVI(J,1)
RVI(J,k)
VI(k)
1406
I(k)
RVI(j+1,1)
RVI(j+1,p)
RVI(j,1)
RVI(j,p)
1404
VH1
1402
DRI
RVI(1,1)
RVI(1,p)
1408(1,1) -
1408(j,1)
1409(1,1) -
1409(j,1)

FIG. 15

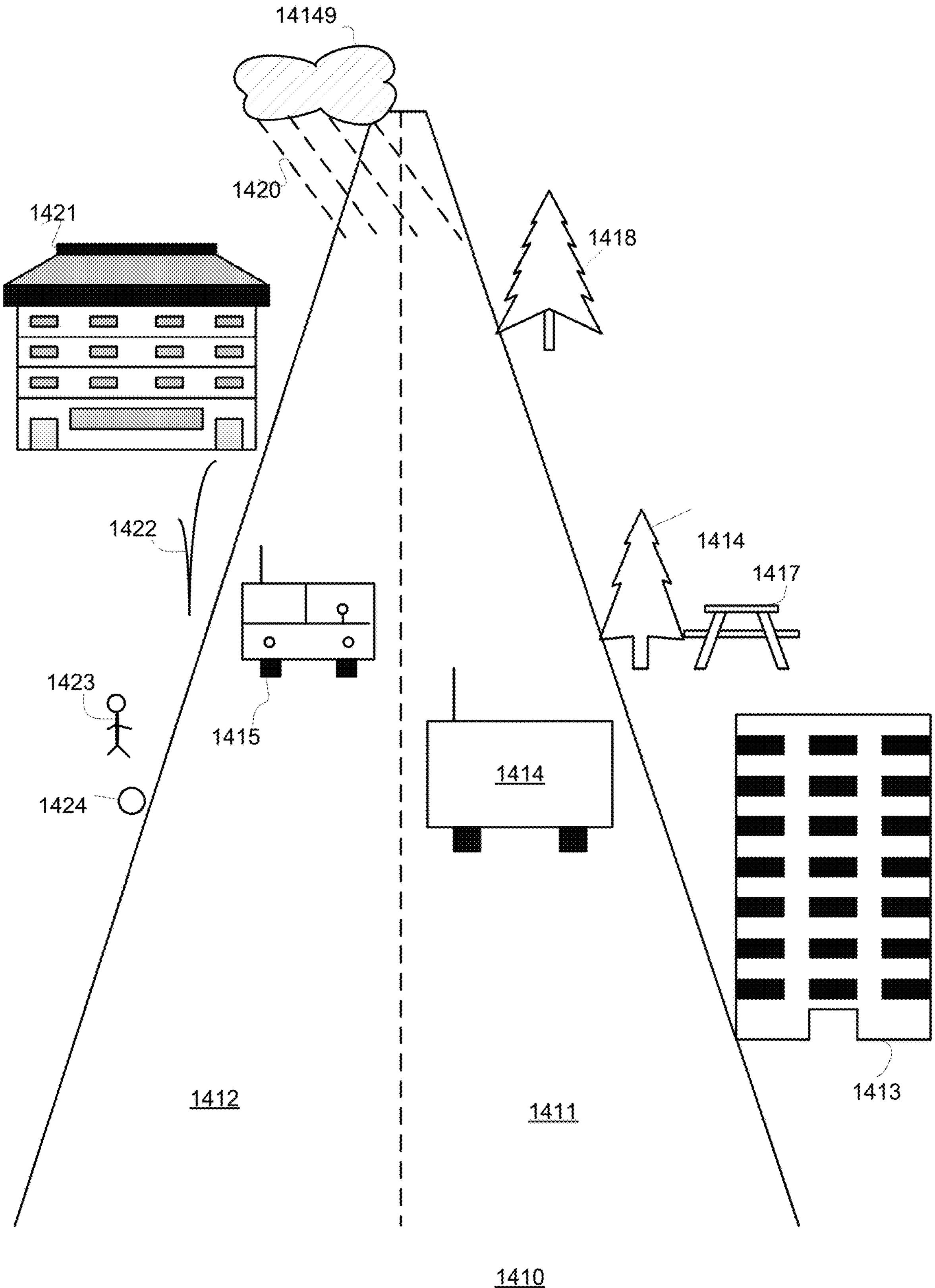

14149
1420
1421
1418
1422
1414
1417
1423
1415
1414
1424
1413
1412
1411
1410

FIG. 16

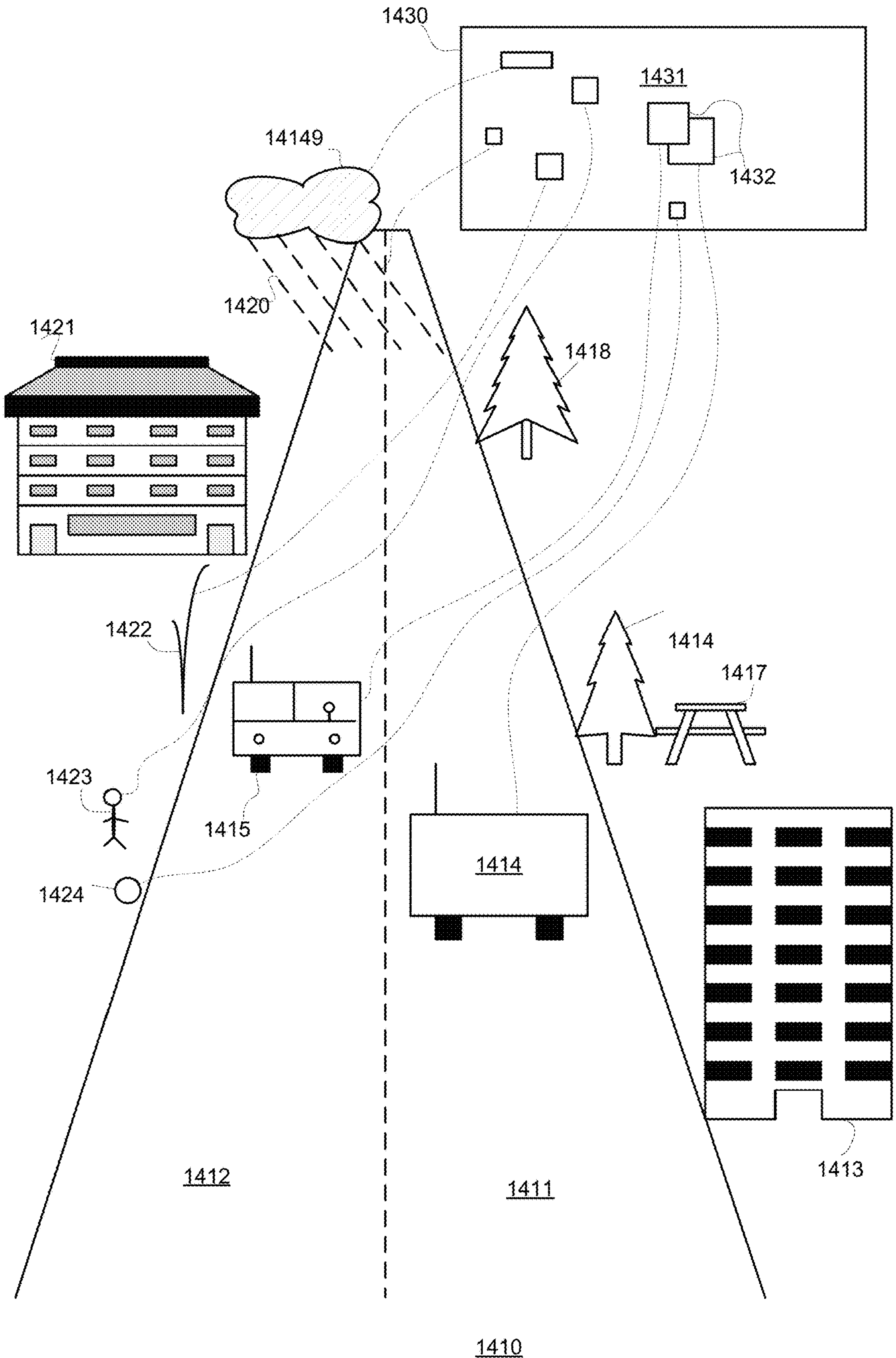

1430
1431
1432
14149
1420
1421
1418
1422
1414
1417
1423
1415
1414
1424
1413
1412
1411
1410

FIG. 17

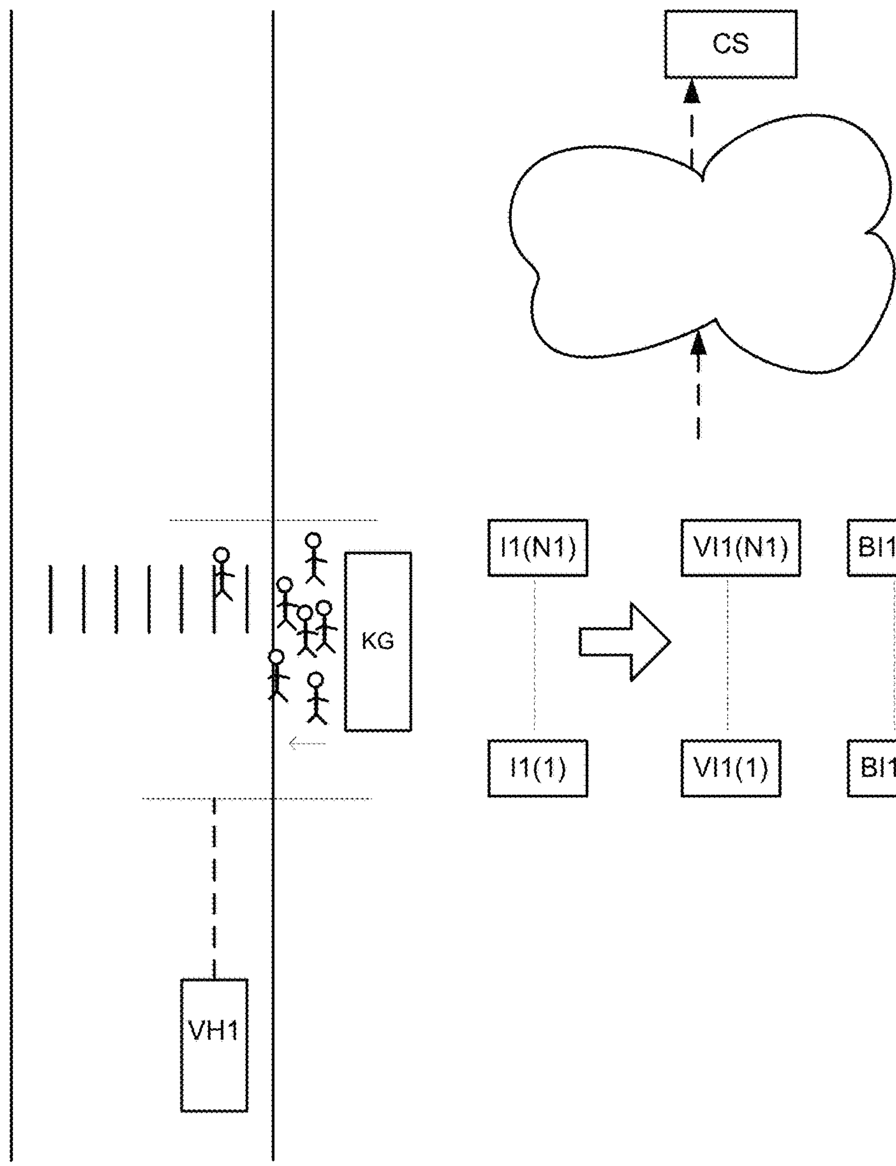


FIG. 18

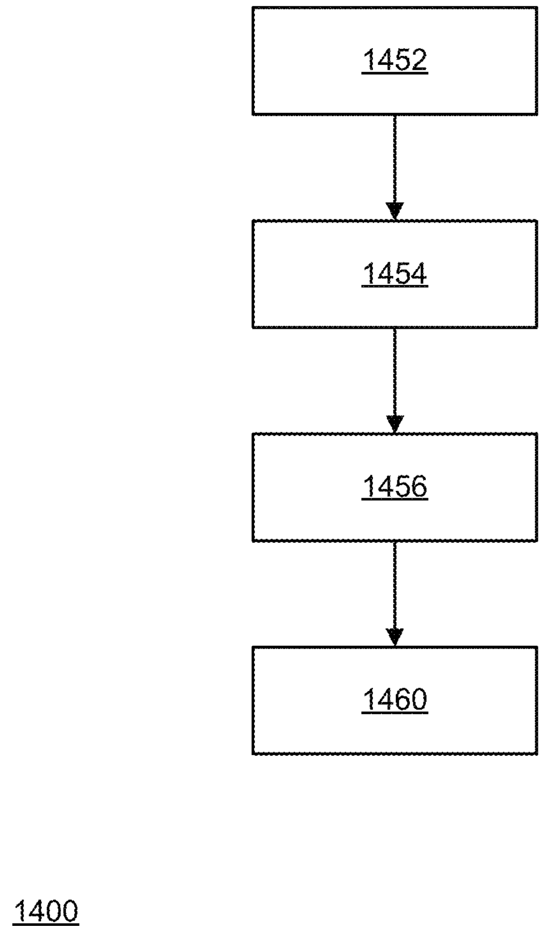


FIG. 19

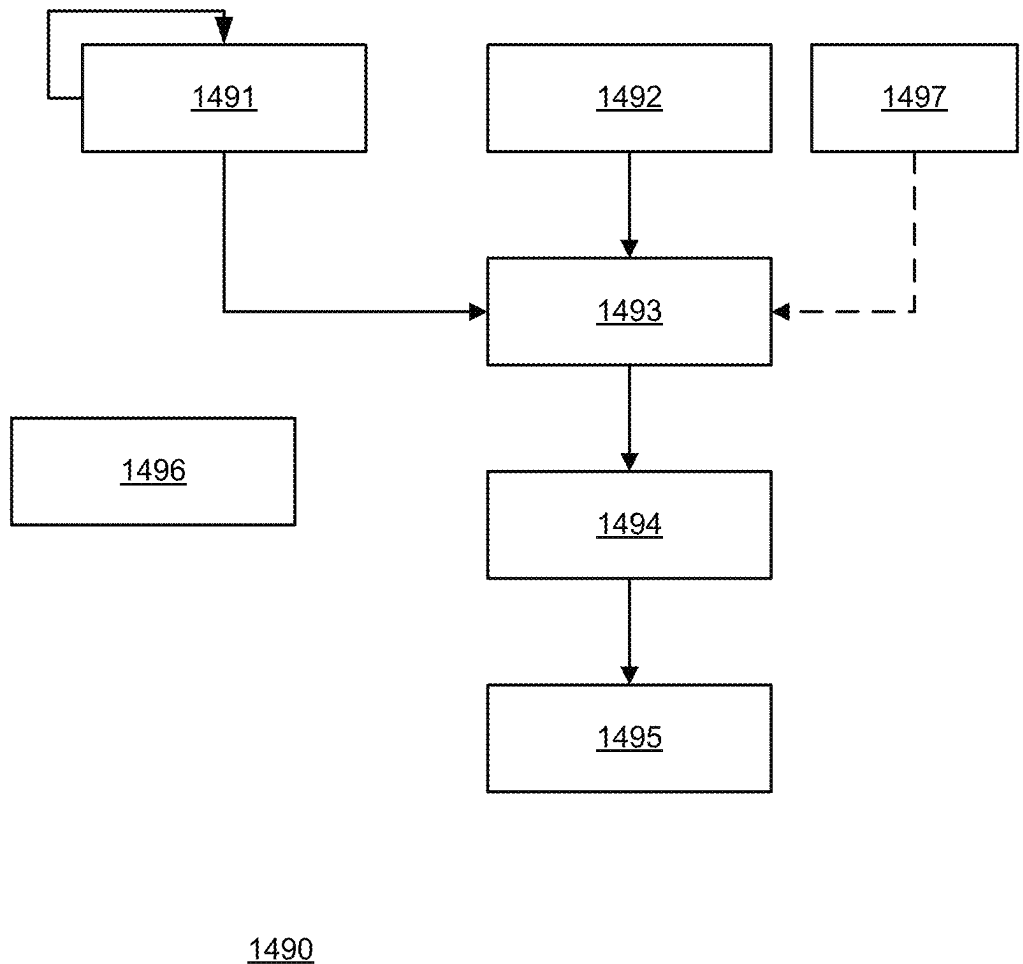


FIG. 20

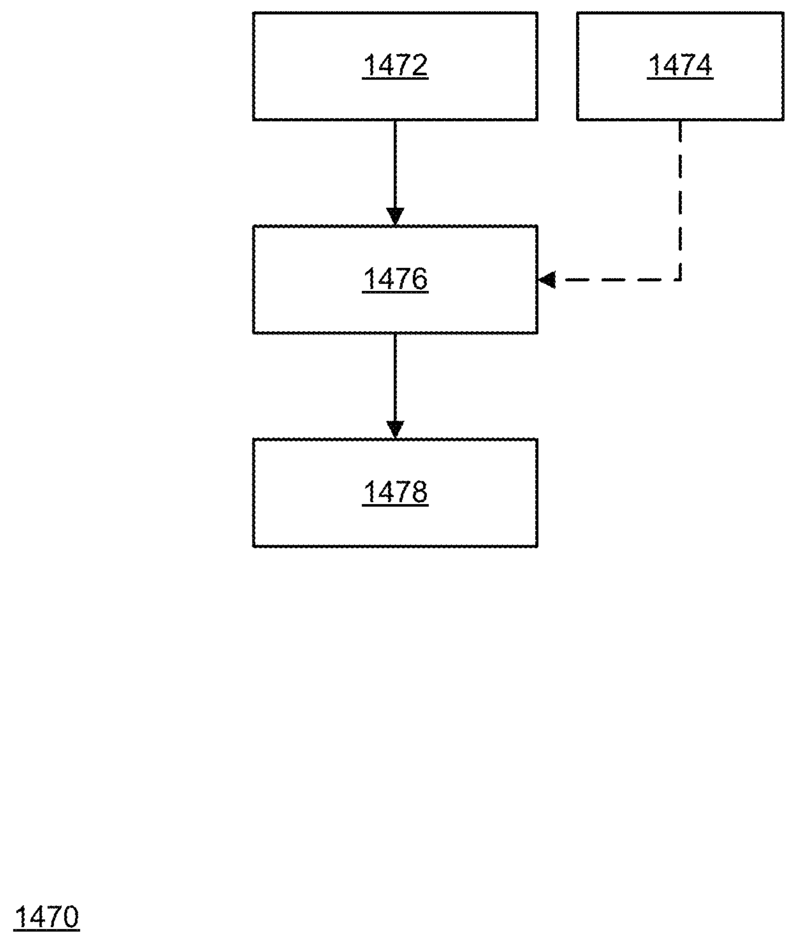


FIG. 21

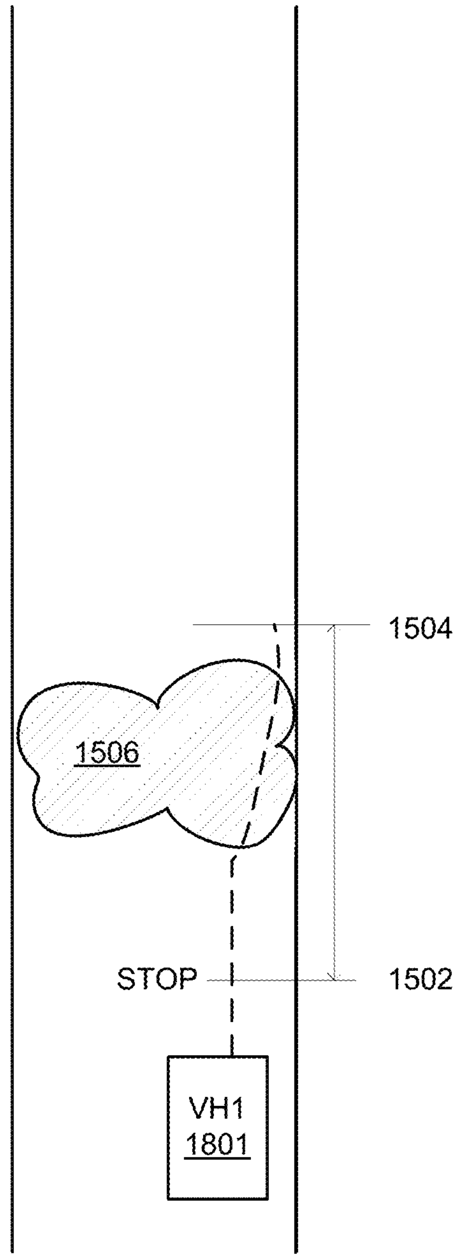


FIG. 22

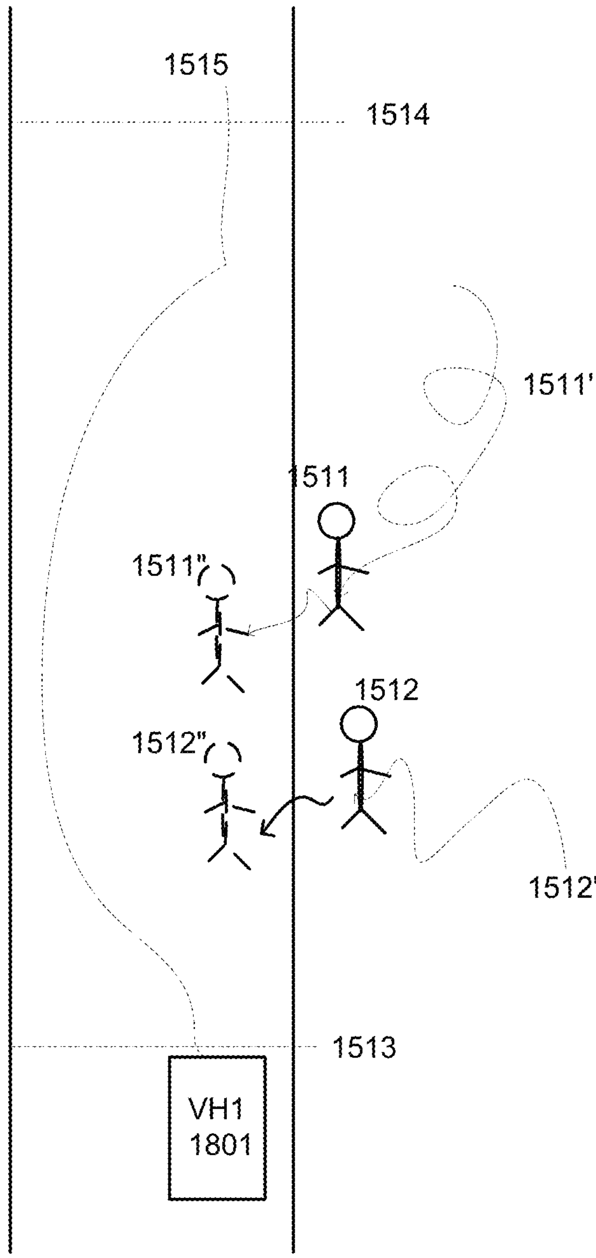


FIG. 23

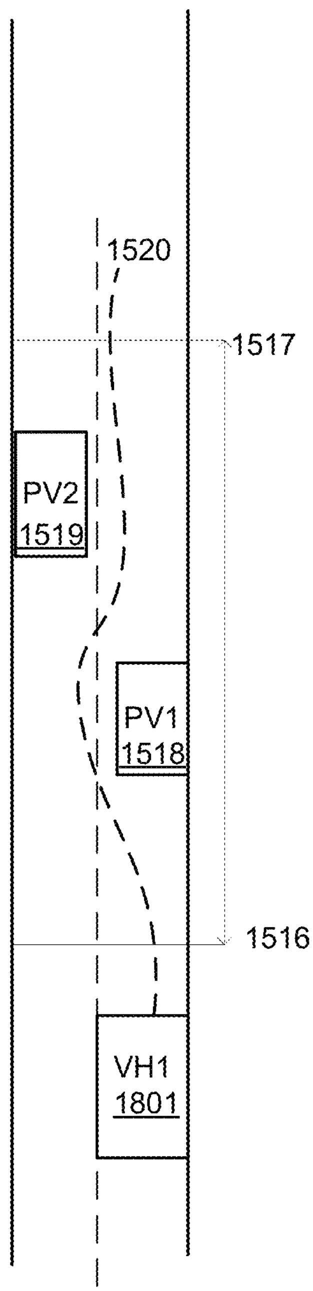


FIG. 24

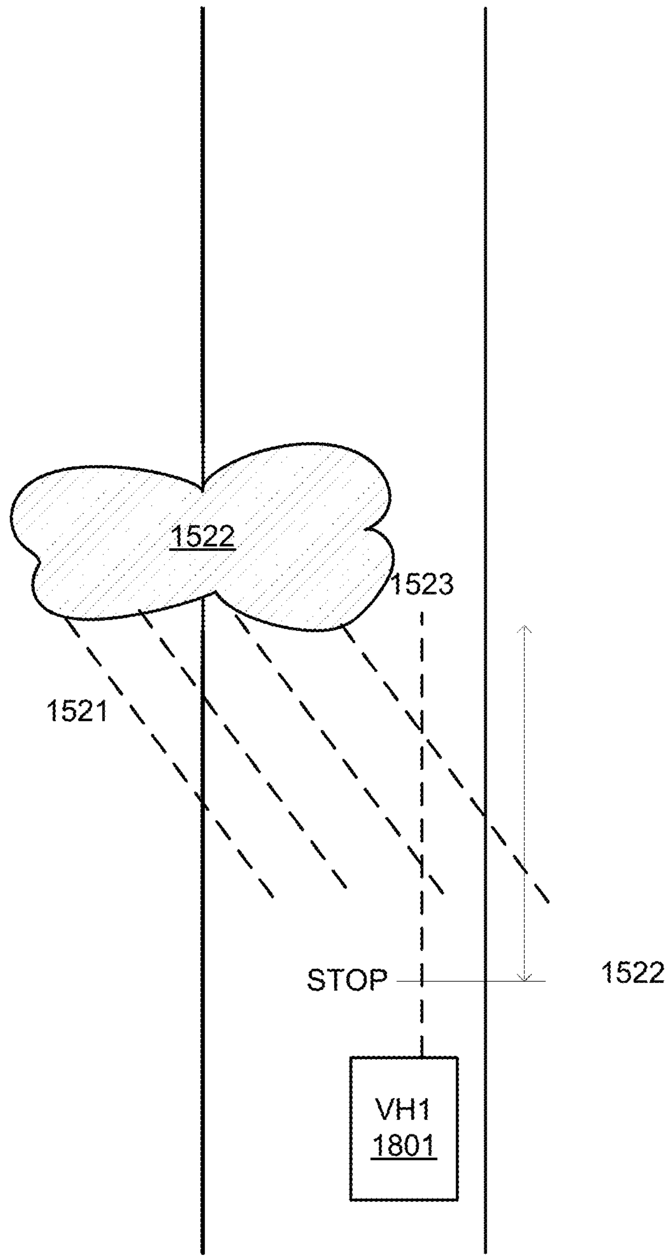


FIG. 25

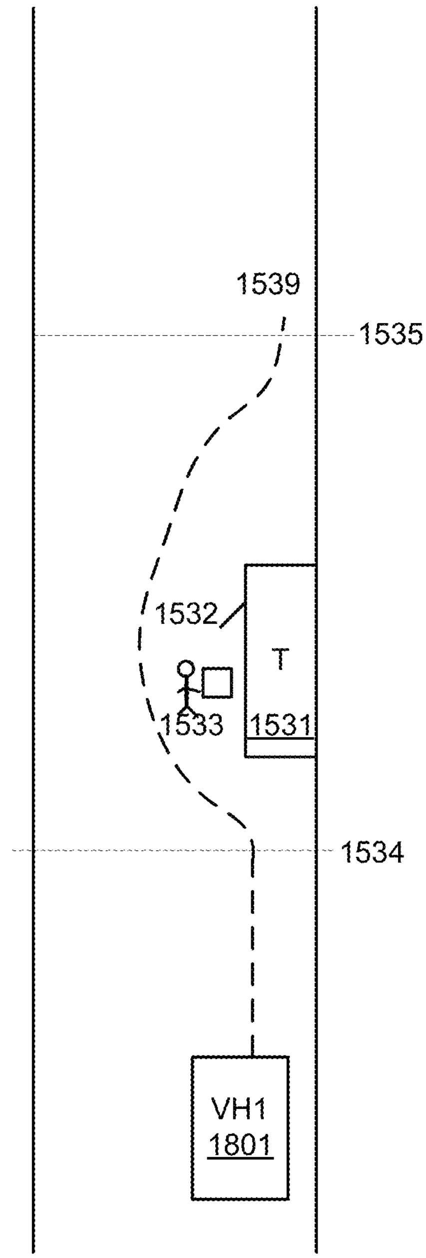


FIG. 26

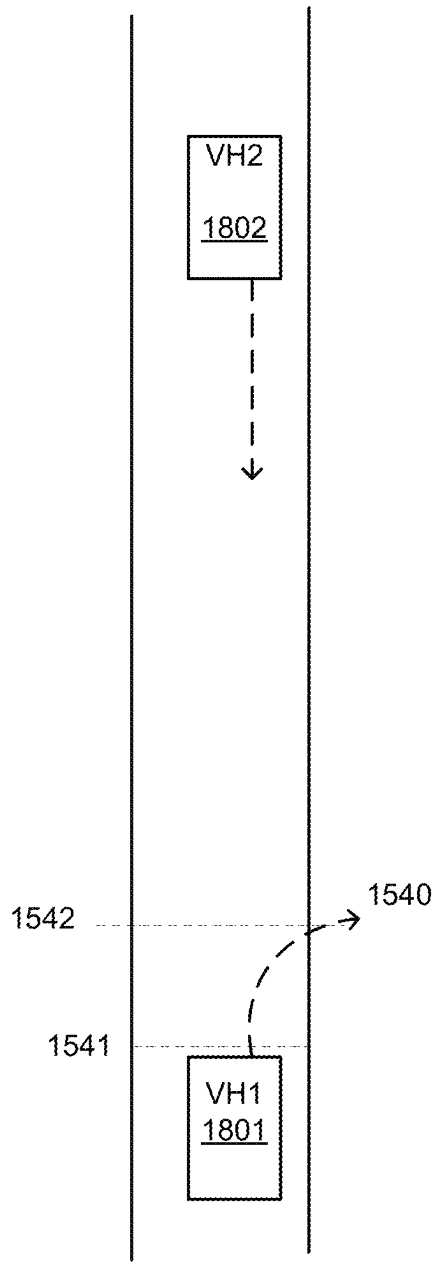
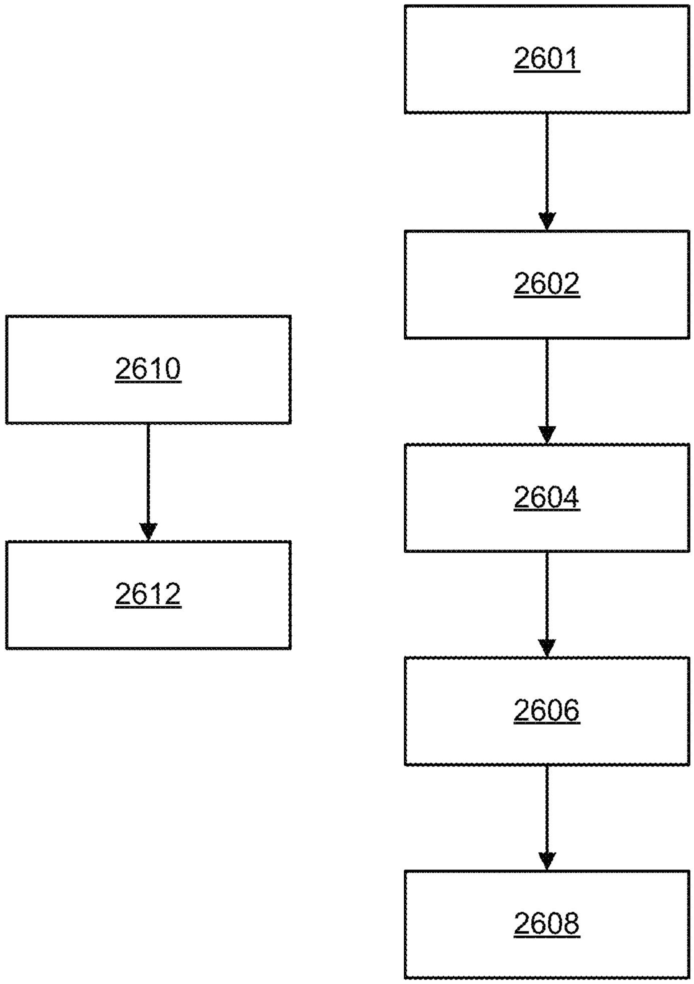


FIG. 27



2600

FIG. 28

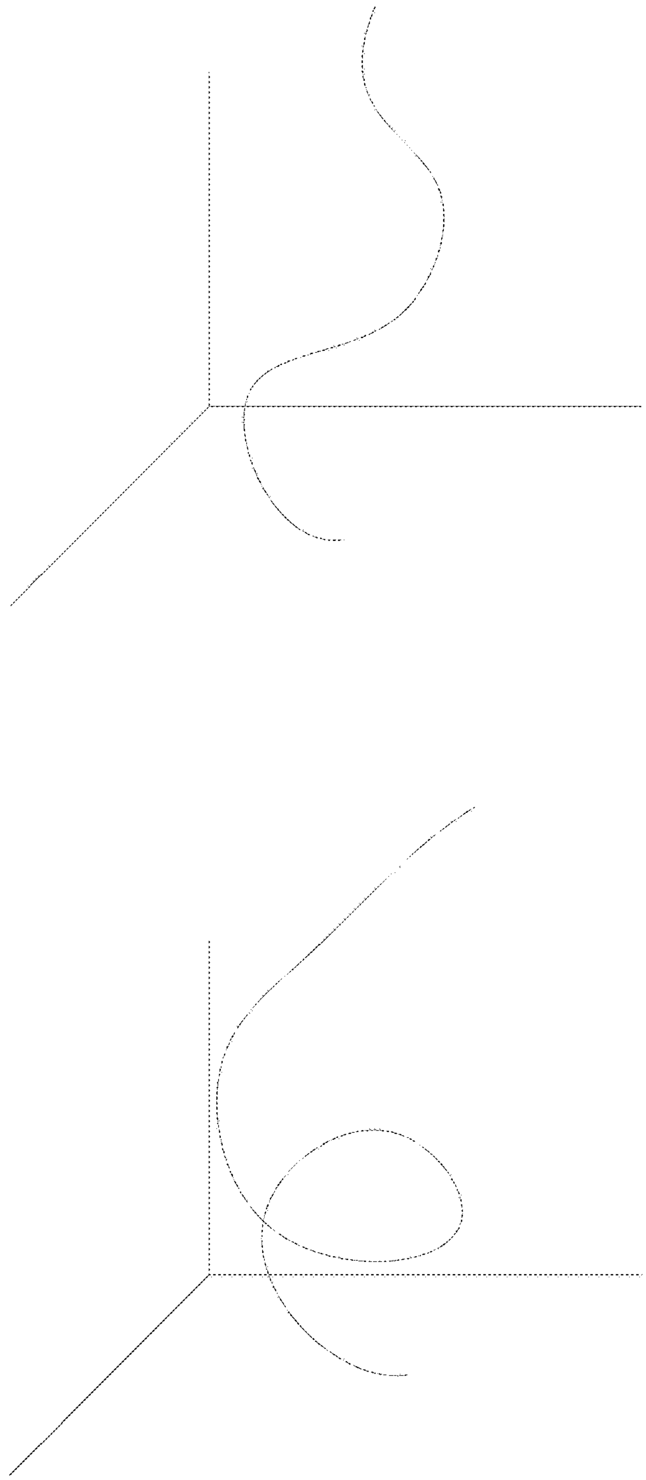
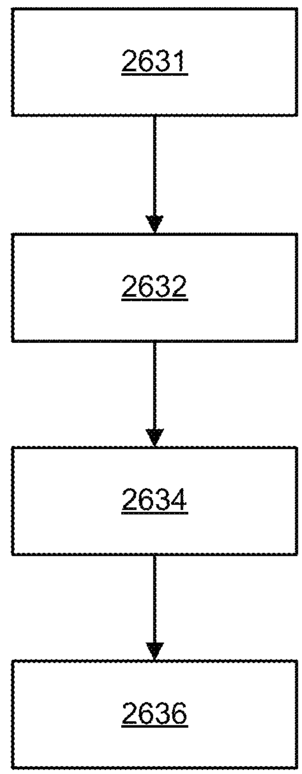
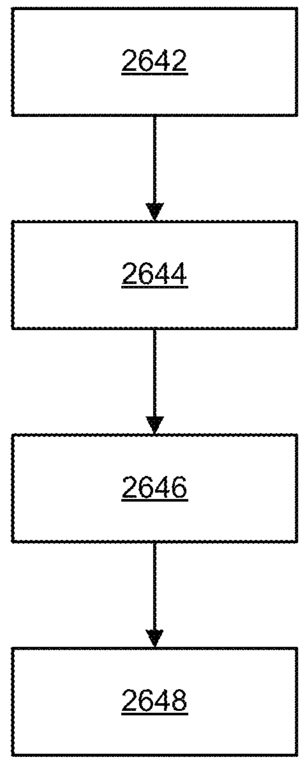


FIG. 29



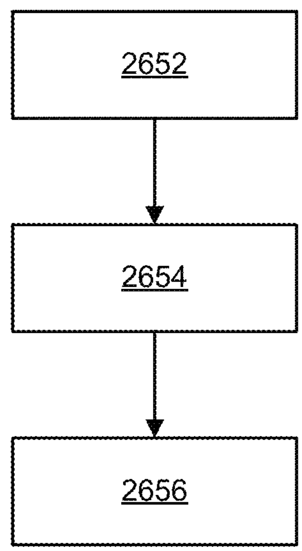
2630

FIG. 30



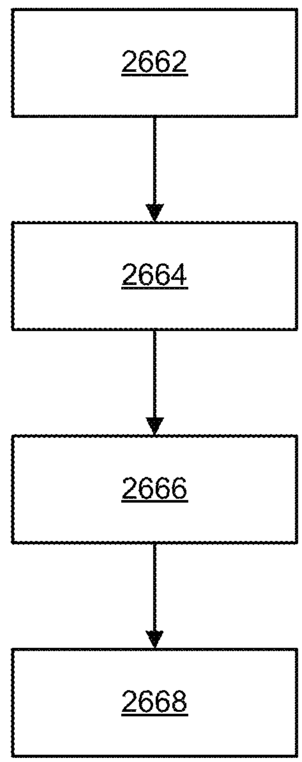
2640

FIG. 31



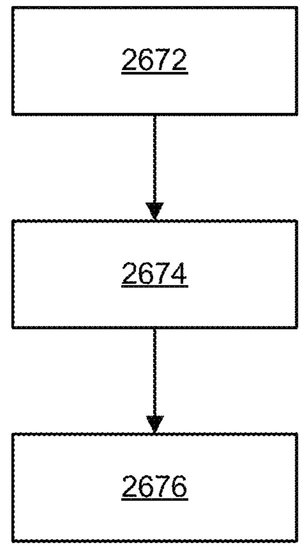
2650

FIG. 32



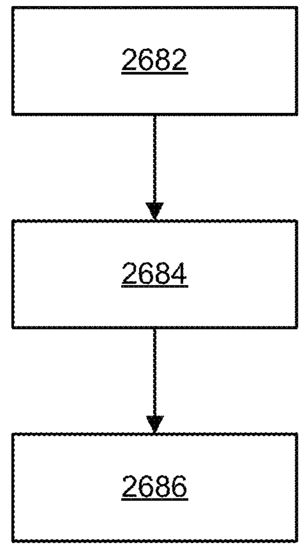
2660

FIG. 33



2670

FIG. 34



2680

FIG. 35

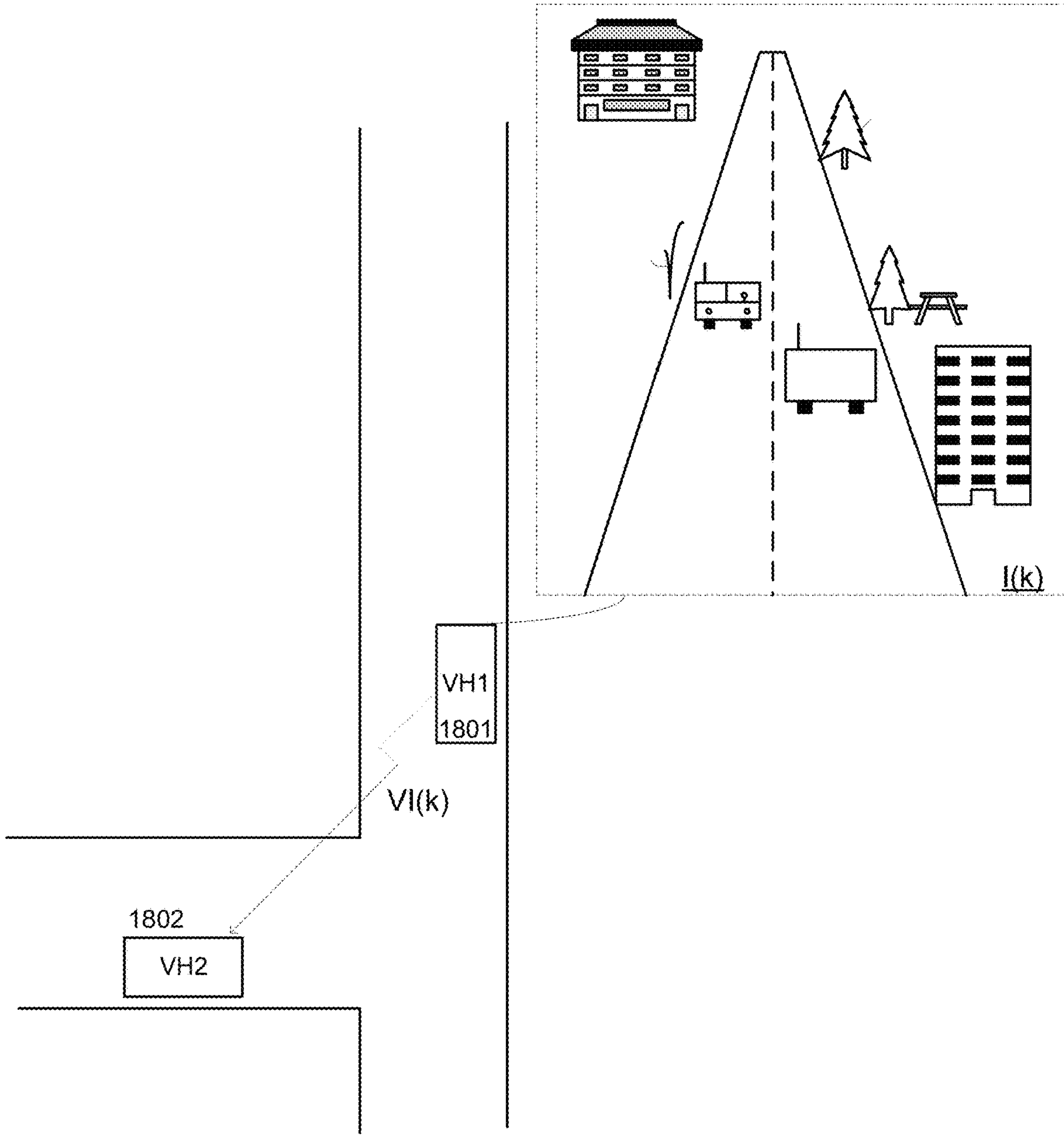
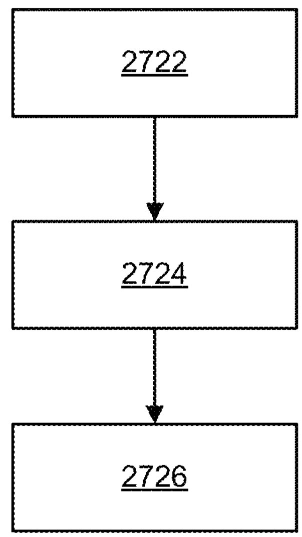


FIG. 36



2720

FIG. 37

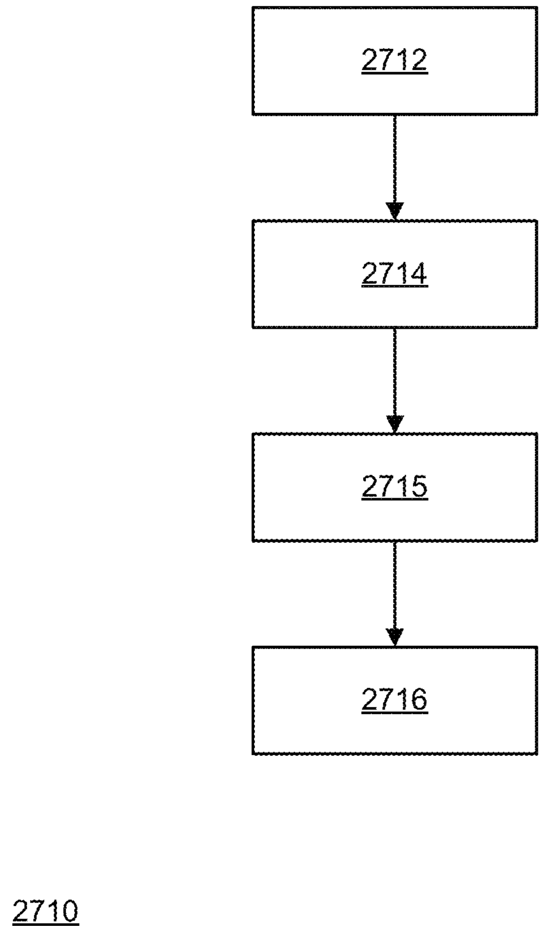


FIG. 38

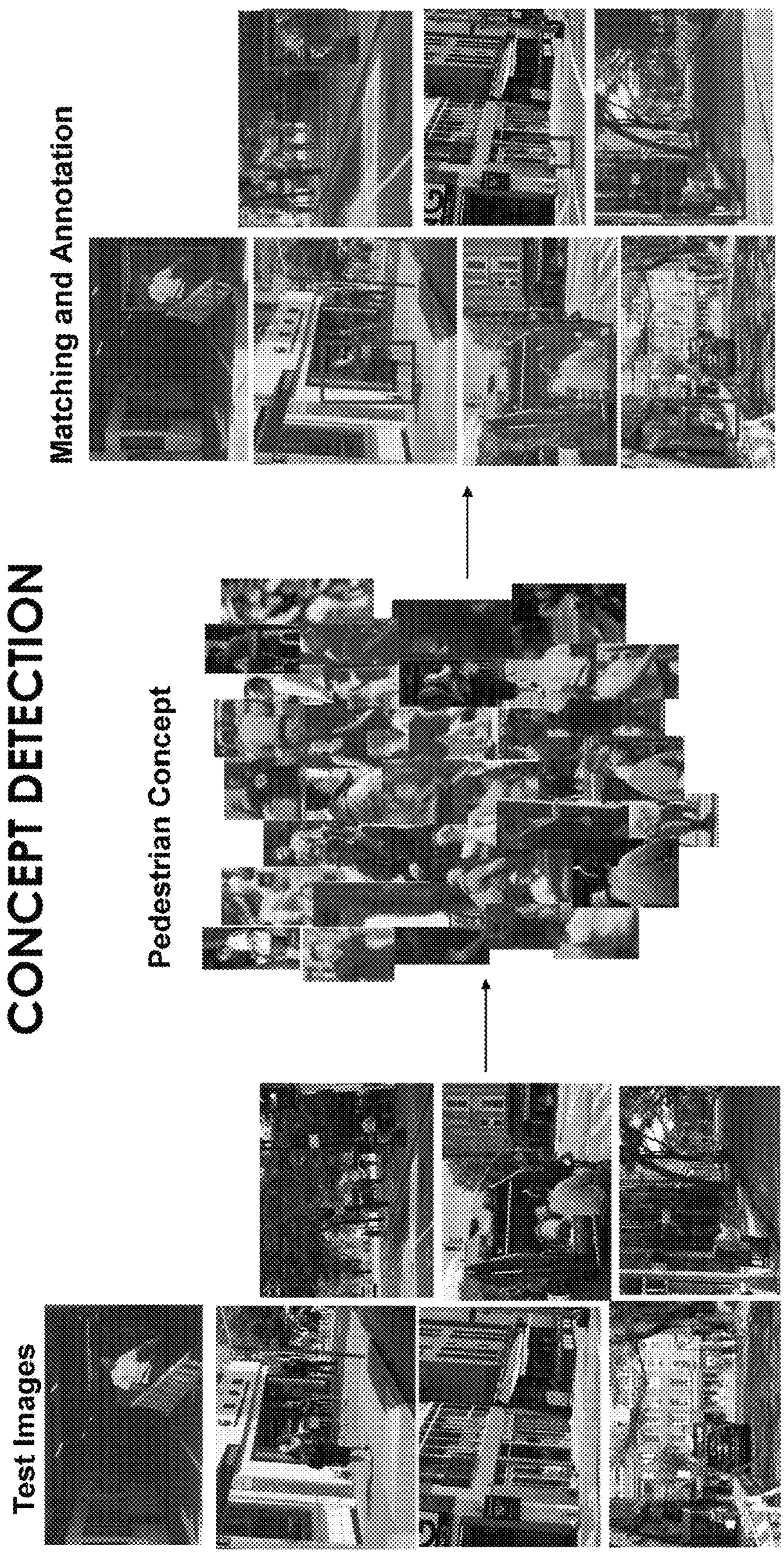

CONCEPT DETECTION
Test Images
Pedestrian Concept
Matching and Annotation

FIG. 39

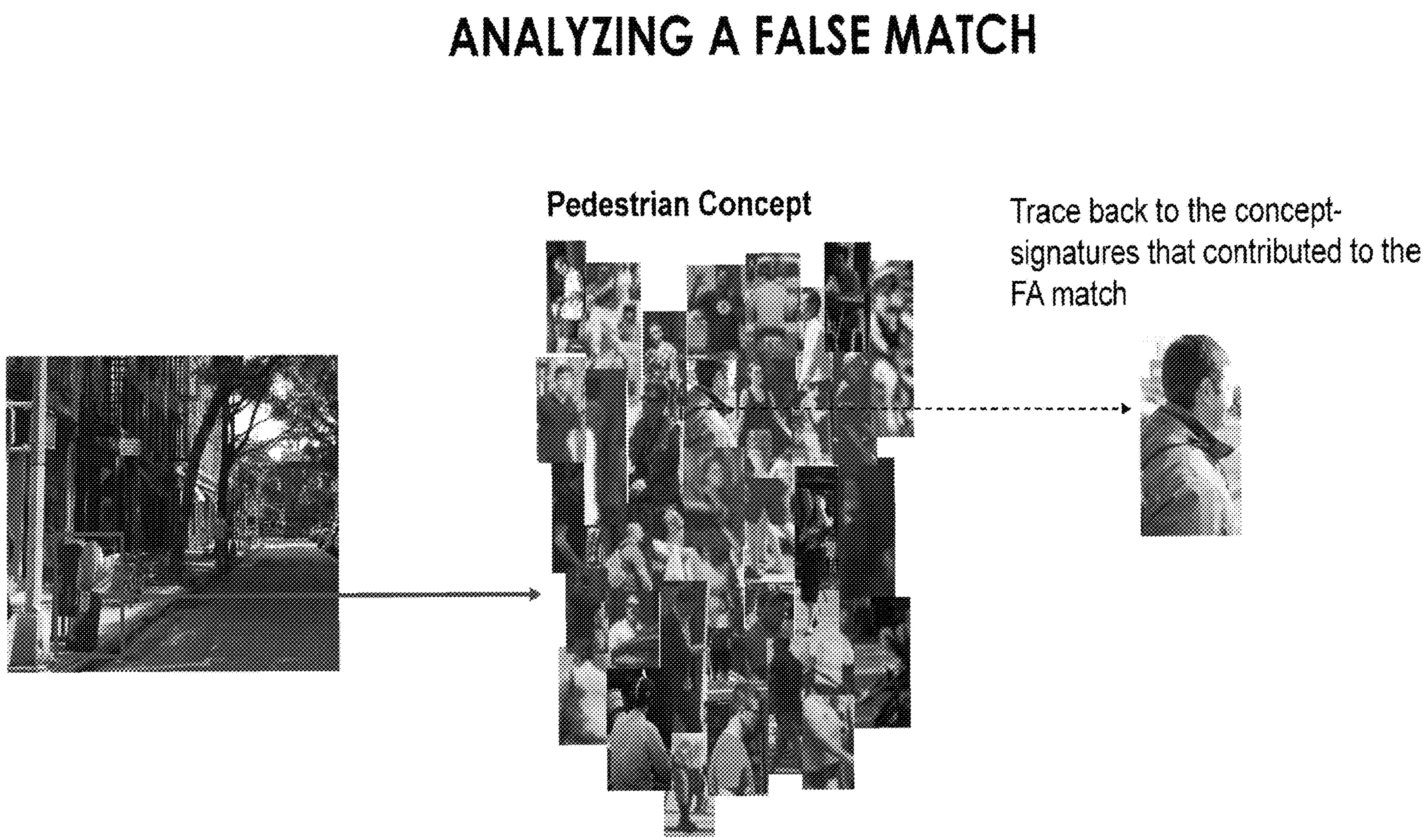

ANALYZING A FALSE MATCH
Pedestrian Concept
Trace back to the concept-signatures that contributed to the FA match

FIG. 40

GENERALIZATION THE FA

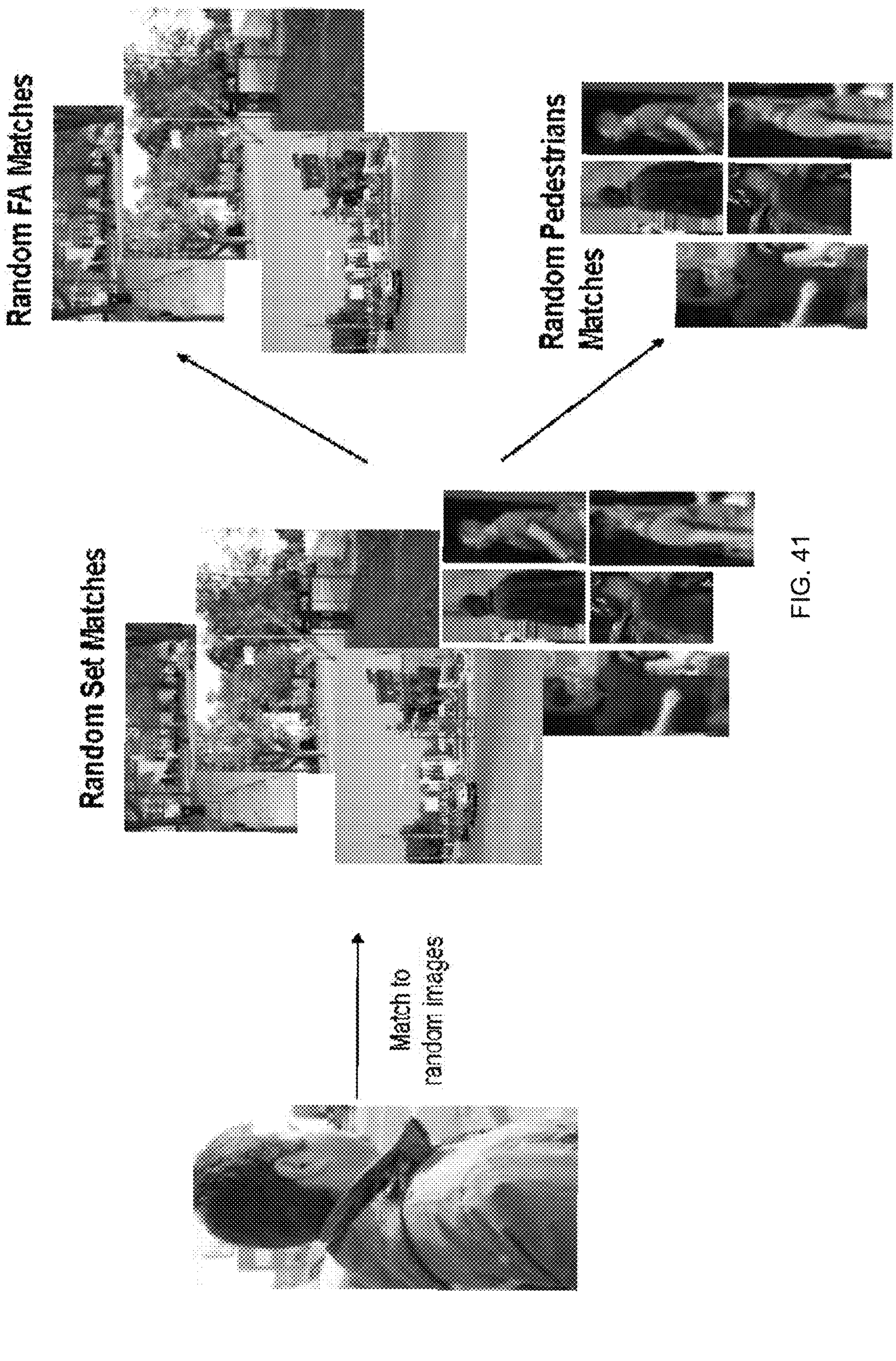


FIG. 41

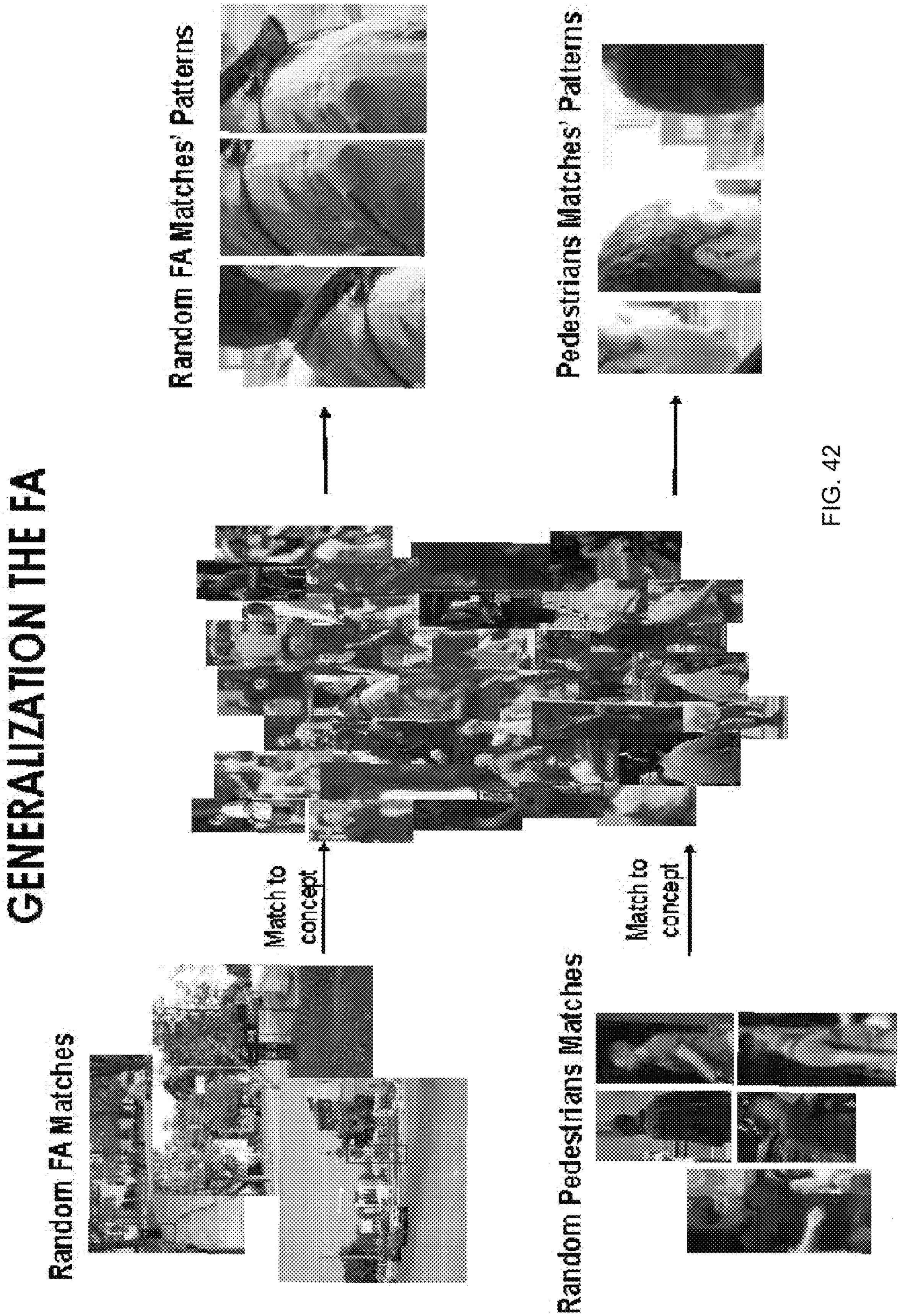

GENERALIZATION THE FA
Random FA Matches
Match to concept
Random FA Matches' Patterns
Random Pedestrians Matches
Match to concept
Pedestrians Matches' Patterns

FIG. 42

FA SIGNATURES CLUSTERING AND REDUCTION

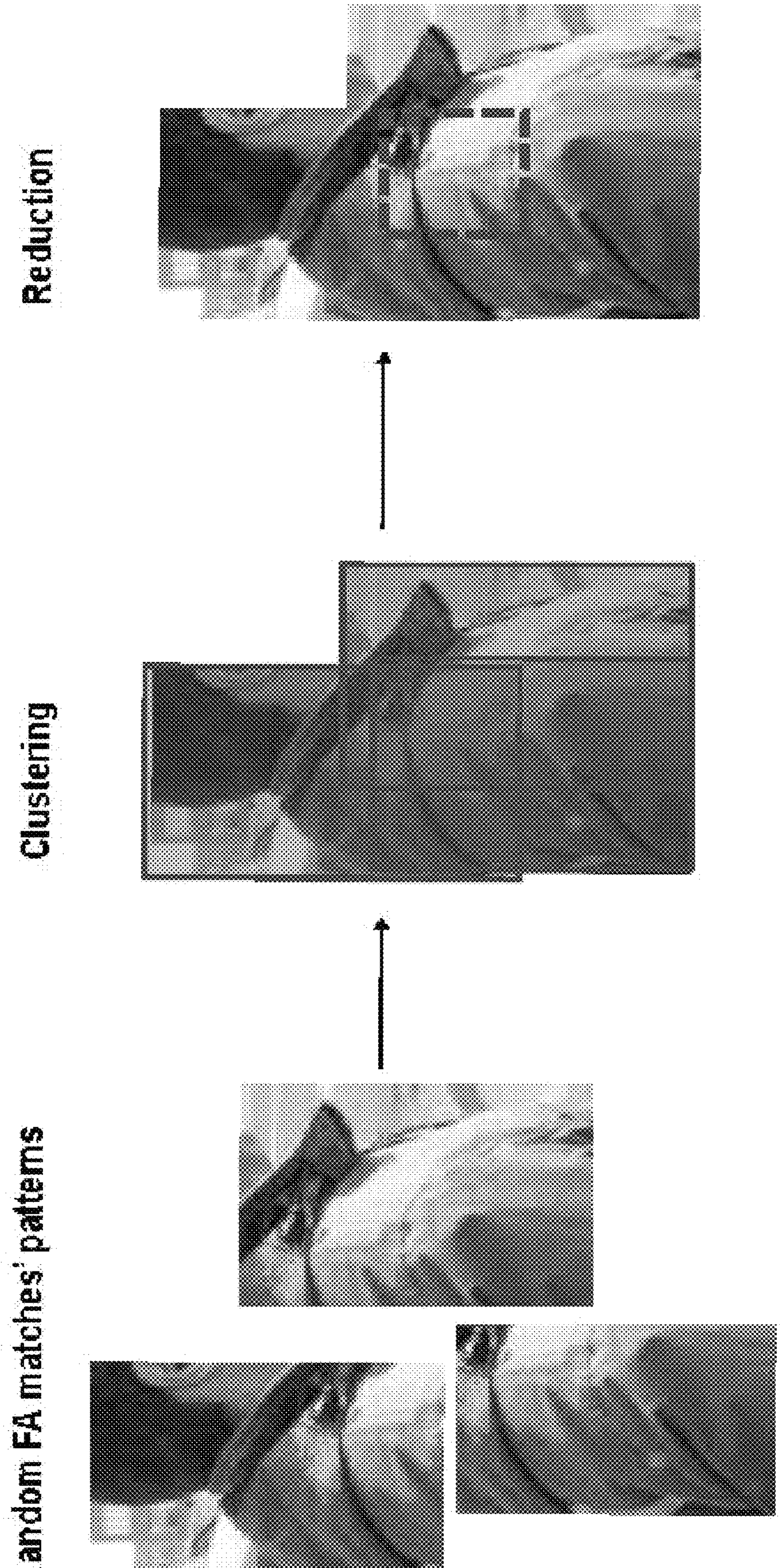


FIG. 43

CONCEPT CLEANING – DELETION OF THE FA REDUCED SIGNATURE

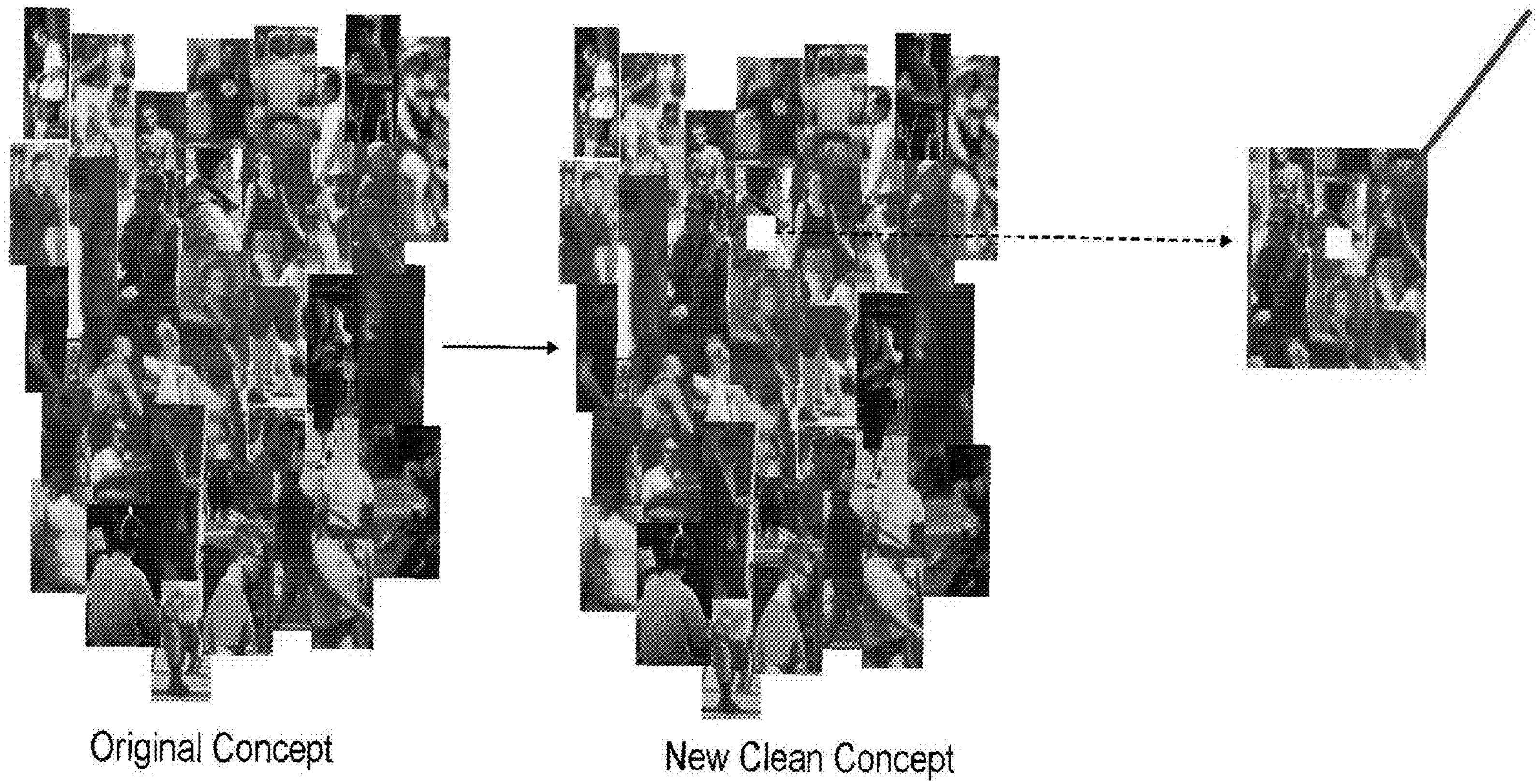


FIG. 44

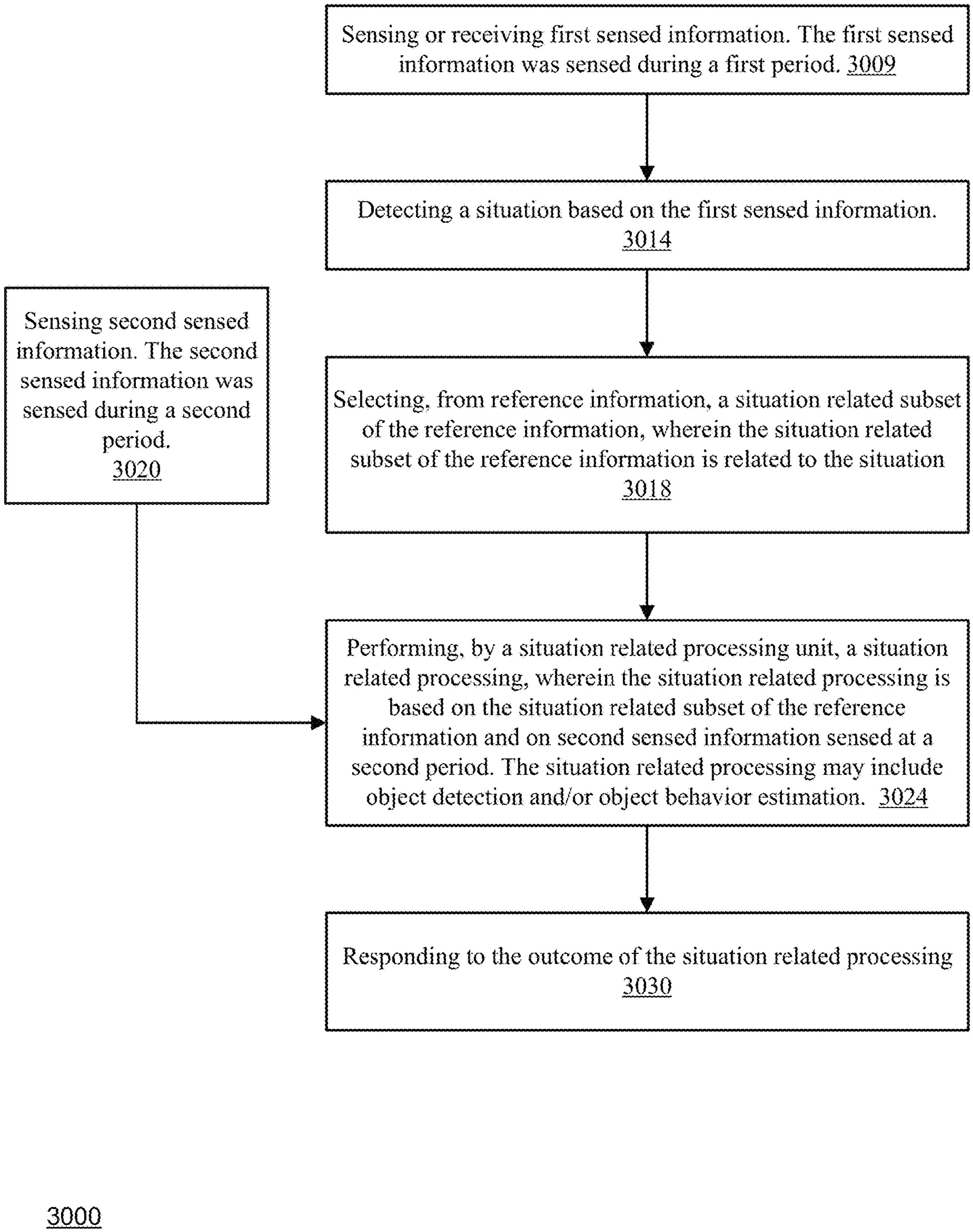

Sensing or receiving first sensed information. The first sensed information was sensed during a first period. 3009
Detecting a situation based on the first sensed information. 3014
Sensing second sensed information. The second sensed information was sensed during a second period. 3020
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information sensed at a second period. The situation related processing may include object detection and/or object behavior estimation. 3024
Responding to the outcome of the situation related processing 3030
3000

FIG. 45

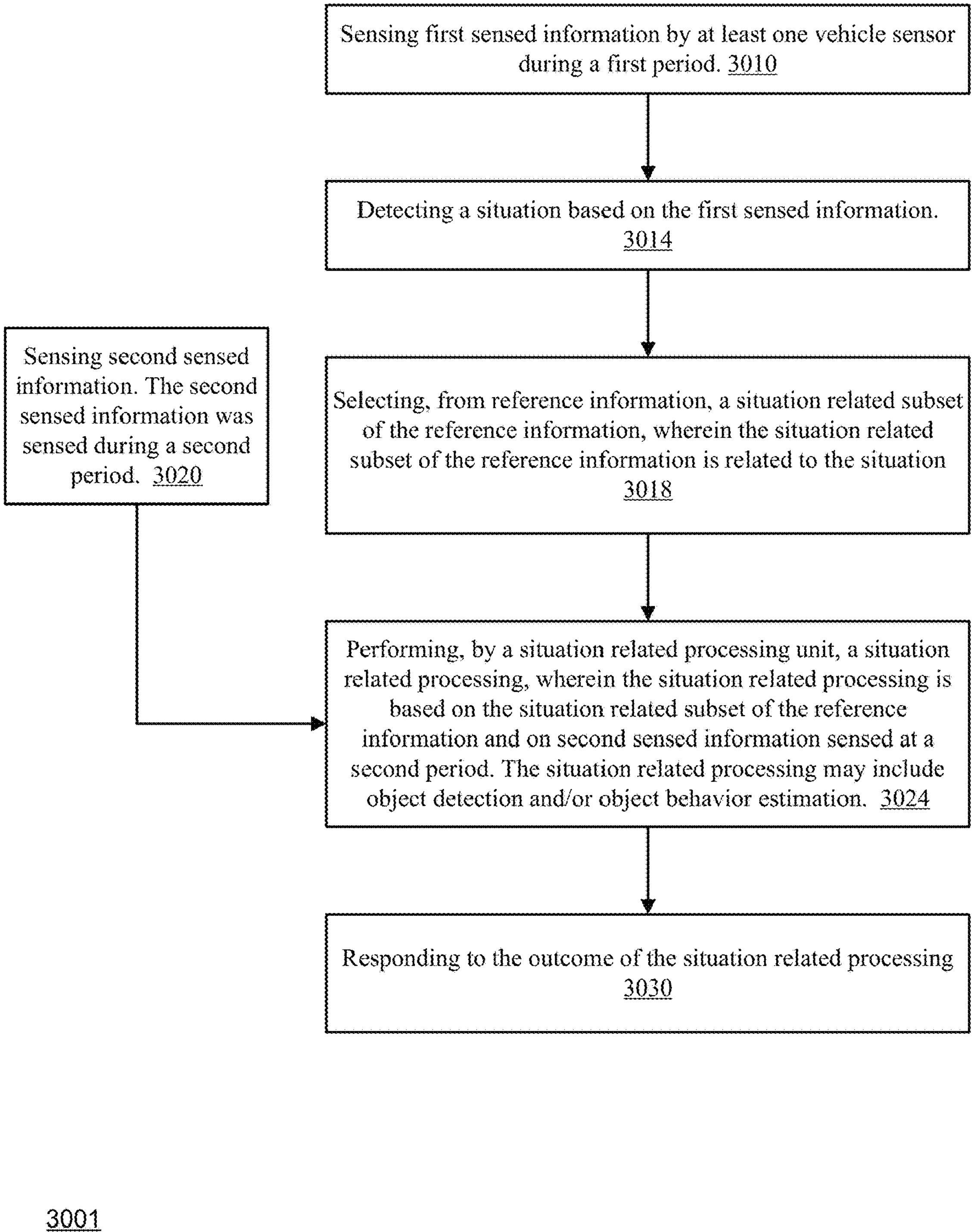

Sensing first sensed information by at least one vehicle sensor during a first period. 3010
Detecting a situation based on the first sensed information. 3014
Sensing second sensed information. The second sensed information was sensed during a second period. 3020
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information sensed at a second period. The situation related processing may include object detection and/or object behavior estimation. 3024
Responding to the outcome of the situation related processing 3030
3001

FIG. 46

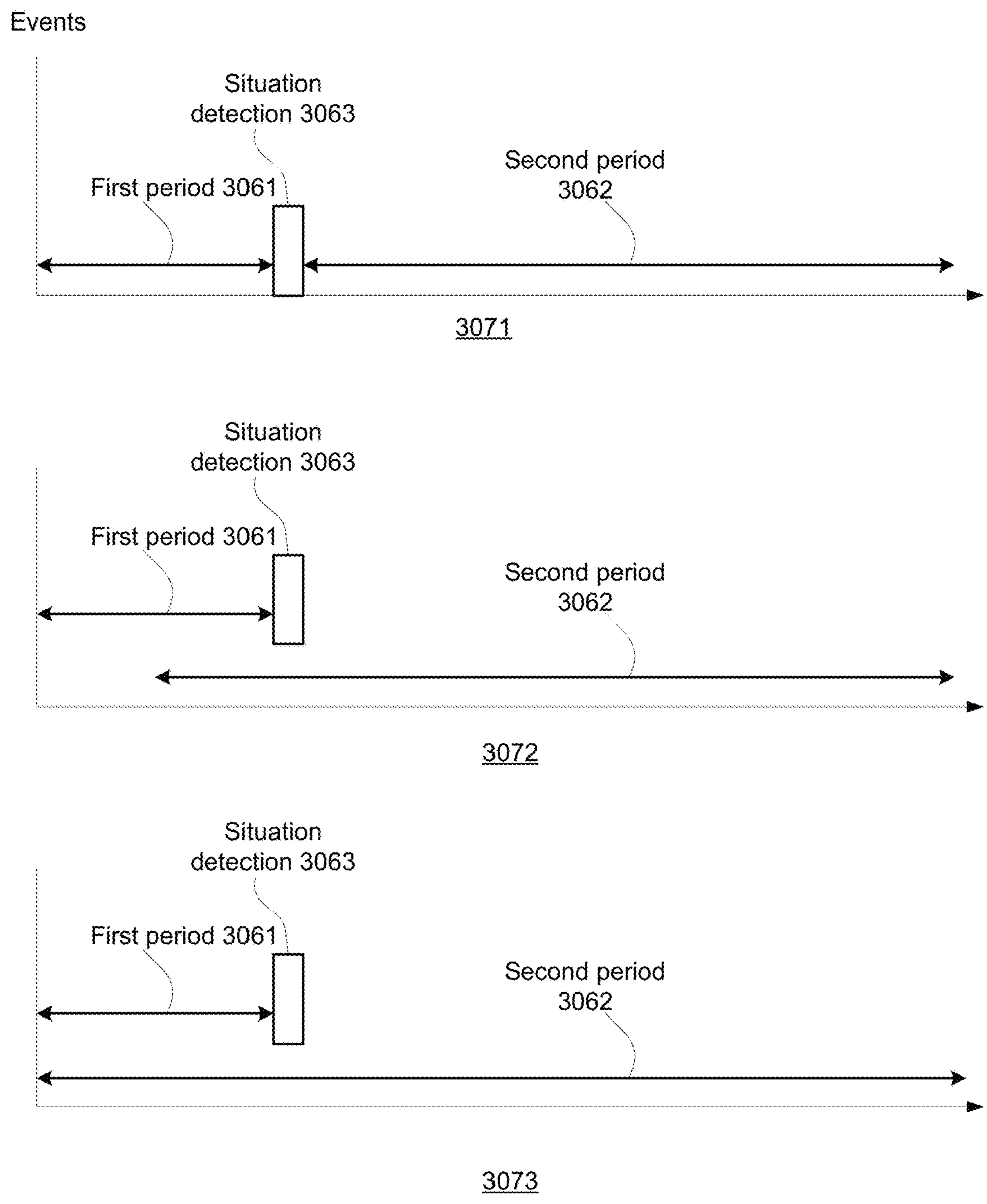

Events
Situation detection 3063
First period 3061
Second period 3062
3071
Situation detection 3063
First period 3061
Second period 3062
3072
Situation detection 3063
First period 3061
Second period 3062
3073

FIG. 47

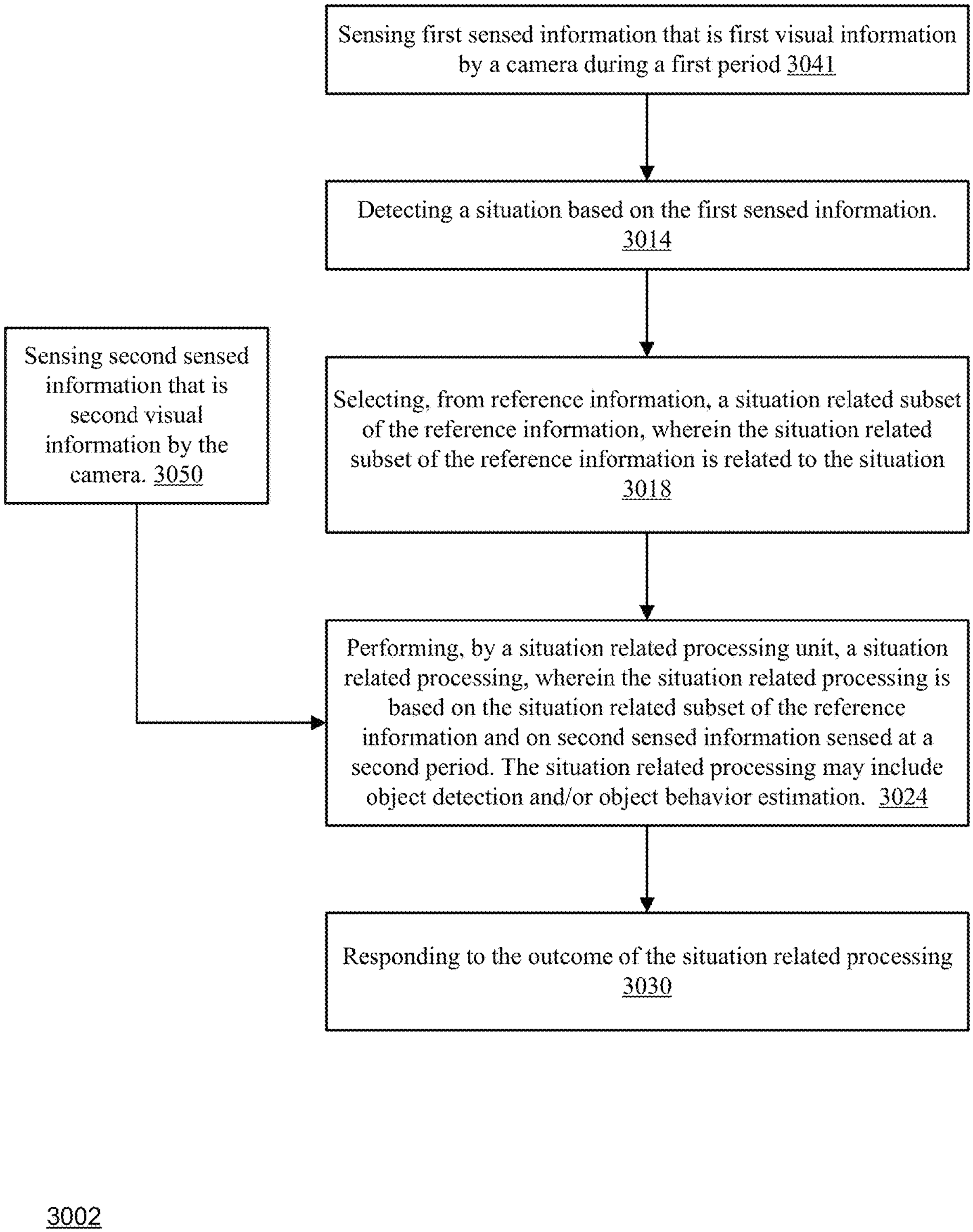

Sensing first sensed information that is first visual information by a camera during a first period 3041
Detecting a situation based on the first sensed information. 3014
Sensing second sensed information that is second visual information by the camera. 3050
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information sensed at a second period. The situation related processing may include object detection and/or object behavior estimation. 3024
Responding to the outcome of the situation related processing 3030
3002

FIG. 48

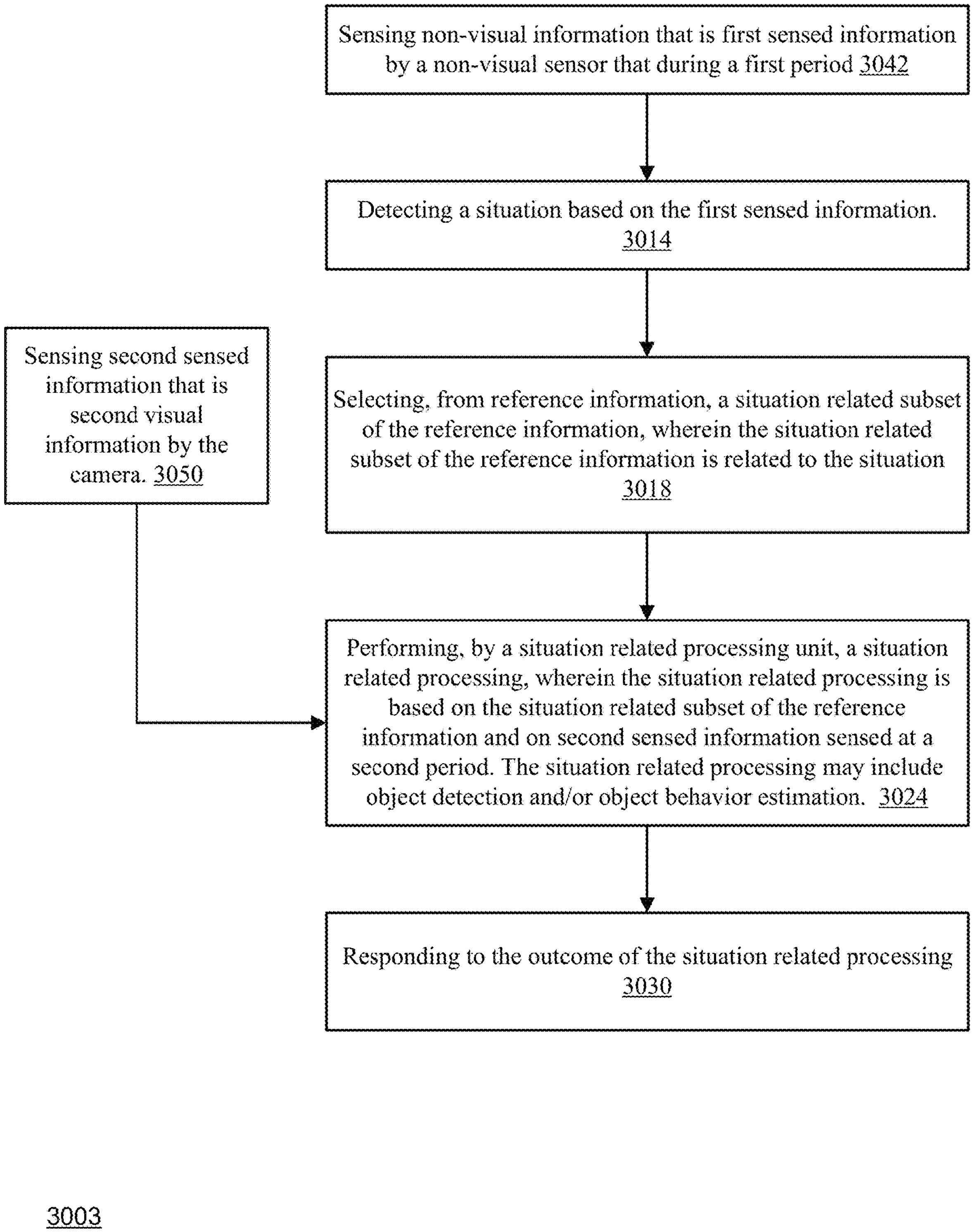

Sensing non-visual information that is first sensed information by a non-visual sensor that during a first period 3042
Detecting a situation based on the first sensed information. 3014
Sensing second sensed information that is second visual information by the camera. 3050
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information sensed at a second period. The situation related processing may include object detection and/or object behavior estimation. 3024
Responding to the outcome of the situation related processing 3030
3003

FIG. 49

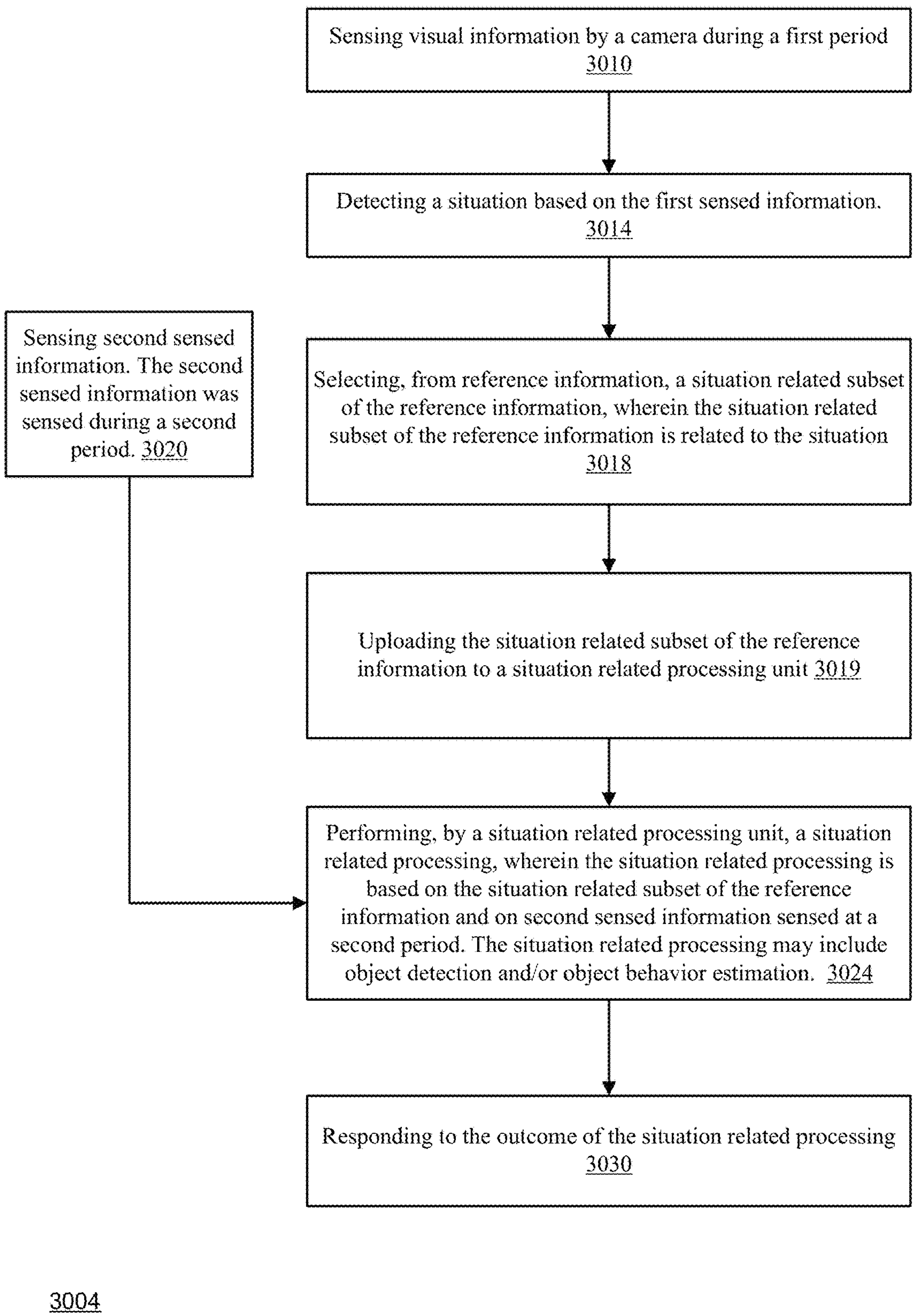

Sensing visual information by a camera during a first period 3010
Detecting a situation based on the first sensed information. 3014
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Uploading the situation related subset of the reference information to a situation related processing unit 3019
Sensing second sensed information. The second sensed information was sensed during a second period. 3020
Performing, by a situation related processing unit, a situation related processing, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information sensed at a second period. The situation related processing may include object detection and/or object behavior estimation. 3024
Responding to the outcome of the situation related processing 3030
3004

FIG. 50

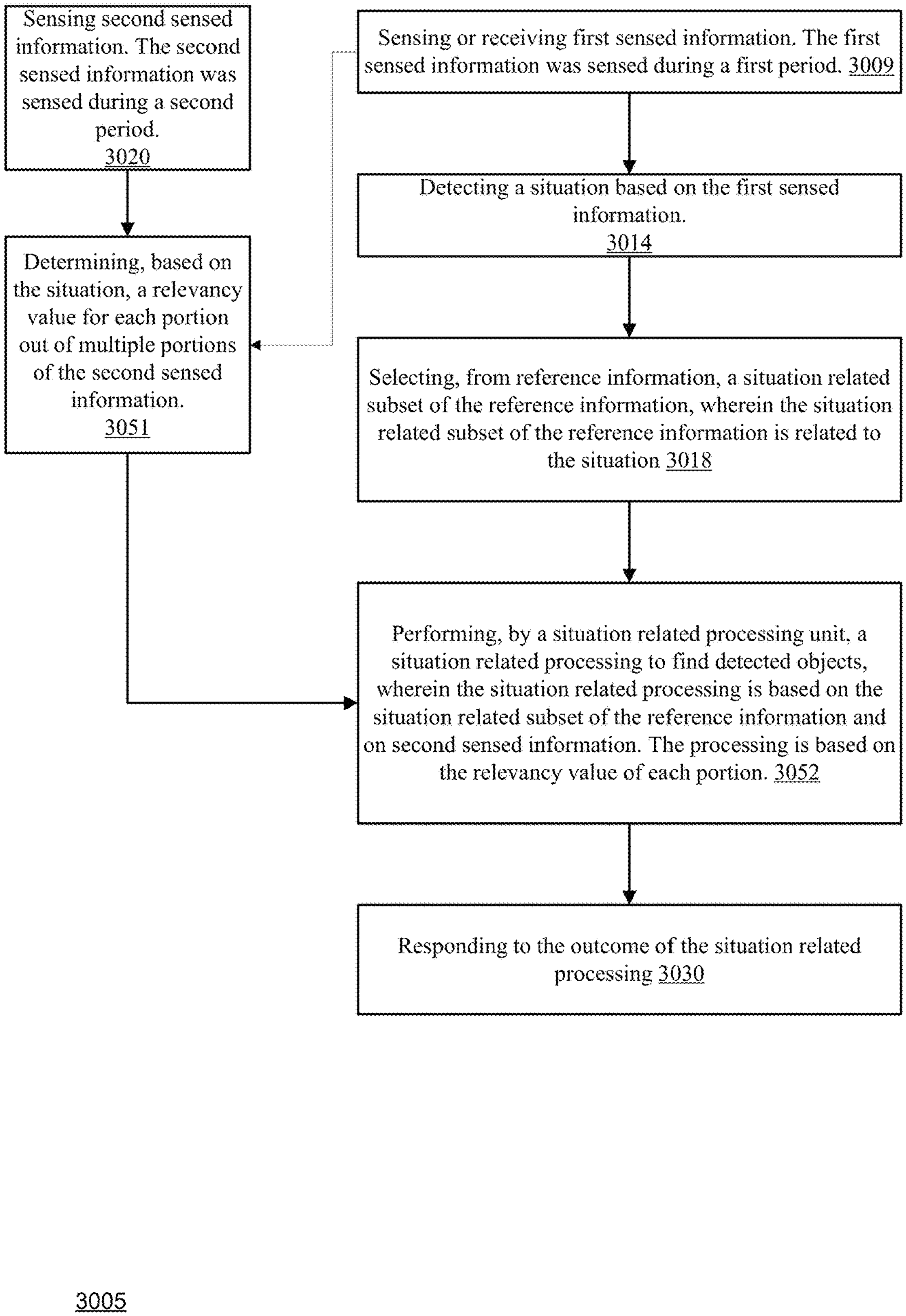

Sensing second sensed information. The second sensed information was sensed during a second period. 3020
Determining, based on the situation, a relevancy value for each portion out of multiple portions of the second sensed information. 3051
Sensing or receiving first sensed information. The first sensed information was sensed during a first period. 3009
Detecting a situation based on the first sensed information. 3014
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing to find detected objects, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information. The processing is based on the relevancy value of each portion. 3052
Responding to the outcome of the situation related processing 3030
3005

FIG. 51

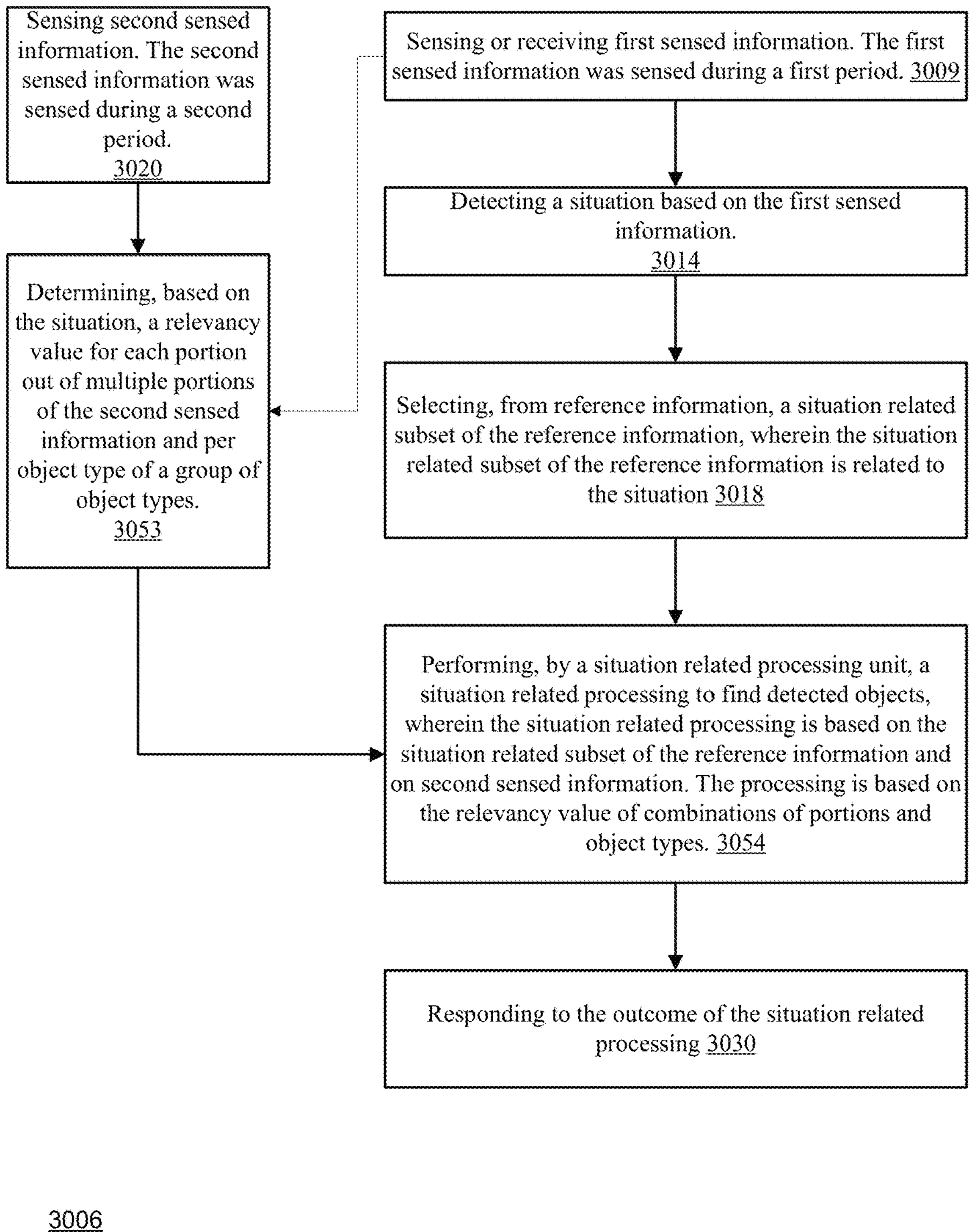

Sensing second sensed information. The second sensed information was sensed during a second period. 3020
Sensing or receiving first sensed information. The first sensed information was sensed during a first period. 3009
Detecting a situation based on the first sensed information. 3014
Determining, based on the situation, a relevancy value for each portion out of multiple portions of the second sensed information and per object type of a group of object types. 3053
Selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information is related to the situation 3018
Performing, by a situation related processing unit, a situation related processing to find detected objects, wherein the situation related processing is based on the situation related subset of the reference information and on second sensed information. The processing is based on the relevancy value of combinations of portions and object types. 3054
Responding to the outcome of the situation related processing 3030
3006

FIG. 52

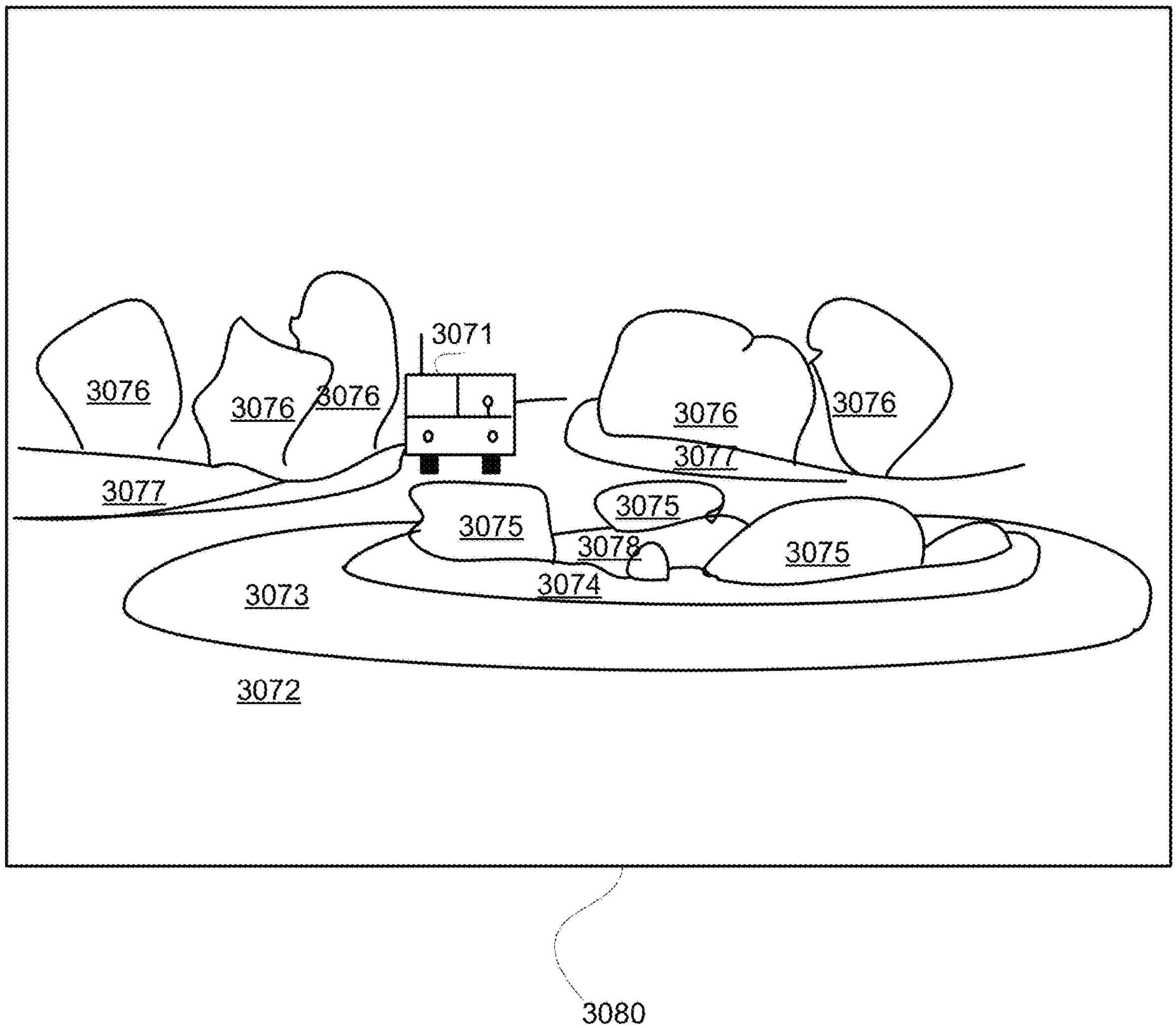

3071
3076
3076
3076
3076
3076
3077
3077
3075
3075
3075
3078
3074
3073
3072
3080

FIG. 53

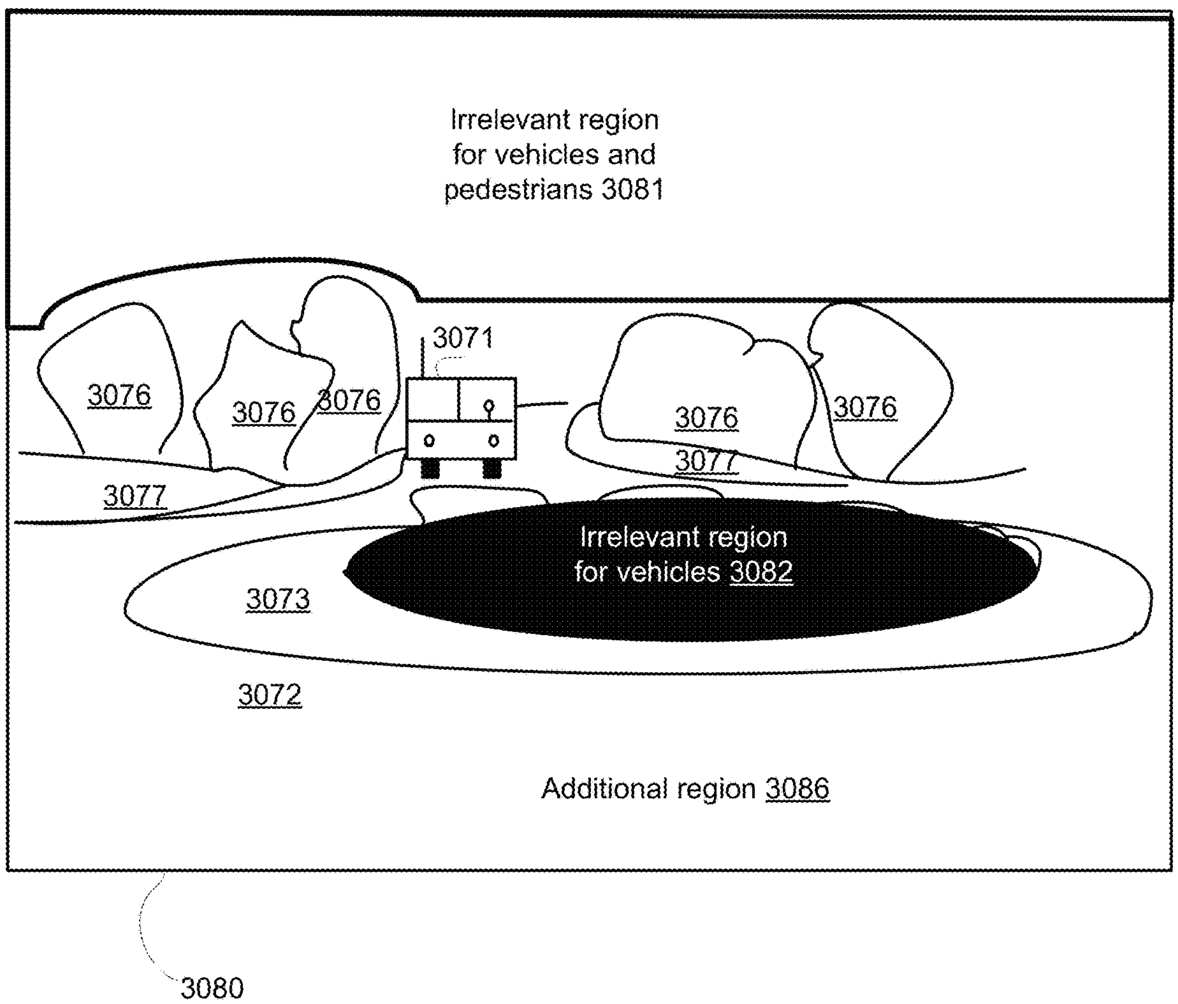

Irrelevant region
for vehicles and
pedestrians 3081
3071
3076
3076
3076
3076
3076
3077
3077
Irrelevant region
for vehicles 3082
3073
3072
Additional region 3086
3080

FIG. 54

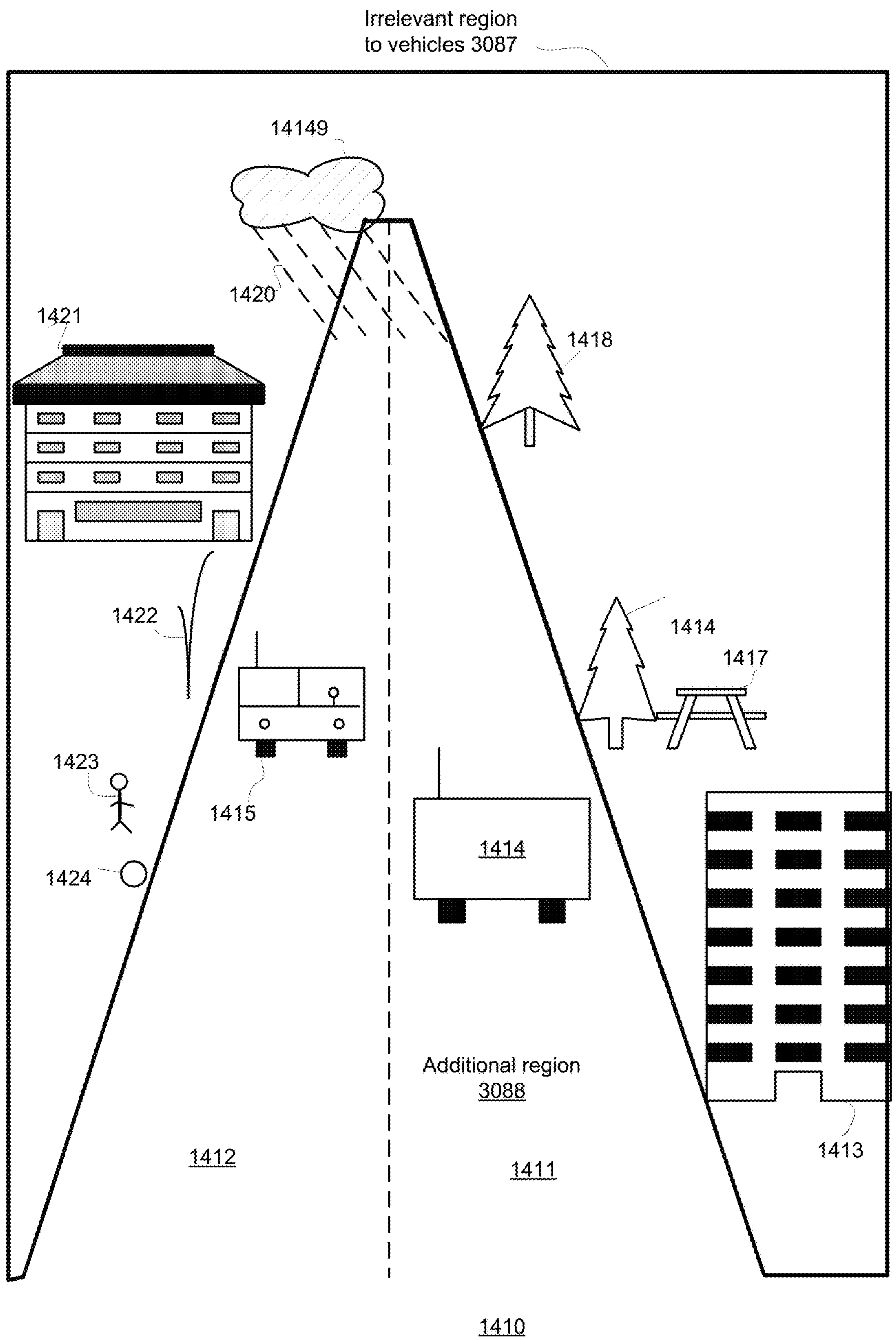

Irrelevant region
to vehicles 3087
14149
1420
1421
1418
1422
1414
1417
1423
1415
1414
1424
Additional region
3088
1412
1411
1413
1410

FIG. 55

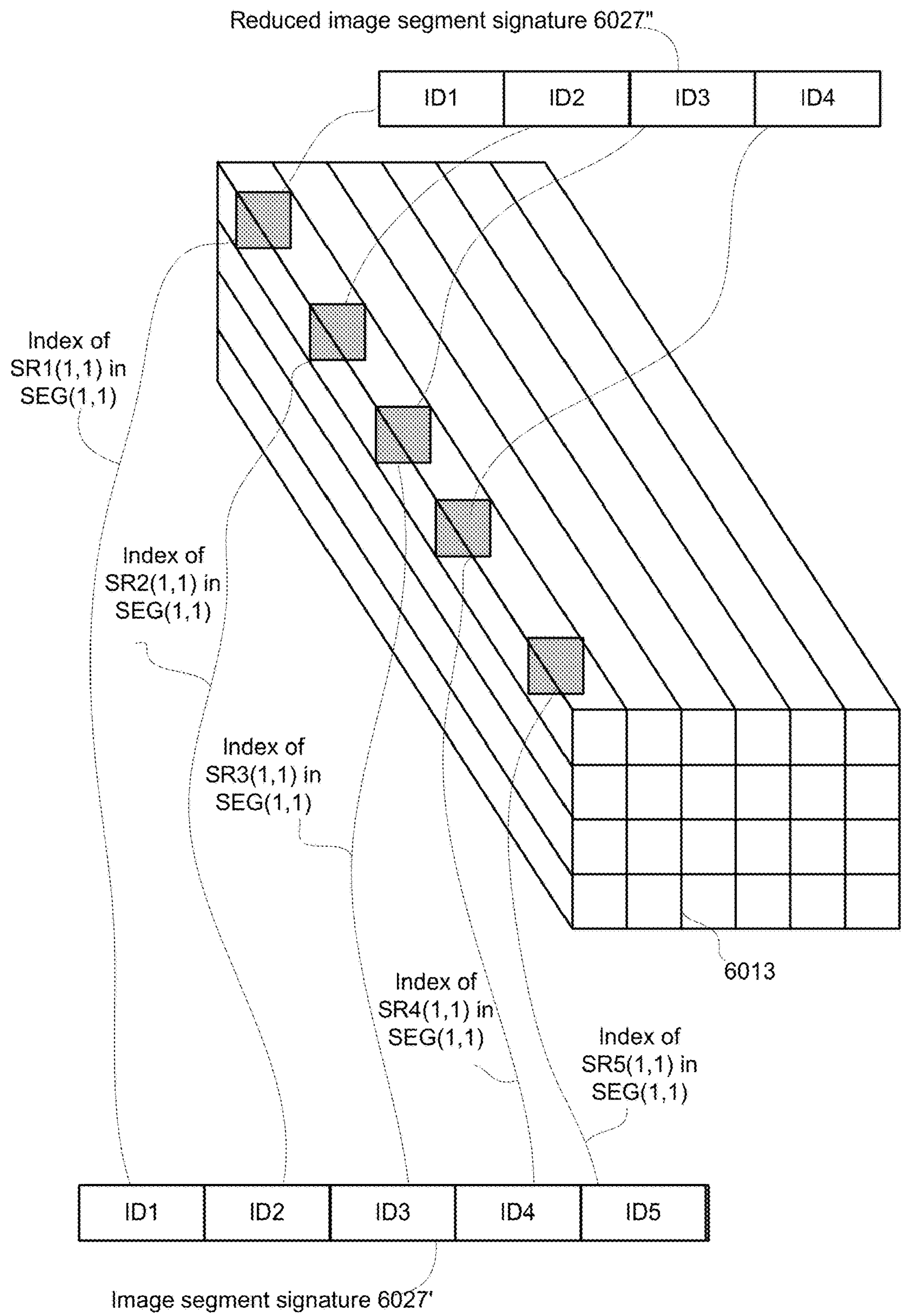

Reduced image segment signature 6027"
ID1
ID2
ID3
ID4
Index of SR1(1,1) in SEG(1,1)
Index of SR2(1,1) in SEG(1,1)
Index of SR3(1,1) in SEG(1,1)
Index of SR4(1,1) in SEG(1,1)
Index of SR5(1,1) in SEG(1,1)
6013
ID1
ID2
ID3
ID4
ID5
Image segment signature 6027'

FIG. 56

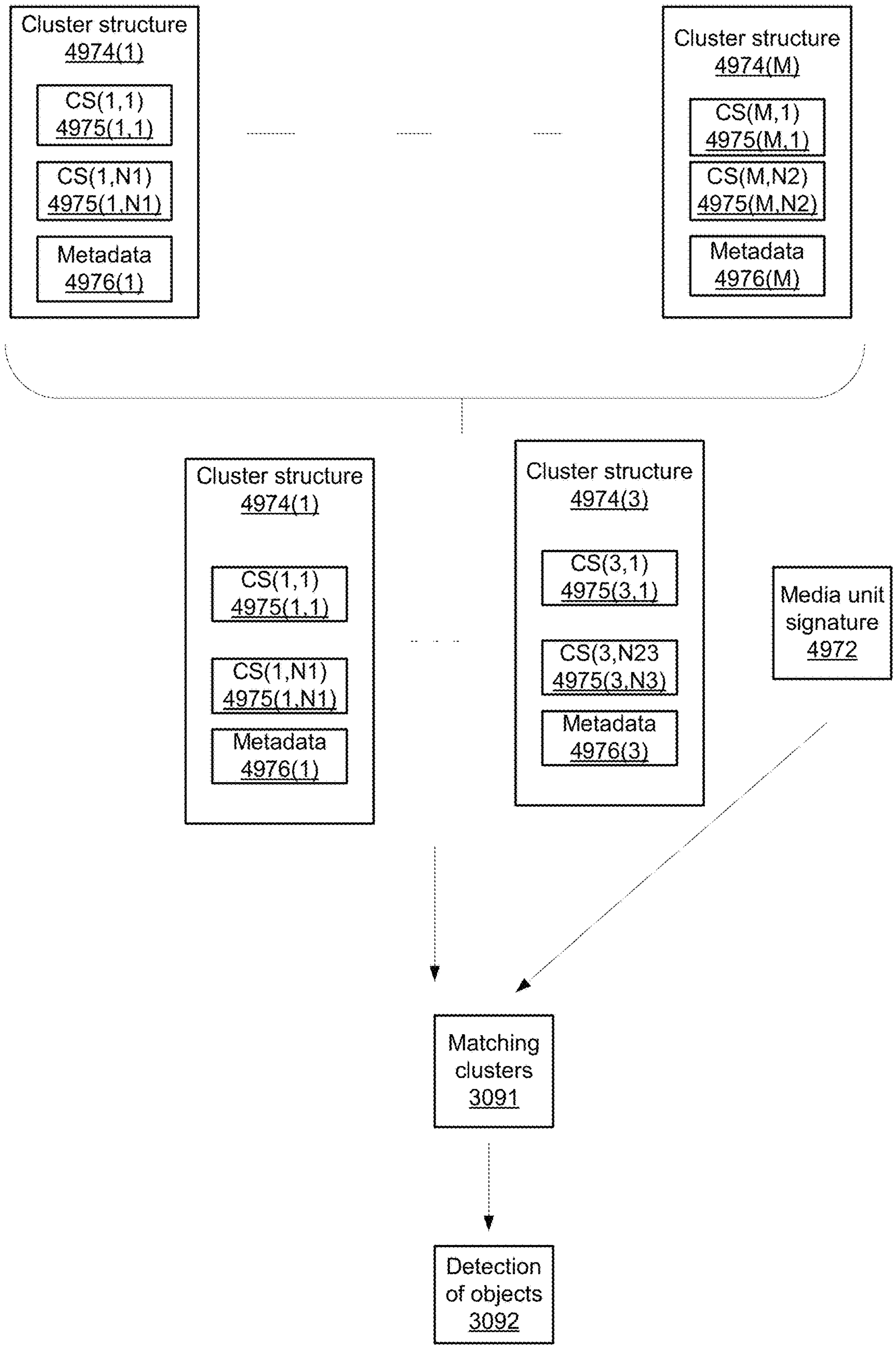

Cluster structure
4974(1)
CS(1,1)
4975(1,1)
CS(1,N1)
4975(1,N1)
Metadata
4976(1)
Cluster structure
4974(M)
CS(M,1)
4975(M,1)
CS(M,N2)
4975(M,N2)
Metadata
4976(M)
Cluster structure
4974(1)
CS(1,1)
4975(1,1)
CS(1,N1)
4975(1,N1)
Metadata
4976(1)
Cluster structure
4974(3)
CS(3,1)
4975(3,1)
CS(3,N23
4975(3,N3)
Metadata
4976(3)
Media unit
signature
4972
Matching
clusters
3091
Detection
of objects
3092

FIG. 57

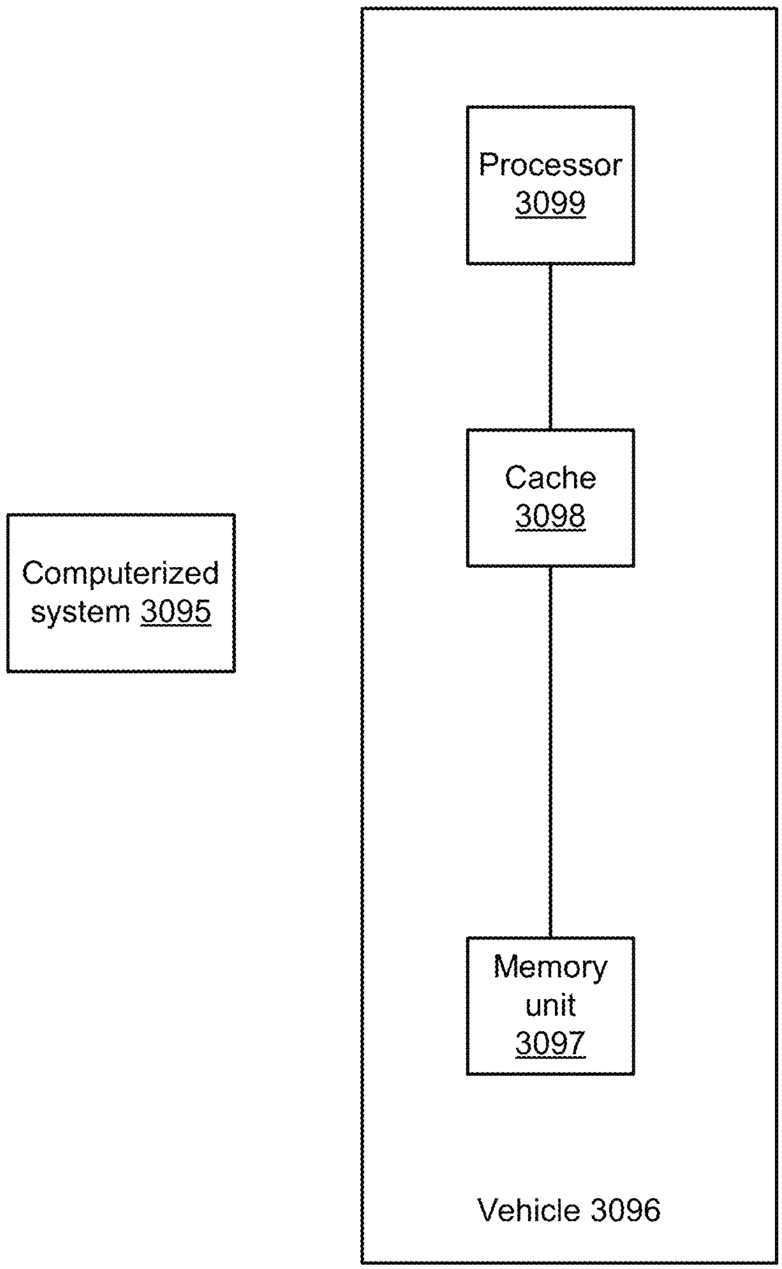
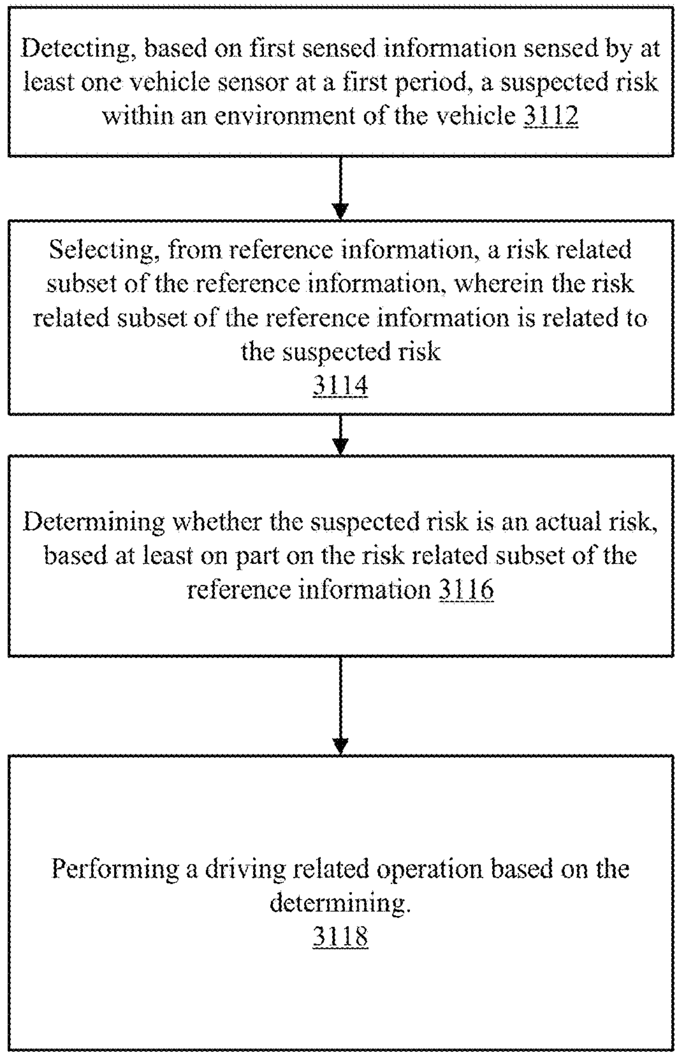
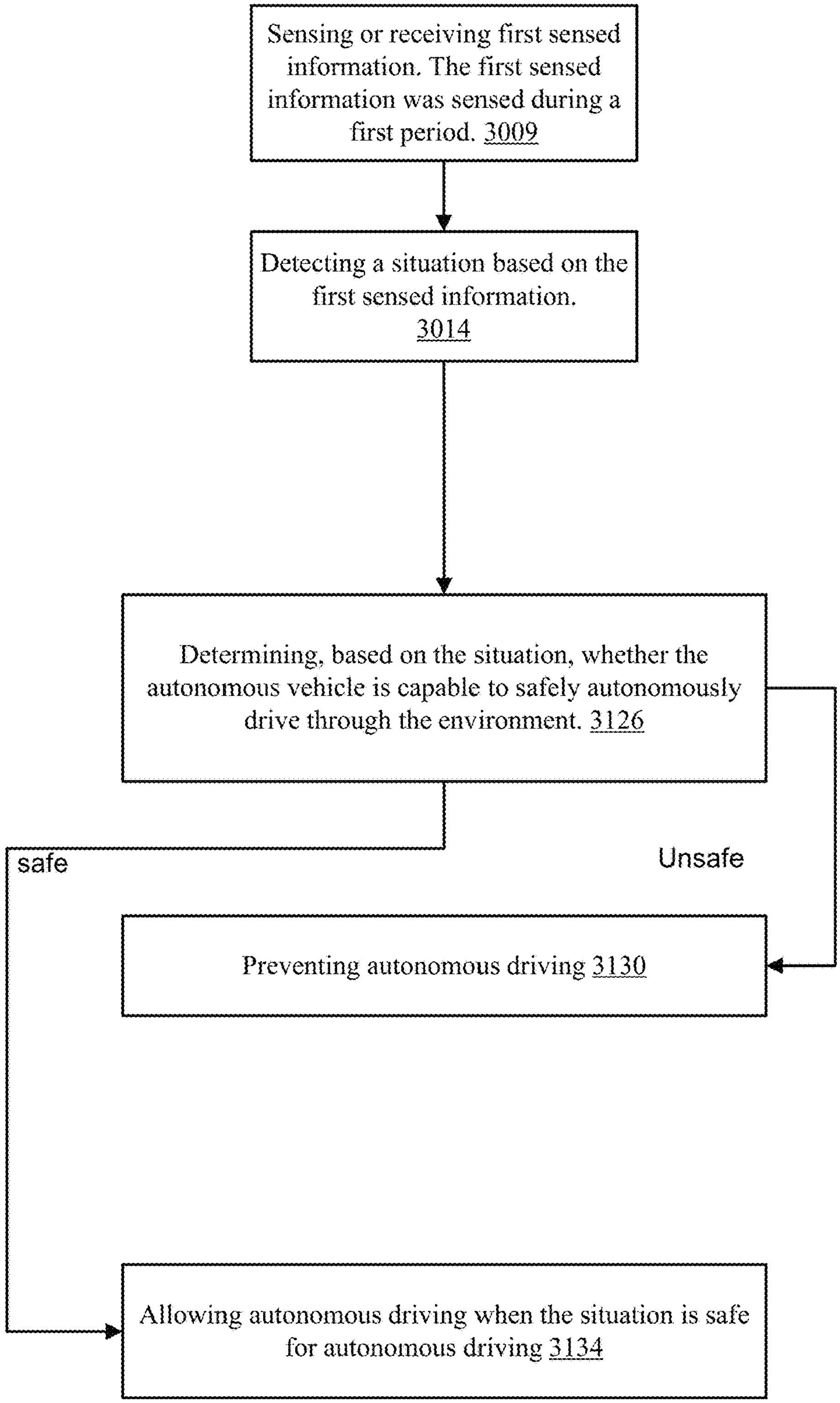

Processor 3099
Cache 3098
Memory unit 3097
Vehicle 3096
Computerized system 3095

FIG. 58


Detecting, based on first sensed information sensed by at least one vehicle sensor at a first period, a suspected risk within an environment of the vehicle 3112
Selecting, from reference information, a risk related subset of the reference information, wherein the risk related subset of the reference information is related to the suspected risk 3114
Determining whether the suspected risk is an actual risk, based at least on part on the risk related subset of the reference information 3116
Performing a driving related operation based on the determining. 3118

3100

FIG. 59


Sensing or receiving first sensed information. The first sensed information was sensed during a first period. 3009
Detecting a situation based on the first sensed information. 3014
Determining, based on the situation, whether the autonomous vehicle is capable to safely autonomously drive through the environment. 3126
safe
Unsafe
Preventing autonomous driving 3130
Allowing autonomous driving when the situation is safe for autonomous driving 3134

3120

FIG. 60

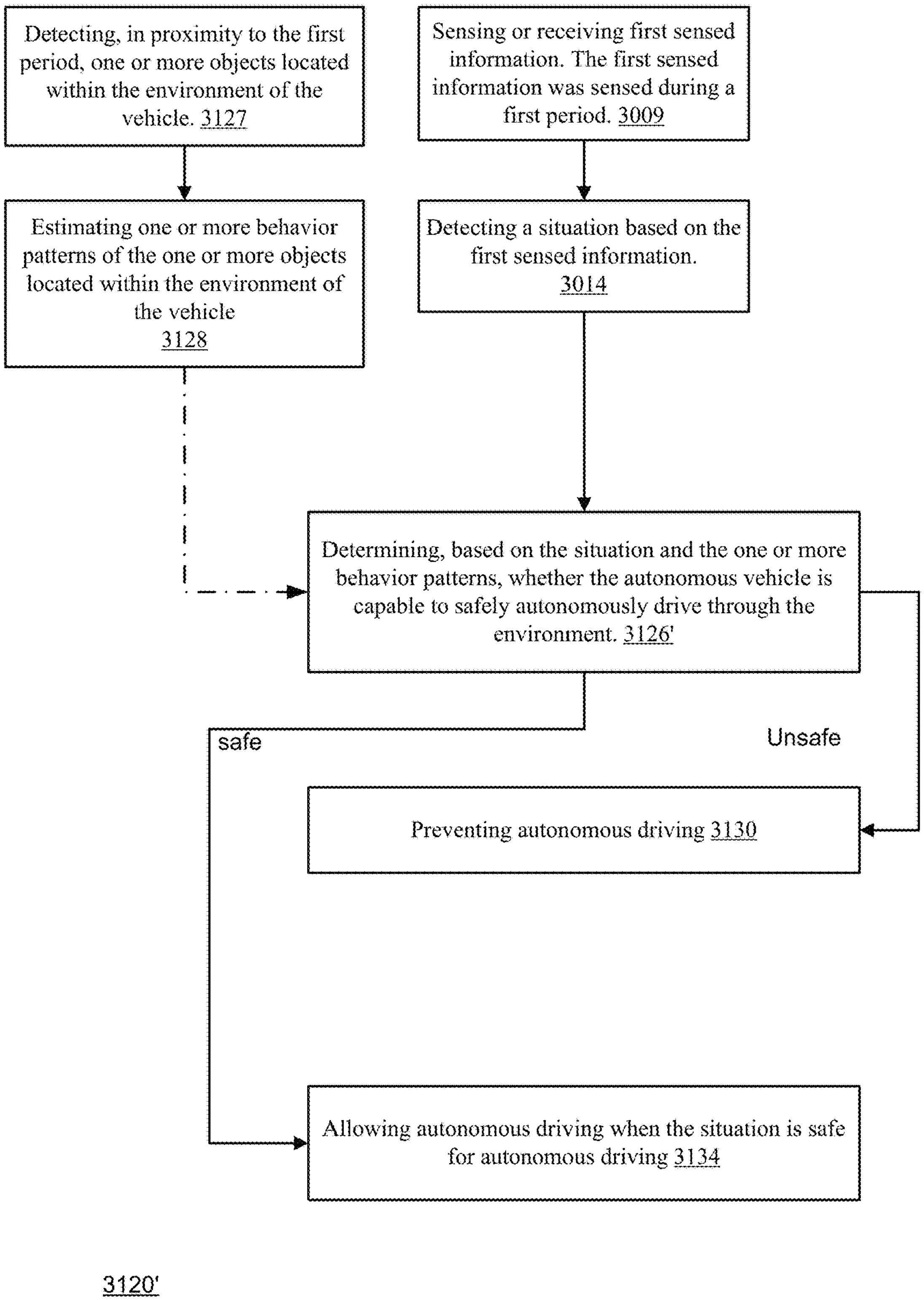

Detecting, in proximity to the first period, one or more objects located within the environment of the vehicle. 3127
Sensing or receiving first sensed information. The first sensed information was sensed during a first period. 3009
Estimating one or more behavior patterns of the one or more objects located within the environment of the vehicle 3128
Detecting a situation based on the first sensed information. 3014
Determining, based on the situation and the one or more behavior patterns, whether the autonomous vehicle is capable to safely autonomously drive through the environment. 3126'
safe
Unsafe
Preventing autonomous driving 3130
Allowing autonomous driving when the situation is safe for autonomous driving 3134
3120'

FIG. 61

SAFE TRANSFER BETWEEN MANNED AND AUTONOMOUS DRIVING MODES

CROSS REFERENCE

This application is a continuation of PCT/M2019/058207 filing date Sep. 27, 2019 which in turn claims priority from U.S. provisional Ser. No. 62/747,147 filing date Oct. 18, 2018.

This application claims priority from U.S. provisional patent Ser. No. 62/827,122 filing date Mar. 31, 2019, both patent applications are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure generally relates to detecting and avoiding obstacles in an autonomous driving environment.

BACKGROUND

Autonomous vehicle and vehicle equipped with driver assistance modules are expected to process quickly and accurately sensed information. The processing may include comparing acquired sensed information to reference information related to a vast number of events.

The store of such reference information may consume substantial memory resources and the processing of such reference information may require substantial processing resources.

There is a growing need to provide a more effective processing of sensed information.

SUMMARY

There may be provided method, devices, and non-transitory computer readable medium as illustrated in the application.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of the disclosure will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:

FIG. **1**A illustrates an example of a method;

FIG. **1**B illustrates an example of a signature;

FIG. **1**C illustrates an example of a dimension expansion process;

FIG. **1**D illustrates an example of a merge operation;

FIG. **1**E illustrates an example of hybrid process;

FIG. **1**F illustrates an example of a first iteration of the dimension expansion process;

FIG. **1**G illustrates an example of a method;

FIG. **1**H illustrates an example of a method;

FIG. **1**I illustrates an example of a method;

FIG. **1**J illustrates an example of a method;

FIG. **1**K illustrates an example of a method;

FIG. **1**L illustrates an example of a method;

FIG. **1**M illustrates an example of a method;

FIG. **1**N illustrates an example of a matching process and a generation of a higher accuracy shape information;

FIG. **1**O illustrates an example of an image and image identifiers;

FIG. **1**P illustrates an example of an image, approximated regions of interest, compressed shape information and image identifiers;

FIG. **1**Q illustrates an example of an image, approximated regions of interest, compressed shape information and image identifiers;

FIG. **1**R illustrates an example of an image;

FIG. **1**S illustrates an example of a method;

FIG. **2**A illustrates an example of images of different scales;

FIG. **2**B illustrates an example of images of different scales;

FIG. **2**C illustrates an example of a method;

FIG. **2**D illustrates an example of a method;

FIG. **2**E illustrates an example of a method;

FIG. **2**F illustrates an example of a method;

FIG. **2**G illustrates an example of different images;

FIG. **2**H illustrates an example of a method;

FIG. **2**I illustrates an example of a method;

FIG. **2**J illustrates an example of a method;

FIG. **2**K illustrates an example of different images acquisition angles;

FIG. **2**L illustrates an example of a method;

FIG. **2**M illustrates an example of a method;

FIG. **2**N illustrates an example of a system;

FIG. **3**A is a partly-pictorial, partly-block diagram illustration of an exemplary obstacle detection and mapping system, constructed and operative in accordance with embodiments described herein;

FIG. **3**B is a block diagram of an exemplary autonomous driving system to be integrated in the vehicle of FIG. **3**A;

FIG. **3**C is a flowchart of an exemplary process to be performed by the autonomous driving system of FIG. **3**B;

FIG. **4** is a block diagram of an exemplary obstacle avoidance server of FIG. **3**A;

FIG. **5** is a flowchart of an exemplary process to be performed by the obstacle avoidance server of FIG. **4**;

FIG. **6** is an example of a method;

FIG. **7** is an example of a method;

FIG. **8** is an example of a driving scenario;

FIG. **9** is an example of a driving scenario;

FIG. **10** is an example of a driving scenario;

FIG. **11** is an example of a method;

FIG. **12** is an example of a method;

FIG. **13** is an example of a method;

FIG. **14** is an example of a driving scenario;

FIG. **15** is an example of a driving scenario;

FIG. **16** is an example of a scene;

FIG. **17** is an example of a scene;

FIG. **18** is an example of a driving scenario;

FIG. **19** is an example of a method;

FIG. **20** is an example of a method;

FIG. **21** is an example of a method;

FIG. **22** is an example of a driving scenario;

FIG. **23** is an example of a driving scenario;

FIG. **24** is an example of a driving scenario;

FIG. **25** is an example of a driving scenario;

FIG. **26** is an example of a driving scenario;

FIG. **27** is an example of a driving scenario;

FIG. **28** is an example of a method;

FIG. **29** is an example of entity movement functions;

FIG. **30** is an example of a method;

FIG. **31** is an example of a method;

FIG. **32** is an example of a method;

FIG. **33** is an example of a method;

FIG. **34** is an example of a method;

FIG. **35** is an example of a method;

FIG. **36** is an example of a driving scenario;

FIG. **37** is an example of a method;

FIG. **38** is an example of a method;

FIGS. **39-44** illustrate various data structures including a concept, test images and matching results, as well as various processes related to the data structures;

FIG. **45** is an example of a method;

FIG. **46** is an example of a method;

FIG. **47** is a timing diagram;

FIG. **48** is an example of a method;

FIG. **49** is an example of a method;

FIG. **50** is an example of a method;

FIG. **51** is an example of a method;

FIG. **52** is an example of a method;

FIG. **53** is an example of an image;

FIG. **54** is an example of an image;

FIG. **55** is an example of an image;

FIG. **56** illustrates examples of two signatures of different resolutions;

FIG. **57** illustrates an example of various data structures and processes;

FIG. **58** illustrates an example of a vehicle and a system;

FIG. **59** is an example of a method;

FIG. **60** is an example of a method; and

FIG. **61** illustrates an example of a method.

DESCRIPTION OF EXAMPLE EMBODIMENTS

The specification and/or drawings may refer to an image. An image is an example of a media unit. Any reference to an image may be applied mutatis mutandis to a media unit. A media unit may be an example of sensed information. Any reference to a media unit may be applied mutatis mutandis to any type of natural signal such as but not limited to signal generated by nature, signal representing human behavior, signal representing operations related to the stock market, a medical signal, financial series, geodetic signals, geophysical, chemical, molecular, textual and numerical signals, time series, and the like. Any reference to a media unit may be applied mutatis mutandis to sensed information. The sensed information may be of any kind and may be sensed by any type of sensors—such as a visual light camera, an audio sensor, a sensor that may sense infrared, radar imagery, ultrasound, electro-optics, radiography, LIDAR (light detection and ranging), etc. The sensing may include generating samples (for example, pixel, audio signals) that represent the signal that was transmitted, or otherwise reach the sensor.

The specification and/or drawings may refer to a spanning element. A spanning element may be implemented in software or hardware. Different spanning element of a certain iteration are configured to apply different mathematical functions on the input they receive. Non-limiting examples of the mathematical functions include filtering, although other functions may be applied.

The specification and/or drawings may refer to a concept structure. A concept structure may include one or more clusters. Each cluster may include signatures and related metadata. Each reference to one or more clusters may be applicable to a reference to a concept structure.

The specification and/or drawings may refer to a processor. The processor may be a processing circuitry. The processing circuitry may be implemented as a central processing unit (CPU), and/or one or more other integrated circuits such as application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), full-custom integrated circuits, etc., or a combination of such integrated circuits.

Any combination of any steps of any method illustrated in the specification and/or drawings may be provided.

Any combination of any subject matter of any of claims may be provided.

Any combinations of systems, units, components, processors, sensors, illustrated in the specification and/or drawings may be provided.

Low Power Generation of Signatures

The analysis of content of a media unit may be executed by generating a signature of the media unit and by comparing the signature to reference signatures. The reference signatures may be arranged in one or more concept structures or may be arranged in any other manner. The signatures may be used for object detection or for any other use.

The signature may be generated by creating a multidimensional representation of the media unit. The multidimensional representation of the media unit may have a very large number of dimensions. The high number of dimensions may guarantee that the multidimensional representation of different media units that include different objects is sparse—and that object identifiers of different objects are distant from each other—thus improving the robustness of the signatures.

The generation of the signature is executed in an iterative manner that includes multiple iterations, each iteration may include an expansion operations that is followed by a merge operation. The expansion operation of an iteration is performed by spanning elements of that iteration. By determining, per iteration, which spanning elements (of that iteration) are relevant—and reducing the power consumption of irrelevant spanning elements—a significant amount of power may be saved.

In many cases, most of the spanning elements of an iteration are irrelevant—thus after determining (by the spanning elements) their relevancy—the spanning elements that are deemed to be irrelevant may be shut down a/or enter an idle mode.

FIG. **1**A illustrates a method **5000** for generating a signature of a media unit.

Method **5000** may start by step **5010** of receiving or generating sensed information.

The sensed information may be a media unit of multiple objects.

Step **5010** may be followed by processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation.

The processing may include:

Step **5020** of performing a k'th iteration expansion process (k may be a variable that is used to track the number of iterations).

Step **5030** of performing a k'th iteration merge process.

Step **5040** of changing the value of k.

Step **5050** of checking if all required iterations were done—if so proceeding to step **S060** of completing the generation of the signature. Else—jumping to step **5020**.

The output of step **5020** is a k'th iteration expansion results **5120**.

The output of step **5030** is a k'th iteration merge results **5130**.

For each iteration (except the first iteration)—the merge result of the previous iteration is an input to the current iteration expansion process.

At least some of the K iterations involve selectively reducing the power consumption of some spanning elements (during step **5020**) that are deemed to be irrelevant.

FIG. **1**B is an example of an image signature **6027** of a media unit that is an image **6000** and of an outcome **6013** of the last (K'th) iteration.

The image **6001** is virtually segments to segments **6000** (i,k). The segments may be of the same shape and size, but this is not necessarily so.

Outcome **6013** may be a tensor that includes a vector of values per each segment of the media unit. One or more objects may appear in a certain segment. For each object—an object identifier (of the signature) points to locations of significant values, within a certain vector associated with the certain segment.

For example—a top left segment (**6001**(**1**,**1**)) of the image may be represented in the outcome **6013** by a vector V(**1**,**1**) **6017**(**1**,**1**) that has multiple values. The number of values per vector may exceed 100, 200, 500, 1000, and the like.

The significant values (for example—more than 10, 20, 30, 40 values, and/or more than 0.1%, 0.2%. 0.5%, 1%, 5% of all values of the vector and the like) may be selected. The significant values may have the values—but may be selected in any other manner.

FIG. **1**B illustrates a set of significant responses **6015**(**1**,**1**) of vector V(**1**,**1**) **6017**(**1**,**1**). The set includes five significant values (such as first significant value SV**1**(**1**,**1**) **6013**(**1**,**1**,**1**), second significant value SV**2**(**1**,**1**), third significant value SV**3**(**1**,**1**), fourth significant value SV**4**(**1**,**1**), and fifth significant value SV**5**(**1**,**1**) **6013**(**1**,**1**,**5**).

The image signature **6027** includes five indexes for the retrieval of the five significant values—first till fifth identifiers ID**1**-ID**5** are indexes for retrieving the first till fifth significant values.

FIG. **1**C illustrates a k'th iteration expansion process.

The k'th iteration expansion process start by receiving the merge results **5060'** of a previous iteration.

The merge results of a previous iteration may include values are indicative of previous expansion processes—for example—may include values that are indicative of relevant spanning elements from a previous expansion operation, values indicative of relevant regions of interest in a multi-dimensional representation of the merge results of a previous iteration.

The merge results (of the previous iteration) are fed to spanning elements such as spanning elements **5061**(**1**)-**5061**(J).

Each spanning element is associated with a unique set of values. The set may include one or more values. The spanning elements apply different functions that may be orthogonal to each other. Using non-orthogonal functions may increase the number of spanning elements—but this increment may be tolerable.

The spanning elements may apply functions that are decorrelated to each other—even if not orthogonal to each other.

The spanning elements may be associated with different combinations of object identifiers that may "cover" multiple possible media units. Candidates for combinations of object identifiers may be selected in various manners—for example based on their occurrence in various images (such as test images) randomly, pseudo randomly, according to some rules and the like. Out of these candidates the combinations may be selected to be decorrelated, to cover said multiple possible media units and/or in a manner that certain objects are mapped to the same spanning elements.

Each spanning element compares the values of the merge results to the unique set (associated with the spanning element) and if there is a match—then the spanning element is deemed to be relevant. If so—the spanning element completes the expansion operation.

If there is no match—the spanning element is deemed to be irrelevant and enters a low power mode. The low power mode may also be referred to as an idle mode, a standby mode, and the like. The low power mode is termed low power because the power consumption of an irrelevant spanning element is lower than the power consumption of a relevant spanning element.

In FIG. **1**C various spanning elements are relevant (**5061**(**1**)-**5061**(**3**)) and one spanning element is irrelevant (**5061**(J)).

Each relevant spanning element may perform a spanning operation that includes assigning an output value that is indicative of an identity of the relevant spanning elements of the iteration. The output value may also be indicative of identities of previous relevant spanning elements (from previous iterations).

For example—assuming that spanning element number fifty is relevant and is associated with a unique set of values of eight and four—then the output value may reflect the numbers fifty, four and eight—for example one thousand multiplied by (fifty+forty) plus forty. Any other mapping function may be applied.

FIG. **1**C also illustrates the steps executed by each spanning element:

Checking if the merge results are relevant to the spanning element (step **5091**).

If-so—completing the spanning operation (step **5093**).

If not—entering an idle state (step **5092**).

FIG. **1**D is an example of various merge operations.

A merge operation may include finding regions of interest. The regions of interest are regions within a multidimensional representation of the sensed information. A region of interest may exhibit a more significant response (for example a stronger, higher intensity response).

The merge operation (executed during a k'th iteration merge operation) may include at least one of the following:

Step **5031** of searching for overlaps between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the overlaps.

Step **5032** of determining to drop one or more region of interest, and dropping according to the determination.

Step **5033** of searching for relationships between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the relationship.

Step **5034** of searching for proximate regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the proximity. Proximate may be a distance that is a certain fraction (for example less than 1%) of the multi-dimensional space, may be a certain fraction of at least one of the regions of interest that are tested for proximity.

Step **5035** of searching for relationships between regions of interest (of the k'th iteration expansion operation results) and define regions of interest that are related to the relationship.

Step **5036** of merging and/or dropping k'th iteration regions of interest based on shape information related to shape of the k'th iteration regions of interest.

The same merge operations may be applied in different iterations.

Alternatively, different merge operations may be executed during different iterations.

FIG. **1**E illustrates an example of a hybrid process and an input image **6001**.

The hybrid process is hybrid in the sense that some expansion and merge operations are executed by a convolutional neural network (CNN) and some expansion and merge operations (denoted additional iterations of expansion and merge) are not executed by the CNN— but rather by a process that may include determining a relevancy of spanning elements and entering irrelevant spanning elements to a low power mode.

In FIG. **1**E one or more initial iterations are executed by first and second CNN layers **6010**(**1**) and **6010**(**2**) that apply first and second functions **6015**(**1**) and **6015**(**2**).

The output of these layers provided information about image properties. The image properties may not amount to object detection. Image properties may include location of edges, properties of curves, and the like.

The CNN may include additional layers (for example third till N'th layer **6010**(N)) that may provide a CNN output **6018** that may include object detection information. It should be noted that the additional layers may not be included.

It should be noted that executing the entire signature generation process by a hardware CNN of fixed connectivity may have a higher power consumption—as the CNN will not be able to reduce the power consumption of irrelevant nodes.

FIG. **1**F illustrates an input image **6001**, and a single iteration of an expansion operation and a merge operation.

In FIG. **1**F the input image **6001** undergoes two expansion operations.

The first expansion operation involves filtering the input image by a first filtering operation **6031** to provide first regions of interest (denoted 1) in a first filtered image **6031'**.

The first expansion operation also involves filtering the input image by a second filtering operation **6032** to provide first regions of interest (denoted 2) in a second filtered image **6032'**,

The merge operation includes merging the two images by overlaying the first filtered image on the second filtered image to provide regions of interest 1, 2, 12 and 21. Region of interest 12 is an overlap area shared by a certain region of interest 1 and a certain region of interest 2. Region of interest 21 is a union of another region of interest 1 and another region of interest 2.

FIG. **1**G illustrates method **5200** for generating a signature.

Method **5200** may include the following sequence of steps:

- Step **5210** of receiving or generating an image.
- Step **5220** of performing a first iteration expansion operation (which is an expansion operation that is executed during a first iteration)
- Step **5230** of performing a first iteration merge operation.
- Step **5240** of amending index k (k is an iteration counter). In FIG. **7** in incremented by one—this is only an example of how the number of iterations are tracked.
- Step **5260** of performing a k'th iteration expansion operation on the (k−1)'th iteration merge results.
- Step **5270** of performing a k'th iteration merge operation (on the k'th iteration expansion operation results.
- Step **5280** of changing the value of index k.
- Step **5290** of checking if all iteration ended (k reached its final value—for example K).

If no—there are still iterations to be executed—jumping from step **5290** to step **5260**.

If yes—jumping to step **5060** of completing the generation of the signature. This may include, for example, selecting significant attributes, determining retrieval information (for example indexes) that point to the selected significant attributes.

Step **5220** may include:

- Step **5222** of generating multiple representations of the image within a multi-dimensional space of f(1) dimensions. The expansion operation of step **5220** generates a first iteration multidimensional representation of the first image. The number of dimensions of this first iteration multidimensional representation is denoted f(1).
- Step **5224** of assigning a unique index for each region of interest within the multiple representations. For example, referring to FIG. **6**—indexes 1 and indexes 2 are assigned to regions of interests generated during the first iteration expansion operations **6031** and **6032**.

Step **5230** may include:

- Step S**232** of searching for relationships between regions of interest and define regions of interest that are related to the relationships. For example—union or intersection illustrated in FIG. **6**.
- Step **5234** of assigning a unique index for each region of interest within the multiple representations. For example—referring to FIG. **6**—indexes 1, 2, 12 and 21.

Step **5260** may include:

- Step **5262** of generating multiple representations of the merge results of the (k−1)'th iteration within a multi-dimensional space of f(k) dimensions. The expansion operation of step **5260** generates a k'th iteration multidimensional representation of the first image. The number of dimensions of this kth iteration multidimensional representation is denoted f(k).
- Step **5264** of assigning a unique index for each region of interest within the multiple representations.

Step **5270** may include

- Step **5272** of searching for relationships between regions of interest and define regions of interest that are related to the relationships.
- Step **5274** of Assigning a unique index for each region of interest within the multiple representations.

FIG. **1**H illustrates a method **5201**. In method **5201** the relationships between the regions of interest are overlaps.

Thus—step **5232** is replaced by step **5232'** of searching for overlaps between regions of interest and define regions of interest that are related to the overlaps.

Step **5272** is replaced by step **5272'** of searching for overlaps between regions of interest and define regions of interest that are related to the overlaps.

FIG. **1**I illustrates a method **7000** for low-power calculation of a signature.

Method **7000** starts by step **7010** of receiving or generating a media unit of multiple objects.

Step **7010** may be followed by step **7012** of processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation.

The applying of the dimension expansion process of an iteration may include (a) determining a relevancy of the spanning elements of the iteration; and (b) completing the dimension expansion process by relevant spanning elements of the iteration and reducing a power consumption of irrelevant spanning elements until, at least, a completion of the applying of the dimension expansion process.

The identifiers may be retrieval information for retrieving the significant portions.

The at least some of the multiple iterations may be a majority of the multiple iterations.

The output of the multiple iteration may include multiple property attributes for each segment out of multiple segments of the media unit; and wherein the significant portions of an output of the multiple iterations may include more impactful property attributes.

The first iteration of the multiple iteration may include applying the dimension expansion process by applying different filters on the media unit.

The at least some of the multiple iteration exclude at least a first iteration of the multiple iterations. See, for example, FIG. **1**E.

The determining the relevancy of the spanning elements of the iteration may be based on at least some identities of relevant spanning elements of at least one previous iteration.

The determining the relevancy of the spanning elements of the iteration may be based on at least some identities of relevant spanning elements of at least one previous iteration that preceded the iteration.

The determining the relevancy of the spanning elements of the iteration may be based on properties of the media unit.

The determining the relevancy of the spanning elements of the iteration may be performed by the spanning elements of the iteration.

Method **7000** may include a neural network processing operation that may be executed by one or more layers of a neural network and does not belong to the at least some of the multiple iterations. See, for example, FIG. **1**E.

The at least one iteration may be executed without reducing power consumption of irrelevant neurons of the one or more layers.

The one or more layers may output information about properties of the media unit, wherein the information differs from a recognition of the multiple objects.

The applying, by spanning elements of an iteration that differs from the first iteration, the dimension expansion process may include assigning output values that may be indicative of an identity of the relevant spanning elements of the iteration. See, for example, FIG. **1**C.

The applying, by spanning elements of an iteration that differs from the first iteration, the dimension expansion process may include assigning output values that may be indicative a history of dimension expansion processes until the iteration that differs from the first iteration.

The each spanning element may be associated with a subset of reference identifiers. The determining of the relevancy of each spanning elements of the iteration may be based a relationship between the subset of the reference identifiers of the spanning element and an output of a last merge operation before the iteration.

The output of a dimension expansion process of an iteration may be a multidimensional representation of the media unit that may include media unit regions of interest that may be associated with one or more expansion processes that generated the regions of interest.

The merge operation of the iteration may include selecting a subgroup of media unit regions of interest based on a spatial relationship between the subgroup of multidimensional regions of interest. See, for example, FIGS. **3** and **6**.

Method **7000** may include applying a merge function on the subgroup of multidimensional regions of interest. See, for example, FIGS. **1**C and **1**F.

Method **7000** may include applying an intersection function on the subgroup of multidimensional regions of interest. See, for example, FIGS. **1**C and **1**F.

The merge operation of the iteration may be based on an actual size of one or more multidimensional regions of interest.

The merge operation of the iteration may be based on relationship between sizes of the multidimensional regions of interest. For example—larger multidimensional regions of interest may be maintained while smaller multidimensional regions of interest may be ignored of.

The merge operation of the iteration may be based on changes of the media unit regions of interest during at least the iteration and one or more previous iteration.

Step **7012** may be followed by step **7014** of determining identifiers that are associated with significant portions of an output of the multiple iterations.

Step **7014** may be followed by step **7016** of providing a signature that comprises the identifiers and represents the multiple objects.

Localization and Segmentation

Any of the mentioned above signature generation method provides a signature that does not explicitly includes accurate shape information. This adds to the robustness of the signature to shape related inaccuracies or to other shape related parameters.

The signature includes identifiers for identifying media regions of interest.

Each media region of interest may represent an object (for example a vehicle, a pedestrian, a road element, a human made structure, wearables, shoes, a natural element such as a tree, the sky, the sun, and the like) or a part of an object (for example—in the case of the pedestrian—a neck, a head, an arm, a leg, a thigh, a hip, a foot, an upper arm, a forearm, a wrist, and a hand). It should be noted that for object detection purposes a part of an object may be regarded as an object.

The exact shape of the object may be of interest.

FIG. **1**J illustrates method **7002** of generating a hybrid representation of a media unit.

Method **7002** may include a sequence of steps **7020**, **7022**, **7024** and **7026**.

Step **7020** may include receiving or generating the media unit.

Step **7022** may include processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation.

Step **7024** may include selecting, based on an output of the multiple iterations, media unit regions of interest that contributed to the output of the multiple iterations.

Step **7026** may include providing a hybrid representation, wherein the hybrid representation may include (a) shape information regarding shapes of the media unit regions of interest, and (b) a media unit signature that includes identifiers that identify the media unit regions of interest.

Step **7024** may include selecting the media regions of interest per segment out of multiple segments of the media unit. See, for example, FIG. **2**.

Step **7026** may include step **7027** of generating the shape information.

The shape information may include polygons that represent shapes that substantially bound the media unit regions of interest. These polygons may be of a high degree.

In order to save storage space, the method may include step **7028** of compressing the shape information of the media unit to provide compressed shape information of the media unit.

FIG. **1**K illustrates method **5002** for generating a hybrid representation of a media unit.

Method **5002** may start by step **5011** of receiving or generating a media unit.

Step **5011** may be followed by processing the media unit by performing multiple iterations, wherein at least some of the multiple iterations comprises applying, by spanning elements of the iteration, dimension expansion process that are followed by a merge operation.

The processing may be followed by steps **5060** and **5062**.

The processing may include steps **5020**, **5030**, **5040** and **5050**.

Step **5020** may include performing a k'th iteration expansion process (k may be a variable that is used to track the number of iterations).

Step **5030** may include performing a k'th iteration merge process.

Step **5040** may include changing the value of k.

Step **5050** may include checking if all required iterations were done—if so proceeding to steps **5060** and **5062**. Else—jumping to step **5020**.

The output of step **5020** is a k'th iteration expansion result.

The output of step **5030** is a k'th iteration merge result.

For each iteration (except the first iteration)—the merge result of the previous iteration is an input to the current iteration expansion process.

Step **5060** may include completing the generation of the signature.

Step **5062** may include generating shape information regarding shapes of media unit regions of interest. The signature and the shape information provide a hybrid representation of the media unit.

The combination of steps **5060** and **5062** amounts to a providing a hybrid representation, wherein the hybrid representation may include (a) shape information regarding shapes of the media unit regions of interest, and (b) a media unit signature that includes identifiers that identify the media unit regions of interest.

FIG. **1**L illustrates method **5203** for generating a hybrid representation of an image.

Method **5200** may include the following sequence of steps:

- Step **5210** of receiving or generating an image.
- Step **5230** of performing a first iteration expansion operation (which is an expansion operation that is executed during a first iteration)
- Step **5240** of performing a first iteration merge operation.
- Step **5240** of amending index k (k is an iteration counter). In FIG. **1**L in incremented by one—this is only an example of how the number of iterations are tracked.
- Step **5260** of performing a k'th iteration expansion operation on the (k−1)'th iteration merge results.
- Step **5270** of Performing a k'th iteration merge operation (on the k'th iteration expansion operation results.
- Step **5280** of changing the value of index k.
- Step **5290** of checking if all iteration ended (k reached its final value—for example K).

If no—there are still iterations to be executed—jumping from step **5290** to step **5260**.

If yes—jumping to step **5060**.

Step **5060** may include completing the generation of the signature. This may include, for example, selecting significant attributes, determining retrieval information (for example indexes) that point to the selected significant attributes.

Step **5062** may include generating shape information regarding shapes of media unit regions of interest. The signature and the shape information provide a hybrid representation of the media unit.

The combination of steps **5060** and **5062** amounts to a providing a hybrid representation, wherein the hybrid representation may include (a) shape information regarding shapes of the media unit regions of interest, and (b) a media unit signature that includes identifiers that identify the media unit regions of interest.

Step **5220** may include:

- Step **5222** of generating multiple representations of the image within a multi-dimensional space of f(k) dimensions.
- Step **5224** of assigning a unique index for each region of interest within the multiple representations. (for example, referring to FIG. **1**F—indexes 1 and indexes 2 following first iteration expansion operations **6031** and **6032**.

Step **5230** may include

- Step **5226** of searching for relationships between regions of interest and define regions of interest that are related to the relationships. For example—union or intersection illustrated in FIG. **1**F.
- Step **5228** of assigning a unique index for each region of interest within the multiple representations. For example—referring to FIG. **1**F—indexes 1, 2, 12 and 21.

Step **5260** may include:

- Step **5262** of generating multiple representations of the merge results of the (k−1)'th iteration within a multi-dimensional space of f(k) dimensions. The expansion operation of step **5260** generates a k'th iteration multidimensional representation of the first image. The number of dimensions of this kth iteration multidimensional representation is denoted f(k).
- Step **5264** of assigning a unique index for each region of interest within the multiple representations.

Step **5270** may include

- Step **5272** of searching for relationships between regions of interest and define regions of interest that are related to the relationships.
- Step **5274** of assigning a unique index for each region of interest within the multiple representations.

FIG. **1**M illustrates method **5205** for generating a hybrid representation of an image.

Method **5200** may include the following sequence of steps:

- Step **5210** of receiving or generating an image.
- Step **5230** of performing a first iteration expansion operation (which is an expansion operation that is executed during a first iteration)
- Step **5240** of performing a first iteration merge operation.
- Step **5240** of amending index k (k is an iteration counter). In FIG. **1**M in incremented by one—this is only an example of how the number of iterations are tracked.
- Step **5260** of performing a k'th iteration expansion operation on the (k−1)'th iteration merge results.
- Step **5270** of performing a k'th iteration merge operation (on the k'th iteration expansion operation results.
- Step **5280** of changing the value of index k.
- Step **5290** of checking if all iteration ended (k reached its final value—for example K).

If no—there are still iterations to be executed—jumping from step **5290** to step **5260**.

If yes—jumping to steps **5060** and **5062**.

Step **5060** may include completing the generation of the signature. This may include, for example, selecting significant attributes, determining retrieval information (for example indexes) that point to the selected significant attributes.

Step **5062** may include generating shape information regarding shapes of media unit regions of interest. The signature and the shape information provide a hybrid representation of the media unit.

The combination of steps **5060** and **5062** amounts to a providing a hybrid representation, wherein the hybrid representation may include (a) shape information regarding shapes of the media unit regions of interest, and (b) a media unit signature that includes identifiers that identify the media unit regions of interest.

Step **5220** may include:

Step **5221** of filtering the image with multiple filters that are orthogonal to each other to provide multiple filtered images that are representations of the image in a multi-dimensional space of f(1) dimensions. The expansion operation of step S**220** generates a first iteration multidimensional representation of the first image. The number of filters is denoted f(1).

Step **5224** of assigning a unique index for each region of interest within the multiple representations. (for example, referring to FIG. **1**F—indexes 1 and indexes 2 following first iteration expansion operations **6031** and **6032**.

Step **5230** may include

Step **5226** of searching for relationships between regions of interest and define regions of interest that are related to the relationships. For example—union or intersection illustrated in FIG. **1**F.

Step **5228** of assigning a unique index for each region of interest within the multiple representations. For example—referring to FIG. **1**F—indexes 1, 2, 12 and 21.

Step **5260** may include:

Step **5262** of generating multiple representations of the merge results of the (k−1)'th iteration within a multi-dimensional space of f(k) dimensions. The expansion operation of step **5260** generates a k'th iteration multi-dimensional representation of the first image. The number of dimensions of this kth iteration multidimensional representation is denoted f(k).

Step **5264** of assigning a unique index for each region of interest within the multiple representations.

Step **5270** may include

Step **5272** of searching for relationships between regions of interest and define regions of interest that are related to the relationships.

Step **5274** of assigning a unique index for each region of interest within the multiple representations.

The filters may be orthogonal may be non-orthogonal—for example be decorrelated. Using non-orthogonal filters may increase the number of filters—but this increment may be tolerable.

Object Detection Using Compressed Shape Information

Object detection may include comparing a signature of an input image to signatures of one or more cluster structures in order to find one or more cluster structures that include one or more matching signatures that match the signature of the input image.

The number of input images that are compared to the cluster structures may well exceed the number of signatures of the cluster structures. For example—thousands, tens of thousands, hundreds of thousands (and even more) of input signature may be compared to much less cluster structure signatures. The ratio between the number of input images to the aggregate number of signatures of all the cluster structures may exceed ten, one hundred, one thousand, and the like.

In order to save computational resources, the shape information of the input images may be compressed.

On the other hand—the shape information of signatures that belong to the cluster structures may be uncompressed—and of higher accuracy than those of the compressed shape information.

When the higher quality is not required—the shape information of the cluster signature may also be compressed.

Compression of the shape information of cluster signatures may be based on a priority of the cluster signature, a popularity of matches to the cluster signatures, and the like.

The shape information related to an input image that matches one or more of the cluster structures may be calculated based on shape information related to matching signatures.

For example—a shape information regarding a certain identifier within the signature of the input image may be determined based on shape information related to the certain identifiers within the matching signatures.

Any operation on the shape information related to the certain identifiers within the matching signatures may be applied in order to determine the (higher accuracy) shape information of a region of interest of the input image identified by the certain identifier.

For example—the shapes may be virtually overlaid on each other and the population per pixel may define the shape.

For example—only pixels that appear in at least a majority of the overlaid shaped should be regarded as belonging to the region of interest.

Other operations may include smoothing the overlaid shapes, selecting pixels that appear in all overlaid shapes.

The compressed shape information may be ignored of or be considered.

FIG. **1**N illustrates method **7003** of determining shape information of a region of interest of a media unit.

Method **7003** may include a sequence of steps **7030**, **7032** and **7034**.

Step **7030** may include receiving or generating a hybrid representation of a media unit. The hybrid representation includes compressed shape information.

Step **7032** may include comparing the media unit signature of the media unit to signatures of multiple concept structures to find a matching concept structure that has at least one matching signature that matches to the media unit signature.

Step **7034** may include calculating higher accuracy shape information that is related to regions of interest of the media unit, wherein the higher accuracy shape information is of higher accuracy than the compressed shape information of the media unit, wherein the calculating is based on shape information associated with at least some of the matching signatures.

Step **7034** may include at least one out of:

Determining shapes of the media unit regions of interest using the higher accuracy shape information.

For each media unit region of interest, virtually overlaying shapes of corresponding media units of interest of at least some of the matching signatures.

FIG. **1**O illustrates a matching process and a generation of a higher accuracy shape information.

It is assumed that there are multiple (M) cluster structures **4974**(**1**)-**4974**(M). Each cluster structure includes cluster signatures, metadata regarding the cluster signatures, and shape information regarding the regions of interest identified by identifiers of the cluster signatures.

For example—first cluster structure **4974**(**1**) includes multiple (N**1**) signatures (referred to as cluster signatures CS) CS(**1**,**1**)-CS(**1**,N**1**) **4975**(**1**,**1**)-**4975**(**1**,N**1**), metadata **4976**(**1**), and shape information (Shapeinfo **4977**(**1**)) regarding shapes of regions of interest associated with identifiers of the CSs.

Yet for another example—M'th cluster structure **4974**(M) includes multiple (N**2**) signatures (referred to as cluster signatures CS) CS(M,**1**)-CS(M,N**2**) **4975**(M,**1**)-**4975**(M, N**2**), metadata **4976**(M), and shape information (Shapeinfo **4977**(M)) regarding shapes of regions of interest associated with identifiers of the CSs.

The number of signatures per concept structure may change over time—for example due to cluster reduction attempts during which a CS is removed from the structure to provide a reduced cluster structure, the reduced structure is checked to determine that the reduced cluster signature may still identify objects that were associated with the (non-reduced) cluster signature—and if so the signature may be reduced from the cluster signature.

The signatures of each cluster structures are associated to each other, wherein the association may be based on similarity of signatures and/or based on association between metadata of the signatures.

Assuming that each cluster structure is associated with a unique object—then objects of a media unit may be identified by finding cluster structures that are associated with said objects. The finding of the matching cluster structures may include comparing a signature of the media unit to signatures of the cluster structures—and searching for one or more matching signature out of the cluster signatures.

In FIG. **1**O—a media unit having a hybrid representation undergoes object detection. The hybrid representation includes media unit signature **4972** and compressed shape information **4973**.

The media unit signature **4972** is compared to the signatures of the M cluster structures—from CS(**1**,**1**) **4975**(**1**,**1**) till CS(M,N**2**) **4975**(M,N**2**).

We assume that one or more cluster structures are matching cluster structures.

Once the matching cluster structures are found the method proceeds by generating shape information that is of higher accuracy then the compressed shape information.

The generation of the shape information is done per identifier.

For each j that ranges between 1 and J (J is the number of identifiers per the media unit signature **4972**) the method may perform the steps of:

Find (step **4978**(*j*)) the shape information of the j'th identifier of each matching signature—or of each signature of the matching cluster structure.

Generate (step **4979**(*j*)) a higher accuracy shape information of the j'th identifier.

For example—assuming that the matching signatures include CS(**1**,**1**) **2975**(**1**,**1**), CS(**2**,**5**) **2975**(**2**,**5**), CS(**7**,**3**) **2975** (**7**,**3**) and CS(**15**,**2**) **2975**(**15**,**2**), and that the j'th identifier is included in CS(**1**,**1**) **2975**(**1**,**1**), CS(**7**,**3**) **2975**(**7**,**3**) and CS(**15**,**2**) **2975**(**15**,**2**)—then the shape information of the j'th identifier of the media unit is determined based on the shape information associated with CS(**1**,**1**) **2975**(**1**,**1**),CS(**7**,**3**) **2975**(**7**,**3**) and CS(**15**,**2**) **2975**(**15**,**2**).

FIG. **1**P illustrates an image **8000** that includes four regions of interest **8001**, **8002**, **8003** and **8004**. The signature **8010** of image **8000** includes various identifiers including ID**1** **8011**, ID**2** **8012**, ID**3** **8013** and ID**4** **8014** that identify the four regions of interest **8001**, **8002**, **8003** and **8004**.

The shapes of the four regions of interest **8001**, **8002**, **8003** and **8004** are four polygons. Accurate shape information regarding the shapes of these regions of interest may be generated during the generation of signature **8010**.

FIG. **1**Q illustrates the compressing of the shape information to represent a compressed shape information that reflects simpler approximations (**8001'**, **8002'**, **8003'** and **8004'**) of the regions of interest **8001**, **8002**, **8003** and **8004**. In this example simpler may include less facets, fewer values of angles, and the like.

The hybrid representation of the media unit, after compression represent a media unit with simplified regions of interest **8001'**, **8002'**, **8003'** and **8004'**—as shown in FIG. **1**R.

Scale Based Bootstrap

Objects may appear in an image at different scales. Scale invariant object detection may improve the reliability and repeatability of the object detection and may also use fewer number of cluster structures—thus reduced memory resources and also lower computational resources required to maintain fewer cluster structures.

FIG. **1**S illustrates method **8020** for scale invariant object detection.

Method **8020** may include a first sequence of steps that may include step **8022**, **8024**, **8026** and **8028**.

Step **8022** may include receiving or generating a first image in which an object appears in a first scale and a second image in which the object appears in a second scale that differs from the first scale.

Step **8024** may include generating a first image signature and a second image signature.

The first image signature includes a first group of at least one certain first image identifier that identifies at least a part of the object. See, for example image **8000'** of FIG. **2**A. The person is identified by identifiers ID**6** **8016** and ID**8** **8018** that represent regions of interest **8006** and **8008**.

The second image signature includes a second group of certain second image identifiers that identify different parts of the object.

See, for example image **8000** of FIG. **19**. The person is identified by identifiers ID**1** **8011**, ID**2** **8012**, ID**3** **8013**, and ID**4** **8014** that represent regions of interest **8001**, **8002**, **8003** and **8004**.

The second group is larger than first group—as the second group has more members than the first group.

Step **8026** may include linking between the at least one certain first image identifier and the certain second image identifiers.

Step **8026** may include linking between the first image signature, the second image signature and the object.

Step **8026** may include adding the first signature and the second signature to a certain concept structure that is associated with the object. For example, referring to FIG. **1**O, the signatures of the first and second images may be included in a cluster concept out of **4974**(**1**)-**4974**(M).

Step **8028** may include determining whether an input image includes the object based, at least in part, on the linking. The input image differs from the first and second images.

The determining may include determining that the input image includes the object when a signature of the input image includes the at least one certain first image identifier or the certain second image identifiers.

The determining may include determining that the input image includes the object when the signature of the input image includes only a part of the at least one certain first image identifier or only a part of the certain second image identifiers.

The linking may be performed for more than two images in which the object appears in more than two scales.

For example, see FIG. **2**B in which a person appears at three different scales—at three different images.

In first image **8051** the person is included in a single region of interest **8061** and the signature **8051'** of first image **8051** includes an identifier ID**61** that identifies the single region of interest—identifies the person.

In second image **8052** the upper part of the person is included in region of interest **8068**, the lower part of the person is included in region of interest **8069** and the signature **8052'** of second image **8052** includes identifiers ID**68** and ID**69** that identify regions of interest **8068** and **8069** respectively.

In third image **8053** the eyes of the person are included in region of interest **8062**, the mouth of the person is included in region of interest **8063**, the head of the person appears in region of interest **8064**, the neck and arms of the person appear in region of interest **8065**, the middle part of the person appears in region of interest **8066**, and the lower part of the person appears in region of interest **8067**. Signature **8053'** of third image **8053** includes identifiers ID**62**, ID**63**, ID**64**, ID**65**, ID**55** and ID**67** that identify regions of interest **8062**-**8067** respectively.

Method **8020** may link signatures **8051'**, **8052'** and **8053'** to each other. For example—these signatures may be included in the same cluster structure.

Method **8020** may link (i) ID**61**, (ii) signatures ID**68** and ID**69**, and (ii) signature ID**62**, ID**63**, ID**64**, ID**65**, ID**66** and ID**67**.

FIG. **2**C illustrates method **8030** for object detection.

Method **8030** may include the steps of method **8020** or may be preceded by steps **8022**, **8024** and **8026**.

Method **8030** may include a sequence of steps **8032**, **8034**, **8036** and **8038**.

Step **8032** may include receiving or generating an input image.

Step **8034** may include generating a signature of the input image.

Step **8036** may include comparing the signature of the input image to signatures of a certain concept structure. The certain concept structure may be generated by method **8020**.

Step **8038** may include determining that the input image comprises the object when at least one of the signatures of the certain concept structure matches the signature of the input image.

FIG. **2**D illustrates method **8040** for object detection.

Method **8040** may include the steps of method **8020** or may be preceded by steps **8022**, **8024** and **8026**.

Method **8040** may include a sequence of steps **8041**, **8043**, **8045**, **8047** and **8049**.

Step **8041** may include receiving or generating an input image.

Step **8043** may include generating a signature of the input image, the signature of the input image comprises only some of the certain second image identifiers; wherein the input image of the second scale.

Step **8045** may include changing a scale of the input image to the first scale to a provide an amended input image.

Step **8047** may include generating a signature of the amended input image.

Step **8049** may include verifying that the input image comprises the object when the signature of the amended input image comprises the at least one certain first image identifier.

FIG. **2**E illustrates method **8050** for object detection.

Method **8050** may include the steps of method **8020** or may be preceded by steps **8022**, **8024** and **8026**.

Method **8050** may include a sequence of steps **8052**, **8054**, **8056** and **8058**.

Step **8052** may include receiving or generating an input image.

Step **8054** may include generating a signature of the input image.

Step **8056** may include searching in the signature of the input image for at least one of (a) the at least one certain first image identifier, and (b) the certain second image identifiers.

Step **8058** may include determining that the input image comprises the object when the signature of the input image comprises the at least one of (a) the at least one certain first image identifier, and (b) the certain second image identifiers.

It should be noted that step **8056** may include searching in the signature of the input image for at least one of (a) one or more certain first image identifier of the at least one certain first image identifier, and (b) at least one certain second image identifier of the certain second image identifiers.

It should be noted that step **8058** may include determining that the input image includes the object when the signature of the input image comprises the at least one of (a) one or more certain first image identifier of the at least one certain first image identifier, and (b) the at least one certain second image identifier.

Movement Based Bootstrapping

A single object may include multiple parts that are identified by different identifiers of a signature of the image. In cases such as unsupervised learning, it may be beneficial to link the multiple object parts to each other without receiving prior knowledge regarding their inclusion in the object.

Additionally or alternatively, the linking can be done in order to verify a previous linking between the multiple object parts.

FIG. **2**F illustrates method **8070** for object detection.

Method **8070** is for movement based object detection.

Method **8070** may include a sequence of steps **8071**, **8073**, **8075**, **8077**, **8078** and **8079**.

Step **8071** may include receiving or generating a video stream that includes a sequence of images.

Step **8073** may include generating image signatures of the images. Each image is associated with an image signature that comprises identifiers. Each identifier identifiers a region of interest within the image.

Step **8075** may include generating movement information indicative of movements of the regions of interest within consecutive images of the sequence of images. Step **8075** may be preceded by or may include generating or receiving location information indicative of a location of each region of interest within each image. The generating of the movement information is based on the location information.

Step **8077** may include searching, based on the movement information, for a first group of regions of interest that follow a first movement. Different first regions of interest are associated with different parts of an object.

Step **8078** may include linking between first identifiers that identify the first group of regions of interest.

Step **8079** may include linking between first image signatures that include the first linked identifiers.

Step **8079** may include adding the first image signatures to a first concept structure, the first concept structure is associated with the first image.

Step **8079** may be followed by determining whether an input image includes the object based, at least in part, on the linking

An example of various steps of method **8070** is illustrated in FIG. **2**H.

FIG. **2**G illustrates three images **8091**, **8092** and **8093** that were taken at different points in time.

First image **8091** illustrates a gate **8089**' that is located in region of interest **8089** and a person that faces the gate. Various parts of the person are located within regions of interest **8081**, **8082**, **8083**, **8084** and **8085**.

The first image signature **8091**' includes identifiers ID**81**, ID**82**, ID**83**, ID**84**, ID**85** and ID**89** that identify regions of interest **8081**, **8082**, **8083**, **8084**, **8085** and **8089** respectively.

The first image location information **8091**" includes the locations L**81**, L**82**, L**83**, L**84**, L**85** and L**89** of regions of interest **8081**, **8082**, **8083**, **8084**, **8085** and **8089** respectively. A location of a region of interest may include a location of the center of the region of interest, the location of the borders of the region of interest or any location information that may define the location of the region of interest or a part of the region of interest.

Second image **8092** illustrates a gate that is located in region of interest **8089** and a person that faces the gate. Various parts of the person are located within regions of interest **8081**, **8082**, **8083**, **8084** and **8085**. Second image also includes a pole that is located within region of interest **8086**. In the first image that the pole was concealed by the person.

The second image signature **8092**' includes identifiers ID**81**, ID**82**, ID**83**, ID**84**, ID**85**, ID**86**, and ID**89** that identify regions of interest **8081**, **8082**, **8083**, **8084**, **8085**, **8086** and **8089** respectively.

The second image location information **8092**" includes the locations L**81**, L**82**, L**83**, L**84**, L**85**, L**86** and L**89** of regions of interest **8081**, **8082**, **8083**, **8084**, **8085**, **8086** and **8089** respectively.

Third image **8093** illustrates a gate that is located in region of interest **8089** and a person that faces the gate. Various parts of the person are located within regions of interest **8081**, **8082**, **8083**, **8084** and **8085**. Third image also includes a pole that is located within region of interest **8086**, and a balloon that is located within region of interest **8087**.

The third image signature **8093**' includes identifiers ID**81**, ID**82**, ID**83**, ID**84**, ID**85**, ID**86**, ID**87** and ID**89** that identify regions of interest **8081**, **8082**, **8083**, **8084**, **8085**, **8086**, **8087** and **8089** respectively.

The third image location information **8093**" includes the locations L**81**, L**82**, L**83**, L**84**, L**85**, L**86**, L**87** and L**89** of regions of interest **8081**, **8082**, **8083**, **8084**, **8085**, **8086**, **8086** and **8089** respectively.

The motion of the various regions of interest may be calculated by comparing the location information related to different images. The movement information may consider the different in the acquisition time of the images.

The comparison shows that regions of interest **8081**, **8082**, **8083**, **8084**, **8085** move together and thus they should be linked to each other—and it may be assumed that they all belong to the same object.

FIG. **2**H illustrates method **8100** for object detection.

Method **8100** may include the steps of method **8070** or may be preceded by steps **8071**, **8073**, **8075**, **8077** and **8078**.

Method **8100** may include the following sequence of steps:

Step **8102** of receiving or generating an input image.

Step **8104** of generating a signature of the input image.

Step **8106** of comparing the signature of the input image to signatures of a first concept structure. The first concept structure includes first identifiers that were linked to each other based on movements of first regions of interest that are identified by the first identifiers.

Step **8108** of determining that the input image includes a first object when at least one of the signatures of the first concept structure matches the signature of the input image.

FIG. **2**I illustrates method **8110** for object detection.

Method **8110** may include the steps of method **8070** or may be preceded by steps **8071**, **8073**, **8075**, **8077** and **8078**.

Method **8110** may include the following sequence of steps:

Step **8112** of receiving or generating an input image.

Step **8114** of generating a signature of the input image.

Step **8116** of searching in the signature of the input image for at least one of the first identifiers.

Step **8118** of determining that the input image comprises the object when the signature of the input image comprises at least one of the first identifiers.

Object Detection that is Robust to Angle of Acquisition

Object detection may benefit from being robust to the angle of acquisition—to the angle between the optical axis of an image sensor and a certain part of the object. This allows the detection process to be more reliable, use fewer different clusters (may not require multiple clusters for identifying the same object from different images).

FIG. **2**J illustrates method **8120** that includes the following steps:

Step **8122** of receiving or generating images of objects taken from different angles.

Step **8124** of finding images of objects taken from different angles that are close to each other. Close enough may be less than 1, 5, 10, 15 and 20 degrees—but the closeness may be better reflected by the reception of substantially the same signature.

Step **8126** of linking between the images of similar signatures. This may include searching for local similarities. The similarities are local in the sense that they are calculated per a subset of signatures. For example—assuming that the similarity is determined per two images—then a first signature may be linked to a second signature that is similar to the first image. A third signature may be linked to the second image based on the similarity between the second and third signatures—and even regardless of the relationship between the first and third signatures.

Step **8126** may include generating a concept data structure that includes the similar signatures.

This so-called local or sliding window approach, in addition to the acquisition of enough images (that will statistically provide a large angular coverage) will enable to generate a concept structure that include signatures of an object taken at multiple directions.

FIG. **2**K illustrates a person **8130** that is imaged from different angles (**8131**, **8132**, **8133**, **8134**, **8135** and **8136**). While the signature of a front view of the person (obtained from angle **8131**) differs from the signature of the side view of the person (obtained from angle **8136**), the signature of images taken from multiple angles between angles **8141** and

8136 compensates for the difference—as the difference between images obtained from close angles are similar (local similarity) to each other.

Signature Tailored Matching Threshold

Object detection may be implemented by (a) receiving or generating concept structures that include signatures of media units and related metadata, (b) receiving a new media unit, generating a new media unit signature, and (c) comparing the new media unit signature to the concept signatures of the concept structures.

The comparison may include comparing new media unit signature identifiers (identifiers of objects that appear in the new media unit) to concept signature identifiers and determining, based on a signature matching criteria whether the new media unit signature matches a concept signature. If such a match is found, then the new media unit is regarded as including the object associated with that concept structure.

It was found that by applying an adjustable signature matching criteria, the matching process may be highly effective and may adapt itself to the statistics of appearance of identifiers in different scenarios. For example—a match may be obtained when a relatively rear but highly distinguishing identifier appears in the new media unit signature and in a cluster signature, but a mismatch may be declared when multiple common and slightly distinguishing identifiers appear in the new media unit signature and in a cluster signature.

FIG. **2**L illustrates method **8200** for object detection.

Method **8200** may include:

Step **8210** of receiving an input image.

Step **8212** of generating a signature of the input image.

Step **8214** of comparing the signature of the input image to signatures of a concept structure.

Step **8216** of determining whether the signature of the input image matches any of the signatures of the concept structure based on signature matching criteria, wherein each signature of the concept structure is associated within a signature matching criterion that is determined based on an object detection parameter of the signature.

Step **8218** of concluding that the input image comprises an object associated with the concept structure based on an outcome of the determining.

The signature matching criteria may be a minimal number of matching identifiers that indicate of a match. For example—assuming a signature that include few tens of identifiers, the minimal number may vary between a single identifier to all of the identifiers of the signature.

It should be noted that an input image may include multiple objects and that a signature of the input image may match multiple cluster structures. Method **8200** is applicable to all of the matching processes- and that the signature matching criteria may be set for each signature of each cluster structure.

Step **8210** may be preceded by step **8202** of determining each signature matching criterion by evaluating object detection capabilities of the signature under different signature matching criteria.

Step **8202** may include:

Step **8203** of receiving or generating signatures of a group of test images.

Step **8204** of calculating the object detection capability of the signature, for each signature matching criterion of the different signature matching criteria.

Step **8206** of selecting the signature matching criterion based on the object detection capabilities of the signature under the different signature matching criteria.

The object detection capability may reflect a percent of signatures of the group of test images that match the signature.

The selecting of the signature matching criterion comprises selecting the signature matching criterion that once applied results in a percent of signatures of the group of test images that match the signature that is closets to a predefined desired percent of signatures of the group of test images that match the signature.

The object detection capability may reflect a significant change in the percent of signatures of the group of test images that match the signature. For example—assuming, that the signature matching criteria is a minimal number of matching identifiers and that changing the value of the minimal numbers may change the percentage of matching test images. A substantial change in the percentage (for example a change of more than 10, 20, 30, 40 percent) may be indicative of the desired value. The desired value may be set before the substantial change, proximate to the substantial change, and the like.

For example, referring to FIG. **1**O, cluster signatures CS(**1**,**1**), CS(**2**,**5**), CS(**7**,**3**) and CS(**15**,**2**) match unit signature **4972**. Each of these matches may apply a unique signature matching criterion.

FIG. **2**M illustrates method **8220** for object detection.

Method **8220** is for managing a concept structure.

Method **8220** may include:

Step **8222** of determining to add a new signature to the concept structure. The concept structure may already include at least one old signature. The new signature includes identifiers that identify at least parts of objects.

Step **8224** of determining a new signature matching criterion that is based on one or more of the identifiers of the new signature. The new signature matching criterion determines when another signature matches the new signature. The determining of the new signature matching criterion may include evaluating object detection capabilities of the signature under different signature matching criteria.

Step **8224** may include steps **8203**, **8204** and **8206** (include din step **8206**) of method **8200**.

Examples of Systems

FIG. **22**N illustrates an example of a system capable of executing one or more of the mentioned above methods.

The system include various components, elements and/or units.

A component element and/or unit may be a processing circuitry may be implemented as a central processing unit (CPU), and/or one or more other integrated circuits such as application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), full-custom integrated circuits, etc., or a combination of such integrated circuits.

Alternatively, each component element and/or unit may be implemented in hardware, firmware, or software that may be executed by a processing circuitry.

System **4900** may include sensing unit **4902**, communication unit **4904**, input **4911**, processor **4950**, and output **4919**. The communication unit **4904** may include the input and/or the output.

Input and/or output may be any suitable communications component such as a network interface card, universal serial bus (USB) port, disk reader, modem or transceiver that may be operative to use protocols such as are known in the art to communicate either directly, or indirectly, with other elements of the system.

Processor **4950** may include at least some out of
- Multiple spanning elements **4951**(q).
- Multiple merge elements **4952**(r).
- Object detector **4953**.
- Cluster manager **4954**.

Controller **4955**.
- Selection unit **4956**.
- Object detection determination unit **4957**.
- Signature generator **4958**.
- Movement information unit **4959**.
- Identifier unit **4960**.

Obstacle Avoidance

It will be appreciated that in "normal", non-assisted driving scenarios, a driver may encounter driving obstacles that are not necessarily indicated in commonly available roadmaps. For example, it is not uncommon for a roadway to have potholes that may complicate the driving process. Some potholes are large enough to cause damage to the wheels or undercarriage of the vehicle. Accordingly, when encountering a pothole, the driver may swerve to avoid it. However, depending on the layout of the roadway and/or traffic conditions, it may not be feasible for the driver to react in such a manner. It is also possible that depending on current visibility, the speed of the vehicle, and/or the alertness of the driver, the driver may not even see the pothole in time to avoid it.

It will be appreciated that such issues are also relevant in an assisted/autonomous driving scenario. As discussed hereinabove, such systems may use one or more sensors to acquire information about the current driving environment in order to determine how to drive along the roadway as per a stored map. It is possible, that the vehicle's sensor(s) may not detect an obstacle, e.g., a pothole, sufficiently in advance in order to enable the system to determine that evasive action should be taken to avoid the obstacle and to perform that action in time. It will be appreciated that such a scenario may be further complicated by the presence of other vehicles and/or pedestrians in the immediate vicinity of the vehicle.

Reference is now made to FIG. **3**A, which is a partly-pictorial, partly-block diagram illustration of an exemplary obstacle detection and mapping system **10** (hereinafter referred to also as "system **10**") constructed and operative in accordance with embodiments described herein. As described herein, system **10** may be operative to process sensor data to detect and map obstacles on a roadway in order to enable an assisted/autonomous driving system to predict and avoid obstacles in the path of a vehicle.

System **10** comprises vehicle **100** and obstacle avoidance server **400** which may be configured to communicate with each other over a communications network such as, for example, the Internet. In accordance with the exemplary embodiment of FIG. **3**A, vehicle **100** may be configured with an autonomous driving system (not shown) operative to autonomously provide driving instructions to vehicle **100** without the intervention of a human driver. It will be appreciated that the embodiments described herein may also support the configuration of vehicle **100** with an assisted (or "semi-autonomous") driving system where in at least some situations a human driver may take control of vehicle **100** and/or where in at least some situations the semi-autonomous driving system provides warnings to the driver without necessarily directly controlling vehicle **100**.

In accordance with the exemplary embodiment of FIG. **3**A, vehicle **100** may be configured with at least one sensor **130** to provide information about a current driving environment as vehicle **100** proceeds along roadway **20**. It will be appreciated that while sensor **130** is depicted in FIG. **3**A as a single entity, in practice, as will be described hereinbelow, there may be multiple sensors **130** arrayed on, or inside of, vehicle **130**. In accordance with embodiments described herein, sensor(s) **130** may be implemented using a conventional camera operative to capture images of roadway **20** and objects in its immediate vicinity. It will be appreciated that sensor **130** may be implemented using any suitable imaging technology instead of, or in addition to, a conventional camera. For example, sensor **130** may also be operative to use infrared, radar imagery, ultrasound, electro-optics, radiography, LIDAR (light detection and ranging), etc. Furthermore, in accordance with some embodiments, one or more sensors **130** may also be installed independently along roadway **20**, where information from such sensors **130** may be provided to vehicle **100** and/or obstacle avoidance server **400** as a service.

In accordance with the exemplary embodiment of FIG. **3**A, static reference points **30**A and **30**B (collectively referred to hereinafter as static reference points **30**) may be located along roadway **20**. For example, static reference point **30**A is depicted as a speed limit sign, and static reference point **30**B is depicted as an exit sign. In operation, sensor **130** may capture images of static reference points **30**. The images may then be processed by the autonomous driving system in vehicle **100** to provide information about the current driving environment for vehicle **100**, e.g., the speed limit or the location of an upcoming exit.

Obstacle **40**, e.g., a pothole, may be located on roadway **20**. In accordance with embodiments described herein, sensor **130** may also be operative to capture images of obstacle **40**. The autonomous driving system may be operative to detect the presence of obstacle **40** in the images provided by sensor **130** and to determine an appropriate response, e.g., whether or not vehicle **100** should change speed and/or direction to avoid or minimize the impact with obstacle **40**. For example, if the autonomous driving system determines that obstacle **40** is a piece of paper on roadway **20**, no further action may be necessary. However, if obstacle **40** is a pothole, the autonomous driving system may instruct vehicle **100** to slow down and/or swerve to avoid obstacle **40**.

It will be appreciated that there may not be sufficient processing time for the autonomous driving system to determine an appropriate response; depending on the speed of vehicle **100** and the distance at which sensor **130** captures an image of obstacle **40**, vehicle **100** may drive over/through obstacle **40** (or collide with it) before obstacle **40** is detected and/or an appropriate response may be determined by the autonomous driving system. In accordance with embodiments described herein, in order to provide additional processing time for obstacle detection and response determination, vehicle **100** may exchange obstacle information with obstacle avoidance server **400**. The autonomous driving system may be operative to upload the imagery received from sensor **130** and/or the determination of the nature of obstacle **40** to obstacle avoidance server **400**, and obstacle avoidance server **400** may be operative to provide obstacle information received in such manner from other vehicles **100** in order to enable the autonomous driving system to anticipate and respond to obstacles **40** in advance of, or in parallel to, receiving the relevant imagery from sensor **130**.

Depending on the configuration of system **10** and vehicle **100**, obstacle avoidance server **400** and vehicle **100** may exchange obstacle information in real-time (or near real-time) and/or in a burst mode, where the obstacle information may be provided either periodically and/or when conditions are suitable for a data exchange between obstacle avoidance server **400** and vehicle **100**. For example, vehicle **100** may be configured to perform regular uploads/downloads of obstacle information when the engine is turned on (or off, with a keep alive battery function to facilitate communication with obstacle avoidance server **400**). Alternatively, or in addition, vehicle **100** may be configured to perform such uploads/downloads when stationary (e.g., parked, or waiting at a traffic light) and a sufficiently reliable wireless connection is detected. Alternatively, or in addition, the uploads/downloads may be triggered by location, where obstacle information may be downloaded from obstacle avoidance server **400** when vehicle **100** enters an area for which it does not have up-to-date obstacle information.

It will also be appreciated that in some situations, the autonomous driving system may not detect obstacle **40** at all; it is possible that vehicle **100** may pass by/over/through obstacle **40** without detecting it. The autonomous driving system may have finite resources available to analyze imagery from sensor **130** in a finite period of time in order to determine a currently relevant driving policy. It is understandable that if, for whatever reason (e.g., driving speed, visibility, etc.), the autonomous driving system does not detect obstacle **40** in real-time or near-real time there may be little benefit to be realized from continuing processing the imagery on vehicle **100**. However, the detection of a "missed" obstacle **40** may be of benefit to another vehicle **100** on roadway **20**.

Accordingly, in some configurations of system **10**, vehicle **100** may also be configured to upload raw driving data from sensor **130** for processing by obstacle avoidance server **400** to detect obstacles **40**. Obstacle information associated with obstacles **40** detected in such manner may then be provided by obstacle avoidance server **400** as described hereinabove.

It will also be appreciated that it may be possible to use non-image data to detect obstacles **40** from the raw driving data. For example, at least one sensor **130** may be implemented as an electronic control unit (ECU) controlling the shock absorbers of vehicle **100**. Obstacle avoidance server **400** may be configured to determine that one or more shocks of a certain magnitude and/or pattern registered by the shock absorber ECU may be indicative of a pothole.

In accordance with some embodiments described herein, the sources for raw driving data to be used by obstacle avoidance server **400** to detect obstacles **40** may not be limited to just vehicles **100** with an autonomous driving system. Obstacle avoidance server **400** may also receive raw driving data from vehicles that are being driven manually and/or with the assistance of a semi-autonomous driving system. It will be appreciated that in for such vehicles, the raw driving data may include indications of driver recognition of obstacles **40**. For example, when a driver detects obstacle **40**, e.g., sees a pothole, the driver may slow down in approach of obstacle **40**, swerve to avoid it, and/or even change lanes. Accordingly, the raw driving data processed by obstacle avoidance server **400** may also include raw data uploaded from vehicles without an autonomous driving system.

Obstacle avoidance server **400** may use supervised and/or unsupervised learning methods to detect obstacles **40** in the raw driving data; reference data sets for the detection of obstacles **40** and the determination of associated driving policies may be prepared based on unsupervised analysis of raw driving data and/or using manual labelling and extraction.

It will be appreciated that it may not be a trivial task to pinpoint a location for obstacle **40**. While it may generally be assumed that vehicle **100** is configured with a location-based service, e.g., global positioning satellite (GPS) system, such systems may not necessarily be accurate enough to provide a sufficient resolution regarding the expected location for obstacle **40**. For example, one possible driving policy for evasive action may entail avoiding obstacle **40** by swerving or changing lanes on roadway **20**. If the location-based service is only accurate to within five feet, swerving within the lane may actually increase the likelihood of vehicle **100** hitting the pothole. Similarly, it may not be possible to determine in which lane obstacle **40** is located.

In accordance with embodiments described herein, static reference points **30** may be used in system **10** to more accurately determine a location for obstacle **40**; the location of obstacle **40** may be defined as an offset from a fixed location for one of static reference points **30**. For example, in the exemplary embodiment of FIG. **3**A, the location of obstacle **40** may be defined as "x" feet past static reference point **30**A, where the distance in feet may be calculated as a function of the frames per second of the camera used for sensor **130** and the speed of vehicle **100**. Alternatively, or in addition, two sensors **130** may be employed to use triangulation to determine the distance between static reference point **30**A and obstacle **40**. Alternatively, or in addition, the distance may be calculated using both reference points **30**A and **30**B to triangulate.

It will be appreciated that as described hereinabove, obstacle **40** may be a "persistent obstacle" in that it may be reasonably expected to remain on roadway **20** for an extended period of time. For example, potholes typically remain (and even increase in size) for several months before being repaired. However, some obstacles may be more transient in nature. For example, a tree fallen on the road or debris from a traffic accident may typically be removed from roadway **20** within a few hours. Such transient obstacles may, at least in some cases, be easier to detect than persistent obstacles, e.g., a fallen tree is easier to identifying from an approaching vehicle than a pothole.

It will therefore be appreciated that the driving policies for transient obstacles may be different than for persistent obstacles. For example, in a case where obstacle **40** is a pothole that may be assumed to be persistent, the associated driving policy may entail the autonomous driving system "erring on the side of caution" by performing evasive action (e.g., swerving or changing lanes) even if the pothole isn't detected in the data provided by sensor **130** when vehicle **100** reaches the pothole's expected location. However, in a case where obstacle **40** is a tree on roadway **20**, the associated driving policy may entail the autonomous driving system performing preventative action, e.g., slowing down, prior to reaching the tree's expected location, but not to perform evasive action if the tree isn't actually detected according to the data provided by sensor **130**.

It will also be appreciated that some obstacles may be "recurring obstacles" where transient obstacles recur in the same location. For example, recurring obstacle **50** represents a slight depression in roadway **20**. The impact on vehicle **100** as it drives over a slight depression in roadway **20** may be minimal. However, in some cases it may be observed that the slight depression may fill with water or mud when it rains, or intermittently ice over during the winter. Accordingly, the driving policies for recurring obstacles **50** may more heavily weight other environmental factors, e.g., season or weather, than the driving policies for obstacles **40**.

Reference is now made to FIG. **3**B which is a block diagram of an exemplary autonomous driving system **200** (hereinafter also referred to as system **200**), constructed and implemented in accordance with embodiments described herein. Autonomous driving system **200** comprises processing circuitry **210**, input/output (I/O) module **220**, camera **230**, telemetry ECU **240**, shock sensor **250**, autonomous driving manager **260**, and obstacle database **270**. Autonomous driving manager **260** may be instantiated in a suitable memory for storing software such as, for example, an optical storage medium, a magnetic storage medium, an electronic storage medium, and/or a combination thereof. It will be appreciated that autonomous driving system **200** may be implemented as an integrated component of an onboard computer system in a vehicle, such as, for example, vehicle **100** from FIG. **3**A. Alternatively, system **200** may be implemented and a separate component in communication with the onboard computer system. It will also be appreciated that in the interests of clarity, while autonomous driving system **200** may comprise additional components and/or functionality e.g., for autonomous driving of vehicle **100**, such additional components and/or functionality are not depicted in FIG. **3**B and/or described herein.

Processing circuitry **210** may be operative to execute instructions stored in memory (not shown). For example, processing circuitry **210** may be operative to execute autonomous driving manager **260**. It will be appreciated that processing circuitry **210** may be implemented as a central processing unit (CPU), and/or one or more other integrated circuits such as application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), full-custom integrated circuits, etc., or a combination of such integrated circuits. It will similarly be appreciated that autonomous driving system **200** may comprise more than one instance of processing circuitry **210**. For example, one such instance of processing circuitry **210** may be a special purpose processor operative to execute autonomous driving manager **260** to perform some, or all, of the functionality of autonomous driving system **200** as described herein.

I/O module **220** may be any suitable communications component such as a network interface card, universal serial bus (USB) port, disk reader, modem or transceiver that may be operative to use protocols such as are known in the art to communicate either directly, or indirectly, with other elements of system **10** (FIG. **3**A) and/or system **200**, such as, for example, obstacle avoidance server **400** (FIG. **3**A), camera **230**, telemetry ECU **240**, and/or shock sensor **250**. As such, I/O module **220** may be operative to use a wired or wireless connection to connect to obstacle avoidance server **400** via a communications network such as a local area network, a backbone network and/or the Internet, etc. I/O module **220** may also be operative to use a wired or wireless connection to connect to other components of system **200**, e.g., camera **230**, telemetry ECU **240**, and/or shock sensor **250**. It will be appreciated that in operation I/O module **220** may be implemented as a multiplicity of modules, where different modules may be operative to use different communication technologies. For example, a module providing mobile network connectivity may be used to connect to obstacle avoidance server **400**, whereas a local area wired connection may be used to connect to camera **230**, telemetry ECU **240**, and/or shock sensor **250**.

In accordance with embodiments described herein, camera **230**, telemetry ECU **240**, and shock sensor **250** represent implementations of sensor(s) **130** from FIG. **3**A. It will be appreciated that camera **230**, telemetry ECU **240**, and/or shock sensor **250** may be implemented as integrated components of vehicle **100** (FIG. **3**A) and may provide other functionality that is the interests of clarity is not explicitly described herein. As described hereinbelow, system **200** may use information about a current driving environment as received from camera **230**, telemetry ECU **240**, and/or shock sensor **250** to determine an appropriate driving policy for vehicle **100**.

Autonomous driving manager **260** may be an application implemented in hardware, firmware, or software that may be executed by processing circuitry **210** to provide driving instructions to vehicle **100**. For example, autonomous driving manager **260** may use images received from camera **230** and/or telemetry data received from telemetry ECU **240** to determine an appropriate driving policy for arriving at a given destination and provide driving instructions to vehicle **100** accordingly. It will be appreciated that autonomous driving manager **260** may also be operative to use other data sources when determining a driving policy, e.g., maps of potential routes, traffic congestion reports, etc.

As depicted in FIG. **3**B, autonomous driving manager **260** comprises obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268**. It will be appreciated that the depiction of obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268** as integrated components of autonomous driving manager **260** may be exemplary. The embodiments described herein may also support implementation of obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268** as independent applications in communication with autonomous driving manager **260**, e.g., via I/O module **220**.

Obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268** may be implemented in hardware, firmware, or software and may be invoked by autonomous driving manager **260** as necessary to provide input to the determination of an appropriate driving policy for vehicle **100**. For example, obstacle detector **265** may be operative to use information from sensor(s) **130** (FIG. **3**A), e.g., camera **230**, telemetry ECU **240**, and/or shock sensor **250** to detect obstacles in (or near) the driving path of vehicle **100**, e.g., along (or near) roadway **20** (FIG. **3**A). Obstacle predictor **262** may be operative to use obstacle information received from obstacle avoidance server **400** to predict the location of obstacles along or near roadway **20** before, or in parallel to their detection by obstacle detector **265**. Obstacle avoidance module **268** may be operative to determine an appropriate driving policy based at least on obstacles detected/predicted (or not detected/predicted) by obstacle detector **265** and/or obstacle predictor **262**.

Autonomous driving manager **260** may store obstacle information received from obstacle avoidance server **400** in obstacle database **270** for use by obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268** as described herein. It will be appreciated that obstacle information, including driving policies to avoid detected obstacles may also be stored in obstacle database **270** for use by obstacle detector **265**, obstacle predictor **262**, and obstacle avoidance module **268**.

Reference is now made also to FIG. **3**C which is a flowchart of an exemplary obstacle detection and avoidance process **300** (hereinafter also referred to as process **300**) to be performed by system **200** during (or preceding) a driving session for vehicle **100** (FIG. **3**A). The components of systems **100** and **200** will be referred to herein as per the reference numerals in FIGS. **3**A and **3**B.

In accordance with some embodiments described herein, system **200** may receive (step **310**) an obstacle warning from obstacle avoidance server **400**, e.g. obstacle information received from other vehicles **100** using system **10** and/or and other suitable sources as described with reference to FIG. **3**A. It will be appreciated that the obstacle warning may be received by I/O module **220**. Depending on the configuration of system **100**, the obstacle warning may be received in a batch update process, either periodically and/or triggered by an event, e.g., when vehicle **100** is turned on, when vehicle **100** enters a new map area, when vehicle **100** enters an area with good wireless reception, etc. It will be appreciated that the obstacle warning received in step **310** may also include a description of the obstacle, e.g., a pothole.

Autonomous driving manager **260** may invoke obstacle predictor **262** to estimate (step **320**) a location for the obstacle associated with the obstacle warning. It will be appreciated that, as discussed hereinabove, while the obstacle warning may include GPS coordinates for the obstacle, the coordinates may not be accurate enough to provide the location with sufficient precision to be of use in an automated. Furthermore, even if the coordinates for the obstacle, e.g., obstacle **40** (FIG. **3**A) are accurate, vehicle **10** will likely be in movement and it may not be possible to accurately determine the GPS coordinates of vehicle **10** as it approaches the obstacle.

In accordance with embodiments described herein, the obstacle information in the obstacle warning may include landmarks to be used for calculating the obstacle location. For example, the location of the associated obstacle may be indicated in the obstacle warning as an offset from a local landmark, e.g., one or more of static reference points **30** (FIG. **3**A). It will be appreciated that static reference points **30** may be visible from a greater distance than obstacle **40**; a signpost typically extends vertically from the ground such that static reference points **30** may be captured by sensor(s) **130**, e.g., camera **230**, in advance of detection of obstacle **40**. Obstacle predictor may then use static reference points **30** to predict a location for the associated obstacle **40** as an offset from the observed location of one or more static reference points **30** before sensor(s) **130** may actually provide an indication of the location. Distance from a given static reference point **30** may be estimated as a function of current speed for vehicle **10** and a frames-per-second setting for camera **230**. Alternatively, or in addition, multiple static reference points may be used to locate obstacle **40** by triangulation, e.g., a distance from both static reference points **30**A and **30**B may be estimated and then used to predict the location of obstacle **40**.

It will be appreciated that in some embodiments, sensor(s) **130** may be operative to provide an estimate for distance. For example, sensor(s) may be operative to use radar imagery, LIDAR, etc.

Autonomous driving manager **260** may invoke obstacle avoidance module **268** to enact (step **330**) preventative measures to avoid colliding with, or running over, obstacle **40**. Obstacle avoidance module **268** may enact the preventative measures in accordance with driving policies stored in obstacle database **270**. It will be appreciated that the obstacle warning may include a description of obstacle **40**, e.g., a pothole. Using the obstacle description and other available information (e.g., distance to obstacle **40** as estimated in step **320**, current speed, road conditions, etc.), obstacle avoidance module **268** may select an appropriate driving policy from obstacle database **270** for implementation. For example, the selected driving policy may entail a reduction in speed, switching lanes (if possible, under current conditions), swerving within a lane, etc. It will be appreciated that the driving policies in obstacle database may be pre-stored in obstacle database **270**. It will further be appreciated that the driving policies may be derived based on supervised or unsupervised analysis of imagery from actual driving sessions. Some, or all, of the driving policies may also be defined manually.

In some cases, the selected driving policy may entail more extreme preventative measures. For example, obstacle **40** may be described in the obstacle warning as a very large obstacle that may effectively block traffic on roadway **20**, e.g., a rockfall, or debris from a multi-vehicle accident. In such a case, the driving policy may entail actions such as, for example, pulling over on the shoulder, stopping in place, or changing routes.

Autonomous driving manager may invoke obstacle detector **265** to determine if an obstacle, e.g., obstacle **40**, has been detected (step **340**). For example, obstacle detector **265** may use imagery received from sensor(s) **130** to detect obstacle **40** in roadway **20**.

It will be appreciated that in typical operation of process **300**, steps **310-330** may not always be performed; autonomous driving system **200** may not always receive an obstacle warning in advance of encountering obstacle **40**. For example, vehicle **100** may be the first vehicle associated with system **10** to encounter obstacle **40**. It will therefore be appreciated that autonomous driving manager **260** may continuously execute process **300** in an attempt to detect and avoid obstacles as vehicle **100** progresses along roadway **20**. Accordingly, step **340** may be performed independently of whether or not an obstacle warning has been received for a given area on roadway **20**. It will, however, be appreciated, that in the event that an obstacle warning was indeed received, the relevant obstacle information may be used by obstacle detector **265** to determine whether or not the indicated obstacle, e.g., obstacle **40**, is indeed in place on roadway **20** as indicated in the obstacle warning.

Alternatively, or in addition, obstacle detector **265** may be detected in response to information received from shock sensor **250**. Shock sensor **250** may be a sensor operative to monitor the operation of the shock absorbers on vehicle **100**. It will be appreciated that when running over a pothole, shock absorbers may absorb the ensuing shock and stabilize vehicle **100**. It will also be appreciated that in some situations, potholes may not be detected until vehicle **100** actually drives over them. In such a situation, obstacle detector **265** may use information from shock sensor **250** to detect obstacle **40**, albeit after vehicle **100** has already hit obstacle **40**.

If obstacle **40** is detected in step **340**, autonomous driving manager **260** may invoke obstacle avoidance module **268** to determine whether obstacle **40** is in the path of vehicle **100** (step **350**). It will be appreciated that some obstacles **40** on roadway **20** may not be in the direct path of vehicle **100**; some obstacles **40** may be in a different lane than vehicle **100**. Additionally, if step **330** was performed, vehicle **100** may have already switched lanes in advance of obstacle **40** being detected in step **340**. Furthermore, if obstacle **40** was detected using shock sensor **250**, vehicle **100** may have already run over obstacle **40** such that the result of step **350** may be "no". If obstacle **40** is not detected in step **340**, process control may flow through to step **380**.

If, as per the result of step **350**, obstacle **40** is in the path of vehicle **100**, obstacle avoidance module **268** may determine whether obstacle **40** is avoidable (step **360**). It will be appreciated that obstacle **40** may not be detected in step **340** until vehicle **100** is very close. For example, visibility may be poor, obstacle **40** may be located immediately around a bend in roadway **20**, obstacle **40** may be effectively camouflaged by its surroundings (e.g., some potholes are not distinguishable from a distance), etc. Accordingly, by the time obstacle **40** is detected, there may not be enough time to enact avoidance measures prior to impact. If, as per the result of step **350**, obstacle **40** is not avoidable, process control may flow through to step **380**.

If, as per the result of step **360**, obstacle **40** is avoidable, obstacle avoidance module **268** may instruct vehicle **100** to avoid (step **370**) obstacle **40**. For example, obstacle avoidance module **268** may instruct vehicle **100** to straddle obstacle **40**, swerve within the lane, or switch lanes altogether.

If, as per the result of step **360**, obstacle **40** is not avoidable, obstacle avoidance module **268** may instruct vehicle **100** to minimize (step **365**) impact with obstacle **40**. For example, obstacle avoidance module **268** may instruct vehicle **100** to reduce speed and/or to swerve to avoid hitting the center of obstacle **40**. Process control may then flow through to step **380**.

Autonomous driving manager **380** may use I/O module **220** to send an obstacle report to obstacle avoidance server **400**. The obstacle report may, for example, include telemetry data and sensor imagery that may be used by obstacle avoidance server **400** to determine a location for obstacle **40**. For example, the obstacle report may include camera images (from camera **230**) and telemetry data (from telemetry ECU **240**) from a period staring a given number of seconds before obstacle **40** was detected in step **340**. Alternatively, or in addition, the telemetry data and camera images may be from a period defined according to a physical distance from obstacle **40**. The obstacle report may also include data from shock sensor **250** where applicable.

Depending on the configuration of systems **100** and **200**, step **380** may be performed in real time or near real time after obstacle **40** is detected in step **340**. Alternatively, or in addition, step **380** may be performed in a periodic and/or event triggered batch process.

It will be appreciated that obstacle reports may also be provided for "no" results in process **300**. If obstacle **40** is not detected in step **340**, an obstacle report may be provided to obstacle avoidance server **400** indicating that an obstacle was not detected at the associated location. It will be appreciated that in some cases, a "no" result in step **340** may indicate that an obstacle associated with an obstacle warning may no longer be found on roadway **20**, e.g., the pothole may have been filled in. In other cases, where an obstacle warning was not received, the "no" result may indicate that the default condition, e.g., "no obstacles", is in effect for roadway **20**.

Reference is now made to FIG. **4** which is a block diagram of an exemplary obstacle avoidance server **400** (hereinafter also referred to as server **400**), constructed and implemented in accordance with embodiments described herein. Server **400** comprises processing circuitry **410**, input/output (I/O) module **420**, obstacle avoidance manager **460**, and obstacle database **470**. Obstacle avoidance manager **460** may be instantiated in a suitable memory for storing software such as, for example, an optical storage medium, a magnetic storage medium, an electronic storage medium, and/or a combination thereof.

Processing circuitry **410** may be operative to execute instructions stored in memory (not shown). For example, processing circuitry **410** may be operative to execute obstacle avoidance manager **260**. It will be appreciated that processing circuitry **410** may be implemented as a central processing unit (CPU), and/or one or more other integrated circuits such as application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), full-custom integrated circuits, etc., or a combination of such integrated circuits. It will similarly be appreciated that server **400** may comprise more than one instance of processing circuitry **410**. For example, one such instance of processing circuitry **410** may be a special purpose processor operative to execute obstacle avoidance manager **460** to perform some, or all, of the functionality of server **400** as described herein.

I/O module **420** may be any suitable communications component such as a network interface card, universal serial bus (USB) port, disk reader, modem or transceiver that may be operative to use protocols such as are known in the art to communicate either directly, or indirectly, with other elements of system **10** (FIG. **3**A) such as, for example, system **200** (FIG. **3**B). As such, I/O module **420** may be operative to use a wired or wireless connection to connect to system **200** via a communications network such as a local area network, a backbone network and/or the Internet, etc. It will be appreciated that in operation I/O module **220** may be implemented as a multiplicity of modules, where different modules may be operative to use different communication technologies. For example, a module providing mobile network connectivity may be used to connect wirelessly to one instance of system **200**, e.g., one vehicle **100** (FIG. **3**A), whereas a local area wired connection may be used to connect to a different instance of system **100**, e.g., a different vehicle **100**.

Obstacle avoidance manager **460** may be an application implemented in hardware, firmware, or software that may be executed by processing circuitry **410** to provide obstacle warnings and/or driving policies to vehicles **100**. For example, obstacle avoidance manager **460** may use obstacle information in obstacle reports received from vehicles **100** to provide obstacle warnings in real time, near real time, and/or batch mode to vehicles **100** in system **10** (FIG. **3**A). It will be appreciated that obstacle avoidance manager **460** may also be operative to use other data sources when detecting obstacles and determining driving policies, e.g., maps of potential routes, traffic congestion reports, etc.

As depicted in FIG. **4**, obstacle avoidance manager **460** comprises obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468**. It will be appreciated that the depiction of obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468** as integrated components of obstacle avoidance manager **460** may be exemplary. The embodiments described herein may also support implementation of obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468** as independent applications in communication with obstacle avoidance manager **460**, e.g., via I/O module **420**.

Obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468** may be implemented in hardware, firmware, or software and may be invoked by obstacle avoidance manager **460** as necessary to provide obstacle warnings and associated driving policies to vehicles **100**. For example, obstacle extractor **462** may be operative to extract obstacle information from obstacle reports received from vehicles **100**. Obstacle categorizer **464** may be operative to use categorize obstacles in terms of danger level and/or persistence characteristics. Obstacle policy manager **466** may be operative to determine an appropriate driving policy for a given obstacle or category of obstacle. Obstacle timer **468** may be operative to set an expiration timer for obstacles according to their associated obstacle categories.

Obstacle avoidance manager **460** may store obstacle information received from vehicles **100** in obstacle database **270** for use by obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468** as described herein. It will be appreciated that obstacle database **470** may also be used to store obstacle information, including driving policies, derived by obstacle extractor **462**, obstacle categorizer **464**, obstacle policy manager **466**, and obstacle timer **468**.

Reference is now made also to FIG. **5** which is a flowchart of an exemplary obstacle detection and warning process **500** (hereinafter also referred to as process **500**) to be performed by server **400** to support driving sessions for vehicles **100** (FIG. **3**A). The components of systems **10** and **200** will be referred to herein as per the reference numerals in FIGS. **3**A and **3**B. **1** and **2**. Similarly, the components of server **400** will be referred to herein as per the reference numerals in FIG. **4**.

In accordance with some embodiments described herein, server **400** may receive (step **510**) an obstacle report from vehicle **100**. It will be appreciated that the obstacle warning may be received by I/O module **420**. Depending on the configuration of system **10**, the obstacle report may be received in real time or near real time after vehicle **100** encounters obstacle **40**. Alternatively, or in addition, the obstacle report may be received in a batch update process, either periodically and/or triggered by an event, e.g., when vehicle **100** is turned on, when vehicle **100** enters a new map area, when vehicle **100** enters an area with good wireless reception, etc. It will be appreciated that the obstacle report received in step **510** may also include an image of obstacle **40**.

It will be appreciated that as described hereinabove with respect to process **300** (FIG. **3**C), a given obstacle report may not necessarily be triggered by detection of an obstacle by vehicle **100**. The obstacle report may represent a periodic upload of GPS data, camera images, telemetry data, and/or other sensor data that may not be associated with a specific detection of an obstacle. The obstacle report may also be associated with a non-detection of an obstacle that was otherwise indicated by a previously issued obstacle warning. Furthermore, as described with respect to FIG. **3**A, raw driving data from manually and/or semi-autonomously driven vehicles may also be provided to server **400**. In accordance with embodiments described herein, such raw driving data may also be input to process **500** along with obstacle reports.

Accordingly, obstacle avoidance server **400** may invoke obstacle extractor **462** to analyze received obstacle reports and/or raw driving data to determine if a given obstacle report (or raw driving data) includes a reported obstacle (step **520**), i.e., where the obstacle report specifically addresses an obstacle, whether it was detected by vehicle **100** and/or it was included in a previous obstacle warning.

Obstacle extractor **462** may determine if the obstacle report or raw driving data includes at least a reference to a reported obstacle (step **520**). It will be appreciated that as described with respect to process **300**, an obstacle report may be sent in response to an obstacle warning, even if the associated obstacle is not actually detected by autonomous driving manager **360**. Such an obstacle report may still include a reference to the reported obstacle. If the obstacle report or raw driving data includes at least a reference to a reported obstacle (step **520**) processing may continue with step **530**.

If the obstacle report or raw driving data does not include a reference to a reported obstacle (step **520**), obstacle extractor **462** may process (step **525**) the obstacle report (or raw driving data) to extract obstacle information for obstacles. For example, obstacle extractor **464** may be operative to detect an obstacle based on an image of an obstacle, where the image may be matched to a reference image of an obstacle stored in obstacle database **470**. Alternatively, or in addition, obstacle extractor **464** may be operative to detect an obstacle based on telemetry data, where certain driving patterns may be indicative of a driver avoiding an obstacle. Alternatively, or in addition, obstacle extractor **464** may be operative to detect an obstacle based on analysis of shock absorber data, where shock events may be indicative of a vehicle running over a pothole. It will be appreciated that the obstacle information extracted in step **525** may also include reference data that may be used to provide a location for the extracted obstacle, e.g., GPS coordinates, offsets from fixed reference points **30**, radar imagery, etc.

If one or more obstacles are extracted (step **529**) processing control may flow through to step **530**. Otherwise, processing control may return to step **510**.

Obstacle avoidance manager **460** may determine whether a reported obstacle or an extracted obstacle is a "new" obstacle (step **530**), i.e., whether the associated obstacle is already known to server **400** from a previous iteration of process **300**. If the obstacle is a new obstacle, obstacle avoidance manager **460** may invoke obstacle categorizer **464** to categorize the obstacle. Obstacle categorizer **464** may categorize the obstacle based on a level of danger it may represent to vehicle **100**. For example, if the obstacle is a relatively shallow pothole, obstacle categorizer may assign it to a low-risk category. If the obstacle is a patch of ice, obstacle categorizer may assign it to a medium-risk category. If the obstacle is a large slab of concrete, obstacle categorizer may assign it to a high-risk category. It will be appreciated that the examples provided herein for obstacles and their associated categories are not limiting; the embodiments described herein may support other combinations of obstacles and categories.

Obstacle categorizer **464** may also be operative to categorize obstacles based on their expected persistence. For example, a pothole may be a persistent obstacle; it may be expected to remain on roadway **20** for weeks, months, or even years after first being observed. Debris from a traffic accident may be a temporary obstacle; it may be expected to be cleared from roadway **20** within hours or perhaps a day or two after first being observed. Some obstacles may be recurring in nature; such obstacles may be observed intermittently in the same location. For example, puddles or ice patches may tend to recur in the same locations based on the topography of roadway **20**. Some obstacles may have hybrid persistence characteristics. For example, a pothole may also tend to fill with rain/slow and freeze during the winter. The pothole itself may be of a persistent nature, whereas the resulting ice patch (or puddle) may be of a recurring nature. Obstacle categorizer **464** may categorize (step **535**) the obstacle accordingly, e.g. as persistent, temporary, recurring, or hybrid.

Obstacle avoidance manager **460** may invoke obstacle policy manager **466** to set (step **539**), based at least on the obstacle's category, an appropriate driving policy for avoiding (or at least minimizing the risks associated with) the obstacle.

If, per step **530**, the obstacle is not a new obstacle, obstacle avoidance manager **460** may use obstacle information stored in obstacle database **470** to determine whether the current obstacle information is consistent with previously observed obstacle information (step **540**). For example, a pothole may expand in size and/or fill with water/ice compared to a previous observation. Debris from a traffic accident may shift or be partially removed. If the current obstacle information is inconsistent with the stored obstacle information, obstacle avoidance manager **460** may invoke obstacle categorizer **464** to update (step **545**) the category or categories assigned in step **535**, and obstacle policy manager **466** to update (step **549**) the driving policy set in step **539**.

As described hereinabove, even though a given obstacle report may be associated with a previously issued obstacle warning, in some cases the obstacle report may not include sensor data indicative of an observation of the associated obstacle. Vehicle **100** may receive an obstacle warning, but by the time vehicle **100** reaches the location indicated in the obstacle warning, the associated obstacle may no longer be there. For example, based on obstacle information received by obstacle avoidance server **400**, obstacle avoidance manager **460** may determine that debris from traffic accident may be at a given location on roadway **20**, and send an obstacle warning to vehicle **100** for the given location. However, the debris may be cleared by the time that vehicle **100** reaches the given location. As described with reference to process **300**, in such a case, vehicle **100** may provide an obstacle report to server **400** even though it did not actually detect the obstacle associated with the obstacle warning.

Accordingly, obstacle avoidance manager **460** may determine if an obstacle associated with the obstacle report was actually detected by autonomous driving system **200** or obstacle extractor **462** (step **550**). If an obstacle was detected per step **550**, obstacle avoidance manager **460** may use obstacle timer **468** to set (step **555**) a timer for the detected obstacle. The timer may be set according to an anticipated duration for the obstacle. For example, if the obstacle is debris from a traffic accident, the timer may be set to a number of hours; for a pothole the timer may be set for a number of weeks or months.

It will be appreciated that the detected obstacle from step **550** may already have had a timer set in a previous iteration of process **500**, e.g., the obstacle report is associated with a known obstacle for which an obstacle warning was previously generated and sent to vehicle(s) **100**. In such a case, the timer may be reset to extend the anticipated duration of the obstacle per a more recent observation. For example, if the obstacle was debris from a traffic accident, the timer may have originally been set for two hours. If an obstacle report is received ninety minutes later indicating that the debris is still there, obstacle avoidance manager **460** may reset the timer to update the anticipated duration based on the latest information (i.e., the recent obstacle report).

If an obstacle associated with the obstacle report was not detected by autonomous driving system **200** or obstacle extractor **462** (step **550**), obstacle avoidance manager **460** may determine if the obstacle warning has expired (step **560**) according to the timer (re)set in step **555**. If the timer has expired, obstacle avoidance manager **460** may remove (step **465**) the obstacle from a warning list of currently relevant obstacle warnings. Otherwise, processing control may continue to step **570**.

Obstacle avoidance manager **460** may then update (step **570**) the warning list as necessary e.g., adding obstacle warnings, modifying driving policies, removing obstacle warnings, etc. Obstacle avoidance manager **460** may then send (step **580**) the obstacle warning(s) from the warning list to vehicle(s) **100**. Depending on the configuration of systems **10**, **200**, and **400**, step **580** may be performed in real time or near real time after step **570**. Alternatively, or in addition, step **580** may be performed in a periodic and/or event triggered batch process.

Process control may then return to step **510**. It will be appreciated that in some embodiments, steps **520-570** (or combinations, thereof) may be performed iteratively, e.g., if multiple obstacles are extracted from raw driving data.

Obstacle Detection Based on Visual and Non-Visual Information

Obstacles may be viewed as elements that introduce changes in the behavior of a vehicle. Obstacles may be automatically detected by processing images obtained while a vehicle performs maneuvers that are suspected as being obstacle avoidance maneuvers.

FIG. **6** illustrates a method **1600** executed by computerized system for detecting obstacles. The computerized system may be located outside a vehicle, may be distributed between different vehicles, and the like.

Method **1600** may start by step **1610** of receiving, from a plurality of vehicles, and by an I/O module of a computerized system, visual information acquired during executions of vehicle maneuvers that are suspected as being obstacle avoidance maneuvers.

Step **1610** may also include receiving behavioral information regarding behavior of the plurality of vehicles, during an execution of the maneuvers that are suspected as being obstacle avoidance maneuvers. The behavioral information may represent the behavior (speed, direction of propagation, and the like) of the entire vehicle and/or may represent the behavior of parts or components of the vehicle (for example damping a shock by a shock absorber, slowing the vehicle by the breaks, turns of the steering wheel, and the like), and/or may represent the behavior of the driver (the manner in which the driver drives), and the like.

A determination of what constitutes a vehicle maneuver suspected as being an obstacle avoidance maneuver may be detected in a supervised manner or in an unsupervised manner. For example—when in supervised manner method **1600** may include receiving examples of maneuvers that are tagged as obstacle avoidance maneuvers.

Such a maneuver may be an uncommon maneuver in comparison to at least one out of (a) maneuvers of the same vehicle, (b) maneuvers performed at the same location by other drivers, (c) maneuvers performed in other places by other drivers, and the like. The maneuver may be compared to other maneuvers at the same time of day, at other times of day, at the same date, at other dates and the like.

The vehicle maneuver suspected as being an obstacle avoidance maneuver may be detected based on behavioral information that may be obtained from one or more sensors such as an accelerometer, a wheel speed sensor, a vehicle speed sensor, a brake sensor, a steering wheel sensor, a shock absorber sensor, an engine sensor, a driver sensor for sensing one or more physiological parameter of the driver (such as heart beat), or any other sensor—especially any other non-visual sensor.

The vehicle maneuver suspected as being an obstacle avoidance maneuver may involve a change (especially a rapid change) in the direction and/or speed and/or acceleration of the vehicle, a deviation from a lane, a deviation from a previous driving pattern followed at attempt to correct the deviation, and the like.

For example—when applying supervised learning of obstacles—the computerized system may be fed with examples of vehicle behaviors when bypassing known obstacles, when approaching a significant bump, and the like.

For example—the determining may include ruling out common behaviors such as stopping in front of a red traffic light.

For example—a fast and significant change in the speed and direction of a vehicle at a linear road path may indicate that the vehicle encountered an obstacle.

The visual information may be raw image data (such as images) acquired by a visual sensor of the vehicle, processed images, metadata or any other type of visual information that represents the images acquired by visual sensor.

The visual information may include one or more robust signatures of one or more images acquired by one or more visual sensors of the vehicle. An example of a robust signature is illustrated in U.S. Pat. No. 8,326,775 which is incorporated herein by reference.

Step **1610** may be followed by step **1620** of determining, based at least on the visual information, at least one visual obstacle identifier for visually identifying at least one obstacle.

The determining may include clustering visual information based on the objects represented by the visual information, and generating a visual obstacle identifier that represents a cluster. The clusters may be linked to each other and a visual obstacle identifier may represent a set of linked clusters.

The determining may include filtering out objects that are represented in the visual information based on a frequency of appearance of the objects in the visual information. For example—the road may appear in virtually all images and may be ignored.

The clustering may or may not be responsive to the behavior of the vehicle during the executions of vehicle maneuvers that are suspected as being obstacle avoidance maneuvers.

The visual obstacle identifier may be a model of the obstacle, a robust signature of the obstacle, configuration information of a neural network related to the obstacle.

A model of the obstacle may define one or more parameters of the obstacle such as shape, size, color of pixels, grayscale of pixels, and the like.

The configuration information of a neural network may include weights to be assigned to a neural network. Different neural networks may be used to detect different obstacles.

The visual obstacle identifier for visually identifying an obstacle may identify a group of obstacles to which the obstacle belongs. For example—the visual obstacle identifier may identify a concept. See, for example, U.S. Pat. No. 8,266,185 which is incorporated herein by reference.

A visual obstacle identifier that identifies an obstacle may include at least one out of severity metadata indicative of a severity of the obstacle, type metadata indicative of a type of the obstacle (for example—pothole, tree, bump), size metadata indicative of a size of the obstacle, timing metadata indicative of a timing of an existence of the obstacle (for example—persistent, temporary, recurring), and location metadata indicative of a location of the obstacle. See, for example severity metadata **1851**, type metadata **1852**, size metadata **1853**, timing metadata **1854** and location metadata **1855** of FIG. **7**.

Step **1620** may be followed by step **1630** of responding to the outcome of step **1620**. For example—step **1630** may include transmitting to one or more of the plurality of vehicles, the at least one visual obstacle identifier. Additionally or alternatively, step **1630** may include populating an obstacle database. The visual obstacle identifier may be included in an obstacle warning.

The behavioral information is obtained by non-visual sensors of the plurality of vehicles.

Step **1630** may also include verifying, based on at least the visual information, whether the maneuvers that are suspected as being obstacle avoidance maneuvers are actually obstacle avoidance maneuvers.

The verification may include detecting at least a predefined number of indications about a certain obstacle.

The verification may include receiving information from another source (for example input from drivers, information provided from the police, department of transportation or other entity responsible to the maintenance of roads, regarding the existence of obstacles.

Step **1630** may be followed by step **1640** of transmitting to one or more of the plurality of vehicles, verification information indicative of the maneuvers that are actually obstacle avoidance maneuvers. This may help the vehicles to ignore maneuvers that were suspected as obstacle avoidance maneuvers—which are not actual obstacle avoidance maneuvers.

Method **1600** enables a vehicle that receives a visual obstacle identifier to identify an obstacle of known parameters—even if the vehicle is the first one to image that obstacle. Method **1600** may be executed without human intervention and may detect obstacles without prior knowledge of such obstacles. This provides a flexible and adaptive method for detecting obstacles. The visual information and/or the visual obstacle identifier may be very compact—thereby allowing the learning of obstacles and afterwards detecting obstacles to be executed using a relatively small amount of computational and/or storage and/or communication resources.

It should be noted that different vehicle may perform obstacle avoidance maneuvers that may differ from each other by duration, vehicle behavior and the like. An example is illustrated in FIGS. **8-10** in which first, second and third vehicles perform different obstacle avoidance maneuvers and acquire different numbers of images during said obstacle avoidance maneuvers.

FIG. **8** illustrates a first vehicle (VH**1**) **1801** that propagates along a road **1820**. First vehicle **1801** performs a maneuver **1832** suspected as being an obstacle avoidance maneuver when encountered with obstacle **1841**. Maneuver **1832** is preceded by a non-suspected maneuver **1831** and is followed by another non-suspected maneuver **1833**.

First vehicle **1801** acquires a first plurality (N**1**) of images I**1**(**1**)-I**1**(N**1**) **1700**(**1**,**1**)-**1700**(**1**,N**1**) during obstacle avoidance maneuver **1832**.

Visual information V**1**(**1**)-V**1**(N**1**) **1702**(**1**,**1**)-**1702**(**1**,N**1**) is sent from first vehicle **1801** to computerized system (CS) **1710** via network **1720**.

The visual information may be the images themselves. Additionally or alternatively, first vehicle processes the images to provide a representation of the images.

First vehicle **1801** may also transmit behavioral information B**1**(**1**)-B**1**(N**1**) **1704**(**1**,**1**)-**1704**(**1**,N**1**) that represents the behavior of the vehicle during maneuver **1832**.

FIG. **9** illustrates a second vehicle (VH**2**) **1802** that propagates along a road **1820**. Second vehicle **1802** performs a maneuver **1833** suspected as being an obstacle avoidance maneuver when encountered with obstacle **1841**. Maneuver **1832** is preceded by a non-suspected maneuver and is followed by another non-suspected maneuver.

Second vehicle **1802** acquires a second plurality (N**2**) of images I**2**(**1**)-I**2**(N**2**) **1700**(**2**,**1**)-**1700**(**2**,N**2**) during maneuver **1833**.

Visual information V**2**(**1**)-V**2**(N**2**) **1702**(**2**,**1**)-**1702**(**2**,N**2**) is sent from second vehicle **1802** to computerized system (CS) **1710** via network **1720**.

The visual information may be the images themselves. Additionally or alternatively, second vehicle processes the images to provide a representation of the images.

Second vehicle **1802** may also transmit behavioral information (not shown) that represents the behavior of the vehicle during maneuver **1832**.

FIG. **10** illustrates a third vehicle (VH**2**) **1803** that propagates along a road. Third vehicle **1803** performs a maneuver **1834** suspected as being an obstacle avoidance maneuver when encountered with obstacle **1841**. Maneuver **1834** is preceded by a non-suspected maneuver and is followed by another non-suspected maneuver.

Third vehicle **1803** acquires a third plurality (N**3**) of images I**3**(**1**)-I**3**(N**3**) **1700**(**3**,**1**)-**1700**(**3**,N**3**) during maneuver **1834**.

Visual information V**3**(**1**)-V**3**(N**3**) **1702**(**3**,**1**)-**1702**(**3**,N**3**) is sent from third vehicle **1803** to computerized system (CS) **1710** via network **1720**.

The visual information may be the images themselves. Additionally or alternatively, third vehicle processes the images to provide a representation of the images.

Third vehicle **1803** may also transmit behavioral information (not shown) that represents the behavior of the vehicle during maneuver **1832**.

FIG. **11** illustrates a method **1900** for detecting obstacles. The method may be executed by a vehicle.

Method **1900** may start by steps **1910** and **1920**.

Step **1910** may include sensing, by a non-visual sensor of a vehicle, a behavior of a vehicle. The non-visual sensor may be an accelerometer, a shock absorber sensor, a brakes sensor, and the like.

Step **1920** may include acquiring, by a visual sensor of the vehicle, images of an environment of the vehicle. Step **1920** may continue even when such a suspected maneuver is not detected.

Steps **1920** and **1930** may be followed by step **1930** of determining, by a processing circuitry of the vehicle, whether the behavior of the vehicle is indicative of a vehicle maneuver that is suspected as being an obstacle avoidance maneuver.

The vehicle maneuver suspected as being an obstacle avoidance maneuver may be detected based on behavioral information that may be obtained from one or more sensors such as an accelerometer, a wheel speed sensor, a vehicle speed sensor, a brake sensor, a steering wheel sensor, an engine sensor, a shock absorber sensor, a driver sensor for sensing one or more physiological parameter of the driver (such as heart beat), or any other sensor—especially any other non-visual sensor.

The vehicle maneuver suspected as being an obstacle avoidance maneuver may involve a change (especially a rapid change) in the direction and/or speed and/or acceleration of the vehicle, a deviation from a lane, a deviation from a previous driving pattern followed at attempt to correct the deviation, and the like.

Step **1930** may be followed by step **1940** of processing the images of the environment of the vehicle obtained during a vehicle maneuver that is suspected as being the obstacle avoidance maneuver to provide visual information.

The visual information may be raw image data (such as images) acquired by a visual sensor of the vehicle, processed images, metadata or any other type of visual information that represents the images acquired by visual sensor.

The visual information may include one or more robust signatures of one or more images acquired by one or more visual sensors of the vehicle. An example of a robust signature is illustrated in U.S. Pat. No. 8,326,775 which is incorporated herein by reference.

Step **1940** may be followed by step **1950** of transmitting the visual information to a system that is located outside the vehicle.

Method **1900** may also include step **1960** of receiving at least one visual obstacle identifier from a remote computer.

Method **1900** may include step **1970** receiving verification information indicative of whether the maneuver that was suspected being an obstacle avoidance maneuver is actually an obstacle avoidance maneuver. This may be used in the vehicle during step **1930**.

Methods **1900** and **1600** illustrate a learning process. The products of the learning process may be used to detect obstacles—during an obstacle detection process that uses the outputs of the learning process.

The obstacle detection process may include detecting newly detected obstacles that were not detected in the past.

The learning process may continue during the obstacle detection process or may terminate before the obstacle detection process.

FIG. **12** illustrates method **2100** for detecting an obstacle.

Method **2100** may start by step **2110** of receiving, by an I/O module of a vehicle, a visual obstacle identifier for visually identifying an obstacle; wherein the visual obstacle identifier is generated based on visual information acquired by at least one visual sensor during an execution of at least vehicle maneuver that is suspected as being an obstacle avoidance maneuver. Thus step **2110** may include receiving outputs of the learning processes of method **1600**.

Method **2100** may also include step **2120** of acquiring, by a visual sensor of the vehicle, images of an environment of the vehicle.

Step **2120** may be followed by step **2130** of searching, by a processing circuitry of the vehicle, in the images of the environment of the vehicle for an obstacle that is identified by the visual obstacle identifier.

If finding such an obstacle then step **2130** may be followed by step **2140** of responding, by the vehicle, to a detection of an obstacle.

The responding may be based on a driving policy, may be responsive to the mode of controlling the vehicle (any level of automation—starting from fully manual control, following by partial human intervention and ending with fully autonomous).

For example—the responding may include generating an alert perceivable by a human driver of the vehicle, sending an alert to a computerized system located outside the vehicle, handing over the control of the vehicle to a human driver, handing over a control of the vehicle to an autonomous driving manager, sending an alert to an autonomous driving manager, performing an obstacle avoidance maneuver, determining whether the obstacle is a newly detected obstacle, informing another vehicle about the obstacle, and the like.

Determining a Location of a Vehicle

A vehicle, especially but not limited to an autonomous vehicle, has to know its exact location in order to propagate in an efficient manner. For example—the vehicle may receive an obstacle warning that includes the location of the obstacle.

Responding to the obstacle warning may require knowing the exact location of the vehicle. Furthermore—an autonomous driving operation may require a knowledge of the exact location of the vehicle or may be executed in a more optimal manner if the exact location of the vehicle is known. In general, the processing burden associated with autonomous driving decision may be lowered when an autonomous driving manager is aware of the location of the vehicle.

Global positioning systems (GPS) are inaccurate and cannot be solely used for locating the exact location of a vehicle.

There is a growing need to determine the exact location of the vehicle in an efficient manner.

There is a provided an efficient and accurate method for determining the exact location of a vehicle.

The method is image-based in the sense that it is based on images acquired by the vehicle and on a comparison between the acquired images and reference information that was acquired at predefined locations.

The determining of the actual location of the vehicle is of a resolution that is smaller than a distance between adjacent reference images. For example—the resolution may be a fraction (for example between 1-15%, between 20-45%, of between 5-50%) of the distance between adjacent reference images. In the following example it is assumed that the distance between adjacent reference images is about 1 meter and the resolution is about 10 to 40 centimeters.

FIG. **13** illustrates method **1300** for determining a location of a vehicle. receiving reference visual information that represents multiple reference images acquired at predefined locations.

Method **1300** may start by step **1310** of receiving reference visual information that represents multiple reference images acquired at predefined locations.

The multiple reference images may be taken within an area in which the vehicle is located. The area may be of any size and/or dimensions. For example, the area may include a part of a neighborhood, a neighborhood, a part of a city, a city, a part of a state, a state, a part of a country, a country, a part of a continent, a continent, and even the entire world.

A computerized system and the vehicle may exchange reference visual information in real-time (or near real-time) and/or in a burst mode, where the reference visual information may be provided either periodically and/or when conditions are suitable for a data exchange between the computerized system and the vehicle. For example, a vehicle may be configured to perform regular uploads/downloads of reference visual information when the engine is turned on (or off, with a keep alive battery function to facilitate communication with the computerized system). Alternatively, or in addition, the vehicle may be configured to perform such uploads/downloads when stationary (e.g., parked, or waiting at a traffic light) and a sufficiently reliable wireless connection is detected. Alternatively, or in addition, the uploads/downloads may be triggered by location, where the reference visual information may be downloaded from the computerized system when the vehicle enters an area for which it does not have up-to-date reference visual information.

The reference visual information may include the reference images themselves or any other representation of the reference images.

The reference information may include robust signatures of reference images. The robust signature may be robust to at least one out of noise, lighting conditions, rotation, orientation, and the like—which may provide a robust location detection signatures. Furthermore—using robust signatures may simplify the acquisition of the reference images—as there is no need to acquired reference images in different lighting conditions or other conditions to which the robust signature is indifferent.

The comparison between the reference information and information from the acquired image may be even more efficient when the reference visual information and information regarding the acquired image is represented in a cortex representation.

A cortex representation of an image is a compressed representation of the visual information that is generated by applying a cortex function that include multiple compression iteration. A signature of an image may be a map of firing neurons that fire when a neural network is fed with an image. The map undergoes multiple compression iterations during which popular strings are replaced by shorter representations.

Method **1300** may also include step **1320** of acquiring, by a visual sensor of the vehicle, an acquired image of an environment of the vehicle. It should be noted that multiple images may be acquired by the vehicle—but for simplicity of explanation the following text will refer to a single image. The vehicle may repetitively determine its location—and multiple repetitions of steps **1320**, **1330** and **1340** may be repeated multiple times—for different images.

Step **1330** may include generating, based on the acquired image, acquired visual information related to the acquired image.

The location determination may be based on visual image regarding static obstacles. Accordingly—each one of the reference visual information and the acquired visual information may include static visual information related to static objects.

The extraction of the static visual information may include reducing from the visual information any visual information that is related to dynamic objects. The extraction of the static information may or may not require object recognition.

The acquired visual information may include acquired static visual information that may be related to at least one static object located within the environment of the vehicle.

Step **1330** may be followed by step **1340** of searching for a selected reference image out of the multiple reference images. The selected reference image may include selected reference static visual information that best matches the acquired static visual information.

Step **1340** may be followed by step **1350** of determining an actual location of the vehicle based on a predefined location of the selected reference image and to a relationship between the acquired static visual information and to the selected reference static visual information.

The determining of the actual location of the vehicle may be of a resolution that may be smaller than a distance between the selected reference image and a reference image that may be immediately followed by the selected reference image.

The determining of the actual location of the vehicle may include calculating a distance between the predefined location of the selected reference image and the actual location of the vehicle based on a relationship between at least one value of a size parameter of the at least one static object within the selected reference image and at least one value of the size parameter of the at least one static object within the acquired image.

The determining of the actual location of the vehicle may include determining whether the actual location of the vehicle precedes the predefined location of the selected reference image or precedes the predefined location of the selected reference image.

The method may include subtracting the first distance from the predefined location of the selected reference image when determining that the actual location of the vehicle precedes the predefined location of the selected reference image.

The method may include adding the first distance from the predefined location of the selected reference image when determining that the actual location of the vehicle follows the predefined location of the selected reference image.

Each predefined location may be associated with a single reference image.

Alternatively, each reference image may be associated with multiple images that differ from each other by scale. The images that differ from each other by scale may represent different relative distances between the vehicle and the predefined image—larger scale may be related to smaller distances.

Using images of different scale (or visual information related to images of different scale) may reduce the computational load associated with location determination—as it may eliminate the need to manipulate a single reference image to find an exact math that may determine the distance between the vehicle and the reference location.

Using visual information related to images of different scale may also assist when the visual information is a lossy representation of the images in which a representation of an object of certain scale cannot be easily generated based on a representation of the same object of another scale. The visual information may be a signature of an image may be a map of firing neurons that fire when a neural network is fed with an image. See, for example, the signature of U.S. Pat. Nos. 8,326,775 and 8,312,031, assigned to common assignee, which are hereby incorporated by reference.

FIG. **14** illustrates an example of a vehicle VH**1 1402** that acquires an image I(k) **1404** and tries to determine its location based on analysis of visual information it generates from image I(k) and multiple reference images RI(**1**,**1**)-RI (j,**1**) **1402**(**1**)-**1402**(*j*) and RI(j+1,**1**)-RI(j) **1402**(*j*+1,**1**)-**1402** (J,**1**). The front of the vehicle is positioned between the predefined locations associated with reference images RI(j) and RI(j+1)—and one of these images should be associated with reference visual information that best matches the visual information associated with image I(k). The distance between consecutive predefined locations is DRI **1422**—and the location resolution is finer than DRI.

FIG. **15** illustrates an example of a vehicle VH**1 1402** that acquires an image I(k) **1404** and tries to determine its location based on analysis of visual information (**1406**) it generates from image I(k) and visual information related to reference images. In FIG. **15** each predefined location is associated with a plurality of reference images—with a plurality (p) of reference visual information per reference location —**1408**(**1**,**1**)-**1408**(*j*,**1**), **1408**(*j*+1,**1**)-**1408**(J,**1**), **1409**(**1**,*p*)-**1409**(*j*,*p*), and **1409**(*j*+1,p)-**1409**(J,p).

The front of the vehicle is positioned between the j'th and (j+1)'th predefined locations and the vehicle may determine the best matching predefined location and also the distance from the predefined locations based on the p different reference visual information associated with one of these predefined locations.

FIGS. **17** and **18** illustrate an example of an image and static visual information.

FIG. **17** illustrates an example of an image acquired by a vehicle. The image shows two lanes **1411** and **1412** of a road, right building **1413**, vehicles **1414** and **1415**, first tree **1416**, table **1417**, second tree **1418**, cloud **1419**, rain **1420**, left building **1421**, shrubbery **1422**, child **1423** and ball **1424**.

The dynamic objects of this image include vehicles **1414** and **1415**, cloud **1419**, rain **1420**, shrubbery **1422**, child **1423** and ball **1424**. These dynamic objects may be represented by visual information (such as signatures) that describe them as dynamic—and visual information related to these dynamic objects may be removed from the visual information of the images to provide static visual information.

FIG. **18** illustrates the removal (blocks **1432**) of dynamic visual information from the visual information **1430** to provide static visual information **1431**.

Visual information **1430** may be a map of firing neurons of a network that fired when the neural network was fed with image I(k)—or may be a compressed representation (for example a cortex representation) of such map.

Triggering Human Intervention

FIG. **19** illustrates method **1450**.

Method **1450** may be aimed to find at least one trigger for human intervention in a control of a vehicle.

Method **1450** may start by step **1452** of receiving, from a plurality of vehicles, and by an I/O module of a computerized system, visual information acquired during situations that are suspected as situations that require human intervention in the control of at least one of the plurality of vehicles.

Step **1452** may be followed by step **1456** of determining, based at least on the visual information, the at least one trigger for human intervention. The at least one trigger is represented by trigger visual information.

Step **1456** may include at least one out of

Determining a complexity of the situation.

Determining a danger level associated with the situation.

Determining in response to statistics of maneuvers executed by different vehicles during a same situation that is suspected as a situation that requires human intervention in the control of at least one of the plurality of vehicles.

Determining based on generated or received movement information of entities included in the visual information. The movement information may represent entity movement functions of the entities.

The determining may include estimating, based on the entity movement functions, a future movement of the entities.

The determining of step **1456** may be executed in an unsupervised manner or a supervised manner.

The determining of step **1456** is responsive to at least one human intervention policy of the at least one vehicle. The human intervention policy of a vehicle may differ from the human intervention policy of another vehicle.

A human intervention policy of a vehicle may define certain criteria human intervention such as the danger level of a situation, the complexity of the situation (especially complexity of maneuvers required to overcome the situation), the potential damage that may result to the vehicle, driver or surroundings due to the situation. A human intervention policy of a vehicle may also define certain situations that require human intervention—such as specific locations (for example near a cross road of a school), or specific content (near a school bus), and the like.

A trigger may be determined during step **1456** when the situation or one of the attributes of the situation (danger level, location, combination of location and time) match those of the human intervention policy.

Situations that are suspected as requiring human intervention in the control of a vehicle may be viewed as situations that are complex and/or dangerous and/or require changes in the behavior of a vehicle. These situations may result from an obstacle or may result from other reasons.

Situations that are suspected as requiring human intervention in the control of a vehicle may be automatically detected by processing images obtained while a vehicle performs maneuvers that are suspected as resulting from situations that are suspected as requiring human intervention in the control of a vehicle.

Situations that are suspected as requiring human intervention in the control of a vehicle may be detected based on behavioral information regarding behavior of the plurality of vehicles, during an execution of the maneuvers that are suspected as resulting from situations that are suspected as requiring human intervention in the control of a vehicle.

The behavioral information may represent the behavior (speed, direction of propagation, and the like) of the entire vehicle and/or may represent the behavior of parts or components of the vehicle (for example damping a shock by a shock absorber, slowing the vehicle by the breaks, turns of the steering wheel, and the like), and/or may represent the behavior of the driver (the manner in which the driver drives), and the like.

After a situation is tagged as requiring human intervention in the control of a vehicle—the vehicle may trigger human intervention when this situation is visually detected. For example—some events (or some types of events) may be defined (even based on previous recognition of a situation as requiring human intervention in the control of a vehicle) as requiring human intervention. For example—an image of a cross road located near a school, an image of a school bus that transports children, an image of a bar and drunken pedestrians near the bar, an image of multiple people that are near the road, and the like may be flagged as situation as requiring human intervention in the control of a vehicle

A determination of what constitutes a vehicle maneuver suspected as requiring human intervention in the control of a vehicle may be detected in a supervised manner or in an unsupervised manner.

Such a maneuver may be an uncommon maneuver in comparison to at least one out of (a) maneuvers of the same vehicle, (b) maneuvers performed at the same location by other drivers, (c) maneuvers performed in other places by other drivers, and the like. The maneuver may be compared to other maneuvers at the same time of day, at other times of day, at the same date, at other dates and the like.

The vehicle maneuver suspected as related to a situation that requires human intervention in the control of a vehicle may involve a change (especially a rapid change) in the direction and/or speed and/or acceleration of the vehicle, a deviation from a lane, a deviation from a previous driving pattern followed at attempt to correct the deviation, and the like.

For example—when applying supervised learning of obstacles—the computerized system may be fed with examples of vehicle behaviors that require human intervention in the control of a vehicle.

For example—the determining may include ruling out common behaviors such as stopping in front of a red traffic light.

For example—a fast and significant change in the speed and direction of a vehicle at a linear road path may indicate that the vehicle encountered a situation requiring human intervention in the control of a vehicle

The visual information may be raw image data (such as images) acquired by a visual sensor of the vehicle, processed images, metadata or any other type of visual information that represents the images acquired by visual sensor.

The visual information may include one or more robust signatures of one or more images acquired by one or more visual sensors of the vehicle. An example of a robust signature is illustrated in U.S. Pat. No. 8,326,775 which is incorporated herein by reference.

Step **1456** may be followed by step **1460** of transmitting to one or more of the plurality of vehicles, the trigger visual information.

FIG. **20** illustrates a method **1490** for detecting situation that requires human intervention in the control of a vehicle. The method may be executed by a vehicle.

Method **1490** may start by steps **1491** and **1492**.

Step **1491** may include sensing, by a non-visual sensor of a vehicle, a behavior of a vehicle. The non-visual sensor may be an accelerometer, a shock absorber sensor, a brakes sensor, and the like.

Step **1492** may include acquiring, by a visual sensor of the vehicle, images of an environment of the vehicle.

Steps **1491** and **1492** may be followed by step **1493** of determining, by a processing circuitry of the vehicle, whether the behavior of the vehicle is indicative of a vehicle maneuver that is related to a situation that is suspected as requiring human intervention in the control of a vehicle.

The vehicle maneuver related to a situation that is suspected as requiring human intervention in the control of a vehicle may be detected based on behavioral information that may be obtained from one or more sensors such as an accelerometer, a wheel speed sensor, a vehicle speed sensor, a brake sensor, a steering wheel sensor, an engine sensor, a shock absorber sensor, a driver sensor for sensing one or more physiological parameter of the driver (such as heart beat), or any other sensor—especially any other non-visual sensor.

The vehicle maneuver related to a situation that is suspected as requiring human intervention in the control of a vehicle may involve a change (especially a rapid change) in the direction and/or speed and/or acceleration of the vehicle, a deviation from a lane, a deviation from a previous driving pattern followed at attempt to correct the deviation, and the like.

Step **1493** may be followed by step **1494** of processing the images of the environment of the vehicle obtained during a vehicle maneuver related to a situation that is suspected as requiring human intervention in the control of a vehicle to provide visual information.

The visual information may be raw image data (such as images) acquired by a visual sensor of the vehicle, processed images, metadata or any other type of visual information that represents the images acquired by visual sensor.

The visual information may include one or more robust signatures of one or more images acquired by one or more visual sensors of the vehicle. An example of a robust signature is illustrated in U.S. Pat. No. 8,326,775 which is incorporated herein by reference.

Step **1494** may be followed by step **1495** of transmitting the visual information to a system that is located outside the vehicle.

Method **1490** may also include step **1496** of receiving at least one visual obstacle identifier from a remote computer.

Method **1490** may include step **1497** receiving verification information indicative of whether the maneuver related to a situation that is suspected as requiring human intervention in the control of a vehicle maneuver is actually related to a situation that actually requires human intervention in the control of a vehicle. This may be used in the vehicle during step **1493**.

Methods **1450** and **1490** illustrate a learning process. The products of the learning process may be used to detect situations that require human intervention in the control of a vehicle—during a vehicle control process that uses the outputs of the learning process.

The learning process may continue during the vehicle control process or may terminate before the vehicle control process.

FIG. **2**I illustrates method **1470**.

Method **1470** may start by step **1472** of receiving, by an I/O module of a vehicle, trigger visual information for visually identifying situations that require human intervention in the control of a vehicle. The visual obstacle identifier is generated based on visual information acquired by at least one visual sensor during an execution of at least vehicle maneuver that is suspected as being an obstacle avoidance maneuver. Thus step **1472** may include receiving outputs of the learning processes of method **1450**.

Method **1470** may also include step **1474** of acquiring, by a visual sensor of the vehicle, images of an environment of the vehicle.

Step **1474** may be followed by step **1476** of searching, by a processing circuitry of the vehicle, images of the environment of the vehicle for a situation that is identified by the trigger visual information.

If finding such a situation then step **1476** may be followed by step **1478** of triggering human intervention in the control of a vehicle. This may include alerting the driver that human intervention is required, handing the control of the vehicle to the driver- and the like.

Step **1478** may include, for example—generating an alert perceivable by a human driver of the vehicle, sending an alert to a computerized system located outside the vehicle, handing over the control of the vehicle to a human driver, handing over a control of the vehicle from an autonomous driving manager, sending an alert to an autonomous driving manager, informing another vehicle about the trigger, and the like.

FIGS. **22**-**27** illustrate example of situations that are suspected as requiring human intervention in the control of a vehicle.

FIG. **22** illustrates first vehicle VH**1** **1801** as stopping (position **1501**) in front of a puddle **1506** and then passing the puddle (may drive straight or change its direction till ending the maneuver at point **1502**. The maneuver may be indicative that passing the puddle may require human intervention.

Visual information acquired between positions **1501** and **1502** are processed during step **1494**.

FIG. **23** illustrates first vehicle VH**1** **1801** as sensing pedestrians **1511** and **1512**. These pedestrians either are associated with temporal signatures that illustrate their movement (**1511'** and **1512'**) and/or the vehicle may sense the movements of the pedestrians—and the vehicle may send (based on the movement) the future movement and the future location (**1511"** and **1512"**) of the pedestrians and perform a maneuver that may include altering speed (for example stopping at point **1513**) and/or bypassing the pedestrians (maneuver **1515**).

Visual information acquired between positions **1513** and **1514** (end of the maneuver) are processed during step **1494**.

FIG. **24** illustrates first vehicle VH**1** **1801** as sensing parked vehicles PV**1** **1518** and PV**2** **1519** that part on both sides of a double-lane bi-directional road, that require the first vehicle to perform a complex maneuver **1520** that includes changing lanes and changing direction relatively rapidly.

Visual information acquired between positions **1516** (beginning of the maneuver) and **1517** (end of the maneuver) are processed during step **1494**.

FIG. **25** illustrates first vehicle VH**1** **1801** as stopping (position **1522**) in front of wet segment of the road on which rain **1521** (from cloud **1522**) falls. The stop (at location **1522**) and any further movement after moving to another part of the road may be regarded as a maneuver **1523** that is indicative that passing the wet segment may require human intervention.

Visual information acquired between position **1522** (beginning of the maneuver) and the end of the maneuver are processed during step **1494**.

FIG. **26** illustrates first vehicle VH**1** **1801** as stopping (position **1534**) in front of a situation that may be labeled as a packing or unpacking situation—a track **1531** is parked on the road, there is an open door **1532**, and a pedestrian **1533** carries luggage on the road. The first vehicle **1801** bypasses the truck and the pedestrian between locations **1534** and **1535** during maneuver **1539**. The maneuver may be indicative that a packing or unpacking situation may require human intervention.

Visual information acquired between positions **1534** and **1535** are processed during step **1494**.

FIG. **27** illustrates first vehicle VH**1** **1801** as turning away (maneuver **1540**) from the road when sensing that it faces a second vehicle VH**2** **1802** that moves towards VH**1** **1801**. Maneuver **1540** may be indicative that such as potential collision situation may require human intervention.

Visual information acquired between positions **1541** and **1542** (start and stop of maneuver **1540**) are processed during step **1494**.

Tracking after an Entity

There may be beneficial to detect and object in a robust and an efficient manner by considering the temporal behavior of the entity and also its spatial signature. Using signatures of multiple type may increase the reliability of the detection, one signature may verify the other signature. In addition—one signature may assist in detecting the entity when the other signature is missing or of a low quality. Furthermore—estimating the future movement of an entity—based on the spatial signature of the entity (or of those of entities of the same type) may enable to predict the future effect that this entity may have on the vehicle and allow to determine the future driving pattern of the vehicle in advance.

The temporal signature may be compresses—thus saving memory and allowing a vehicle to allocate limited memory resources for tracking each entity—even when tracking multiple entities.

FIG. **28** illustrates a method **2600** for tracking after an entity.

Method **2600** may include steps **2602**, **2604**, **2606** and **2608**.

Tracking, by a monitor of a vehicle, a movement of an entity that appears in various images acquired during a tracking period. S**2602**.

Generating, by a processing circuitry of the vehicle, an entity movement function that represents the movement of the entity during the tracking period. S**2604**.

Generating, by the processing circuitry of the vehicle, a compressed representation of the entity movement function. S**2606**.

Responding to the compressed representation of the entity movement function. S**2608**.

The compressed representation of the entity movement function may be indicative of multiple properties of extremum points of the entity movement function.

The multiple properties of an extremum point of the extremum points, may include a location of the extremum point, and at least one derivative of the extremum point.

The multiple properties of an extremum point of the extremum points may include a location of the extremum point, and at least two derivative of at least two different orders of the extremum point.

The multiple properties of an extremum point of the extremum points, may include a curvature of the function at a vicinity of the extremum point.

The multiple properties of an extremum point of the extremum points, may include a location and a curvature of the function at a vicinity of the extremum point.

Method **2600** may include acquiring the images by a visual sensor of the vehicle. S**2601**.

At least one image of the various images may be acquired by an image sensor of another vehicle.

Step **2608** may include at least one out of:

- Storing, in a memory unit of the vehicle, the compressed representation of the entity movement function.
- Transmitting the compressed representation of the entity movement function to a system that may be located outside the vehicle.
- Estimating, by a processing circuitry of the vehicle, a future movement of the entity, based on the compressed representation of the entity movement function.
- Generating a profile of the entity, by a processing circuitry of the vehicle, based on the compressed representation of the entity movement function.
- Predicting an effect of a future movement of the entity on a future movement of the vehicle, wherein the predicting may be executed by a processing circuitry of the vehicle, and may be based on the compressed representation of the entity movement function.
- searching for a certain movement pattern within the movement of the entity, by a processing circuitry of the vehicle, based on the compressed representation of the entity movement function.

Method **2600** may include receiving a compressed representation of another entity movement function, the other entity movement function may be generated by another vehicle and may be indicative of the movement of the entity during at least a subperiod of the tracking period. S**2610**.

Method **2600** may include amending the compressed representation of the entity movement function based on the compressed representation of the other entity movement function. S**2612**.

Method **2600** may include determining a duration of the tracking period. S**2603**. The duration may be determine based on the certainty level of the prediction, based on memory and/or computational resources allocated to the tracking, and the like.

FIG. **29** illustrates examples of entity movement functions. One axis is time. Other axes are spatial axes (x-axis, y-axis and the like).

FIG. **30** illustrates method **2630**.

Method **2630** may include steps **2631**, **2632**, **2634** and **2636**.

Method **2630** may include calculating or receiving an entity movement function that represents a movement of the entity during a tracking period. S**2632**.

Searching, by a search engine, for a matching reference entity movement function. S**2634**

Identifying the entity using reference identification information that identifies a reference entity that exhibits the matching reference entity movement function. S**2636**.

The reference identification information may be a signature of the entity.

The method may include acquiring a sequence of images by an image sensor; and calculating the entity movement based on the sequence of images. S**2631**

FIG. **31** illustrates a method **2640**.

Method **2630** may include steps **2642**, **2644**, **2646** and **2648**.

Method **2640** may start by calculating or receiving multiple entity movement functions that represent movements of multiple entities. S**2642**.

Clustering the multiple entity movement functions to clusters. S**2644**.

For each cluster, searching, by a search engine, for a matching type of reference entity movement functions. S**2646**.

Identifying, for each cluster, a type of entity, using reference identification information that identifies a type of reference entities that exhibits the matching type of reference entity movement functions. S**2648**.

FIG. **32** illustrates a method **2650**.

Method **2650** may include steps **2652**, **2654** and **2656**.

Method **2650** may start by calculating or receiving (a) an entity movement function that represents a movement of an entity, and (b) a visual signature of the entity. S**2652**.

Comparing the entity movement function and the visual signature to reference entity movement functions and reference visual signatures of multiple reference objects to provide comparison results. S**2654**.

Classifying the object as one of the reference objects, based on the comparison results. S**2656**.

FIG. **33** illustrates a method **2660**.

Method **2660** may include steps **2662**, **2664**, **2666** and **2668**.

Step **2662** may include calculating or receiving an entity movement function that represents a movement of an entity.

Step **2664** may include comparing the entity movement function to reference entity movement functions to provide comparison results.

Step **2666** may include classifying the object as a selected reference object of the reference objects, based on the comparison results.

Step **2668** may include verifying the classifying of the object as the selected reference object by comparing a visual signature of the object to a reference visual signature of the reference object.

FIG. **34** illustrates a method **2670**.

Method **2670** may include steps **2672**, **2674**, **2676** and **2678**.

Step **2672** may include calculating or receiving a visual signature of the object.

Step **2674** may include comparing a visual signature of the object to reference visual signatures of multiple reference objects to provide comparison results.

Step **2644** may include classifying the object as a selected reference object of the reference objects, based on the comparison results.

Step **2678** may include verifying the classifying of the object as the selected reference object by comparing an entity movement function that represents a movement of the entity to a reference entity movement functions to provide comparison results.

FIG. **35** illustrates a method **2680**.

Method **2680** may include steps **2682**, **2684** and **2686**.

Method **2680** is for generating a signature of an object.

Step **2682** may include calculating or receiving a visual signature of the object.

Step **2684** may include calculating or receiving an entity movement function that represents a movement of the object.

Step **2686** may include generating a spatial-temporal signature of the object that represents the visual signature and the entity movement function of the object.

FIG. **36** illustrates a method **2700**.

Method **2700** may include steps **2702**, **2704** and **2706**.

Method **2700** is for driving a first vehicle based on information received from a second vehicle.

The exchange of information—especially compact and robust signatures between vehicles may improve the driving of the vehicle and at a modest cost—especially when the vehicles exchange compact signatures between themselves.

Step **2702** may include receiving, by the first vehicle, acquired image information regarding (a) a signature of an acquired image that was acquired by the second vehicle, (b) a location of acquisition of the acquired image.

Step **2704** may include extracting, from the acquired image information, information about objects within the acquired image.

Step **2706** may include preforming a driving related operation of the first vehicle based on the information about objects within the acquired image.

The acquired image information regarding the robust signature of the acquired image may be the robust signature of the acquired image.

The acquired image information regarding the robust signature of the acquired image may be a cortex representation of the signature.

The method may include acquiring first vehicle images by the first vehicle; extracting, from the first vehicle acquired images, information about objects within the first vehicle acquired images; and preforming the driving related operation of the first vehicle based on the information about objects within the acquired image and based on the information about objects within the first vehicle acquired images.

The image information may represent data regarding neurons of a neural network, of the second vehicle, that fired when the neural network was fed with the acquired image.

The extracting, of the information about objects within the acquired image may include (a) comparing the signature of the acquired image to concept signatures to provide comparison results; each concept signature represents a type of objects; and (b) determining types of objects that may be included in the acquired image based on the comparison results.

FIG. **37** illustrates one vehicle that updates another vehicle—especially with objects that are not currently seen by the other vehicle—thereby allowing the other vehicle to calculate his driving path based (also) on the object it currently does not see.

Concept Update

A concept structure includes multiple signatures that are related to each other and metadata related to these signatures.

The concepts may be updated by removing signatures—and even parts of the signatures that are more "costly" to keep than maintain.

The "cost" may represent one or more factors such as false detection probability, a robustness of the concept, an accuracy of the concept, and the like.

One or more vehicles may decide to update a concept—and then send the update (or an indication about the update) to other vehicles—therefore improving the detection process by using updated concepts.

FIG. **38** illustrates a method **2710**.

Method **2710** may include steps **2712**, **2714** and **2726**.

Method **2710** is for concept update.

Method **2710** may include:

Step **2712** of detecting that a certain signature of an object causes a false detection (see, for example FIGS. **39** and **40**). The certain signature belongs to a certain concept structure that may include multiple signatures. The false detection may include determining that the object may be represented by the certain concept structure while the object may be of a certain type that may be not related to the certain concept structure. For example—a concept of a pedestrian may classify (by error) a mail box as a pedestrian.

Step **2714** of searching for an error inducing part (see, for example FIGS. **41-43**) of the certain signature that induced the false detection.

Step **2715** of determining whether to remove the error inducing part of the certain signature. This may involve calculating a cost related to a removing the error inducing part from the concept structure—and removing the error inducing part when the cost may be within a predefined range.

Step **2726** of removing (see, for example FIG. **44**) from the concept structure the error inducing part to provide an updated concept structure.

Each signature may represent a map of firing neurons of a neural network.

Step **2714** may include:

Generating or receiving a test concept structure that includes (a) first signatures of images that may include one or more objects of the certain type (that should not belong to the concept), and (b) second signatures of second images that may include one or more objects of a given type that may be properly associated with the concept structure. The images of both types may be selected in any manner—for example may be randomly selected.

Comparing the certain signature to the test concept structure to provide matching first signatures and matching second signatures.

Comparing the matching first signatures and matching second signatures to find parts that causes false errors and parts that result in positive detection.

Defining the error inducing part of the certain signature based on an overlap between the matching parts of the certain signature.

The updated concept may be shared between vehicles.

Situation Aware Processing

Reference information such as but not limited to signatures, cluster structures and the like may be related to a vast number of situations. It has been found that situation aware processing may be much more effective than processing that is not situation aware.

Situation aware processing includes various steps, some of which may include detecting a situation and then performing a situation related processing based on the situation. The situation related processing may include, for example, object detection, object behavior estimation, responding to an outcome of the object detection, responding to an outcome of an object behavior estimation, and the like.

Various examples of responses to object detection and/or responding to an outcome of an object behavior estimation are provided in FIGS. **3A-44**. For example—the responding may include at least one out of (i) performing an obstacle avoidance movement in response to the outcome of the object detection and/or the object behavior estimation, (ii) autonomously driving an autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, (iii) assisting a driver in response to the outcome of the object detection and/or the object behavior estimation, (iv) determining which entity (human or non-human) will control the autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, (v) communicate with other vehicles in response to the outcome of the object detection and/or the object behavior estimation, and the like.

Various examples of a situation may be at least one of (a) a location of the vehicle, (b) one or more weather conditions, (c) one or more contextual parameters, (d) a road condition, (e) a traffic parameter.

Various examples of a road condition may include the roughness of the road, the maintenance level of the road, presence of potholes or other related road obstacles, whether the road is slippery, covered with snow or other particles.

Various examples of a traffic parameter and the one or more contextual parameters may include time (hour, day, period or year, certain hours at certain days, and the like), a traffic load, a distribution of vehicles on the road, the behavior of one or more vehicles (aggressive, calm, predictable, unpredictable, and the like), the presence of pedestrians near the road, the presence of pedestrians near the vehicle, the presence of pedestrians away from the vehicle, the behavior of the pedestrians (aggressive, calm, predictable, unpredictable, and the like), risk associated with driving within a vicinity of the vehicle, complexity associated with driving within of the vehicle, the presence (near the vehicle) of at least one out of a kindergarten, a school, a gathering of people, and the like. A contextual parameter may be related to the context of the sensed information—context may be depending on or relating to the circumstances that form the setting for an event, statement, or idea.

FIG. **45** illustrates an example of method **3000**.

Method **3000** may start by step **3009** of sensing or receiving first sensed information. The first sensed information may be any type of information (visual, radar, sonar, non-visual, and the like).

The first sensed information may be sensed by at least one first sensor of a vehicle.

The first sensed information may be sensed by a sensor that does not belong to a vehicle.

Step **3009** may be followed by step **3014** of detecting a situation based on first sensed information.

Step **3014** may be followed by step **3018** of selecting, from reference information, a situation related subset of the reference information. The situation related subset of the reference information is related to the situation.

Step **3014** of detection of the situation may include object detection and step **3018** of the selection may be responsive to the detected objects. Alternatively, step **3014** may not include object detection and/or step **3018** may not be based on a detection or lack of detection of a certain object. For example—if a vehicle is at a certain inclination than step **2018** may include selecting only a part of sensed information based on the inclination (for example if the front vehicle is in a downwards inclination than only a certain lower part of an image sensor of the vehicle may be processed—assuming that an upper part may include irrelevant information—such a clouds or the sky.

For example—step **3018** may include limiting, without recognizing objects, a processing at certain areas of the picture. (For example—when the vehicle is estimated to be ahead of a T-junction, an only relevant area to process is located at the lower part (for example lower half—or any other part—depending at least in part on the inclination and the field of view of the sensors acquiring the image). The selection may not require identify any objects specifically. The limitation of the processing area is based on situation identification.

The selection may be responsive to statistical information, probabilistic information or any other information related to the chances of an appearance of a situation significant information in different parts of the sensed information—for example the appearance of a pedestrian at different locations near a roundabout, within a roundabout, and the like.

The statistical information, probabilistic information or any other information related to the chances of an appearance of a situation significant information in different parts of the sensed information may be used to adjust, fine tune or otherwise adapt the situation related processing. For example—changing detection rules, changing similarity tests, altering signatures match criteria, changing threshold of similarity for inclusion of a signature within a concept, and the like

The situation related subset of the object related reference information is a subset of reference information required for performing a situation related processing operation that may be relevant to the situation. The reference information may be, for example, information that may be compared to sensed information in order to detect objects, detect movement patterns, and the like. The comparing is merely one example of how the reference information can be used.

Various examples of a situation related subset of the reference information may be related to at least one out of (a) a location of the vehicle, (b) one or more weather conditions, (c) one or more contextual parameters, (d) a road condition, (e) a traffic parameter, (f) a potential risk, and the like.

For example—the situation related subset may include at least one out of (a) signatures and metadata of a cluster structure of roundabouts, (b) signatures and metadata of a cluster structure of a cross road located at a certain city, (c) signatures and metadata of a cluster structure of a highway acquired at sunset, (d) signatures and metadata of a cluster structure of a specific road segment, (e) signatures and metadata of a cluster structure of a specific road segment taken at a certain weather condition, (f) signatures and metadata of a cluster structure of certain roundabout acquired at rush hour, and the like.

The reference information may be stored in a memory unit of the vehicle and the selecting may include informing a processor the locations of the situation related subset, allocating a higher cache or other data retrieval priority to the subset, and the like.

At least a part of the subset may be stored outside the vehicle and the selecting may include retrieving the at least part of the subset to the vehicle.

The retrieval of the situation related subset may include retrieving it to a cache or other memory unit (for example a high priority memory unit) allocated to the processor or included in the processor.

The retrieval may include ignoring reference information not related to the situation—reference information that differs from the situation related subset.

The retrieval of only the situation related subset reduces memory resource consumption—as the situation related subset and not the entire reference information is loaded to the vehicle and/or to the processor and/or to the cache and/to a high priority memory unit.

Method **3000** may also include step **3020** of sensing second sensed information.

The second sensed information is sensed at a second period.

The second period may follow the detecting of the situation, may precede the detection of the situation, may partially overlap (or fully overlap) the detecting of the situation, may follow the detecting of the situation, may precede the detecting of the situation, may partially overlap (or fully overlap) the detecting of the situation.

The first sensed information may be included in the second sensed information, may partially overlap the second sensed information, may not overlap the second sensed information, may differ by type from the first sensed information, and the like.

For example an image that is processed to detect a situation may also be used for image processing, may be a part of images that are processed to detect movement of objects, and the like.

The second sensed information may be sensed by at least one sensor of a vehicle.

The second sensed information may be sensed by a sensor that does not belong to a vehicle.

The first and second sensed information may be sensed by the same sensor or sensors, may be sensed by completely different sensors, may be sensed in part by a same sensor.

Steps **3020** and **3018** may be followed by step **3024** of performing, by a situation related processing unit, a situation related processing. The situation related processing is based on the situation related subset of the reference information and on the second sensed information sensed by the at least one vehicle sensor at a second period.

The situation related processing may include object detection and/or object behavior estimation.

Step **3024** may be executed by any of the method illustrated in the specification.

For example—step **3024** may include calculating one or more second signatures of the second sensed information; and searching, out of the concept structures related to the situation, for at least one matching concept structure that matches the one or more second signatures, and determining at least an identity of a detected object based on metadata of the at least one matching concept structure.

Step **3024** may include, for example, object detection, object behavior estimation, and the like. Step **3024** may include, for example applying any type of machine learning algorithms, not necessarily signatures. e.g. Deep Learning where models are loaded, and input is a direct "pixel information" of the sensor. Step **3024** may be executed without signature calculation.

Step **3014** of detecting the situation and step **3024** may be executed at the same manner—using the same algorithms, applying a same type of operations, exhibit a same or similar levels of complexity, exhibit a same or similar consumption of memory resources, exhibit a same or similar consumption of processing resources, and the like.

Alternatively—step **3014** of detecting the situation and step **3024** may differ from each other by at least one out of manner of execution—for example may differ from each other by at least one out of—algorithms, type of operations, levels of complexity, consumption of memory resources, consumption of processing resources, and the like.

Step **3024** may be executed by searching matching concept structures and step **3014** may be executed without searching for matching concept structures.

Step **3014** may be executed calculating signatures of a type used in the object related processing.

Step **3024** may be followed by step **3030** of responding to the outcome of the object related processing.

Various examples of a responses to object detection and/or responding to an outcome of an object behavior estimation are provided in FIGS. **3A-44**. For example—the responding may include at least one out of (i) performing an obstacle avoidance movement in response to the outcome of the object detection and/or the object behavior estimation, (ii) autonomously driving an autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, (iii) assisting a driver in response to the outcome of the object detection and/or the object behavior estimation, (iv) determining which entity (human or non-human) will control the autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, the determining may be responsive to at least one autonomous driving rule, (v) communicate with other vehicles in response to the outcome of the object detection and/or the object behavior estimation, and the like.

Method **3000** may be executed by different vehicles that cooperate with each other—the cooperation may be made in a distributed manner, in a hierarchical or non-hierarchical manner. For example—one vehicle may process first sensed information from another vehicle, two or more vehicle may share resources to execute step **3024**, and the like.

FIG. **46** illustrates an example of method **3001**.

Method **3001** of FIG. **46** differs from method **3000** of FIG. **45** by having step **3010** instead of step **3009**. Step **3010** includes sensing the first sensed information by at least one vehicle sensor during the first period. The at least one vehicle sensor belongs to the same vehicle that includes the one or more second sensors that sensed the second sensed information.

Step **3010** is followed by step **3014**. Step **3014** is followed by step **3018**. Steps **3020** and **3014** are followed by step **3018**. Step **3018** is followed by step **3024**. Step **3024** is followed by step **3030**.

The at least one first vehicle sensor may differ from the one or more second sensors. The at least one first vehicle sensor may be the one or more second sensors. There may be a partial overlap between the at least one first vehicle sensor and the one or more second sensors.

FIG. **47** includes three timing diagrams **3071**, **3072** and **3073** that illustrates various timing relationships between the first period **3061**, situation detection **3062** and the second period **3062**.

Timing diagram **3071** illustrates the situation detection **3062** as following the first period **3061** and preceding the second period **3062**—without timing overlap.

Timing diagram **3071** illustrates the situation detection **3062** as following the first period **3061**. The second period **3062** partially overlaps the first period **3061** and overlaps the entire situation detection **3062**.

Timing diagram **3073** illustrates the situation detection **3062** as following the first period **3061**. The second period **3062** and the first period **3061** start together and the second period **3061** precedes the situation detection **3062**.

FIG. **48** illustrates an example of method **3003**.

Method **3003** of FIG. **48** differs from method **3000** of FIG. **45** by having step **3041** instead of step **3009**, and having step **3050** instead of step **3020**.

Step **3041** includes sensing first sensed information that is first visual information by a camera during a first period.

Step **3050** includes sensing second sensed information that is second visual information by the camera.

Step **3041** is followed by step **3014**. Step **3014** is followed by step **3018**. Steps **3050** and **3014** are followed by step **3018**. Step **3018** is followed by step **3024**. Step **3024** is followed by step **3030**.

FIG. **49** illustrates an example of method **3003**.

Method **3003** of FIG. **49** differs from method **3000** of FIG. **45** by having step **3042** instead of step **3009**, and having step **3050** instead of step **3020**.

Step **3042** includes sensing non-visual information that is first sensed information by a non-visual sensor that during a first period.

Step **3051** includes sensing second sensed information that is second visual information by the camera.

Step **3042** is followed by step **3014**. Step **3014** is followed by step **3018**. Steps **3050** and **3014** are followed by step **3018**. Step **3018** is followed by step **3024**. Step **3024** is followed by step **3030**.

FIG. **50** illustrates an example of method **3004**.

Method **3003** of FIG. **49** differs from method **3000** of FIG. **45** by having step **3049** of uploading the situation related subset of the reference information to a situation related processing unit.

Step **3010** is followed by step **3014**. Step **3014** is followed by step **3018**. Step **3018** is followed by step **3019**. Steps **3019** and **3020** are followed by step **3018**. Step **3018** is followed by step **3024**. Step **3024** is followed by step **3030**.

Computational resources and/or memory resources can further be saved when the second sensed information is segmented to portions and the processing of the portions is based on their relevancy to the situation.

Thus—less resources are allocated to the processing of irrelevant second sensed information segments, and such irrelevant portions may be ignored of.

There may be more than two relevancy classes and the allocation of the resources may be based on the relevancy level.

The relevancy may also be determined based on the situation and at least one additional parameters—such as the type of the object that is being detected and/or the type of object that its behavior is being monitored.

Any of the mentioned below methods may include determining, based on the situation, a relevancy value for each portion out of multiple portions of the second sensed information.

The determining may be followed by performing the situation related processing based on the relevancy value of the each portion.

The determining of the relevancy value may include finding, for a certain type of object, a relevant portion and an irrelevant portion. The performing of the situation related processing is executed without searching for the certain type of object within the irrelevant portion.

The determining of relevancy value may be followed by determining a resolution of signatures or concept structures based on the relevancy value of the each portion.

The preforming of the situation related processing within the irrelevant portion may be executed without using concept structures that describe the object of the certain type.

FIG. **51** illustrates an example of method **3005**.

Method **3005** may start by step **3009** of sensing or receiving first sensed information. The first sensed information may be any type of information (visual, radar, sonar, non-visual, and the like).

Step **3009** may be followed by step **3014** of detecting a situation based on first sensed information.

Step **3014** may be followed by step **3018** of selecting, from reference information, a situation related subset of the reference information. The situation related subset of the reference information is related to the situation.

Method **3000** may also include step **3020** of sensing second sensed information by the at least one vehicle sensor at a second period.

Step **3030** and step **3009** may be followed by step **3051** of determining, based on the situation, a relevancy value for each portion out of multiple portions of the second sensed information.

Steps **3051** and **3018** may be followed by step **3052** of performing, by a situation related processing unit, a situation related processing to find detected objects, wherein the situation related processing is based on the situation related subset of the reference information and on the second sensed information sensed by the at least one vehicle sensor at a second period. The processing is based on the relevancy value of each portion.

Step **3052** may be executed by any of the method illustrated in the specification—but is based on the relevancy value of each portion.

For example—step **3052** may include calculating one or more second signatures of the second sensed information; and searching, out of the concept structures related to the situation, for at least one matching concept structure that matches the one or more second signatures, and determining at least an identity of a detected object based on metadata of the at least one matching concept structure.

The length of the signatures may be based on the priority of each portion. For example—shorter or lower resolution signatures may be allocated to lower priority portions. The size of the concepts may be based on the priority of each portion.

Step **3052** may include, for example, object detection, object behavior estimation, and the like.

Step **3014** of detecting the situation and step **3052** may be executed at the same manner—using the same algorithms, applying a same type of operations, exhibit a same or similar levels of complexity, exhibit a same or similar consumption of memory resources, exhibit a same or similar consumption of processing resources, and the like.

Step **3024** may be followed by step **3052** of responding to the outcome of the object related processing.

Various examples of a responses to object detection and/or responding to an outcome of an object behavior estimation are provided in FIGS. **3A-44**. For example—the responding may include at least one out of (i) performing an obstacle avoidance movement in response to the outcome of the object detection and/or the object behavior estimation, (ii) autonomously driving an autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, (iii) assisting a driver in response to the outcome of the object detection and/or the object behavior estimation, (iv) determining which entity (human or non-human) will control the autonomous vehicle in response to the outcome of the object detection and/or the object behavior estimation, the determining may be responsive to at least one autonomous driving rule, (v) communicate with other vehicles in response to the outcome of the object detection and/or the object behavior estimation, and the like.

Method **3005** may be executed by different vehicles that cooperate with each other—the cooperation may be made in a distributed manner, in a hierarchical or non-hierarchical manner. For example—one vehicle may process first sensed information from another vehicle, two or more vehicle may share resources to execute step **3024**, and the like.

FIG. **52** illustrates an example of method **3006**.

Method **3006** differs from method **51** of FIG. **51** by determining the priority per a combination of second sensed information portions and object types. One image portion may be relevant to one object type and irrelevant for another object type. One image portion may be relevant to multiple object types and irrelevant to one or more object types.

The relevancy of portion to a type of object may be determined in any manner-supervised or unsupervised.

FIG. **53** illustrates an image **3080** of a roundabout. Image **3080** includes include a curved road **3072** within the roundabout, an ring shaped zone **3073** that surrounds the inner circle **3074** of the roundabout, sand **3078** and few bushes or plants **3075** located in the inner circle **3074**, various trees **3076** that surround the roundabout, a pavement **3075** that surrounds the roundabout, and a vehicle **3071** that is about to enter the roundabout.

FIG. **54** illustrates relevant and irrelevant regions within image **3080**.

Assuming that vehicles are not expected to pass through the inner circle **3074** of the roundabout, sand **3078** and few bushes or plants **3075** located in the inner circle **3074**—the inner circle **3074** and the elements located within the inner circle may be deemed irrelevant for searching for vehicles—as indicated by region **3082**.

Assuming that pedestrians and vehicle do not fly—the sky part of the image may be regarded as irrelevant for pedestrians and vehicle—as can be noted in region **3081**.

Accordingly—the situation related processing may not search for vehicles in regions **3082** and **3081**—and may not retrieve reference concept structures related to vehicles, may not search for matching concept structures from reference concept data structures associated with vehicles, may use, when searching these regions, vehicle signatures of less resolution.

The same applies for searching pedestrians in region **3081**.

FIG. **55** illustrates an example of an image acquired by a vehicle. The image shows two lanes **1411** and **1412** of a road, right building **1413**, vehicles **1414** and **1415**, first tree **1416**, table **1417**, second tree **1418**, cloud **1419**, rain **1420**, left building **1421**, shrubbery **1422**, child **1423** and ball **1424**.

It is assumed that vehicles should not be found outside lanes **1411** and **1412**—and thus a region of irrelevancy (for vehicles) **3087** is defined outside the lanes.

FIG. **56** illustrates two signatures of different resolutions—one signature (**6027**') including five indexes and a second signature (**6027**") include less indexes.

The number of indexes may differ from four of five. The first and second signatures may differ from each other by more than a single index.

FIG. **57** illustrates an example of various data structures and processes.

Assuming that the reference information includes multiple (M) cluster structures **4974**(**1**)-**4974**(M). Each cluster structure includes cluster signatures, metadata regarding the cluster signature

For example—first cluster structure **4974**(**1**) includes multiple (N1) signatures (referred to as cluster signatures CS) CS(**1**,**1**)-CS(**1**,N**1**) **4975**(**1**,**1**)-**4975**(**1**,N**1**), and metadata **4976**(**1**). Y Yet for another example—M'th cluster structure **4974**(M) includes multiple (N2) signatures (referred to as cluster signatures CS) CS(M,**1**)-CS(M,N**2**) **4975**(M,**1**)-**4975**(M,N**2**), and metadata **4976**(M).

It is assumed that there are three cluster structures related to a situation—first cluster structure **4974**(**1**), second cluster structure **4974**(**2**), and first cluster structure **4974**(**3**).

In this scenario only the three cluster structures may be retrieved or processed during a situation related processing.

For example—a media unit signature **4972** may be compared to the three clusters in order to find one or more matching clusters **3091**.

The object or object that represented by the media unit signature **4972** is detected **3092** based on the metadata of the matching concept structure.

FIG. **58** illustrates an example of a vehicle **3096** that include a processor **3099**, a cache **3098**, and memory unit **3097**, as well as other components (not shown). The vehicle may communicate with a computerized system **3095** located outside the vehicle.

The processor **3099** may be a processing unit, may include one or more processing units, and the like. Processor **3099** may be a situation related processing unit and/or may be a risk related processing unit. When processor **3099** performs operations related to situations (for example execute any method of methods **3000**-**3005**) then it may be regarded as a situation related processing unit. When processor **3099** performs operations related to risks (for example execute method **3100**) then it may be regarded as a risk related processing unit. A processing unit may include one or more processing circuits.

Risk Level Assessment

Reference information may be used to detect a large number of scenarios and/or a large number of objects. In such cases the reference information and the processing operation may exhibit a certain level of accuracy and/or a certain response period.

In some situations—especially in risky situations there may be a need to improve the accuracy and/or increase resolution and/or reduce the response time.

For example—detecting a pedestrian in relatively low visibility (for example dark) environment may be very problematic.

Yet for another examples—risky situations may be determined by:

Using contextual understanding of risk—

i. For example driving near a playground located near the road—children might appear.

ii. Driving in crowded places.

Going uphill—and looking for a car that might be overtaking from the other lane in front of me.

Driving inside a roundabout—putting resources in the areas of roundabout entrances looking for cars that will not yield.

4) driving in the forest area—putting resources in looking to the forest on the side roads to verify if no animal suddenly appears.

Driving at situations that caused accidents in the past.

Recognizing a sun dazzle—putting more resources into the road pixels

The accuracy may be improved is more resources are allocated for detecting such a pedestrian.

FIG. **59** is an example of a method **3100** for detecting potential risks in an efficient and accurate manner.

Method **3100** may start by step **3112** detecting, based on first sensed information sensed by at least one vehicle sensor at a first period, a suspected risk within an environment of the vehicle.

A suspected risk may be any object and/or behavior that may damage the vehicle or be damaged by the vehicle.

The detection may or may not include applying at least one of the previously mentioned methods for detecting a situation. The situation may be an occurrence of a suspected risk within an environment of the vehicle.

The detection may include finding potential risk indicators that may be found in various manners—for example—by searching for sensed information of a case that resulted in damage and processing sensed information obtained immediately before the damage occurred to find potential risk indicators.

The detection can be made at situations that are known to be risky or may be executed regardless of prior knowledge.

The first sensed information may be image information, sound information, and the like. For example—the first sensed information may be a group of pixels that may be brighter than the dark surroundings.

Step **3112** may be followed by step **3114** of selecting, from reference information, a risk related subset of the reference information, wherein the risk related subset of the reference information is related to the suspected risk.

The selection, especially if it is followed by a retrieval of such information (without retrieving irrelevant information), may save bandwidth, same memory resources, save power and may speed up the retrieval—as fewer information is retrieved.

Step **3114** may be followed by step **3116** of determining whether the suspected risk is an actual risk, based at least on part on the risk related subset of the reference information.

The risk related subset of reference information may be risks that were detected in the past. For example—the few pixels that are brighter than their dark surrounding may be a part of a face of a pedestrian of an item worn of held by the pedestrian.

Step **3116** may include object detection, searching for similarity of the sensed information to risk related subset of the reference information that represented a real risk or to risk related subset of the reference information representing an absence of such risk. For example—whether the sensed information represents a pedestrian (located within the path of the vehicle) or not.

Step **3116** may be followed by step **3118** of performing a driving based operation based on the determining **3118**. For example—altering the progress of the vehicle to avoid the risk or reduce the changes of a collision or any other risk.

The selecting of step **3114** may be followed by uploading the risk related subset of the reference information to a risk related processing unit.

The determining of step **3116** may include ignoring reference information that is not a part of the risk related subset of the reference information.

The reference information comprises concept structures, wherein each concept structure comprises multiple signatures and metadata related to the multiple signatures; wherein the risk related subset of the reference information comprises concept structures related to the risk.

The method may include calculating one or more second signatures of the second sensed information; and searching, out of the concept structures related to the risk, for at least one matching concept structure that matches the one or more second signatures.

Step **3116** may include determining at least an identity of a detected object based on metadata of the at least one matching concept structure.

Step **3112** may include detecting without or without searching for matching concept structures.

Step **3112** may or may not be executed without calculating signatures of a type used in the object related processing.

Step **3116** may include determining, based on the risk related subset, a relevancy value for each portion out of multiple portions of the second sensed information.

Step **3116** may be based on the relevancy value of the each portion.

Step **3116** may include finding, for a certain type of risk, a relevant portion and an irrelevant portion; and wherein the determining is executed without searching for the certain type of risk related object within the irrelevant portion.

All the examples in the specification are non-limiting examples. Any reference to “may be” is merely an example. May be indicates that a statement following the “may be” phrase may also may not be.

Safe transfer between manned and autonomous driving modes

The autonomous driving capabilities of autonomous vehicles are gradually increasing.

No existing autonomous vehicle may successfully drive in all conditions.

There is a need for a method that will enable autonomous vehicles to determine in a safe manner whether autonomous driving may be applicable in a certain situation.

The following situation enables an autonomous vehicle to determine by itself whether autonomous driving may be applied in a certain situation.

At least some of the steps of the following methods (and/or of any of the mentioned above methods) may be executed by a vehicle, by multiple vehicles, by one or more remote computerized systems located outside vehicles, by a combination of processing units that perform distributed processing (the processing units may be located within or outside a vehicle), and the like.

Sensed information may be sensed by one or more sensors located within one or more vehicles, one or more sensors located outside a vehicle, a combination of sensors within and outside sensors.

FIG. **60** illustrates an example of method **3120** of safe transfer between manned and autonomous driving modes.

Method **3120** may start by step **3009** of sensing or receiving first sensed information.

The first sensed information may be any type of information (visual, radar, sonar, non-visual, and the like).

The first sensed information may be sensed by at least one first sensor of a vehicle.

The first sensed information may be sensed by a sensor that does not belong to a vehicle.

Step **3009** may be followed by step **3014** of detecting a situation based on first sensed information.

The situation may be one out of (a) a location of the vehicle, (b) one or more weather conditions, (c) one or more contextual parameters, (d) a road condition, (e) a traffic parameter, (f) a potential risk, and the like.

Step **3014** may be followed by step **3126** of determining, based on the situation, whether the autonomous vehicle is capable to safely autonomously drive through the environment.

Situations can be determined to be safe based on successful testing (testing the situation to be safe for autonomous driving).

Alternatively—the situation may be deemed to be safe is at least a predefined number of similar situations were successfully tested (testing the situation to be safe for autonomous driving).

Similar situation can be defined by using any similarity test and/or similarity test rules. A test may involve actually driving through a situation, processing (for example evaluating whether autonomous driving was safe) sensed information acquired from vehicle while driving, simulation the driving through the situation, a combination of actual driving and simulation or any type of testing.

A group of scenarios may be represented by scenario signatures that represent the scenarios and belong to a certain cluster of signatures. Signatures may be generated in any manner—including the manners illustrated in the current specification.

If a predefined number or a predefined percent of the situations are tested to be safe (or otherwise declared to be safe)—the entire scenarios of the certain cluster are deemed to be safe—even if not all have been actually tested.

Thus—some (but not all) of the scenarios of the cluster may be selected (in any manner)—and are tested for safe autonomous driving—and if succeeding then all the cluster members are deemed to be safe.

It should be noted that multiple clusters that include situation may be virtually linked to each other in the sense that a group of clusters may be specific examples of a lager cluster—and that once all of the members of the group are safe—the larger cluster may be regarded safe as well.

When searching whether an untested scenario is similar to another scenario (known or assumed to be safe)—the other scenario should not be riskier that the untested scenario. One or more risk factors may include at least one out of (i) illumination (for example darker environments may be deemed to be risky that more illuminated scenarios, on the other hand driving in a scenario that is too bright (for example when the sum may blind the driver) may also be more risky that slightly more dark scenarios, (ii) traffic load (for example driving over an empty lane is less risky than driving in a heavily crowded lane), (iii) weather conditions (visibility, rain, snow, ice), and the like.

Similar scenarios may differ from each other by at least one parameter that defines the situation. For example—differ from each other by at least one out of (a) a location of the vehicle, (b) one or more weather conditions, (c) one or more contextual parameters, (d) a road condition, (e) a traffic parameter, (f) a potential risk, and the like.

Each parameter of (a)-(e) may be associated (learnt or estimated) risk factor—that may be used for determining which situation (tested situation or untested scenario).

Examples of relationships between tested situation or untested scenario may include, for example:

- Tested situation differs by being tested at a later or earlier hour (or more generally slightly different lighting conditions).
- Tested situation differs by being tested at different location (although the road segments are the same between the situations)—straight empty one lane, one directional urban road, but in a different place
- Tested situation differs by being tested at different traffic loads (although the locations are the same between the situations).
- Tested situations may differ from each other by degree of predictability of movement of road users should be taken into a—for example driving in a presence of many bikes, motorized kick scooters, and the like may be deemed to be more risky that driving at the presence of cars—due to the lower level of predictability of the bikes, motorized kick scooters.

The determining may be based at least in part of test results of autonomous driving at similar situations. If enough test results executed in the situation were successful then the situation may be deemed safe.

If safe—then allowing autonomous driving (step **3134**), if unsafe—preventing autonomous driving (step **3130**).

The similarity may be based on supervised or unsupervised learning. For example—assuming that a cluster is safe then the distance between the safe cluster and other untested clusters may be calculated and where other clusters that are within a certain distance from the safe cluster may be deemed safe. The same applies when there are multiple safe clusters—distances between one or more other clusters that are safe to untested clusters may be taken into account.

Any distance between clusters may be calculated—for example Euclidian distance between centers of clusters, between the borders of a cluster.

The similarity between clusters may be defined, for example, based on a number of similar signatures within the clusters, may be based on relationship between matching and non-matching signatures of different clusters, and the like.

Similarity between situations (for example one simulation known to be safe and the other is evaluated) may also be evaluated—for example by comparing between the similar and/or different signatures of different situations. The comparison may also take into account the location (within the sensed information) of the objects represented by the signatures.

An example of a similarity between situations: a safe situation involves driving a certain road segment at a certain time of the day without a vehicle in site. An evaluated situation includes driving at the certain road segment at the certain time of the day but at a presence of another vehicle—the difference between the safe situation and the evaluated situation can be easily be detected—as the portion of the images that include the vehicle differ from the corresponding images of the safe situation. The same similarity evaluation may be applied to the absence of presence of any other object or variable—for example the presence or absence of an obstacle.

FIG. **61** illustrates an example of method **3120'** of controlling an autonomous vehicle.

Method **3120'** may start by step **3009** of sensing or receiving first sensed information.

The first sensed information may be any type of information (visual, radar, sonar, non-visual, and the like).

The first sensed information may be sensed by at least one first sensor of a vehicle.

The first sensed information may be sensed by a sensor that does not belong to a vehicle.

Step **3009** may be followed by step **3014** of detecting a situation based on first sensed information.

The situation may be one out of (a) a location of the vehicle, (b) one or more weather conditions, (c) one or more contextual parameters, (d) a road condition, (e) a traffic parameter, (f) a potential risk, and the like.

Method **3120'** may also include step **3127** of detecting, in proximity to the first period, objects located within the environment of the vehicle.

Step **3127** may be followed by step **3128** of estimating behavior patterns of the objects located within the environment of the vehicle.

Steps **3014** and **3128** may be followed by step **3126'** of determining, based on the situation and the one or more behavior patterns, whether the autonomous vehicle is capable to safely autonomously drive through the environment.

If safe—then allowing autonomous driving (step **3134**), if unsafe—preventing autonomous driving (step **3130**).

There may be provided a method for situation aware processing, the method may include detecting a situation, based on first sensed information at a first period; selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; and performing, by a situation related processing unit, a situation related processing, wherein the situation related processing may be based on the situation related subset of the reference information and on second sensed information sensed at a second period, wherein the situation related processing may include at least one out of object detection and object behavior estimation.

There may be provided a system for situation aware processing, the system may include a situation related processing unit and a memory unit; wherein the memory unit may be configured to store first sensed information sensed at a first period; wherein the situation related processing unit may be configured to detect a situation, based on the first sensed information, select, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; and perform a situation related processing to find detected objects, wherein the situation related processing may be based on the situation related subset of the reference information and on second sensed information sensed by one or more second vehicle sensors at a second period.

There may be provided a non-transitory computer readable medium that may store instructions for detecting a situation, based on first sensed information at a first period; selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; and performing, by a situation related processing unit, a situation related processing, wherein the situation related processing may be based on the situation related subset of the reference information and on second sensed information sensed at a second period, wherein the situation related processing may include at least one out of object detection and object behavior estimation.

There may be provided a method for safe transfer between manned and autonomous driving modes, the method may include detecting, based on first sensed information sensed during a first period, a situation related to an environment of the autonomous vehicle; searching for one or more matching concepts of a group of reference concepts, to which the situation belongs, each reference concept of the group represents a plurality of situations and has a reference concept safety level; wherein for each reference concept of at least a sub-group of the group of reference concepts the safety level of the reference concept may be based on a tested success level of at only some of the plurality of scenarios represented by the reference concept; determining, based on an outcome of the searching, whether the vehicle may be capable to safely autonomously drive through the environment; performing a human assisted driving of the vehicle when determining that the autonomous vehicle may be not capable to safely autonomously drive through the environment; and performing an autonomous driving of the vehicle when determining that the autonomous vehicle may be capable to safely autonomously drive through the environment.

A system for safe transfer between manned and autonomous driving modes, the system may include a processing unit and a memory unit; wherein the memory unit may be configured to store first sensed information sensed during a first period; wherein the processing unit may be configured to: detect, based on the first sensed information, a situation related to an environment of the autonomous vehicle; search for one or more matching concepts of a group of reference concepts, to which the situation belongs, wherein each reference concept of the group represents a plurality of situations and has a reference concept safety level, for each reference concept of at least a sub-group of the group of reference concepts the safety level of the reference concept may be based on a tested success level of at only some of the plurality of scenarios represented by the reference concept; and determine, based on an outcome of the searching, whether the vehicle may be capable to safely autonomously drive through the environment.

A non-transitory computer readable medium that may store instructions for: detecting, based on first sensed information sensed during a first period, a situation related to an environment of the autonomous vehicle; searching for one or more matching concepts of a group of reference concepts, to which the situation belongs, each reference concept of the group represents a plurality of situations and has a reference concept safety level, for each reference concept of at least a sub-group of the group of reference concepts the safety level of the reference concept may be based on success levels of multiple attempts to autonomously drive one or more autonomous vehicles at only some of the plurality of scenarios represented by the reference concept; and determine, based on an outcome of the searching, whether the vehicle may be capable to safely autonomously drive through the environment

A method for risk based processing, the method may include: detecting, based on first sensed information sensed at a first period, a suspected risk within an environment of a vehicle; selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; selecting, from reference information, a suspected risk related subset reference information, wherein the potential risk related subset of the reference information may be related to the potential risk; and determining whether the suspected risk may be an actual risk, based at least on part on the suspected risk related subset reference information.

A non-transitory computer readable medium that may store instructions for: detecting, based on first sensed information sensed at a first period, a suspected risk within an environment of a vehicle; selecting, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; selecting, from reference information, a suspected risk related subset reference information, wherein the potential risk related subset of the reference information may be related to the potential risk; and determining whether the suspected risk may be an actual risk, based at least on part on the suspected risk related subset reference information.

A system for risk based processing, the system may include a processing unit and a memory unit; wherein the memory unit may be configured to store first sensed information sensed during a first period; wherein the processing unit may be configured to: detecting, based on the first sensed information, a suspected risk within an environment of a vehicle; select, from reference information, a situation related subset of the reference information, wherein the situation related subset of the reference information may be related to the situation; select, from reference information, a suspected risk related subset reference information, wherein the potential risk related subset of the reference information may be related to the potential risk; and determine whether the suspected risk may be an actual risk, based at least on part on the suspected risk related subset reference information.

Any reference in the specification to a method should be applied mutatis mutandis to a system capable of executing the method and should be applied mutatis mutandis to a non-transitory computer readable medium that may store instructions that once executed by a computer result in the execution of the method.

Any reference in the specification to a system and any other component should be applied mutatis mutandis to a method that may be executed by a system and should be applied mutatis mutandis to a non-transitory computer readable medium that may store instructions that may be executed by the system.

Any reference in the specification to a non-transitory computer readable medium should be applied mutatis mutandis to a system capable of executing the instructions stored in the non-transitory computer readable medium and should be applied mutatis mutandis to method that may be executed by a computer that reads the instructions stored in the non-transitory computer readable medium.

Any combination of any module or unit listed in any of the figures, any part of the specification and/or any claims may be provided. Especially any combination of any claimed feature may be provided.

Any reference to the term "comprising" or "having" should be interpreted also as referring to "consisting" of "essentially consisting of". For example—a method that comprises certain steps can include additional steps, can be limited to the certain steps or may include additional steps that do not materially affect the basic and novel characteristics of the method—respectively.

The invention may also be implemented in a computer program for running on a computer system, at least including code portions for performing steps of a method according to the invention when run on a programmable apparatus, such as a computer system or enabling a programmable apparatus to perform functions of a device or system according to the invention. The computer program may cause the storage system to allocate disk drives to disk drive groups.

A computer program is a list of instructions such as a particular application program and/or an operating system. The computer program may for instance include one or more of: a subroutine, a function, a procedure, an object method, an object implementation, an executable application, an applet, a servlet, a source code, an object code, a shared library/dynamic load library and/or other sequence of instructions designed for execution on a computer system.

The computer program may be stored internally on a computer program product such as non-transitory computer readable medium. All or some of the computer program may be provided on non-transitory computer readable media permanently, removably or remotely coupled to an information processing system. The non-transitory computer readable media may include, for example and without limitation, any number of the following: magnetic storage media including disk and tape storage media; optical storage media such as compact disk media (e.g., CD-ROM, CD-R, etc.) and digital video disk storage media; nonvolatile memory storage media including semiconductor-based memory units such as FLASH memory, EEPROM, EPROM, ROM; ferromagnetic digital memories; MRAM; volatile storage media including registers, buffers or caches, main memory, RAM, etc. A computer process typically includes an executing (running) program or portion of a program, current program values and state information, and the resources used by the operating system to manage the execution of the process. An operating system (OS) is the software that manages the sharing of the resources of a computer and provides programmers with an interface used to access those resources. An operating system processes system data and user input, and responds by allocating and managing tasks and internal system resources as a service to users and programs of the system. The computer system may for instance include at least one processing unit, associated memory and a number of input/output (I/O) devices. When executing the computer program, the computer system processes information according to the computer program and produces resultant output information via I/O devices.

In the foregoing specification, the invention has been described with reference to specific examples of embodiments of the invention. It will, however, be evident that various modifications and changes may be made therein without departing from the broader spirit and scope of the invention as set forth in the appended claims.

Moreover, the terms "front," "back," "top," "bottom," "over," "under" and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

Those skilled in the art will recognize that the boundaries between logic blocks are merely illustrative and that alternative embodiments may merge logic blocks or circuit elements or impose an alternate decomposition of functionality upon various logic blocks or circuit elements. Thus, it is to be understood that the architectures depicted herein are merely exemplary, and that in fact many other architectures may be implemented which achieve the same functionality.

Any arrangement of components to achieve the same functionality is effectively "associated" such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality may be seen as "associated with" each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being "operably connected," or "operably coupled," to each other to achieve the desired functionality.

Furthermore, those skilled in the art will recognize that boundaries between the above described operations merely illustrative. The multiple operations may be combined into a single operation, a single operation may be distributed in additional operations and operations may be executed at least partially overlapping in time. Moreover, alternative embodiments may include multiple instances of a particular operation, and the order of operations may be altered in various other embodiments. Also for example, in one embodiment, the illustrated examples may be implemented as circuitry located on a single integrated circuit or within a same device. Alternatively, the examples may be implemented as any number of separate integrated circuits or separate devices interconnected with each other in a suitable manner.

Also for example, the examples, or portions thereof, may implemented as soft or code representations of physical circuitry or of logical representations convertible into physical circuitry, such as in a hardware description language of any appropriate type.

Also, the invention is not limited to physical devices or units implemented in non-programmable hardware but can also be applied in programmable devices or units able to perform the desired device functions by operating in accordance with suitable program code, such as mainframes, minicomputers, servers, workstations, personal computers, notepads, personal digital assistants, electronic games, automotive and other embedded systems, cell phones and various other wireless devices, commonly denoted in this application as 'computer systems'.

However, other modifications, variations and alternatives are also possible. The specifications and drawings are, accordingly, to be regarded in an illustrative rather than in a restrictive sense.

In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. The word 'comprising' does not exclude the presence of other elements or steps then those listed in a claim. Furthermore, the terms "a" or "an," as used herein, are defined as one or more than one. Also, the use of introductory phrases such as "at least one" and "one or more" in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an." The same holds true for the use of definite articles. Unless stated otherwise, terms such as "first" and "second" are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The mere fact that certain measures are recited in mutually different claims does not indicate that a combination of these measures cannot be used to advantage.

While certain features of the invention have been illustrated and described herein, many modifications, substitutions, changes, and equivalents will now occur to those of ordinary skill in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the invention.

We claim:

1. A method for safe transfer between manned and autonomous driving modes, the method comprises:

detecting, by using a machine learning process, a situation related to an environment of an autonomous vehicle, based on first sensed information sensed during a first period using a situation signature generated by limiting, without applying a recognition process to the first sensed information, a processing of the first sensed information according to statistical probabilities of appearances of situation pertaining information within different parts of the first sensed information, such that the situation signature is generated in association with the first sensed information; wherein the limiting, without applying the recognition process to the first sensed information, is (a) also based on an inclination of the autonomous vehicle, and (b) further comprises determining whether to process only a lower part of the first sensed information, wherein the lower part of the first sensed information is defined based on the inclination of the autonomous vehicle and according to an estimation that the autonomous vehicle is located ahead of a T-junction;

searching, based on the generated situation signature, a matching concept that is related to the situation, the matching concept belongs to a group of reference concept structures associated with the first sensed information;

each reference concept structure of the group represents a plurality of situations and is associated with a reference concept structure safety level; wherein each reference concept structure comprises a reference concept structure signature, reference signatures and reference metadata related to the reference signatures;

wherein for each reference concept structure of at least a sub-group of the group of reference concept structures the reference concept structure safety level of the reference concept structure is based on a tested success level of only some of the plurality of scenarios represented by the reference concept structure;

determining, based on an outcome of the searching, whether the autonomous vehicle is capable to safely autonomously drive through the environment; and

outputting, based on the determining, a transition control instruction, wherein the transition control instruction is related to a transition of control of the autonomous vehicle from autonomous driving to human assisted driving of the autonomous vehicle when determining that the autonomous vehicle is not capable to safely autonomously drive through the environment and when the determining whether the autonomous vehicle is capable to safely autonomously drive through the environment occurred while autonomously driving the autonomous vehicle.

2. The method according to claim **1**, wherein the method comprises generating the situation signature, and searching for the one or more matching concept structures by comparing the situation signature to at least one type of signatures out of reference concept structure signatures and reference signatures of the group of reference concept structures.

3. The method according to claim **1** wherein at least some of the reference concept structures differ from each other by complexity of maneuvers required to overcome one or more situations of at least some of the reference concept structures.

4. The method according to claim **1** wherein at least some of the reference concept structures differ from each other by resolution, wherein a resolution of a reference concept structure equals a length of a reference signature that belongs to the reference concept structure, wherein the length of the reference signature that belongs to the reference concept structure is a number of indexes included in the reference signature.

5. The method according to claim **1**, further comprising detecting, in proximity to the first period, one or more objects located within the environment of the autonomous vehicle; and estimating one or more behavior patterns of the one or more objects located within the environment of the autonomous vehicle.

6. The method according to claim **1**, further comprising detecting, in proximity to the first period, one or more objects located within the environment of the autonomous vehicle, wherein the one or more objects are road users and wherein the determining is also responsive to a degree of predictability of movement of road users.

7. A system for safe transfer between manned and autonomous driving modes, the system comprises a processing unit and a memory unit;

wherein the memory unit is configured to store first sensed information sensed during a first period;

wherein the processing unit is configured to:

detect, by using machine a learning process, a situation related to an environment of an autonomous vehicle, based on first sensed information sensed during the first period using a situation signature generated by limiting, without applying a recognition process to the first sensed information, a processing of the first sensed information according to statistical probabilities of appearances of situation pertaining information within different parts of the first sensed information; wherein the limiting, without applying the recognition process to the first sensed information, is (a) also based on an inclination of the autonomous vehicle, and (b) further comprises determining whether to process only a lower part of the first sensed information, wherein the lower part of the first sensed information is defined based on the inclination of the autonomous vehicle and according to an estimation that the autonomous vehicle is located ahead of a T-junction;

search, based on the situation signature, a matching concept that is related to the situation, the matching concept belongs to a group of reference concept structures;

each reference concept structure of the group represents a plurality of situations and is associated with a reference concept structure safety level, for each reference concept structure of at least a sub-group of the group of reference concept structures the reference concept structure safety level is based on a tested success level of only some of the plurality of scenarios represented by the reference concept structure; and

determine, based on an outcome of the searching, whether the autonomous vehicle is capable to safely autonomously drive through the environment; and

outputting, based on the determining, a transition control instruction, wherein the transition control instruction is related to a transition of control of the autonomous vehicle from autonomous driving to human assisted driving of the autonomous vehicle when determining that the autonomous vehicle is not capable to safely autonomously drive through the environment and when the determining of whether the autonomous vehicle is capable to safely autonomously drive through the environment occurred while autonomously driving the autonomous vehicle.

8. The system according to claim 7, wherein the processing unit is configured to generate the situation signature, and search for the one or more matching concept structures by comparing the situation signature to at least one type signatures of reference concept structure signatures and reference signatures of the group of reference concept structures.

9. The system according to claim 7 wherein at least some of the reference concept structures differ from each other by complexity of maneuvers required to overcome one or more situations of at least some of the reference concept structures.

10. The system according to claim 7 wherein at least some of the reference concept structures differ from each other by resolution, wherein a resolution of a reference concept structure equals a length of a reference signature that belongs to the reference concept structure, wherein the length of the reference signature that belongs to the reference concept structure is a number of indexes included in the reference signature.

11. The system according to claim 7 wherein the processing unit is configured to detect, in proximity to the first period, one or more objects located within the environment of the autonomous vehicle.

12. The system according to claim 11 wherein the processing unit is configured to estimate one or more behavior patterns of the one or more objects located within the environment of the autonomous vehicle.

13. The system according to claim 12 wherein the one or more objects are one or more road users and wherein the determining comprises estimating one or more behavior patterns of the one or more road users.

14. A non-transitory computer readable medium that stores instructions for:

detecting, by using a machine learning process, a situation related to an environment of an autonomous vehicle, based on first sensed information sensed during a first period using a situation signature generated by limiting, without applying a recognition process to the first sensed information, a processing of the first sensed information according to statistical probabilities of appearances of situation pertaining information within different parts of the first sensed information; wherein the limiting, without applying the recognition process to the first sensed information, is (a) also based on an inclination of the autonomous vehicle, and (b) further comprises determining whether to process only a lower part of the first sensed information, wherein the lower part of the first sensed information is defined based on the inclination of the autonomous vehicle and according to an estimation that the autonomous vehicle is located ahead of a T-junction;

searching, based on the situation signature, a matching concept that is related to the situation, the matching concept belongs to a group of reference concept structures;

each reference concept structure of the group represents a plurality of situations and is associated with a reference concept structure safety level, for each reference concept structure of at least a sub-group of the group of reference concept structures the reference concept structure safety level is based on success levels of multiple attempts to autonomously drive one or more autonomous vehicles at only some of the plurality of scenarios represented by the reference concept structure; wherein each reference concept structure comprises a reference concept structure signature, reference signatures and reference metadata related to the reference signatures;

determining, based on an outcome of the searching, whether the autonomous vehicle is capable to safely autonomously drive through the environment; and

outputting, based on the determining, a transition control instruction, wherein the transition control instruction is related to a transition of control of the autonomous vehicle from autonomous driving to human assisted driving of the autonomous vehicle when determining that the autonomous vehicle is not capable to safely autonomously drive through the environment and when the determining whether the autonomous vehicle is capable to safely autonomously drive through the environment occurred while autonomously driving the autonomous vehicle.

15. The non-transitory computer readable medium according to claim **14**, wherein the non-transitory computer readable medium further stores instructions for generating the situation signature; and searching for the one or more matching concept structures by comparing the situation signature to at least one type of signatures of reference concept structure signatures and reference signatures of the group of reference concept structures.

16. The non-transitory computer readable medium according to claim **14** wherein at least some of the reference concept structures differ from each other by complexity of maneuvers required to overcome one or more situations of at least some of the reference concept structures.

17. The non-transitory computer readable medium according to claim **14** wherein at least some of the reference concept structures differ from each other by resolution, wherein a resolution of a reference concept structure equals a length of a reference signature that belongs to the reference concept structure, wherein the length of the reference signature that belongs to the reference concept structure is a number of indexes included in the reference signature.

18. The non-transitory computer readable medium according to claim **14**, further storing instructions for detecting, in proximity to the first period, one or more objects located within the environment of the autonomous vehicle.

19. The non-transitory computer readable medium according to claim **18** wherein the one or more objects are one or more road users and wherein the non-transitory computer readable medium stores instructions for estimating one or more behavior patterns of the one or more road users.

20. The non-transitory computer readable medium according to claim **14** wherein the situation comprises a weather condition and a potential risk, wherein the potential risk is based on one or more risk factors that comprises illumination of the environment and traffic load.

21. The non-transitory computer readable medium according to claim **14**, wherein the autonomous driving further comprises driving the autonomous vehicle through the environment.

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